LUPINE SYSTEMS

COMPLETE EIGHT-PART EDITION

ENERGY INFRASTRUCTURE

The visual engineering handbook

First principles. Real measurements.
Reliable control. A company built on evidence.



Conceptual artwork retained from the supplied cover; not an actual Tasmanian site or engineered layout.

8 CONNECTED PARTS

292 STUDY PAGES

WORKED ANSWERS

A first-principles learning and prototype reference for Chris and Lupine Systems.
Hobart, Tasmania / October 2026

Learn the system.
Build your own evidence.

A personal reference for developing technical depth in energy infrastructure, AI data centres and industrial control - and for discovering what Lupine Systems should build.

Your direction

Build a highly technical life and career in Tasmania’s energy and AI-infrastructure sector, with the long-term aim of creating an energy technology company. Begin with electrical, thermal and control fundamentals, then use small, bounded experiments to discover specific engineering problems.

How to use the eight parts

Read in order on the first pass. Follow each figure, predict the worked result, and answer the checks before reading the solution. Use later volumes to connect the physics to Tasmania and to a Lupine prototype. Return to the relevant concept whenever an experiment exposes a gap.

YOUR LEARNING RULE

Make your own prediction, sketch or calculation before asking AI. Run the experiment yourself and explain the result in your own words. Keep the mistake and its correction; that record becomes knowledge you own.

YOUR EVIDENCE RULE

Keep assumed, simulated, published and measured results distinct. A command is not a physical outcome. A hash is not proof of truth. Preserve uncertainty, raw observations and the limits of every verification claim.

Scope: an illustrated foundation and prototype reference, not a complete engineering qualification or a device installation manual. Exact equipment documentation and appropriate engineering review are required before real infrastructure work. Project-status claims belong to their cited publication dates.

Original AI-assisted text and teaching diagrams; no independent professional technical sign-off or field validation is claimed.

Eight parts. One technical direction.

Electricity → components → power systems → heat → storage → reliable control → Tasmania → your own prototype.

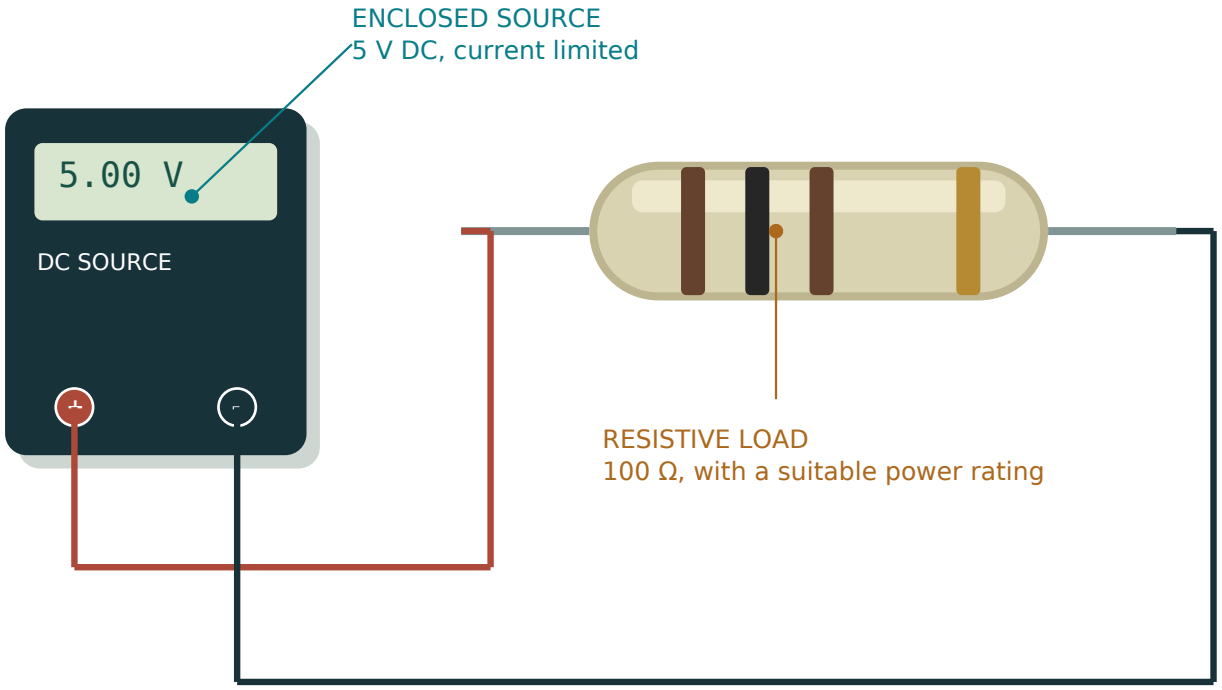
Part	Subject	Pages	Start*
01	Electricity You Can Measure	24	4
02	Components, Sensors and Microcontrollers	32	28
03	AC Power, Machines, Networks and Protection	36	60
04	Heat, Fluid Flow and Cooling	40	96
05	Batteries, Inverters and Power Electronics	40	136
06	Reliable Control, Communications and Evidence	40	176
07	Tasmania’s Grid and AI Data Centres	40	216
08	Lupine Prototype and Field Validation	40	256

*PDF-viewer start pages assume these three opening pages are followed by Parts 01-08 in order, with every original part cover retained. Total: 295 pages (292 study pages + 3 opening pages). The printed 01-01 to 08-40 labels remain the stable part references.

ELECTRICITY YOU CAN MEASURE

A complete first-principles lesson built around one small DC load.

FROM A PHYSICAL CIRCUIT TO A DEFENSIBLE RESULT



Illustrative equipment: the physical arrangement and ratings need review before use.

Voltage • current • resistance • power • energy
Annotated drawings, worked reasoning, a measurement method and answers you can study without another textbook.

Prepared for Chris / Lupine Systems / Hobart
24 pages. Part-specific page labels remain useful after merging.

One subject at a time

This part replaces the electricity introduction in the earlier survey. The remaining volumes are planned as separate, mergeable PDFs.

Order	Part and scope
01	Electricity you can measure / this completed volume
02	Components, sensors and the microcontroller
03	AC, electrical machines, networks and protection
04	Tasmania: hydro, frequency and grid behaviour
05	Batteries, converters, UPS and microgrids
06	Heat, fluid flow and data-centre cooling
07	Compute hardware, workloads and facility power
08	Feedback control, PLCs, SCADA and device protocols
09	Lupine: execution, reconciliation, persistence and evidence
10	Integration projects, validation and technical discovery

Read this part in four sittings

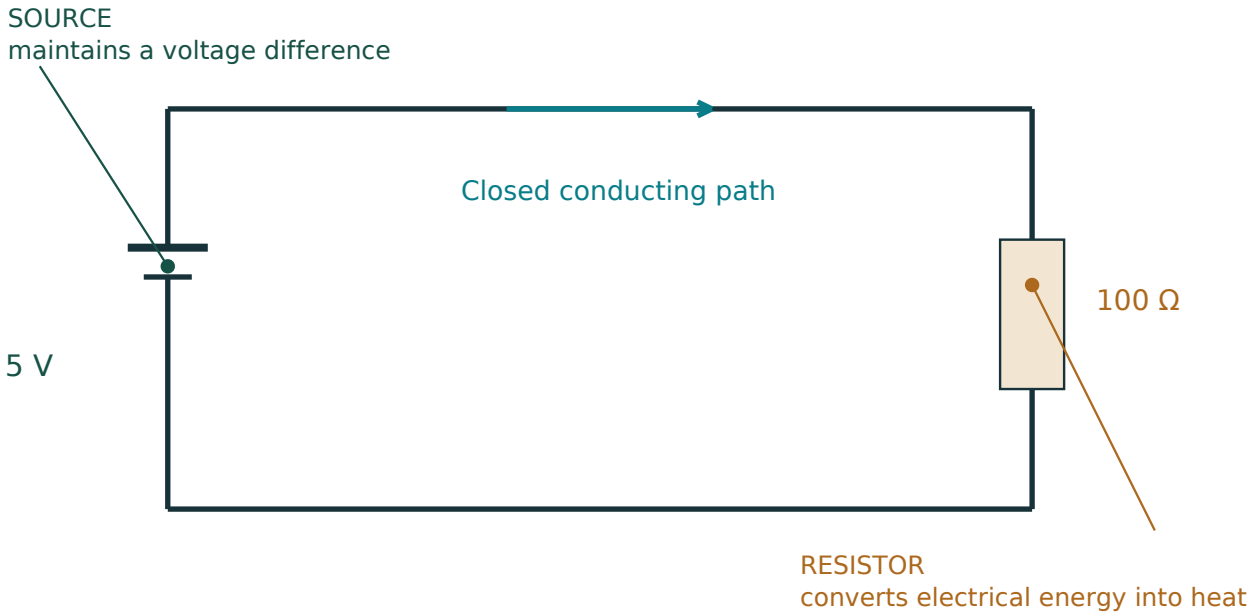
Pages 3-10: build the physical picture and learn V, I and R.
Pages 11-15: connect circuits to power and energy.
Pages 16-21: learn the measurement method and complete a test.
Pages 22-24: practise, correct your work and keep the reference.

HOW THE PARTS WILL JOIN

Files use a numeric prefix. Pages use labels such as 01-08, meaning Part 01, page 8. Merge the parts in numeric order and retain bookmarks if your tool supports them. Other parts are not yet delivered.

Meet the circuit you will understand

One source, two wires and one resistor are enough to connect the main ideas. Begin with a model; measurement comes later.



Five questions, five different quantities

Voltage: how much electrical energy is transferred per unit charge?
Current: how much charge crosses a point each second?
Resistance: how strongly does this component limit current for a given voltage?
Power: how quickly is energy transferred?
Energy: how much is transferred during the chosen time?

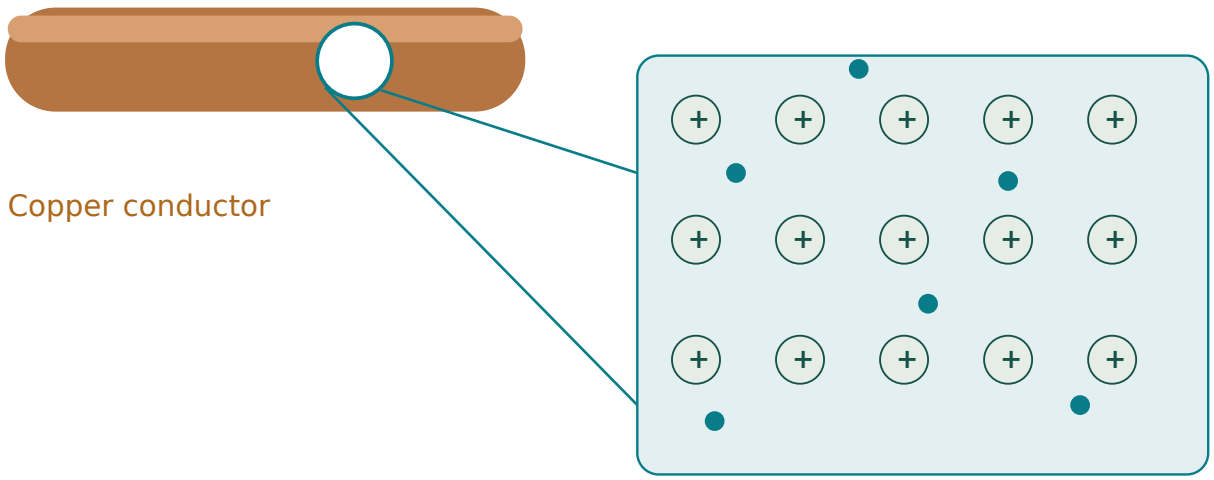
OUR MODEL

Assume steady 5.00 V across an ideal 100 Ω resistor, negligible wire resistance and no start-up transient. These are teaching assumptions, not measurements. We will predict 0.050 A, 0.250 W and 0.250 Wh in one hour.

The same accounting later applies to a rack, converter or hydro station, but those systems require more detailed models and different instruments.

Charge is already in the wire

The source does not need to send a new batch of electrons along an empty pipe before the far end can respond.



Simplified metal model, not an atomic-scale picture.

Moving charge and a spreading electrical disturbance

Metals contain mobile electrons as well as a lattice of positive ion cores. Closing a circuit changes the electric field through the connected system. The field drives a small average drift of mobile charge. The electrical response is not limited by the slow drift of one electron travelling around the whole circuit.

Why current is not “used up”

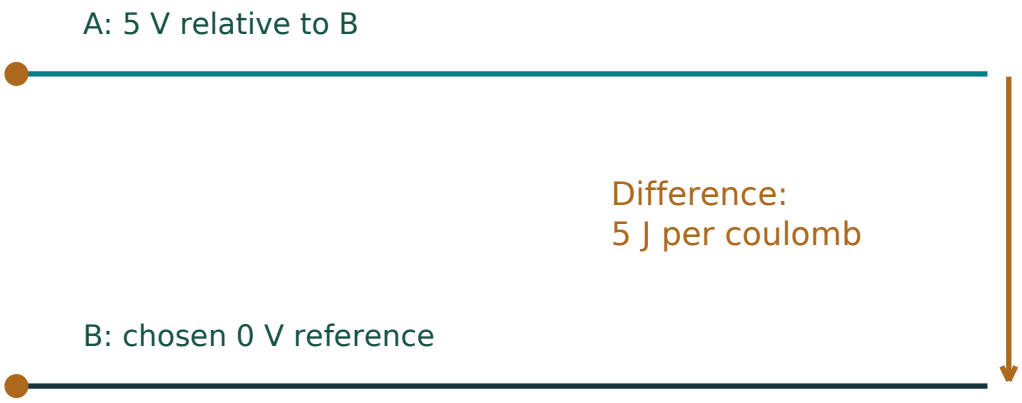
In a steady one-path loop, charge entering the resistor is balanced by charge leaving it. Energy is transferred to the resistor, while charge continues around the circuit. If your explanation says the resistor consumes electrons, it is confusing charge with energy.

KEEP THE DISTINCTION

Charge is measured in coulombs (C). Energy is measured in joules (J). A load can transfer energy without consuming the circulating charge.

Voltage means energy per charge

A voltage is a difference between two points. Always name both points or name a shared reference.



An energy-account diagram. Vertical position is not a physical height.

Unpack “5 volts”

One volt means one joule of energy transferred per coulomb of charge: **1 V = 1 J/C**. Across a 5 V resistor, 1 C of conventional positive charge corresponds to 5 J transferred to that component. Voltage is not current and does not tell you how many coulombs pass each second.

WORK IT THROUGH

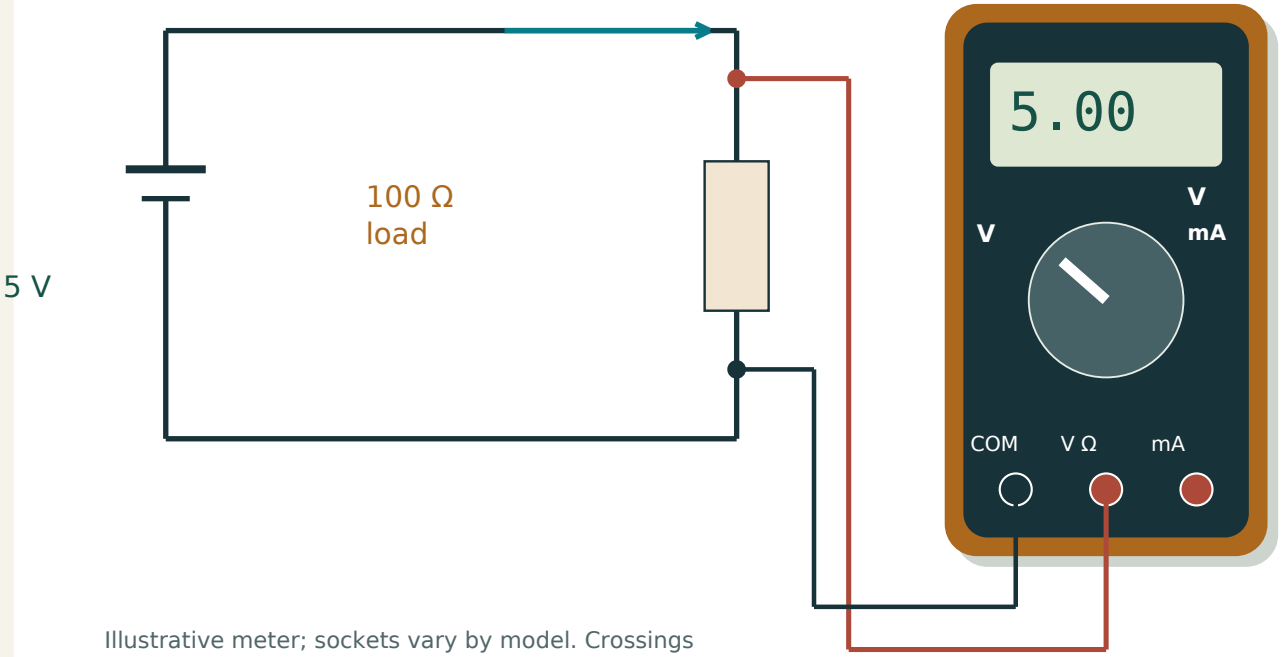
If 2 C corresponds to 10 J transferred across a component, V = E/Q = 10 J ÷ 2 C = 5 V. If point A is 5 V and point C is 2 V relative to B, A relative to C is 5 - 2 = 3 V.

A reference is a convention

“0 V” identifies the chosen comparison node. It does not automatically mean building earth, absence of charge or absence of energy.

Measure voltage across two points

A voltmeter forms a parallel measurement branch. It should draw little current compared with the circuit being observed.



Illustrative meter; sockets vary by model. Crossings without dots are not electrical junctions.

Connect with power disconnected

Use the meter’s voltage mode and the correct voltage-input socket according to its manual. In this 5 V learning circuit, place the probes across the resistor: red at its higher-potential end, black at its lower-potential end. The source and load remain connected; the meter observes their difference.

Interpret the sign and the boundary

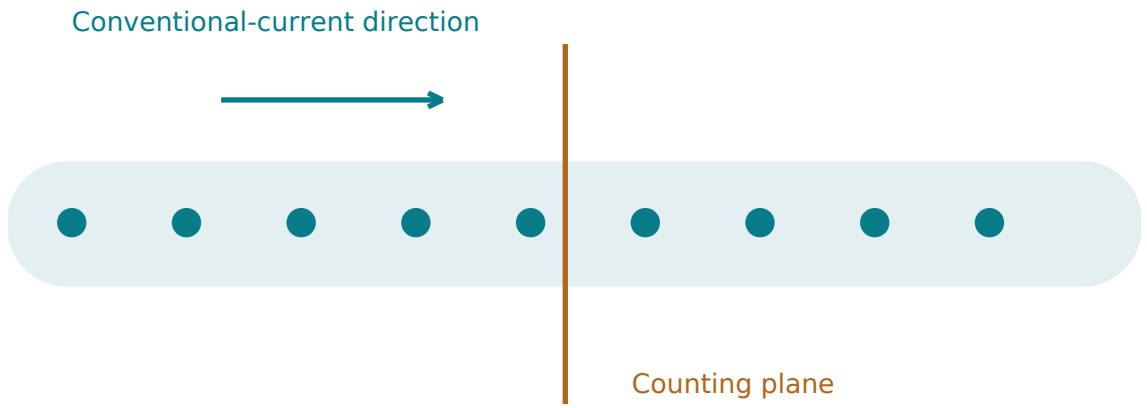
Swapping the probes reverses the sign. The magnitude should stay approximately the same in this steady case. A reading across the source can differ from one across the load if the wires or contacts have a meaningful voltage drop.

BEFORE POWER

Check mode, lead sockets and the actual manual. This illustration is for a small current-limited 5 V circuit, not for mains measurements.

Current counts charge per second

At one point in the wire, imagine a counting plane. Current describes how quickly charge crosses that plane.



Dots stand for charge amounts, not literal electron paths or spacing.

What does 0.050 A mean?

One ampere is one coulomb per second: **1 A = 1 C/s**. Therefore 0.050 A means 0.050 C passes the counting plane each second. Conventional current follows the direction positive charge would move. Electron drift in a metal is opposite that convention.

COUNT OVER A MINUTE

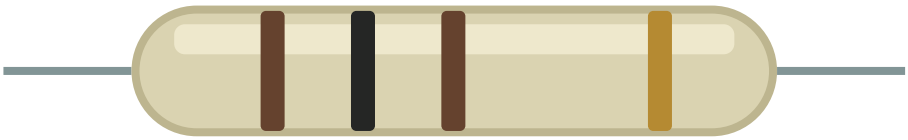
At 0.050 C/s for 60 s, transferred charge Q = It = 0.050 × 60 = 3.0 C. This is charge crossing one point during that minute. It is not an amount of charge permanently removed from the wire.

The same current on both sides

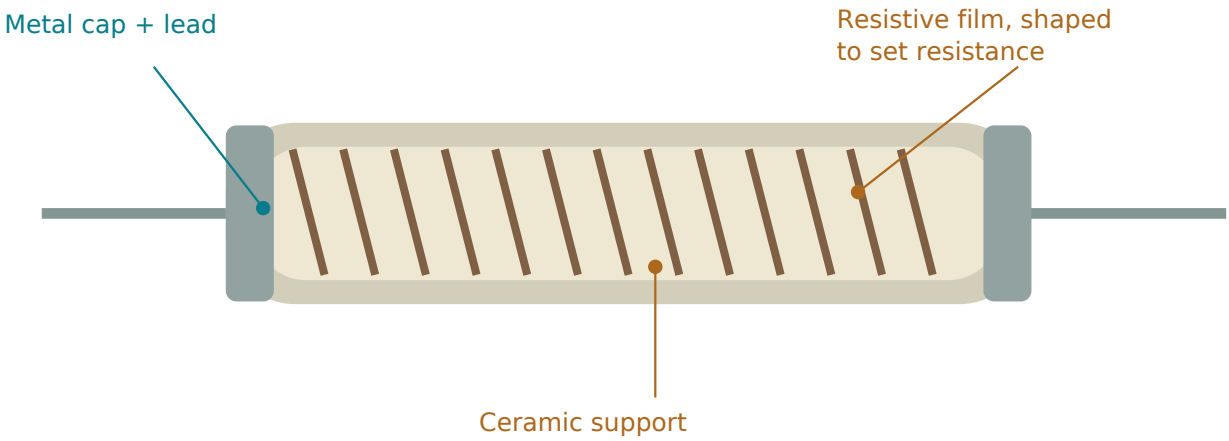
For our steady series loop, 0.050 A enters the resistor and 0.050 A leaves it. The resistor transfers energy, not a net accumulation of charge. Transients and branching need additional accounting.

Look inside a film resistor

A real component has materials, construction, tolerances and heat limits. Its resistance number is only one specification.



Example four-band code: brown, black, brown, gold → 100 Ω ±5%.



Schematic cutaway: coating and hidden wraparound film are partly omitted.

Resistance and heat belong together

A film resistor uses a resistive layer on an insulating support. Leads and end caps connect that layer to the circuit. A protective coating covers it. Different resistor technologies use other constructions. The colour bands tell you a nominal resistance and tolerance; they do not establish the power rating.

WHAT “100 Ω ±5%” ALLOWS

At the specified reference conditions, the nominal tolerance range is 95-105 Ω. At a fixed 5 V, those endpoints imply about 52.6-47.6 mA. Temperature and other specifications can add further variation. Use the actual datasheet for power rating and derating.

A resistor converts electrical input to heat. Later pages calculate that heat-production rate.

Use Ohm’s law in three directions

For an ohmic resistor under the stated conditions, voltage is proportional to current: V = IR.

$$V = I \times R \qquad I = V \div R \qquad R = V \div I$$

Choose the relationship that leaves your unknown on its own.

Known	Substitute with units	Result
V = 5 V; R = 100 Ω	I = 5 V ÷ 100 Ω	0.050 A
I = 0.050 A; R = 100 Ω	V = 0.050 A × 100 Ω	5.0 V
V = 5 V; I = 0.050 A	R = 5 V ÷ 0.050 A	100 Ω

Check the direction of the result

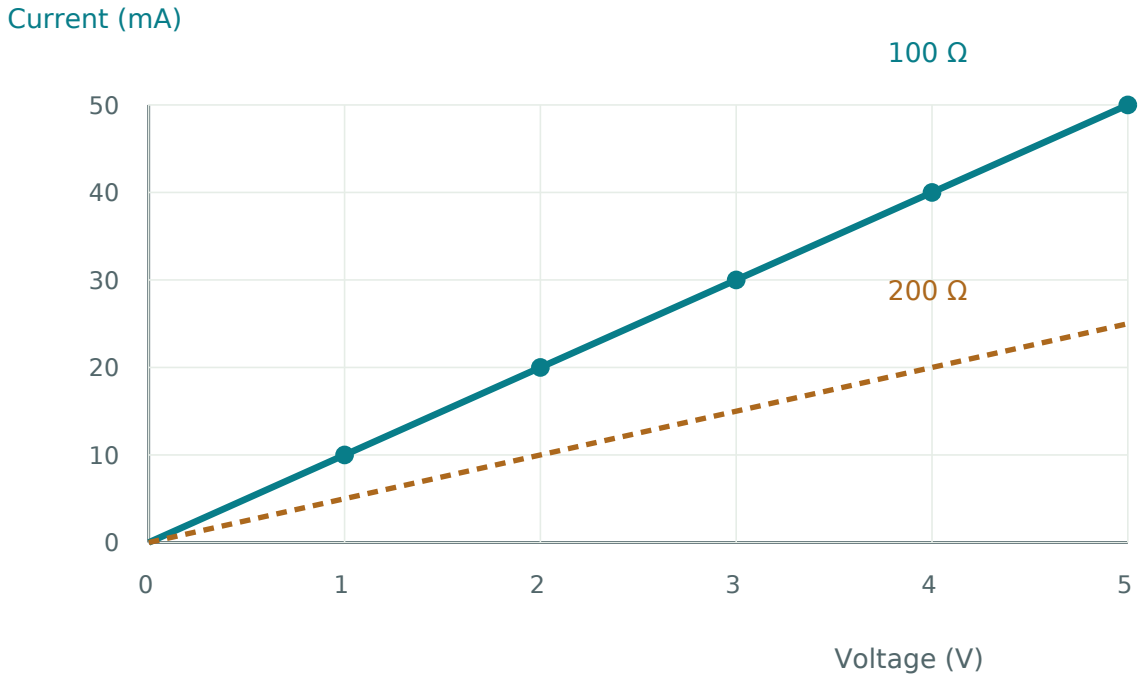
At the same voltage, larger resistance gives smaller current. At the same resistance, larger voltage gives larger current. If your answer predicts the reverse, check the rearrangement and units. Convert milliamperes to amperes before multiplying with ohms.

WHERE THIS MODEL STOPS

An LED, motor or regulated power supply is not generally one fixed resistor. Select a component model that fits the physical question.

See the relationship on a graph

A graph makes the prediction testable. For a fixed ideal resistor, each voltage-current pair lies on one straight line.



Read a point, then read the slope

At 3 V, the 100 Ω line predicts 30 mA. The 200 Ω line predicts 15 mA. On these axes the slope is current divided by voltage, equal to 1/R. A steeper straight line therefore means lower resistance. If you swap the axes, the slope has a different meaning.

A real test can depart from the line

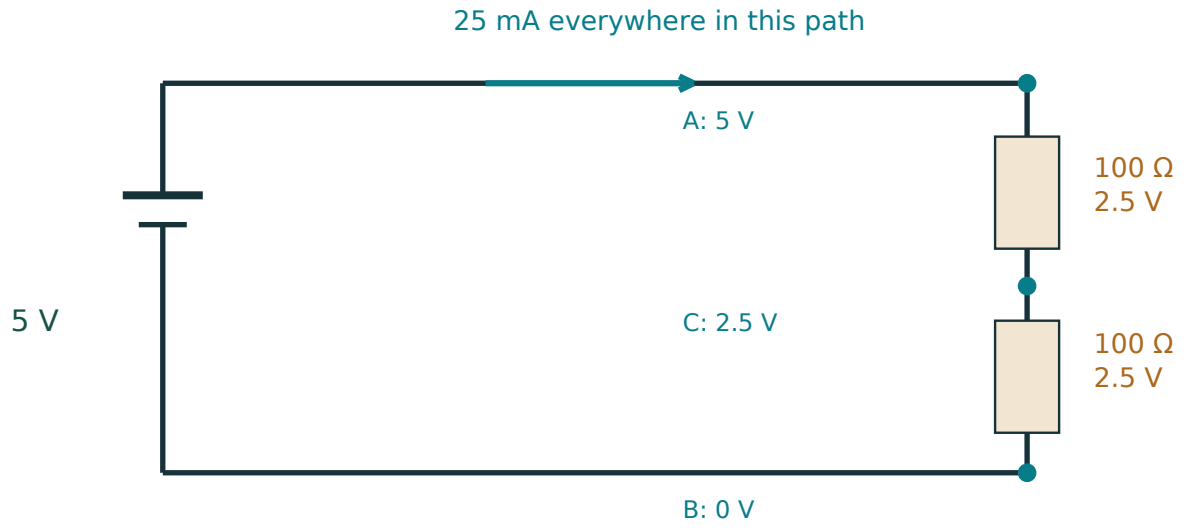
Source variation, resistor tolerance, heating and meter effects can shift readings. A curved relationship may mean a changing resistance or an unsuitable model. Record conditions before concluding that Ohm’s law “failed.”

YOUR TURN

At 4 V, predict current for both lines. Answer: 40 mA for 100 Ω; 20 mA for 200 Ω. Both predictions assume constant resistance.

Series: one current, shared voltage

Two resistors in one unbranched path carry the same steady current. Their voltage drops add to the source voltage.



First find the current for the whole path

Equivalent resistance R = 100 + 100 = 200 Ω. Current I = 5/200 = 0.025 A = 25 mA. Each resistor then drops V = IR = 0.025 × 100 = 2.5 V. The two drops add to 5 V; they do not each receive the full source voltage.

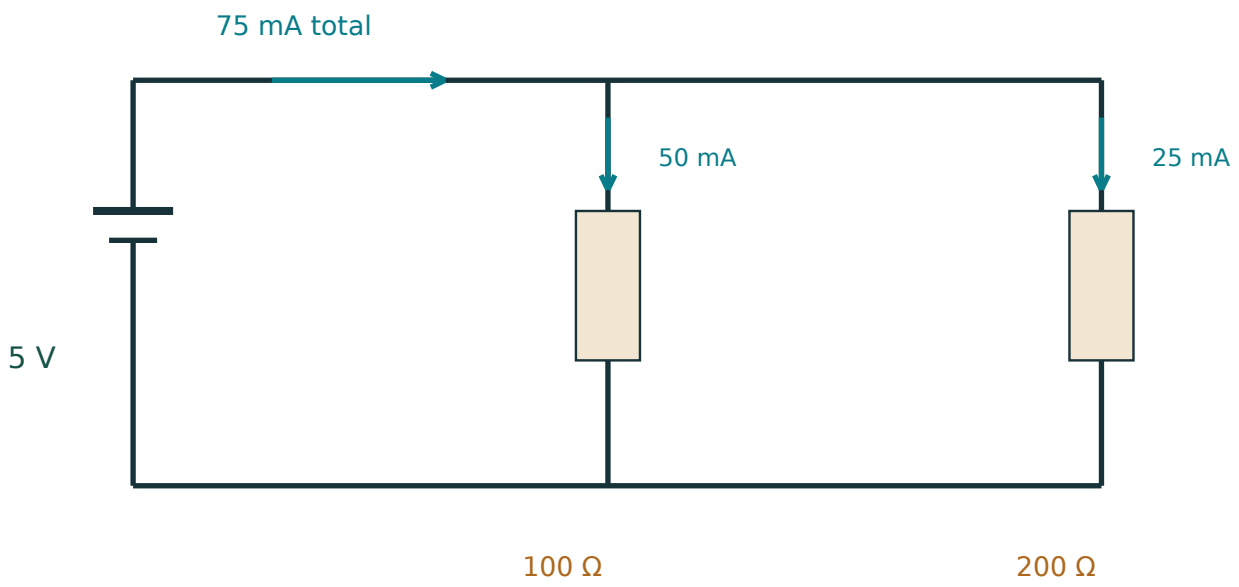
ENERGY PER COULOMB

One coulomb corresponds to 5 J transferred across both resistors together: 2.5 J at the first and 2.5 J at the second. The current is unchanged between them. This is energy sharing, not charge being used up.

If the resistors are unequal, current is still shared but voltage division changes. For 100 Ω + 200 Ω across 5 V: current is 16.7 mA; the drops are about 1.67 V and 3.33 V.

Parallel: one voltage, divided current

Separate branches connected to the same two nodes have the same voltage. Current flows through both finite-resistance branches.



Calculate each branch independently

Both resistors connect between the upper 5 V node and lower 0 V node. Branch currents are 5/100 = 50 mA and 5/200 = 25 mA. Their sum is 75 mA. The source therefore sees an equivalent resistance of 5 V/0.075 A = 66.7 Ω.

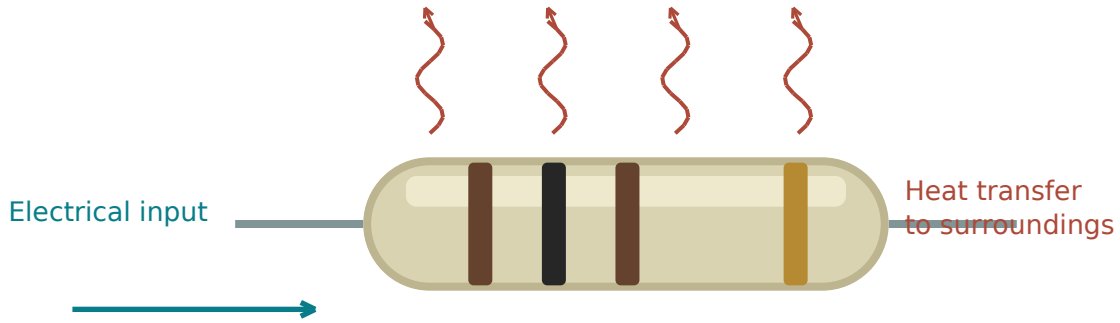
WHY “THE PATH OF LEAST RESISTANCE” IS INCOMPLETE

The lower-resistance branch carries more current, but the other conducting branch still carries current. A useful explanation must account for every path, not assume the entire current chooses only one.

Check charge conservation at the split and return: 75 mA = 50 mA + 25 mA. Opening one branch leaves the other connected in this ideal-source model; actual supplies may respond differently under their limits.

Power tells you how fast energy moves

The resistor’s electrical input becomes heat. Voltage and current together tell you the transfer rate.



0.250 J each second = 0.250 W

Build P = VI from the meanings

Voltage is joules per coulomb. Current is coulombs per second. Multiply them:
(J/C) × (C/s) = J/s = W.
For the reference circuit, 5.00 V × 0.050 A = 0.250 W. This is the energy-transfer rate at the resistor terminals.

Two useful resistor-only rearrangements

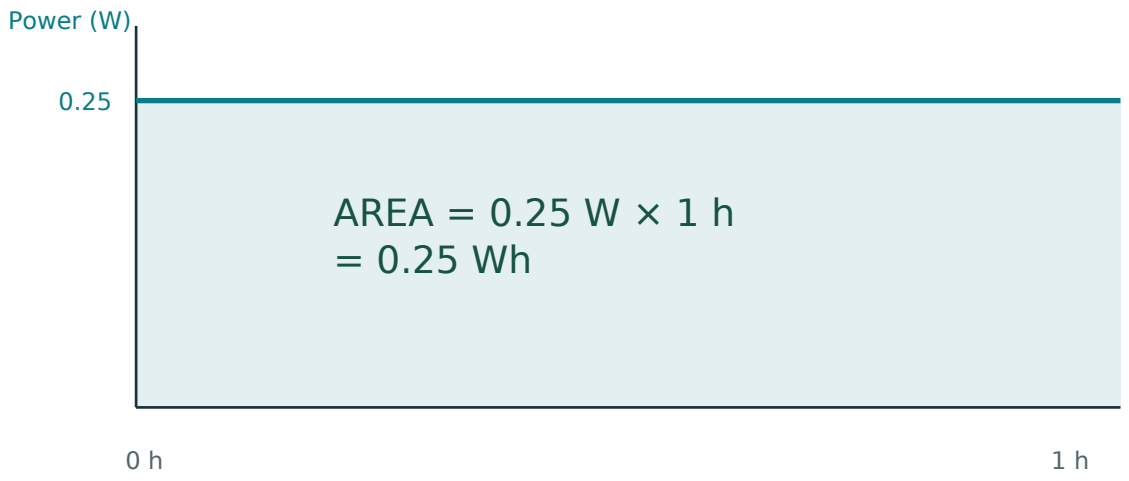
Substitute I = V/R into P = VI to get **$P = V^2/R$**. Substitute V = IR to get **$P = I^2R$**. For our case: 25/100 = 0.250 W, or $0.050^2 \times 100 = 0.250$ W. These forms need the resistor model and consistent operating conditions.

POWER IS NOT TEMPERATURE

Temperature also depends on heat flow, material, mounting, air movement and time. A 0.250 W power calculation does not by itself give the resistor’s surface temperature.

Energy is power accumulated over time

A small rate maintained for a long time can transfer a substantial amount of energy.



The rectangle is the calculation

For constant power, energy E = Pt. The height of this rectangle is power and its width is duration. At 0.250 W for one hour, E = 0.250 Wh. For ten minutes, E = 0.250 × (10/60) = 0.0417 Wh. Do not insert “10” as hours when the duration is ten minutes.

JOULES AND WATT-HOURS ARE BOTH ENERGY

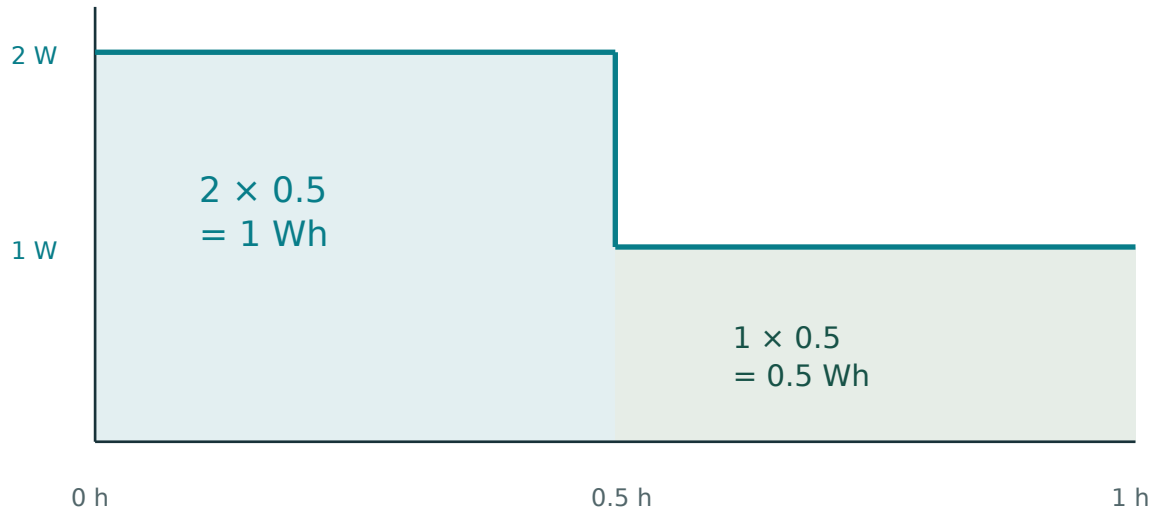
1 Wh = 3,600 J because an hour contains 3,600 seconds. Our one-hour example transfers 0.250 × 3,600 = 900 J. The ten-minute example transfers 0.250 W × 600 s = 150 J.

Read an energy label correctly

A kWh is 1,000 Wh. A kW is 1,000 W. The former is an amount; the latter is a rate. A battery’s Wh capacity also needs usable-capacity and discharge-limit conditions before predicting real runtime.

Changing power needs several areas

Add the energy of each interval. The exact method depends on what the recorded values represent.



Constant intervals: add the rectangles

For the stated step-shaped load: E = (2 W × 0.5 h) + (1 W × 0.5 h) = 1.5 Wh. Its one-hour average power is 1.5 W, while its peak is 2 W. Equal energy does not imply equal peaks or equal demands on equipment.

Point samples are a different case

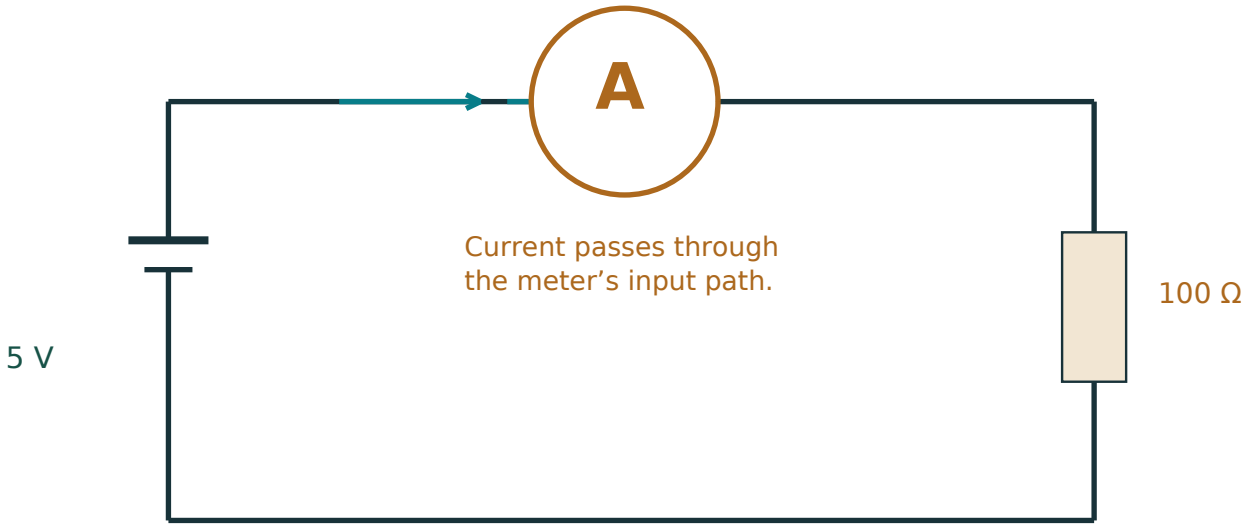
If two readings are instantaneous endpoints, power between them is unknown. A straight-line assumption gives trapezoid energy: average endpoint power × duration. With 1 W and 3 W separated by 4 s, the assumed ramp transfers (1 + 3)/2 × 4 = 8 J. A short hidden spike could change that answer.

KEEP THE DATA DEFINITION

A point sample, an interval mean and a cumulative energy counter require different treatment. Gaps are missing evidence; do not silently integrate across them as though the load were known.

Measure current through the branch

An ammeter is inserted into the current path. Connecting it across the source can create a short circuit.



Conceptual topology. Meter sockets, range and fuse requirements are model-specific.

Make the change while unpowered

Disconnect power. Use the meter’s prescribed DC-current mode, range and fused input. Open one branch connection and insert the meter to restore the path. Review the arrangement before briefly powering the supervised, current-limited circuit. After the test, disconnect and return the red lead to the voltage socket.

The meter changes the circuit slightly

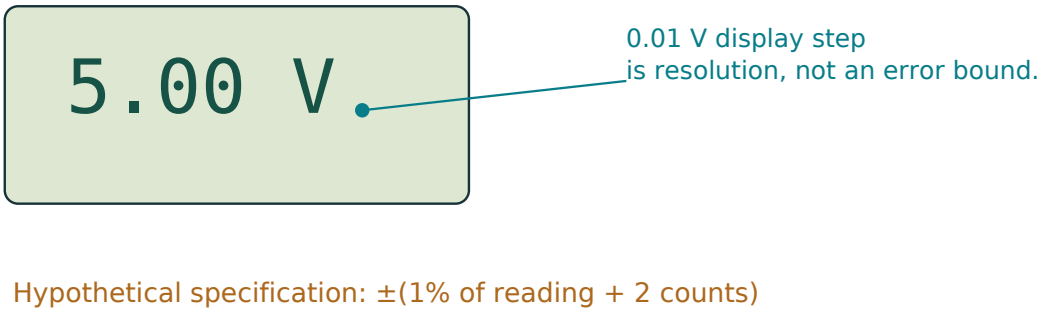
Its input path has resistance and a burden voltage. This can lower the current compared with the circuit without that meter. A source-side reading may also include other branches. Name the branch and measurement method rather than reporting “the current” without a boundary.

ONE METER IS ENOUGH TO BEGIN

You can measure voltage and current in separate supervised runs, rewiring only while unpowered. Record that the readings are sequential: they do not prove the two quantities stayed identical between runs.

Put a bound around the reading

Resolution is not accuracy. A stable final digit does not prove that the indicated number is correct.



Hypothetical specification: ±(1% of reading + 2 counts)

Work through the meter specification

Suppose the selected range displays 0.01 V steps and its specification is ±(1% of reading + 2 counts). At 5.00 V: 1% contributes 0.05 V; two counts contribute 0.02 V. The combined stated bound is ±0.07 V, subject to the manufacturer’s conditions. This is not automatically a 95% confidence interval.

Carry bounds into the power estimate

Also assume a current result of 0.050 A ±0.0025 A. Nominal P = 5.00 × 0.050 = 0.250 W. With both quantities positive, endpoint products give 4.93 × 0.0475 = 0.234175 W and 5.07 × 0.0525 = 0.266175 W. An outward-rounded interval is **0.234-0.267 W**.

WHY THE INTERVAL MATTERS

If your target band is only 0.245-0.255 W, a displayed 0.250 W does not settle the question: the measurement interval extends outside it. Improve the measurement or preserve the ambiguous verdict.

These specifications and readings are invented examples. Use your instrument’s actual range and conditions. A full uncertainty budget can require additional contributors.

Zero, missing and stale are different

Preserve what the instrument actually observed, and when it observed it. Do not replace a missing value with a plausible physical number.

Record	What you know
0.000 A / fresh / valid	The instrument reported zero within its resolution and measurement limits.
No value / missing	No usable current observation is available. Physical current is unknown.
0.050 A / old capture time	The report contains an earlier observation. It may not describe the present circuit.
0.050 A / fresh / wrong branch	The number may be valid for that branch but irrelevant to the requested claim.

Capture time is not receive time

If a sample was captured at 12:00:00 and received at 12:00:10, it is already ten seconds old when it arrives, assuming aligned clocks. Sending it again at 12:00:11 does not make it a new physical sample. Record capture time, receive time and clock uncertainty separately where available.

WHY THIS IS PART OF LUPINE

A request for zero output must not be marked successful because a disconnected sensor was converted to zero. A lost reply must not be marked failed merely because communication ended. Separate the physical quantity, evidence quality and command status.

A quality flag is useful only if the program preserves and checks it. The number alone cannot tell you whether it belongs to the right device, branch or moment.

Compare a model with a constructed record

The example below shows how to report a result. These are invented readings, not a measurement made for this book.

Item	Example value	Interpretation
Nominal resistor	100 Ω ±5%	A specified tolerance
Unpowered resistance check	99.2 Ω	Example indicated value
Load voltage	5.02 V	Example indicated value
Branch current	0.050 A	Example indicated value
Observed input estimate	5.02 × 0.050 = 0.251 W	Product of the readings
Ohmic-model estimate	$5.02^2/99.2 = 0.254$ W	Model using the indicated R

A small disagreement is a question to explain

The two estimates differ by about 0.003 W, roughly 1.2% of the model estimate. That alone does not prove a bad resistor or a superior method. Meter bounds, sequential measurement, temperature, burden voltage and connection changes may account for the difference. Compare the limitations before declaring a contradiction.

A DEFENSIBLE SENTENCE

“In this constructed example, the voltage-current product is 0.251 W at the load boundary. The difference from the simple resistor estimate is unresolved without the actual instrument specifications and test conditions.”

Your first physical measurement

The goal is a small result you can explain and revisit. Complete the paper predictions before assembling anything.

PARTS AND BOUNDARY

A suitable reviewed current-limited 5 V DC source; one 100 Ω resistor rated at least 1 W; suitable insulated leads/connectors; one multimeter with its manual. A 0.10 A source limit is an example for this nominal 0.050 A load. No mains alterations or opened power supplies.

1 / Predict

Write I = 0.050 A and P = 0.250 W for the nominal model. Include the resistor tolerance and source limits.

2 / Inspect unpowered

Check component identity, power rating, lead sockets, meter mode and the whole circuit. Have a competent person review the arrangement before first power.

3 / Measure voltage

Power briefly under supervision and measure across the resistor. Record value, unit, range and time. Disconnect before changing connections.

4 / Measure current

Insert the meter in the branch using its correct fused input. Review the unpowered arrangement, then measure under supervision. Record that this is a separate run.

5 / Explain

Calculate VI. Compare with the model using appropriate bounds. Stop for unexpected heat, unstable readings, damage, smell or any departure from the reviewed setup.

WRITE THE RAW RECORD FIRST

Voltage: ___ V. Current: ___ A. Meter/range: ___. Capture times: ___. Note whether the readings were sequential, what changed and which bounds apply.

A low voltage does not remove short-circuit, heating or mechanical risks. Keep the test supervised and disconnect it when finished.

Troubleshoot without guessing

Disconnect power before changing the circuit. Observe, form a hypothesis, then perform one controlled check.

Observation	Possible explanations	Next safe check
Current displays zero	Open path; wrong input/range; blown meter fuse; source off; a genuine low current.	Check mode and continuity only under the meter’s prescribed unpowered conditions.
Current larger than predicted	Wrong resistor; unexpected parallel path; wrong source voltage; wrong units.	Disconnect. Confirm part identity and the complete path.
Voltage lower than expected	Source limiting; contact/wire drop; overload; measurement or source issue.	Disconnect. Check load and source settings before repeating.
Readings drift	Heating; source variation; moving contacts; environmental or instrument effects.	Record the sequence. Stop for unexpected heating; do not keep adjusting to fit a target.
Numbers disagree with a model	Model assumptions may be wrong; uncertainty may be larger than the discrepancy.	Compare ranges, boundaries, timing and instrument specifications.

DO NOT TURN A HYPOTHESIS INTO A FINDING

“Possibly an open circuit” is a hypothesis. “The connection was open and the prescribed unpowered check confirmed it” is a finding with a method. Keep the distinction in your notes.

Practise without looking back

Write the relationship, substitute the units, calculate and then state the limitation. Worked answers follow on the next page.

1. A 5 V ideal source drives a 200 Ω resistor. Find current and power.

2. The reference circuit carries 0.050 A for 120 s. How much charge crosses one point?

3. A 0.250 W load runs for 20 min. Find energy in Wh and J.

4. Two 100 Ω resistors are in series across 5 V. Find current and each voltage drop.

5. The same two resistors are in parallel across 5 V. Find each current, total current and total power.

6. A load uses 4 W for 15 min, then 2 W for 45 min. Find its one-hour energy and average power.

7. A hypothetical meter displays 2.00 V in 0.01 V steps. Its specified bound is ±(1% of reading + 2 counts). Calculate that bound.

8. A request asks for zero current. The sensor sends no value. Later it sends an old zero reading with a new receive time. What can you conclude at each stage?

Worked answers and reasoning

Check the units and the model, not only the final number. If you made an error, preserve the original attempt beside your correction.

1 I = 5/200 = 0.025 A. P = 5 × 0.025 = **0.125 W**. Assumes a steady ideal source and a constant ohmic resistance.

2 Q = It = 0.050 C/s × 120 s = **6 C**. This counts charge crossing one point, not charge consumed by the resistor.

3 20 min = 1/3 h = 1,200 s. E = 0.250 × 1/3 = **0.0833 Wh**; or 0.250 × 1,200 = **300 J**.

4 Series R = 200 Ω. I = **25 mA**. Each drop is 0.025 × 100 = **2.5 V**. The drops add to 5 V.

5 Each branch carries **50 mA**; total is **100 mA**. Total P = 5 × 0.100 = **0.50 W**. An actual source must be able to supply it within its limits.

6 E = (4 × 0.25) + (2 × 0.75) = **2.5 Wh**. Average P = 2.5 Wh/1 h = **2.5 W**. Peak remains 4 W.

7 1% of 2.00 V = 0.02 V. Two counts = 0.02 V. Bound = **±0.04 V** under the specified conditions, not automatically a confidence interval.

8 Missing means no usable physical observation; old zero is stale evidence. Neither confirms a fresh zero-current outcome. Preserve uncertainty and seek an appropriate fresh sample and command-status evidence.

Keep this beside your first test

Use this page as a compact reference. Every relationship below needs its model, boundary and units.

Quantity	Relationship	Meaning or limit
Voltage	$V = E/Q$	J/C; difference between two points.
Current	$I = \Delta Q/\Delta t$	C/s; name the branch.
Ohmic resistor	$V = IR$	Use the correct component model.
Steady DC power	$P = VI$	W; same relevant operating conditions.
Resistor power	$P = I^2R = V^2/R$	Heat-production rate, not temperature.
Constant-power energy	$E = Pt$	W × h = Wh; W × s = J.
Unit conversions	1 Wh = 3,600 J	1 kW = 1,000 W; 1 kWh = 1,000 Wh.

Minimum useful record

Date and question; circuit sketch; source/current limit; resistor identity/rating; meter identity, sockets, range and specification; measurement points and timing; raw readings; calculations; bounds; explanation; next question.

Sources and editorial scope

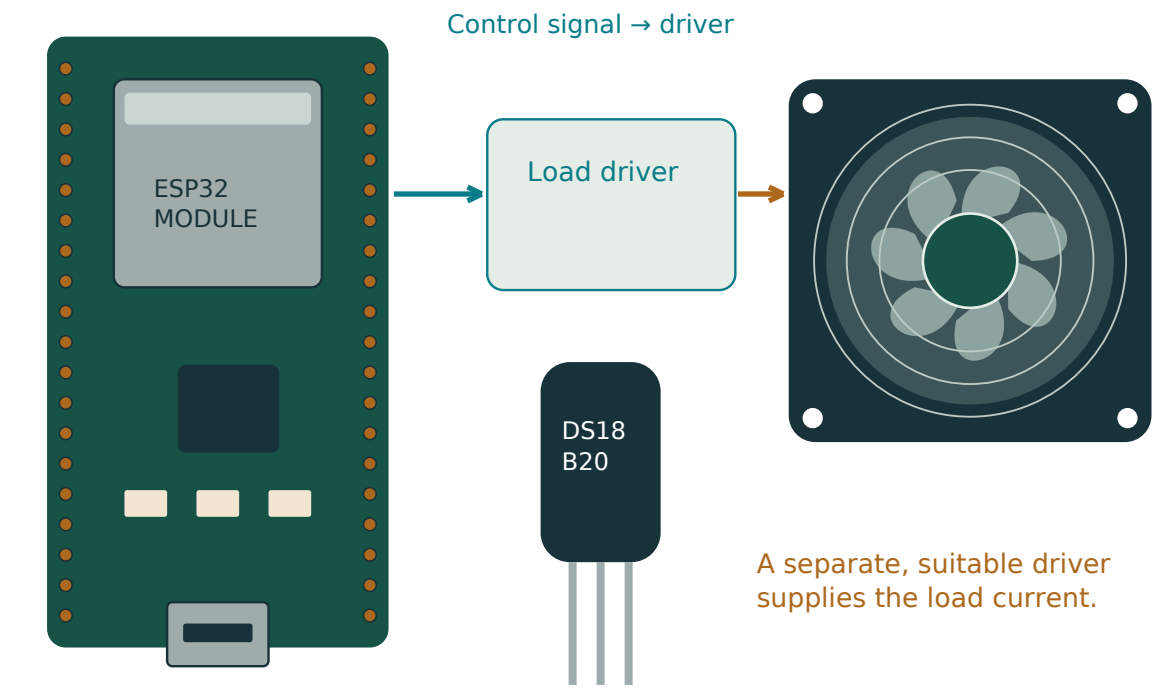
Original text and artwork prepared with ChatGPT. Background: DK, How Technology Works (2019); Steven W. Blume, Electric Power System Basics, 3rd ed. (2025). These are sources, not required reading. Component construction/marking and measurement terminology are supported by the linked primary references below.

Vishay PR01/02/03 construction | Vishay colour chart | NIST TN 1297 terminology

Edition 01 / 17 October 2026. This is a first-pass learning volume, not an independently peer-reviewed engineering manual. All numerical examples are assumed or constructed; no original measurement is claimed. Exact component and meter documentation governs physical use.

COMPONENTS, SENSORS & CONTROL

Part 02 / From an electrical circuit to a measured, programmable system.



Functional illustration. These are not physical pin positions or a wiring diagram.

Learn the complete path

A microcontroller reads a sensor, runs instructions and changes a signal. A driver handles load current. The physical device responds on its own time scale. Feedback tells you what was observed, not automatically why it happened.

WHAT YOU WILL BE ABLE TO EXPLAIN

Board versus chip; 5 V versus 3.3 V; digital and analogue inputs; MOSFET switching; PWM; temperature and speed sensing; firmware timing; and a first ambient-temperature logger.

Follow one system, learn each interface

The fan is a teaching example. The first physical project is an ambient-temperature logger with no fan or power actuator connected.

Pages	Study question
3-5	What is on the board, and what can its pins safely connect to?
6-8	What do resistors, LEDs, capacitors, coils and protection components do?
9-11	How does a small signal control a separate power path?
12-17	How do physical quantities become numbers, with units and limits?
18-20	What do fan power, PWM duty and rotational speed each tell you?
21-24	How does firmware acquire, validate and report a sample?
25-26	How do you build and test the ambient logger?
27-29	Can you work through the problems and explain the answers?
30-32	Reference tables, glossary, implementation checks and sources.

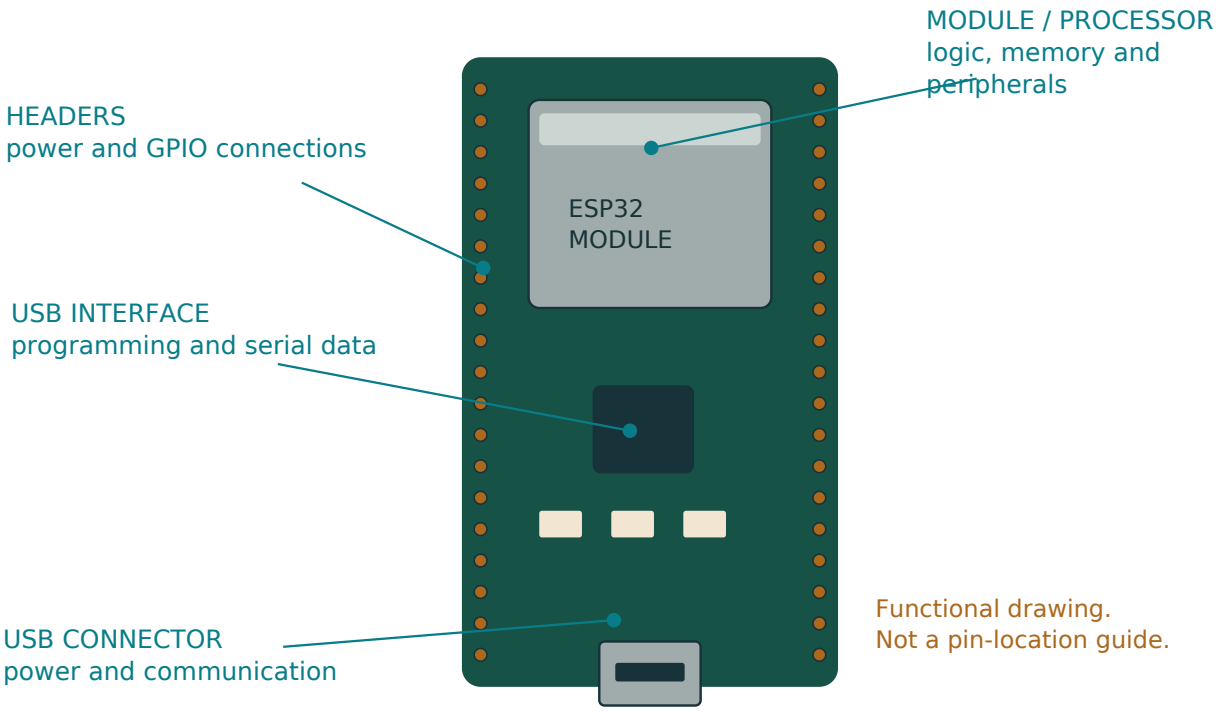
The small amount of Part 01 you need

For a steady DC load, P = VI. For an ohmic resistor, V = IR. Energy accumulates as E = Pt when power is constant. Missing evidence is not a physical zero. Each of these ideas is used directly in the worked examples here.

Use labels such as 02-13 after merging: Part 02, page 13. No return to the two background books is assigned.

A board is more than a processor

The chip runs the program. The development board adds interfaces, power handling, connectors and other supporting parts.



Keep three identities in your notes

Chip: the silicon device, such as the original ESP32 or another family member.
Module: a package that adds items such as flash memory, radio support and an antenna.
Board: the assembled development product with its own connector and header layout.

Choose the exact board before following pin instructions

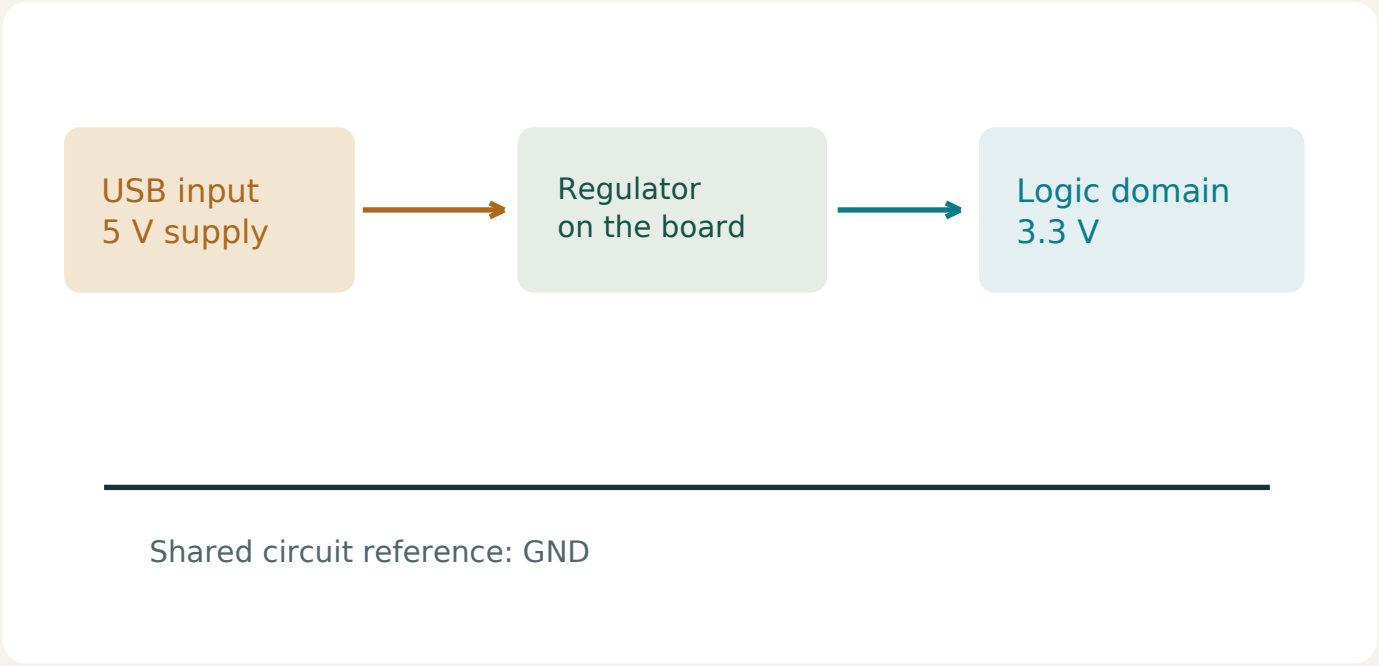
This volume’s optional wiring example names the original ESP32-DevKitC V4 with an ESP32-WROOM-32-series module. It is not a universal ESP32 pin map. A board marked ESP32-C3, S3 or another variant requires its own map and compatible firmware.

RECORD

Manufacturer, exact board/module, revision, firmware release and the documentation used. A product family name alone is not enough.

A 5 V input does not mean 5 V logic

Power-supply rails and signal-voltage limits describe different parts of the electrical system.



Power is converted; the pin limits do not follow USB

A board can receive USB power at 5 V while its microcontroller and external signals use a 3.3 V domain. Connecting a 5 V signal to a 3.3 V input can violate its limits. A label such as “5 V board” does not describe every pin. Check both normal operating and absolute maximum conditions.

A voltage needs a reference

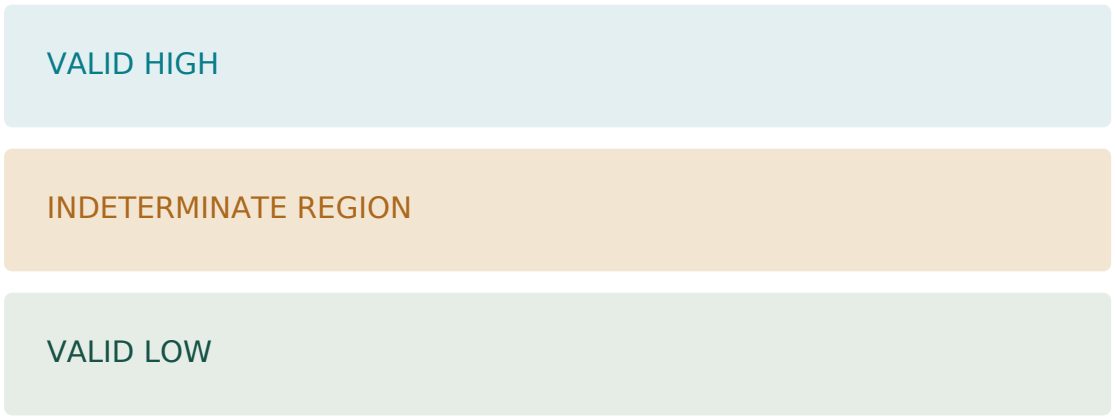
For an ordinary non-isolated interface, connected circuits need an appropriate common reference. If a signal is 3.3 V relative to one circuit’s GND, the receiving device must interpret it relative to its own reference. Isolation is a different interface design; do not add ground connections blindly across isolated or unrelated systems.

ONE POWER ROUTE FOR THE NAMED BOARD

Espressif lists USB, the 5 V header, or the 3V3 header as mutually exclusive supply routes for DevKitC V4. For the logger here, use USB only; do not add a second external supply.

Digital inputs classify voltage

A GPIO input recognises valid low and high regions. It is not an exact voltage measurement.



Thresholds belong to the exact device and supply conditions, not to this drawing.

Do not leave a decision floating

An unconnected input may have no stable defined level. A pull-up resistor connects it weakly to the logic supply; a pull-down connects it weakly to the reference. A switch or open-drain device can then change its state. The resistor sets a default without acting like a short across the supply.

WORKED PULL-UP EXAMPLE

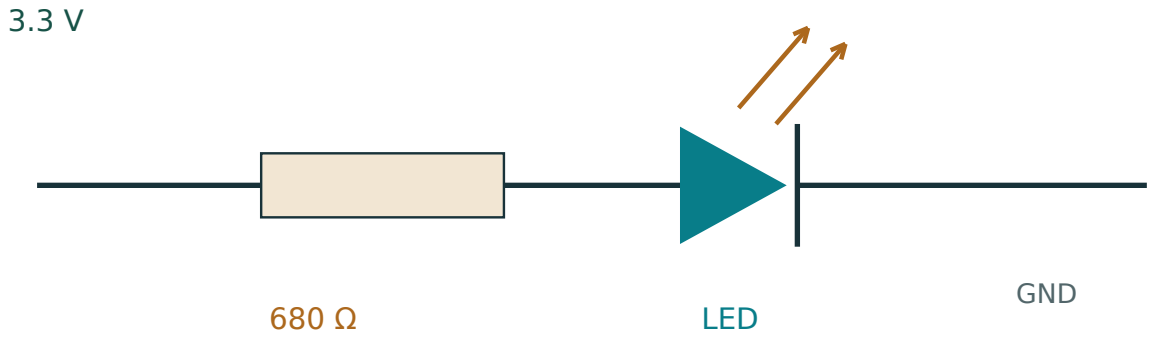
Assume a 10 kΩ resistor from 3.3 V to a signal, and a switch from that signal to GND. Open switch: the signal tends toward high. Closed switch: current is about 3.3/10,000 = 0.330 mA. This ignores input leakage and switch resistance.

Input and output are different modes

A push-pull output actively drives both levels. An open-drain output pulls low or releases the line and needs a suitable pull-up. Avoid connecting outputs that can fight each other. Check boot-time pin functions before assigning a load.

An LED needs current control

A resistor can set an approximate indicator current. The LED is not a fixed resistor.



Illustrative branch; choose an actual LED, polarity and resistor rating before use.

Find the voltage left for the resistor

If the assumed LED forward drop is 2.0 V, the resistor sees 3.3 - 2.0 = 1.3 V. With 680 Ω, approximate current is 1.3/680 = 0.00191 A, or 1.91 mA. Resistor power is about 1.3 × 0.00191 = 0.00249 W. The forward drop changes with device, current and temperature.

A high logic state is not an unlimited supply

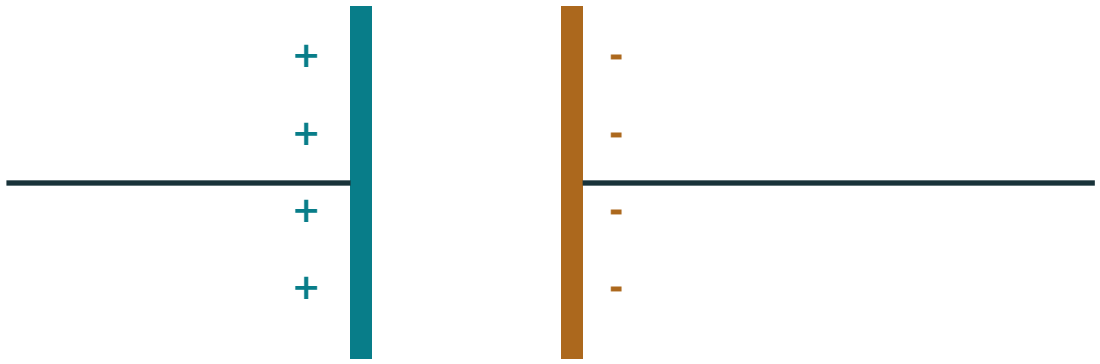
If a GPIO drives the branch, check its permitted current and output-voltage behaviour, including all simultaneously active pins. Use a suitable driver for larger loads. An LED can indicate that software selected a state; it cannot prove a fan rotated or a valve moved.

PREDICT BEFORE CONNECTING

In the same simplified model, doubling the resistor to 1,360 Ω approximately halves the current. Connecting the LED directly across a stiff supply removes the intended current limit.

A capacitor manages changing energy

Two separated conductors store electric-field energy. Their voltage changes when charge is added or removed.



Insulating separation

Use the relationships with their units

Capacitance C = Q/V, in farads, relates charge separation and voltage. Stored energy is E = ½CV². For approximately constant discharge current, ΔV = IΔt/C. A local capacitor can help supply a brief current change, but its value, series resistance, mounting and the regulator still matter.

WORKED TRANSIENT EXAMPLE

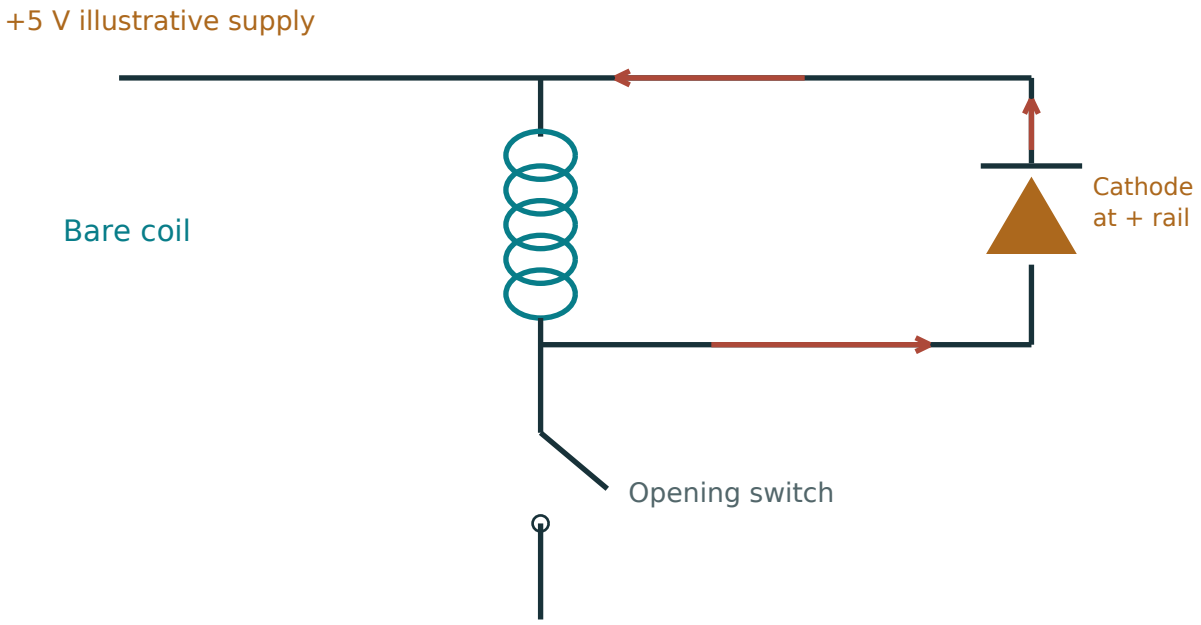
Assume 100 µF supplies an extra 50 mA for 1 ms with no other current contribution. ΔV = 0.050 × 0.001 / 0.000100 = 0.50 V. A capacitor label alone therefore does not guarantee that a 3.3 V rail stays within limits.

Choose more than the capacitance

Check voltage rating, polarity if applicable, leakage, ripple current and temperature. Do not treat a polarised capacitor as reversible or connect it above its allowed voltage.

A coil needs a path when switched off

Inductors store magnetic-field energy. Interrupting their current can produce a voltage above the supply.



Red arrows: recirculating current after opening. A bare-coil example, not a fan pinout.

The diode is normally reverse-biased

When the switch conducts, current flows through the coil and switch. After it opens, the inductor can drive its lower terminal above the supply, forward-biasing the diode. Current then circulates and decays as stored energy is dissipated. Diode orientation, current rating, voltage rating and switching needs must match the design.

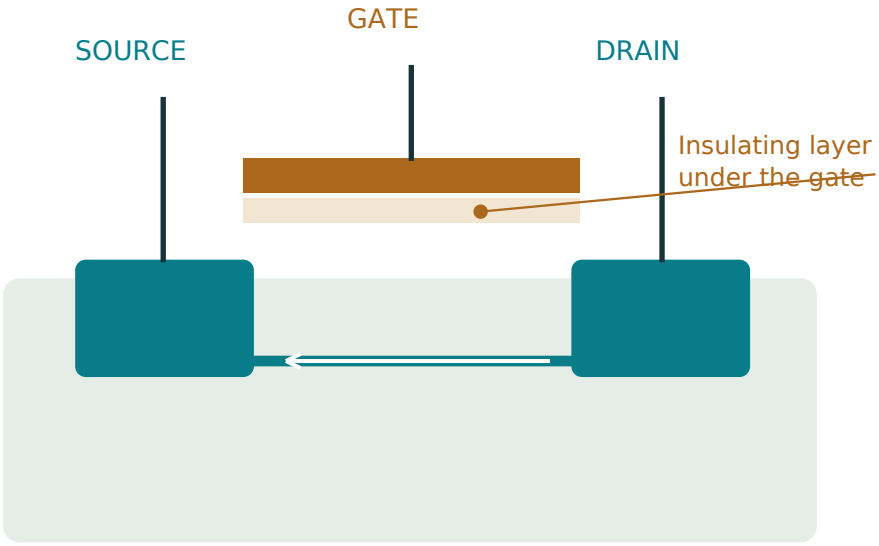
QUANTIFY THE STORED ENERGY

For L = 10 mH and I = 0.10 A, $E = \frac{1}{2}LI^2 = 0.000050$ J. If that current were forced to fall linearly to zero in 50 µs, $|v| = L|\Delta I/\Delta t| = 20$ V in the ideal model. Real protection and parasitic effects determine the actual waveform.

A two-wire brushless fan contains electronics. Follow its manufacturer’s switching guidance rather than assuming it is a bare coil.

A MOSFET separates signal from load current

Gate-to-source voltage controls the channel. The source is the voltage reference for the gate, not just an arbitrary pin name.



Conceptual n-channel section; hidden structures and body diode omitted.

What the signal changes

A gate voltage can establish a conducting channel between drain and source in an enhancement-mode device. The gate draws transient current while charging and discharging, even though its steady leakage may be small. This picture is conceptual; it is not a fabrication drawing or a complete device model.

THE REFERENCE MATTERS

Gate = 3.3 V and source = 0 V gives VGS = 3.3 V. Gate = 3.3 V and source = 3.0 V gives only VGS = 0.3 V. The same voltage relative to GND does not imply the same switching state.

For the usual low-side load arrangement, the source is near the local reference and the drain connects to the load return. High-side arrangements need a different gate-drive analysis.

Threshold is not “fully on”

A gate-threshold number describes a defined low-current test point. Use guaranteed on-resistance at your actual drive voltage.

Check	Why it changes the decision
VGS threshold	Beginning of conduction under test conditions, not proof of low loss.
RDS(on) versus gate voltage	Needs a specification that applies to your available gate drive.
Drain voltage / current	Check operating conditions, transient limits and thermal constraints.
Gate charge	Determines transient drive demand and affects switching time.
Package and thermal path	A current headline alone does not establish permitted heating.

Work a loss estimate

If a device has a justified on-resistance of 0.10 Ω at the actual gate voltage and temperature, a steady 0.20 A gives $I^2R = 0.004$ W conduction loss. This excludes switching loss, gate-drive loss and other effects. A typical plot alone is weaker evidence than an applicable guaranteed limit.

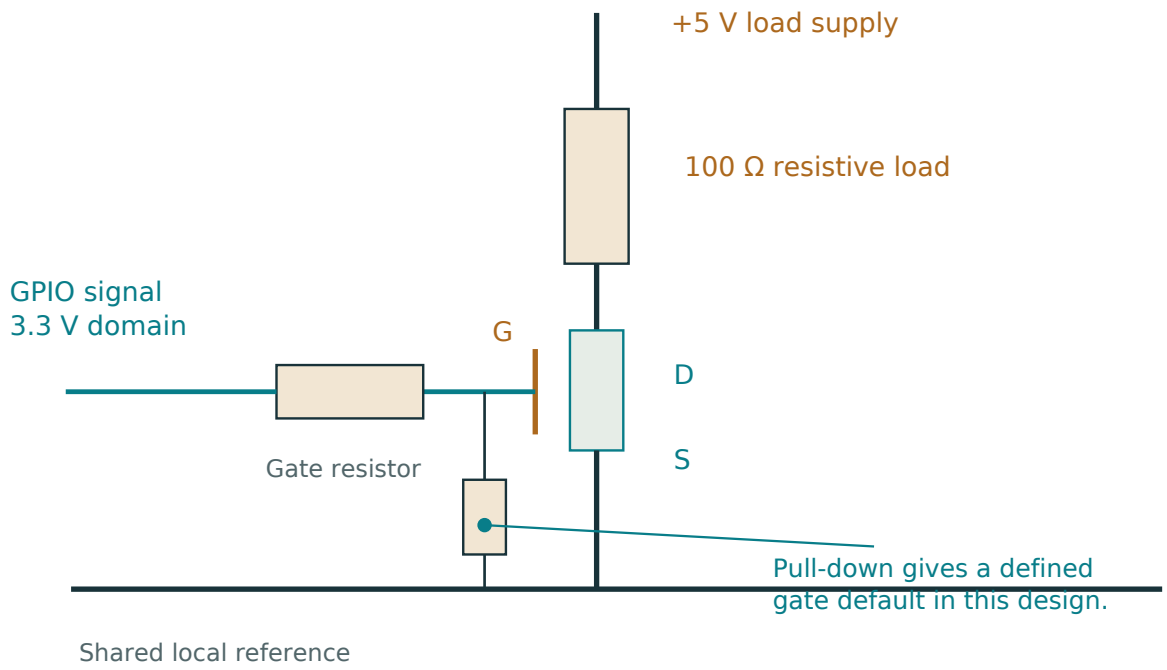
GATE CHARGE NEEDS TIME

A hypothetical 20 nC gate charge supplied at an average 10 mA takes Q/I = 2 µs. Actual current and capacitance vary during switching, so this is an order-of-magnitude estimate, not a precise switching-time prediction.

All device values here are hypothetical. The implementation needs a selected part, its full datasheet and a reviewed circuit.

Trace a complete low-side load path

Follow the load current separately from the gate-drive path. The example uses a resistor so that the load model is explicit.



The high-current loop avoids the GPIO

When the selected MOSFET is on, current flows from the 5 V source through the resistor, drain and source, then returns to the supply. The GPIO controls the gate relative to source. Its gate resistor and pull-down must be selected for the actual pin, device and switching rate. This is a conceptual circuit, not a board-specific assembly recipe.

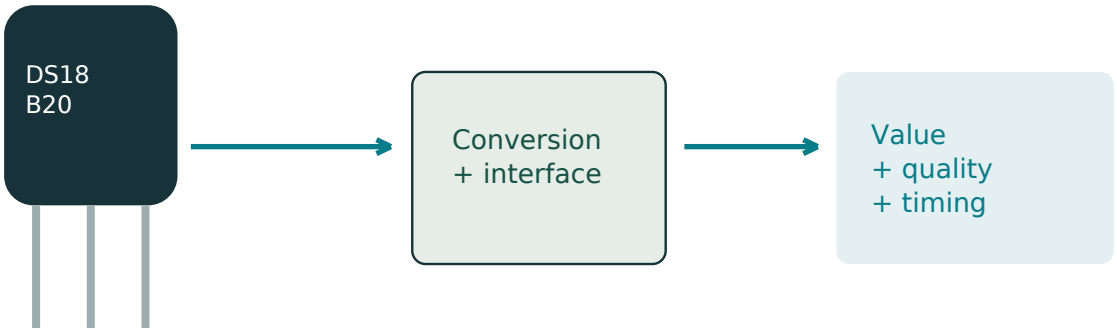
MODEL THE SWITCH LOSS TOO

Assume 100 Ω load and 0.10 Ω on-resistance. Current is 5/(100 + 0.10) = 49.95 mA. The switch dissipates about 0.000250 W. The resistor gets slightly less than 5 V; an ideal-switch model omits that small drop.

A low-side switch raises the disconnected load’s return when off. Do not assume that node remains at GND or use it as a sensor reference.

A sensor is the start of a measurement

The physical location, mounting, electronics and software all contribute to what the reported value means.



A room probe, a chip sensor and a coolant probe measure different locations.

Choose the physical question first

For ambient temperature, place a suitable sensor in the air location you mean to describe. For chip junction temperature, a room probe is not a substitute. For cooling performance, temperature alone may not establish heat removal: flow, inlet/outlet points and other heat paths can matter.

Signal path and response time matter

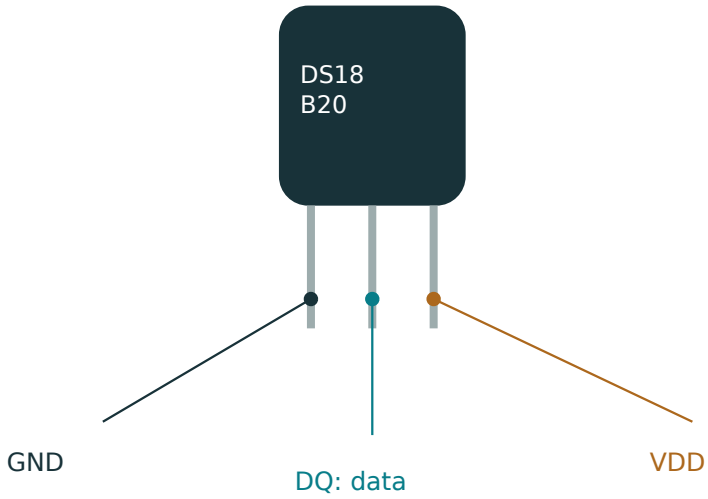
An analogue sensor provides a voltage, current or resistance response that needs conversion. A digital sensor performs some conversion internally and reports data. Digital communication does not eliminate sensor error, mounting effects or lag. A fast logger cannot make a slow probe physically respond faster.

WRITE A MEASUREMENT SENTENCE

“This sensor reports ambient temperature at this location, under this mounting and airflow, with this update interval and these known limits.” That is more useful than simply recording “temperature = 22.”

Meet the temperature sensor for the logger

The optional project uses one externally powered DS18B20 on a 3.3 V 1-Wire bus. It does not use parasite-power mode.



Functional terminal illustration, not a package-orientation or probe-wire-colour guide.

Read the number with its specification

The genuine device supports 9-12-bit resolution. At 12 bits, one temperature step is 0.0625 °C; that does not imply ±0.0625 °C accuracy. Analog Devices specifies ±0.5 °C accuracy from -10 to +85 °C under the datasheet conditions. Probe assembly and mounting add their own limitations.

A CONVERSION TAKES TIME

At 12-bit resolution, allow the specified conversion time of up to 750 ms before reading. Requesting a conversion and immediately reading can return an older value. More displayed decimal places do not repair an old sample.

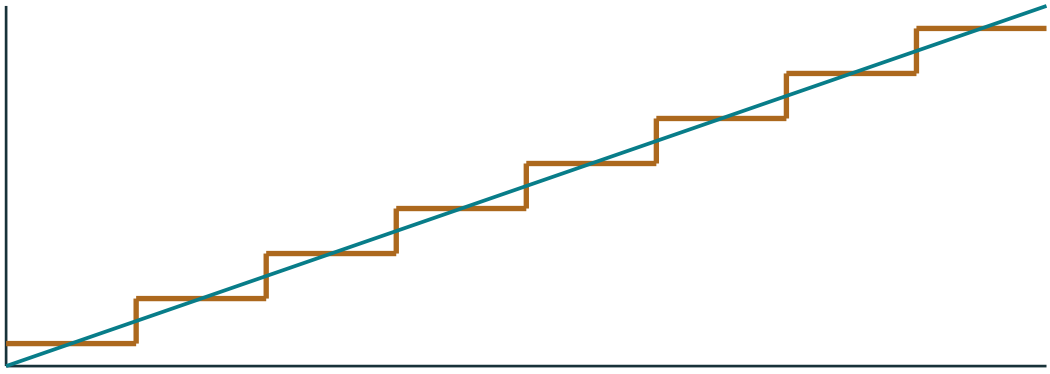
Identify the actual part and its pins

The device has a unique ROM identifier. Sealed probes may use different lead colours, and a listing does not guarantee genuine silicon. Verify the supplier’s pinout and the manufacturer’s package view before connecting power.

An ADC reports a code, not certainty

An analogue-to-digital converter divides an input range into discrete representations. Calibration and input limits remain separate questions.

Coarse quantisation illustration



Eight illustrated bins for clarity; this is not a measured ADC transfer curve.

Calculate ideal resolution, then name its limits

An N-bit converter has 2^N possible codes. A useful ideal code-width estimate is full-scale range/2^N. For a hypothetical 12-bit, 0-3.3 V converter: 3.3/4,096 = 0.000806 V, about 0.806 mV. That is resolution, not an accuracy bound and not an ESP32 voltage-range specification.

An ESP32 needs its own input configuration

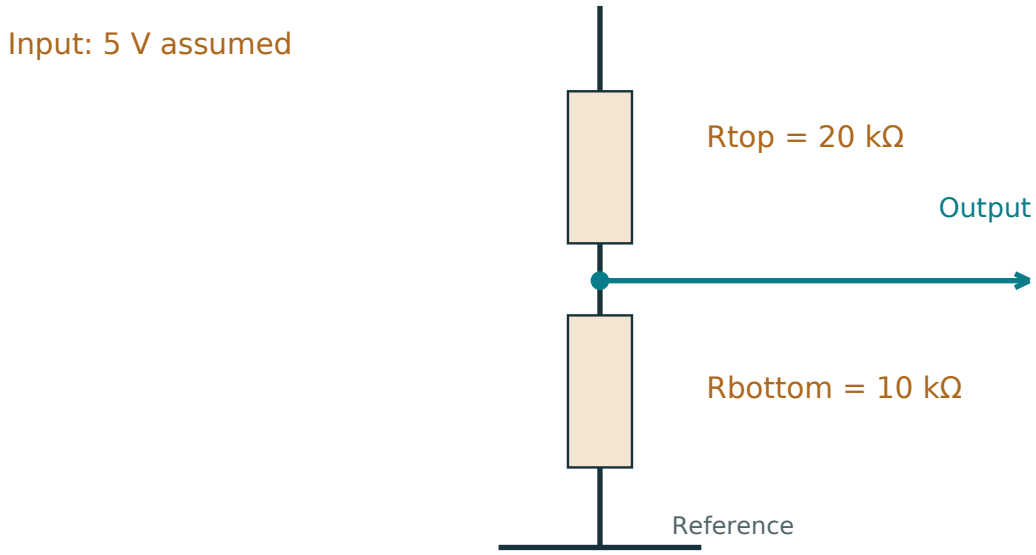
Reference voltage, attenuation, calibration, input loading, noise and chip variant affect conversion. Do not assume its ADC accepts every value up to the board’s 3.3 V rail. Use the exact device documentation and applicable calibration method.

AVERAGING HAS LIMITS

Averaging may reduce some variation. It does not automatically remove offset, gain error, nonlinear response, aliasing or an overloaded input.

Scale a voltage before interpreting it

A resistor divider can reduce a voltage in a model. It is not isolation or complete input protection.



Derive the ratio from one series current

Ignoring input loading, total resistance is 30 kΩ. Current is 5/30,000 = 0.167 mA. The lower resistor then has 0.000167 × 10,000 ≈ 1.667 V across it. More generally, Vout = Vin × Rbottom/(Rtop + Rbottom). Both circuits share the appropriate reference in this model.

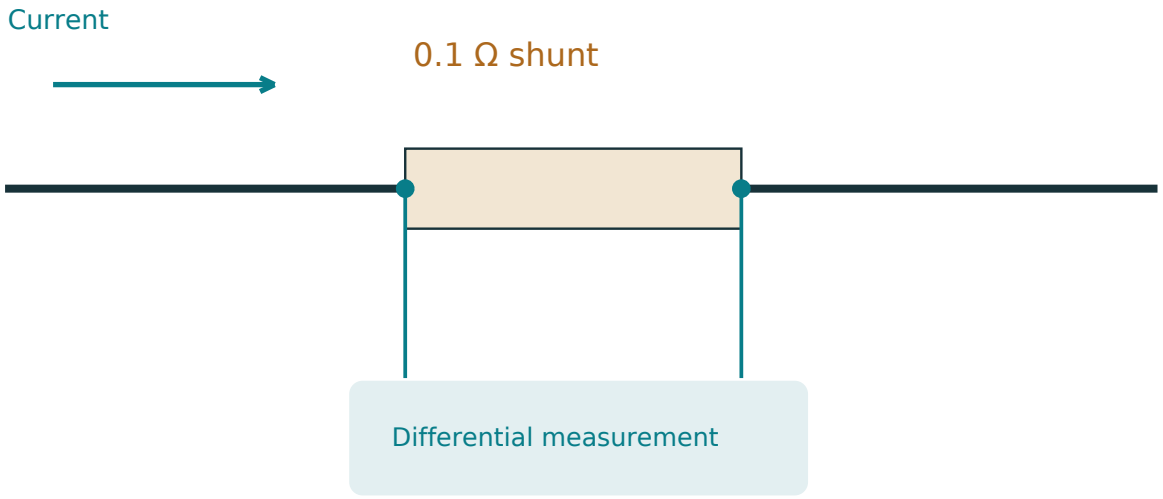
CHECK THE REAL FRONT END

Source extremes, resistor tolerance, ADC input loading, sample-and-hold settling, transients and protection can change the result. A correct nominal ratio does not establish that an input stays within all limits or that the conversion is accurate.

This is not a connection recipe for measuring mains, battery packs or an unknown supply. This series' physical exercises stay with the defined small low-voltage setup.

A shunt makes current observable

A small known resistance produces a voltage drop proportional to branch current. The measurement changes the branch slightly.



Sense both shunt terminals, not an arbitrary distant ground point.

Use the small differential voltage

At 0.20 A through 0.1 Ω, Vshunt = IR = 0.020 V = 20 mV. Dissipation is I^2R = 0.004 W. A current-sense circuit must resolve that small difference while tolerating the voltage of both terminals relative to its own reference. Common-mode range and differential range are different limits.

A digital monitor still needs configuration

Devices such as TI’s INA219 can report shunt/bus information over I^2C and use calibration settings for current and power scaling. The actual shunt value, module wiring, logic voltage, conversion timing and limits must be documented. Do not assume every breakout board implements the same shunt or scaling.

NAME THE BRANCH

A source-input sensor may include board and driver consumption. A fan-only branch sensor excludes other loads. Neither boundary is automatically wrong; they answer different questions.

Power readings need the right time scale

A pair of slowly averaged values can miss the power transferred by a switched load.

STEADY EXAMPLE

5 V × 0.20 A = 1.0 W

SWITCHED EXAMPLE

Average of v(t)i(t) may need faster measurement

Average power is the average of the product

For steady DC, P = VI is straightforward. For changing signals, instantaneous power is v(t)i(t). Averaging voltage and current separately, then multiplying, need not give the same result. A monitor’s internal conversion order, filtering and update rate affect the interpretation.

A SIMPLE COUNTEREXAMPLE

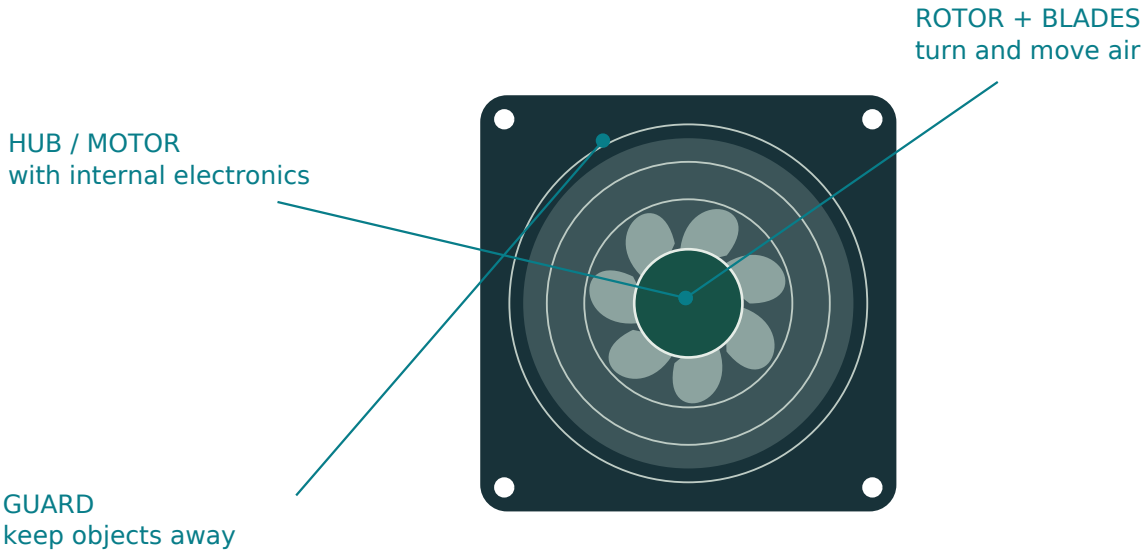
For half the time let v = 5 V and i = 0.10 A; for the other half let both be zero. Average power is 0.25 W. But average v = 2.5 V and average i = 0.050 A; their product is only 0.125 W.

Choose the instrument for the question

A slow input-energy measurement may be useful for total consumption but insufficient for switching peaks or waveform analysis. Record the boundary, sample timing, measurement method and what bandwidth is unresolved. A spreadsheet cannot recover peaks that the measurement never captured.

A brushless fan is a small controlled machine

Electrical input, rotor speed and cooling performance are related but different quantities.



Illustrative guarded axial fan

Do not model the whole fan as a fixed resistor

A brushless fan includes commutation electronics and a motor. Start-up, stall, command interface and supply behaviour depend on its design. Two-wire fans may restrict supply switching; a three-wire fan may add a tachometer output; a four-wire design may add a separate PWM input. Pin functions must come from the actual datasheet.

THREE SEPARATE CLAIMS

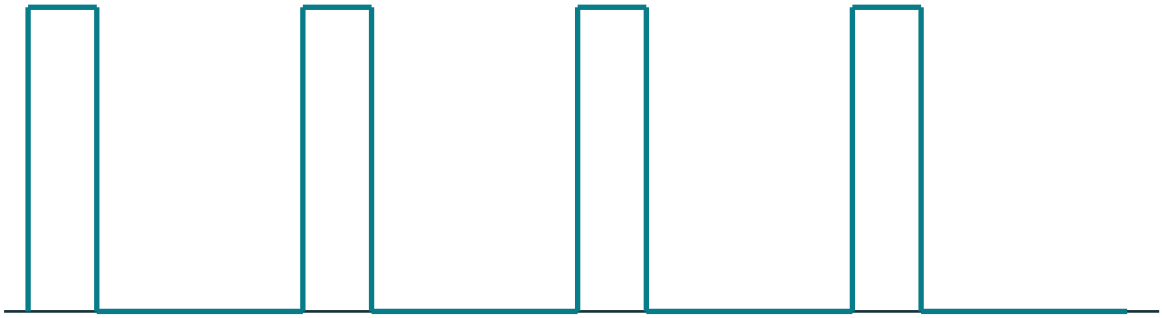
A power reading can support an input-power claim. Tachometer pulses can support a speed estimate. Cooling performance needs suitable thermal/flow evidence under a defined condition. One signal does not automatically establish all three.

The ambient-temperature logger later in this volume connects no fan. Motor switching belongs only in a reviewed setup with a selected fan and driver.

PWM changes the on-time fraction

Duty cycle describes a waveform. The physical load determines how that waveform becomes motion, heat or another response.

Example: 1 kHz, 25% duty



1 ms per cycle; 0.25 ms on, 0.75 ms off

Frequency and duty are independent settings

Frequency is cycles per second; duty is on-time/period. At 1 kHz, the period is 1 ms. A 25% duty waveform is on for 0.25 ms. For an ideal 0-5 V signal its time-average voltage is 1.25 V. The supply itself has not become a steady 1.25 V source.

POWER IS NOT THE SQUARE OF THE AVERAGE VOLTAGE

For a 100 Ω ideal resistor switched between 0 and 5 V at 25% duty, average heating is $0.25 \times 5^2/100 = 0.0625$ W. Using $1.25^2/100$ would incorrectly give 0.0156 W.

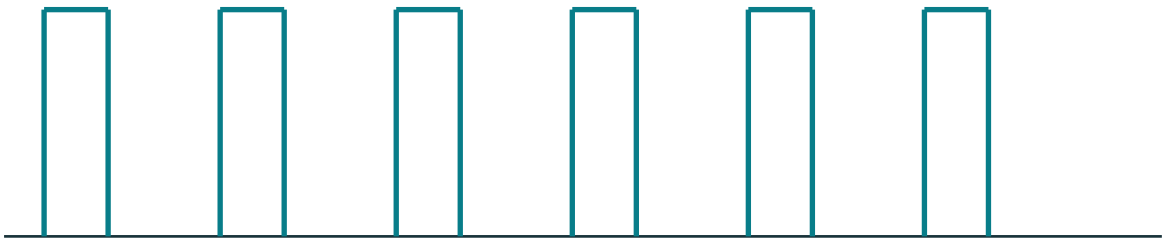
A fan’s allowed waveform is device-specific

Duty is not automatically the same fraction of speed or power. Use the manufacturer’s control input, frequency, voltage and start-up requirements. Do not transfer this resistor calculation into a fan-speed promise.

Count rotation with a tachometer signal

Pulse count becomes speed only after you know the pulses-per-revolution convention and the observation interval.

Illustrative tachometer pulses, not a power waveform



Count edges of one defined polarity over a measured interval.

Convert pulses to revolutions and minutes

Let N be pulses counted, p the specified pulses per revolution and Δt the interval in seconds. Speed = 60N/(pΔt) in revolutions per minute. If 100 pulses occur in 5 s and p = 2, speed is 60 × 100/(2 × 5) = 600 rpm. The result is an interval average, not instantaneous speed.

Check the electrical output before connecting it

A tach output may be open collector/open drain and need a pull-up compatible with the receiving input and fan specifications. Supply voltage, output rating and allowed pull-up are separate matters. A blocked fan may leave the signal stuck high or low; missing pulses are not enough to identify every cause.

NO UNIVERSAL WIRE COLOURS OR PULSE COUNT

Noctua documents two pulses per revolution and open-collector tach outputs for its specified fans. Verify the actual fan: do not assume all products share those characteristics.

Choose the bus the sensor actually uses

A connector, a voltage level and a communication protocol are different interface properties.

Interface	Signals / role	What to check
1-Wire	One shared data line plus reference; external power used in this project.	Pull-up, timing, device identity and conversion delay.
I²C	Clock SCL and bidirectional data SDA, with a shared reference.	Compatible pull-ups, addresses, bus loading and voltage levels.
SPI	Clock plus defined controller/device data and selection lines.	Clock mode, chip select, direction, voltage and timing.
UART	Transmit and receive streams under an agreed framing/baud rate.	Logic compatibility, crossed TX/RX roles and shared reference.
USB	Host/device connection used by the board’s programming or serial interface.	Actual board interface, data-capable cable and supported driver.

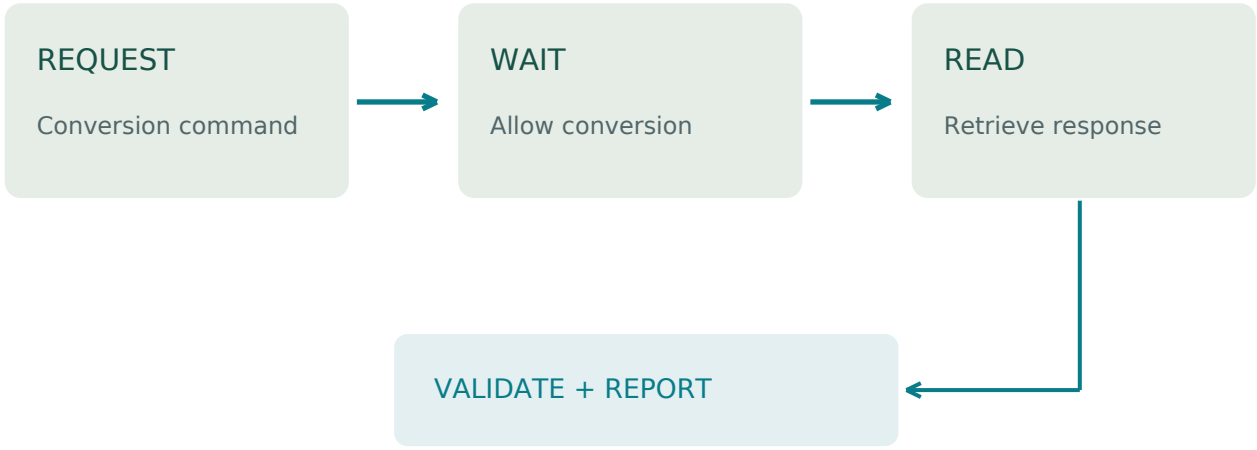
“Digital” does not mean interchangeable

A DS18B20 needs a 1-Wire driver, not an I²C address. A breakout’s power pin may tolerate a voltage that its signal pins do not. A board’s USB serial connection does not make every header pin a USB interface. Match the complete interface before connecting.

Long cables, electrical noise and multiple devices require further signal-integrity and timing checks. Begin with one sensor and short suitable connections.

A sample has a lifecycle too

The program must request conversion, allow completion, read, validate and report. Skipping a step can turn an old value into an apparent new one.



The report must identify the event

A read timestamp identifies when the program retrieved a response. It is not necessarily the exact instant when the sensor sampled temperature internally. Record the conversion request and read times where possible. A local tick clock measures elapsed time, can wrap, and is not a UTC date-time.

Separate acquisition quality from physical truth

A successful driver read and plausible range can support a “read_ok” flag. They do not establish calibration, correct mounting or immunity to every stale-device fault. On an acquisition exception, record no value and an explicit error. Keep the reason rather than reporting a convenient zero.

THIS IS NOT ACTUATOR COMPLETION

A sensor-acquisition state machine describes obtaining evidence. Rail’s execution state machine describes a requested physical action and its outcome. Keep the two responsibilities separate.

Read one sensor, preserve failed readings

Save as logger.py on a compatible MicroPython installation. This program observes temperature only. It drives no fan, relay or power output.

```
# MicroPython; original ESP32-DevKitC V4 example.
# One externally powered DS18B20; wiring on page 02-25.
from machine import Pin
import time, json, onewire, ds18x20

ds = ds18x20.DS18X20(onewire.OneWire(Pin(19)))
roms = ds.scan()
if len(roms) != 1 or roms[0][0] != 0x28:
    raise RuntimeError("Expected exactly one DS18B20")
rom, sequence = roms[0], 0

while True:
    sequence += 1
    record = {"sequence": sequence, "rom": rom.hex(),
              "value": None, "unit": "degC",
              "quality": "acquisition_error",
              "started_ticks_ms": time.ticks_ms(),
              "read_ticks_ms": None}
    try:
        ds.convert_temp()
        time.sleep_ms(750)
        value = ds.read_temp(rom)
        if not -55 <= value <= 125:
            raise ValueError("Outside sensor range")
        record["value"] = value
        record["read_ticks_ms"] = time.ticks_ms()
        record["quality"] = "read_ok"
    except Exception as exc:
        record["error"] = type(exc).__name__
    print(json.dumps(record))
    time.sleep_ms(250)
```

REQUIRES THE NAMED INTERFACE

GPIO19 is a chip GPIO identifier on the stated classic ESP32 example, not “the nineteenth header pin” or a universal ESP32-family assignment. Confirm the board map and correct 3.3 V wiring before use.

Stop with the IDE’s interrupt command. Startup intentionally stops if exactly one expected sensor is not discovered. Record the firmware release used. This example has not been tested on physical equipment for this book.

Understand what each line can claim

The example is short enough to explain completely. A printout is still evidence about an acquisition attempt, not proof of calibration.

Operation	Meaning and limitation
Pin(19) and drivers	Create a 1-Wire interface on the selected GPIO. Electrical compatibility is not checked by Python.
scan() and ROM check	Identify one device of the expected family. An identifier is not proof of genuine manufacture or correct location.
convert_temp() + wait	Request a new conversion and wait 750 ms. Loop duration also includes driver and print time.
read_temp() and range	Retrieve a temperature through the driver and reject values outside the sensor’s range. This is not a full plausibility or freshness test.
except + None	Preserve an acquisition error and absence. A valid 0 °C remains a numeric zero.
json.dumps()	Print one structured record per attempt. The host still needs to capture and preserve the output.

Work with ticks using the intended API

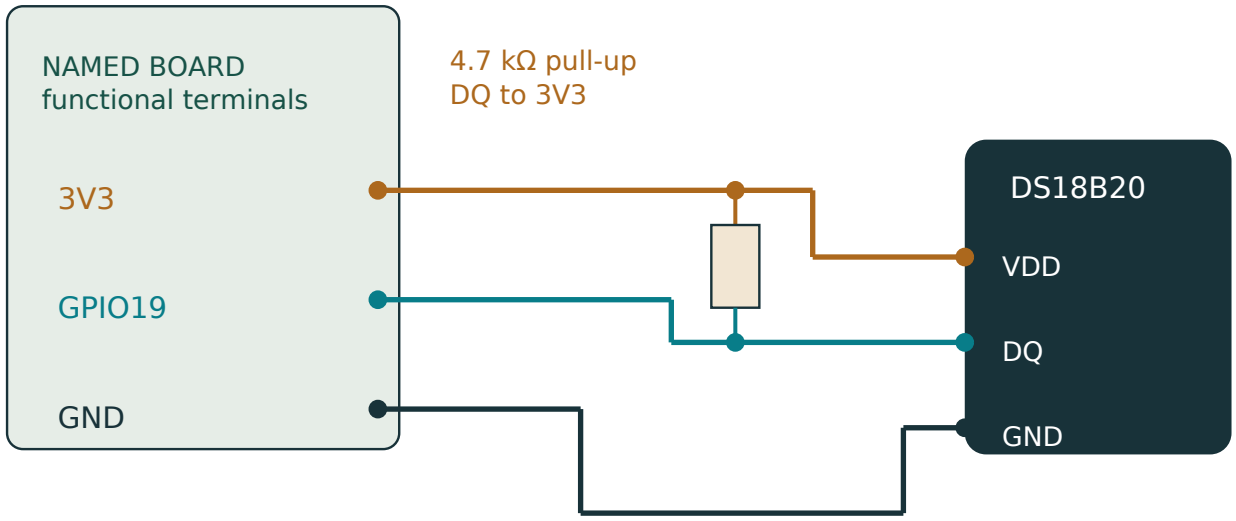
MicroPython tick counters can wrap. For elapsed time between compatible tick values use time.ticks_diff(later, earlier), within the API’s supported interval. Do not treat raw ticks as UTC or subtract them across resets. Add a host session ID and receive time when saving the serial stream.

KNOW THE MISSING FEATURES

The example has no durable file store, network service, actuator, watchdog policy or production recovery design. Those are separate additions to specify and test.

Build the ambient logger first

One board, one sensor and one pull-up are enough to practise a real measurement path. Leave the fan and MOSFET disconnected.



Terminal-to-terminal schematic. It does not establish physical pin order or probe lead colours.

Match names before matching positions

With USB disconnected, connect the verified sensor GND to board GND, sensor VDD to board 3V3, and sensor DQ to the header mapped to GPIO19. Add 4.7 kΩ from DQ to 3V3. A module may already have a pull-up; inspect its schematic rather than adding parts blindly. Use short suitable connections.

CHECK BEFORE POWER

Confirm the exact board, sensor orientation, resistor, supply route and signal voltages. Have the first arrangement reviewed by a competent person. Power this named board by USB only; do not also energise a supply header. Keep the sensor and electronics dry.

At a low DQ level, the 4.7 kΩ pull-up carries about 3.3/4,700 = 0.702 mA in the simple model. This is a data-line current, not a fan current.

Bring up one layer at a time

A calm start-up sequence makes a wiring, firmware or acquisition problem easier to locate.

1 / Identify and install

Use the official MicroPython image for the exact board/chip, and record the release. A free IDE such as Thonny can provide the supported installation/serial workflow. Firmware installation can erase existing board content; use your learning board, not equipment with required firmware.

2 / Confirm the interpreter

Before connecting the sensor, establish the board’s REPL/serial connection using a data-capable USB cable. Run a simple print statement. Disconnect USB before wiring the reviewed sensor circuit.

3 / Run logger.py

Copy the printed program and run it. Confirm one expected sensor is found and inspect several records. Note room location, mounting and the sequence/timing fields. Do not deliberately heat the setup.

4 / Preserve the session

Capture the serial output in the host/IDE, save the text and label the session, board, firmware and method. Reopen the saved record to ensure it contains the output, not only a screenshot.

5 / Test errors safely

Use an offline mock to inject an exception, NaN and a genuine zero into the acquisition function. For a hardware absence test, power down, remove the sensor, then restart: discovery should stop explicitly. Never change wiring while powered.

Do not interpret absence of serial messages as zero temperature. The host should mark a gap or timeout; the device cannot report after it loses power.

Ten checks before the next part

Write the relationship and assumptions. Questions 8-10 require explanations, not invented measurements.

1. A 10 kΩ pull-up goes to 3.3 V. What current flows when the line is held near GND?

2. Use an assumed 2.0 V LED drop, 3.3 V supply and 680 Ω resistor. Estimate current.

3. 100 µF supplies 50 mA for 1 ms. What ideal voltage decrease follows?

4. A MOSFET gate is 3.3 V and its source is 3.0 V relative to GND. Find VGS.

5. A justified 0.10 Ω on-resistance carries 0.20 A. Estimate conduction loss.

6. A divider has 20 kΩ above and 10 kΩ below its output, with 5 V input. Find unloaded output.

7. You count 100 pulses over 5 s from a documented two-pulse-per-revolution tach. Find average speed.

8. Why does a 0.0625 °C step size not establish ±0.0625 °C accuracy?

9. A fresh network message repeats an earlier sensor sample. What event does its arrival time establish?

10. A fan draws the requested power but no tach edges arrive. What can you conclude, and what remains unresolved?

Worked answers: 1-5

Compare the reasoning, units and limitations with your own attempt. Keep the correction so you can recognise the same error later.

1

Pull-up current

I = 3.3/10,000 = 0.000330 A = **0.330 mA**. This assumes a line near GND and ignores leakage and switch resistance. It says nothing about available GPIO load power.

2

LED branch

Resistor drop = 3.3 - 2.0 = 1.3 V. I = 1.3/680 = **1.91 mA** approximately. An actual LED’s forward drop varies, so this is a component-model estimate.

3

Capacitor response

C = 100 µF = 0.000100 F; Δt = 0.001 s. ΔV = IΔt/C = 0.050 × 0.001/0.000100 = **0.50 V**. The estimate assumes the capacitor supplies that entire constant current.

4

Gate reference

VGS = VG - VS = 3.3 - 3.0 = **0.3 V**. Gate voltage relative to GND is insufficient to establish the channel state. Check the actual device behaviour.

5

Conduction loss

P = I^2R = 0.20^2 × 0.10 = **0.004 W = 4 mW**. This applies only if that on-resistance is justified at the actual gate voltage and temperature; switching losses are excluded.

Worked answers: 6-10

Compare the reasoning, units and limitations with your own attempt. Keep the correction so you can recognise the same error later.

6

Divider output

Vout = 5 × 10/(20 + 10) = **1.667 V** approximately. The ratio uses matched resistance units. Input loading, tolerance and source extremes must be considered before a real ADC connection.

7

Tachometer speed

Speed = 60N/(pΔt) = 60 × 100/(2 × 5) = **600 rpm**. This is an average over the count interval, conditional on valid pulses and the stated pulses-per-revolution definition.

8

Resolution versus accuracy

The increment describes the representation’s resolution. Sensor bias, specified accuracy, mounting, calibration and thermal response remain. The genuine DS18B20’s accuracy specification is a separate statement with conditions.

9

Arrival versus capture

The arrival timestamp establishes receipt of that report. It does not establish a new physical observation or reset its age. Preserve capture/conversion context and clock limitations.

10

Power is not speed

You have a scoped input-power observation and no observed tach edges. Possibilities include a stopped fan, wrong interface, disconnected signal or unsuitable acquisition. Do not claim successful speed control or diagnose the exact fault without more evidence.

Reference: quantities and interfaces

A correct calculation and an electrically compatible connection are separate achievements.

Relationship	Use and conditions
Pull-up current ≈ V/R	When a suitable switch/output holds the signal near GND.
LED I ≈ (Vs - Vf)/R	Approximate device model; check pin and component limits.
Capacitor $E = \frac{1}{2}CV^2$	C in F, V in V, energy in J.
$\Delta V \approx I\Delta t/C$	Approximately constant net capacitor current over the interval.
Inductor $E = \frac{1}{2}LI^2$	L in H, I in A, energy in J.
MOSFET VGS = VG - VS	Gate drive is relative to source.
Conduction $P = I^2R_{DS(on)}$	RDS(on) must apply to actual drive and conditions.
Divider $V_{out} = V_{in} R_b/(R_t + R_b)$	Ideal unloaded model; input limits still need checking.
PWM duty = ton/T	T = 1/f; duty does not directly establish motor speed.
Speed = 60N/(pΔt)	N pulses; p pulses/revolution; Δt in seconds; answer rpm.

BEFORE ANY PHYSICAL CONNECTION

Confirm the exact part and pin map, reference, supply and signal ranges, current limits, boot/default state, protection and measurement boundary. Unknown is not compatible by default.

Words and parts you should recognise

Use the names precisely when documenting a circuit or asking an engineer about an interface.

GPIO A configurable digital input/output; capabilities and restrictions depend on the device.

Driver Circuit that provides the electrical drive required by the load or another control stage.

Pull-up A resistor that establishes a high default when no suitable output holds the line low.

MOSFET A transistor whose channel is controlled by gate-to-source voltage, within device limits.

ADC A converter from an analogue input to digital codes; bit depth is not an accuracy guarantee.

Shunt A known resistance used to infer branch current from a small differential voltage.

PWM Pulse-width modulation; a repeated switched waveform with an adjustable on-time fraction.

Tachometer An observation of rotation, often reported as pulses requiring a known conversion convention.

Firmware Program running on the embedded device, including acquisition timing and reporting logic.

REPL Interactive read-evaluate-print loop used to communicate with a compatible interpreter.

MINIMUM LOGGER PARTS

The named ESP32 board and data-capable USB cable; one verified DS18B20; 4.7 kΩ pull-up if not already present; short suitable connections; a host computer and compatible firmware/IDE. The named sensor is powered from the board’s 3V3 rail. No motor, battery pack or external power supply is required.

Sources and the next boundary

These are primary implementation references, not additional textbook assignments. Use the exact device and firmware documentation for a real build.

Espressif / ESP32-DevKitC V4
Board identity, supply routes and header map.

Espressif / ADC calibration
Reference variation, calibrated conversion and limitations.

Analog Devices / DS18B20
Sensor identity, accuracy, resolution and datasheet.

Analog Devices / DS18B20 versus DS18S20
Conversion times and temperature increments.

MicroPython / ESP32 and 1-Wire
Pin identifiers and supported acquisition API.

MicroPython / time
Tick arithmetic and wrap limitations.

Texas Instruments / INA219
Shunt and bus monitoring; settings and conversion timing.

Noctua / PWM specification
Manufacturer-specific fan control and tachometer behaviour.

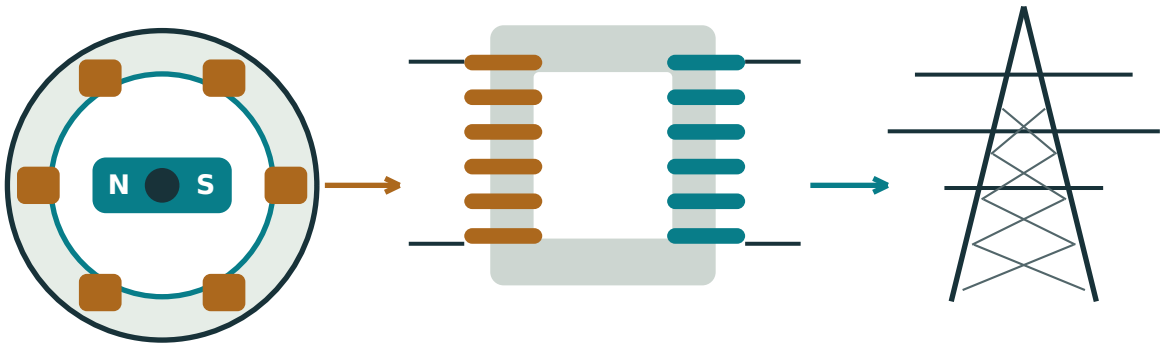
NEXT: AC, MACHINES, NETWORKS AND PROTECTION

Part 03 moves from this low-voltage component model to AC waveforms, three phases, generators, transformers and protective boundaries. Merge in numeric filename order; labels such as 02-25 retain their meaning.

Part 02 / Edition 01 / 16 October 2026. Original text, diagrams and code prepared with ChatGPT; broad component background also draws on the two user-supplied books. Numerical cases are assumed. Code is checked with mocked interfaces, not physical hardware. No independent engineering peer review is claimed.

AC POWER, MACHINES & NETWORKS

Part 03 / Understand the electrical system behind the sensor, the cooling plant and the computing load.



GENERATOR
mechanical to electrical

TRANSFORMER
voltage and current

NETWORK
power transfer

Functional illustrations. Energy moves through fields and circuits; the arrows show the system story.

Move from a board to an infrastructure model

You already know voltage, current, power and measurement quality. Now learn what changes when voltage alternates, when three phases share a network and when rotating machines exchange energy with it. Every calculation states its simplified conditions.

THE PRACTICAL OUTCOME

Read a basic single-line diagram; distinguish kW, kVA and kvar; estimate current and losses; explain grid frequency and protection; and calculate power from synthetic waveform samples.

A route through the electrical system

Read in order on the first pass. The later reference pages let you return directly to a calculation or term.

Pages	Question to answer
3-7	How do waveforms, RMS, phase and impedance describe AC?
8-10	What do active power, reactive power and distortion mean?
11-13	Why three phases, and how do line and phase values differ?
14-18	How do transformers, hydro generators and motors exchange energy?
19-21	How do frequency, inertia and network paths fit together?
22-24	What do faults, protection, earthing and isolation each mean?
25-29	What should you measure, and what can you practise without mains?
30-34	Practice questions, worked answers, formulas and vocabulary.
35-36	Primary references and the complete eight-part series.

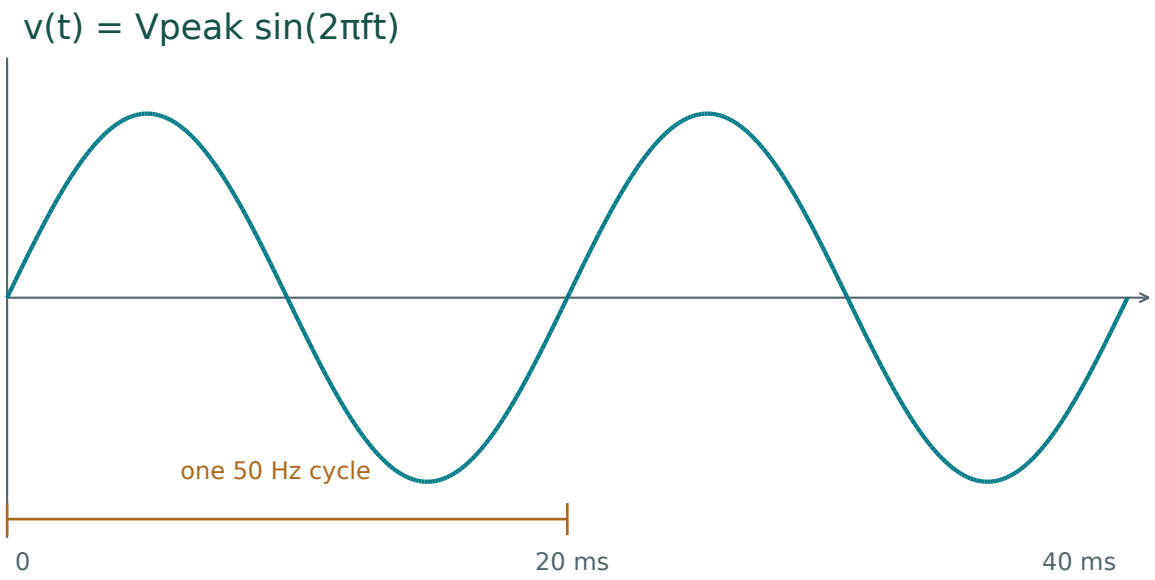
The learning boundary for this volume

Physical work stays with the USB-powered apparatus from Part 02. AC waveforms, faults and machines here are drawings, calculations and computer models. Do not connect an ESP32, breadboard, ordinary oscilloscope or improvised divider to a wall outlet or switchboard.

Page labels such as 03-14 remain usable after you merge the volumes.

AC reverses polarity through time

A sine wave is one useful mathematical model. Real waveforms can contain distortion, noise, offsets and transients.



Give the axes a physical meaning

Voltage is a difference between two stated points. Positive and negative signs describe polarity relative to that reference. A 50 Hz periodic waveform repeats 50 times per second. Its period is T = 1/f = 0.020 s = 20 ms; angular frequency is ω = 2πf = 314.16 rad/s.

Frequency, amplitude and phase are separate

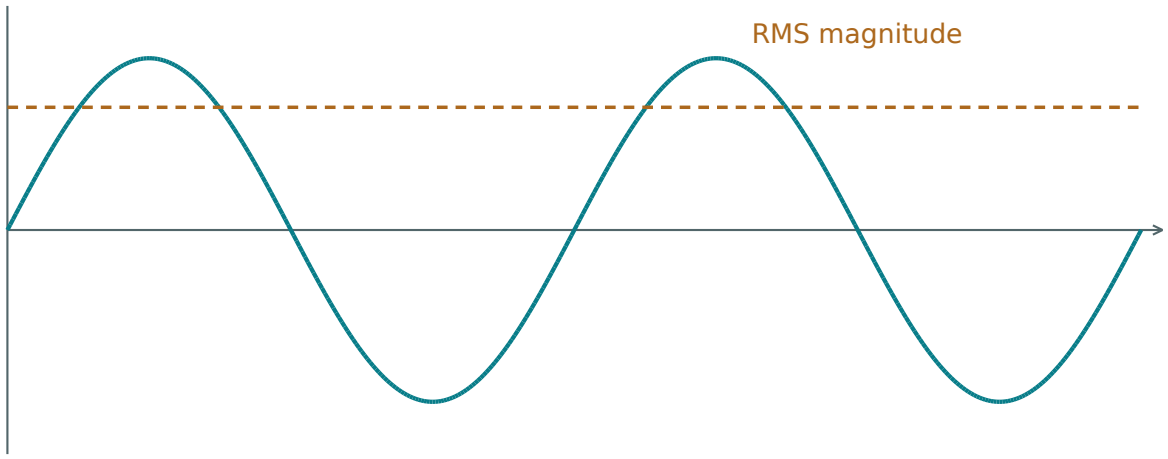
Amplitude changes the height of the waveform. Frequency changes its repetition rate. Phase shifts its position relative to another waveform or a chosen clock. Frequency is not power: a lightly loaded and heavily loaded synchronous system can both operate near 50 Hz.

A MODEL, NOT A TEST INSTRUCTION

All AC voltages in this volume are hypothetical calculation values unless explicitly labelled otherwise. A plotted 230 V waveform is not a measured Hobart supply.

RMS connects a waveform to heating

Root mean square means square the samples, take their mean, then take the square root. Do not average the signed values instead.



The RMS line is a magnitude, not the waveform’s time average.

Start with the definition

For equally spaced samples over a suitable window, $V_{rms} = \sqrt{[\Sigma(v^2)/N]}$. An ideal sine wave with no DC offset has $V_{rms} = V_{peak}/\sqrt{2}$. Its signed mean over complete cycles is zero, but it still heats a resistor. RMS expresses the equivalent DC heating for the same resistance.

WORKED EXAMPLE

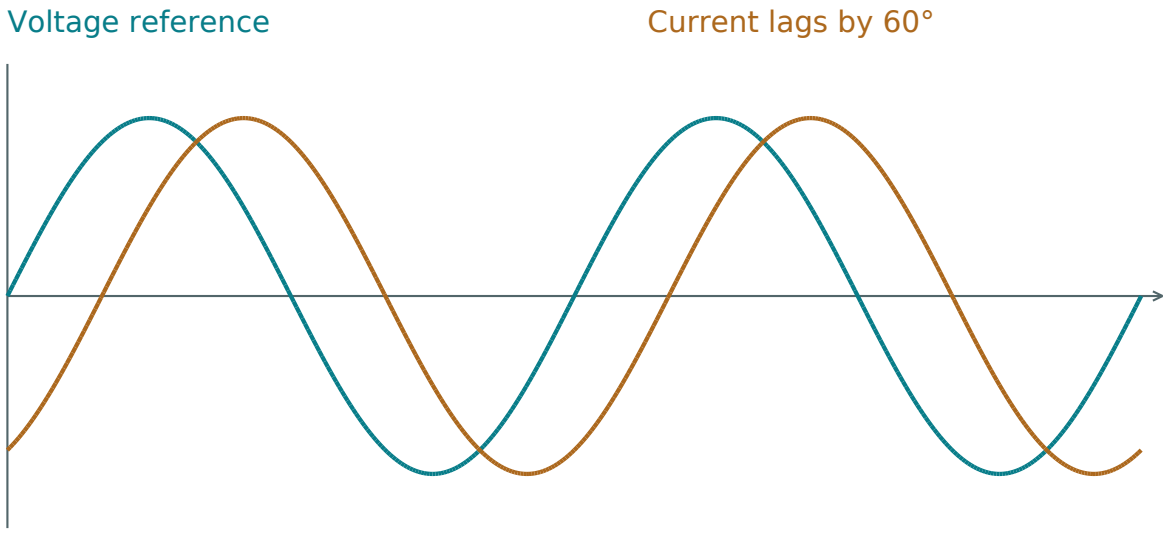
Assume an ideal 230 V RMS sine wave. $V_{peak} = 230\sqrt{2} = 325.27$ V. Across an ideal 100 Ω resistor, $I_{rms} = 2.30$ A and average heating is $V^2/R = 529$ W. These are calculations, not a circuit to assemble.

Keep the condition with the shortcut

Dividing by √2 works for a sine wave. A 10 V peak square wave alternating between +10 V and -10 V has 10 V RMS, not 7.07 V.

Phase describes relative timing

Compare waveforms at the same frequency and with a stated sign convention. A phase angle is not a time delay at every frequency.



Amplitudes normalised for comparison; voltage and current have different units.

Read the word “lags” from left to right

In this example the current reaches each corresponding point later than the voltage. A 60° lag is one sixth of a cycle. At 50 Hz, Δt = (60/360) × 20 ms = 3.33 ms. The same angle at 100 Hz corresponds to 1.67 ms.

A phasor is a compact steady-state description

A phasor encodes the magnitude and phase of one sinusoidal frequency. It helps add voltages with different phase angles. It is not an arrow showing the physical path of electricity. Harmonics need separate frequency components; a single phase angle does not describe an arbitrary waveform.

RECORD THE CONVENTION

Here voltage is the reference and positive φ means current lags voltage. Other software can use other sign conventions. State whether your magnitudes are peak or RMS.

Inductors and capacitors shift the response

Resistance dissipates energy. Ideal inductance and capacitance store and return energy; their AC opposition depends on frequency.

Element	Sinusoidal relation	Current relative to voltage
Resistor R	Magnitude ratio = R	In phase in the ideal model.
Inductor L	XL = 2πfL	Current lags by 90° for a pure ideal inductor.
Capacitor C	XC = 1/(2πfC)	Current leads by 90° for a pure ideal capacitor.

Work in base units first

At 50 Hz with L = 10 mH = 0.010 H, XL = 2π × 50 × 0.010 = 3.142 Ω. With C = 100 µF = 0.000100 F, XC = 1/(2π × 50 × 0.000100) = 31.83 Ω. Doubling the frequency doubles XL and halves XC.

THE TIME-DOMAIN IDEAS STILL APPLY

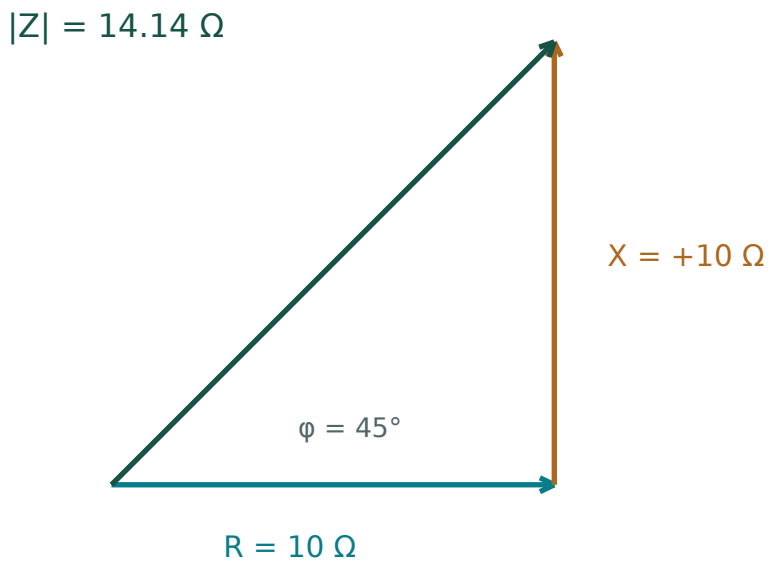
For the ideal inductor v = L di/dt. For the ideal capacitor i = C dv/dt. These equations describe changing signals; the reactance formulas describe their single-frequency sinusoidal steady state.

Real parts add limits

Windings have resistance and magnetic limits; capacitors have loss, voltage limits and parasitic inductance. Reactance alone is not a component-selection specification.

Impedance combines magnitude and phase

Adding ohms as ordinary numbers loses the phase relationship. In a simple series R-L-C model, combine resistance and net reactance.



Separate the two components

Write Z = R + j(XL - XC), where $j^2 = -1$. For this series model, $|Z| = \sqrt{[R^2 + (XL - XC)^2]}$. The phase angle is atan2(XL - XC, R). Positive net reactance gives lagging current under this book’s convention.

WORKED SERIES CASE

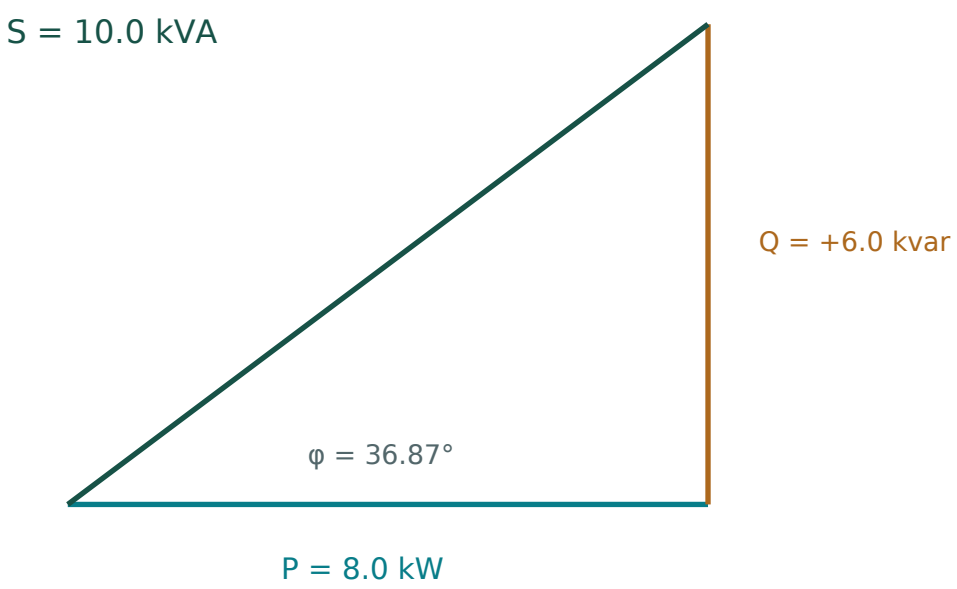
Assume R = 10 Ω, net X = +10 Ω and V = 10 V RMS at one frequency. |Z| = 14.142 Ω; I = 10/14.142 = 0.7071 A RMS. Average resistive power is I^2R = 5.00 W; the current lags by 45°.

Resonance needs a bounded model

When XL = XC, this ideal series impedance becomes R. Internal component voltages can still be large. Real networks include damping, controls and multiple resonances; do not infer equipment safety from cancellation.

Three power quantities answer different questions

For sinusoidal voltage and current, active, reactive and apparent power form a useful triangle. Keep their units separate.



Active power transfers net energy

P = Vrms Irms cosφ in the sinusoidal single-phase case. Its unit is watts (W). Q = Vrms Irms sinφ in var describes the reactive exchange in that same model. Apparent power S = Vrms Irms in VA expresses voltage-current loading. $S^2 = P^2 + Q^2$ here.

FROM POWER TO ENERGY

If P remains 8 kW for 0.5 h, active energy is 4 kWh. Neither 10 kVA × 0.5 h nor 6 kvar × 0.5 h is the same active-energy quantity. In this example power factor P/S = 0.80.

Keep the model boundary visible

The triangle shown assumes sinusoidal conditions. With distorted waveforms use average instantaneous power for P and RMS values for S; the same two-axis triangle need not capture all nonactive components.

Power factor changes the required current

For a given active power and voltage, lower true power factor means more RMS current. It does not mean all the difference is wasted active power.

Assumed load	PF = 1.00	PF = 0.80
Active power P	1,000 W	1,000 W
Voltage V	230 V RMS	230 V RMS
$I = P/(V \times PF)$	4.348 A	5.435 A
S = VI	1,000 VA	1,250 VA
I^2R in 0.50 Ω loop	9.45 W	14.77 W

Name the system boundary

The two hypothetical loads deliver the same active power at their terminals. The lower-PF case loads the upstream equipment more heavily. At an assumed fixed loop resistance, its conductor heating is $1/(0.8^2) = 1.5625$ times the unity-PF case. The listed terminal voltage is held fixed for this comparison.

Correction is an engineering design

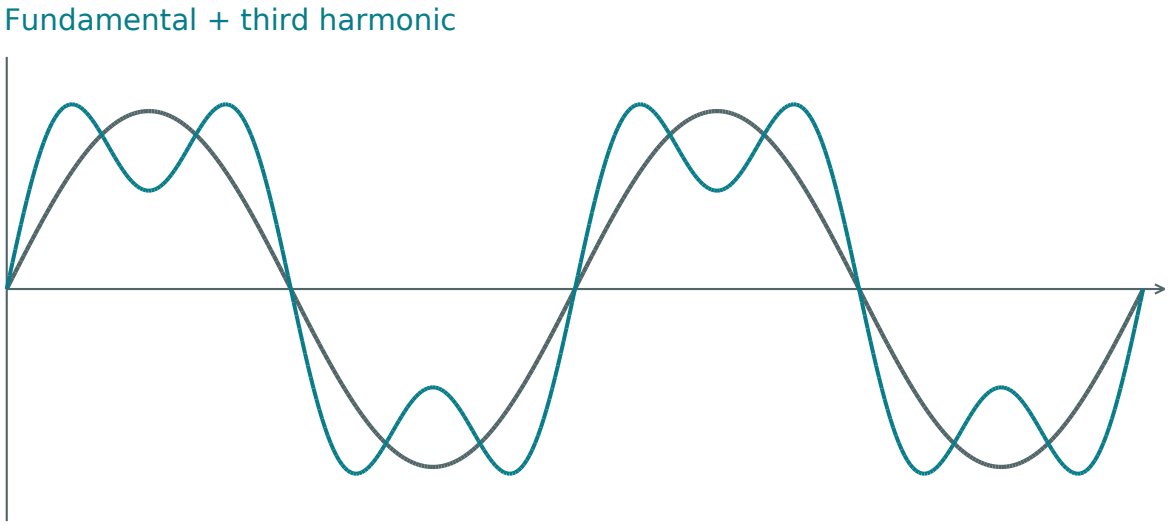
For a sinusoidal inductive load, reactive compensation can reduce upstream current. A nonlinear converter may instead need waveform shaping or filtering. Adding a capacitor without studying resonance, switching, voltage and protection is not a general fix.

A BETTER QUESTION

Ask “what current, heat, voltage drop and waveform result?” rather than interpreting power factor as an appliance’s conversion efficiency.

Distortion breaks the simple cosine shortcut

A nonlinear load can draw a nonsinusoidal current even when voltage is close to sinusoidal. Fundamental phase and true power factor then differ.



Normalised illustrative signals. Not a measured server-power waveform.

Calculate from the waveform

True PF = P/(Vrms Irms). For periodic current, THDi = √(I2² + I3² + ...)/I1, with harmonic RMS magnitudes. If voltage is sinusoidal and the fundamental current is in phase, true PF = 1/√(1 + THDi²). This is not simply cosφ for the total waveform.

A NUMERICAL CHECK

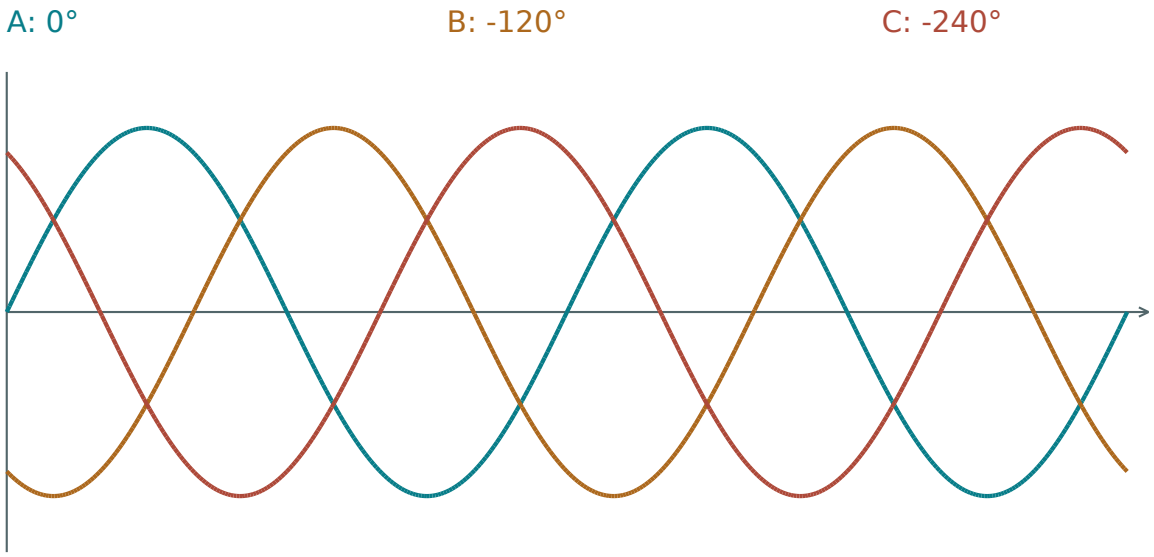
Assume I1 = 10 A RMS and a third harmonic I3 = 5 A RMS, with all others zero. Irms = √(10² + 5²) = 11.18 A. THDi = 50%. With ideal sinusoidal voltage and zero fundamental phase difference, true PF = 0.8944.

The instrument must see the waveform

An average-responding meter or slow telemetry feed cannot establish harmonic content. Record bandwidth, sampling, sensor limits and aggregation windows.

Three phases share one electrical cycle

A balanced positive-sequence set has equal magnitudes separated by 120 electrical degrees. The three waveforms are not three separate frequencies.



Balanced sinusoidal phase-to-neutral voltages, normalised.

Use balance as an assumption, not a guarantee

The instantaneous sum of these three phase voltages is zero. For a balanced load their phase currents also sum to zero, so ideal neutral current is zero. Real installations can be unbalanced, and some harmonics add in the neutral instead of cancelling.

Why it matters for machines and power transfer

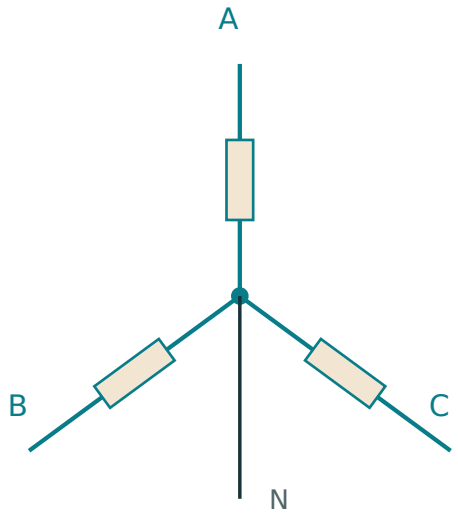
Three spatially arranged windings driven by three phase-shifted currents can create a rotating magnetic field. Balanced sinusoidal three-phase loads also have constant total instantaneous power, even though each phase’s power varies. Motors and transmission exploit these properties.

WHAT A BALANCE CHECK NEEDS

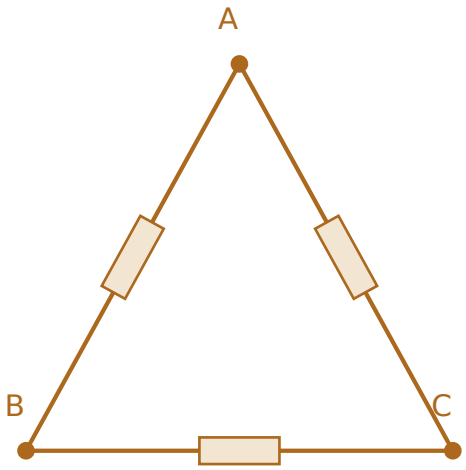
Measure or obtain data for all three phases with compatible windows. A total kW figure alone does not establish phase balance or acceptable neutral current.

Line values are not always winding values

Star and delta describe how three windings or loads connect. The relationships below assume balanced sinusoidal conditions.



STAR / Y



DELTA / Δ

Balanced connection	Voltage	Current
Star	Vline = √3 Vphase	Iline = Iphase
Delta	Vline = Vphase	Iline = √3 Iphase

KEEP THE SUBSCRIPTS

With 230 V phase-to-neutral in a balanced star system, line-to-line voltage is 230√3 = 398.37 V. The commonly rounded pair “230/400 V” is not an exact arithmetic identity. These are assumed examples, not a supply tolerance statement.

Estimate three-phase loading correctly

For a balanced load use line-to-line voltage and line current together. Do not multiply by three again after using the √3 formula.

$$P = \sqrt{3}\, V_{LL}\, I_L\, PF$$

$$S = \sqrt{3}\, V_{LL}\, I_L$$

For sinusoidal loads: $Q = \sqrt{3}\, V_{LL}\, I_L \sin\varphi$

VLL: line-to-line RMS voltage. IL: line RMS current.

Worked load: a hypothetical 100 kW input

Assume 400 V line-to-line, balanced phases and true PF = 0.95. IL = 100,000/(√3 × 400 × 0.95) = 151.93 A. Apparent power is 100/0.95 = 105.26 kVA. If the waveforms are sinusoidal, $|Q| = \sqrt{105.26^2 - 100^2} = 32.87$ kvar.

Scale is not a cable-selection method

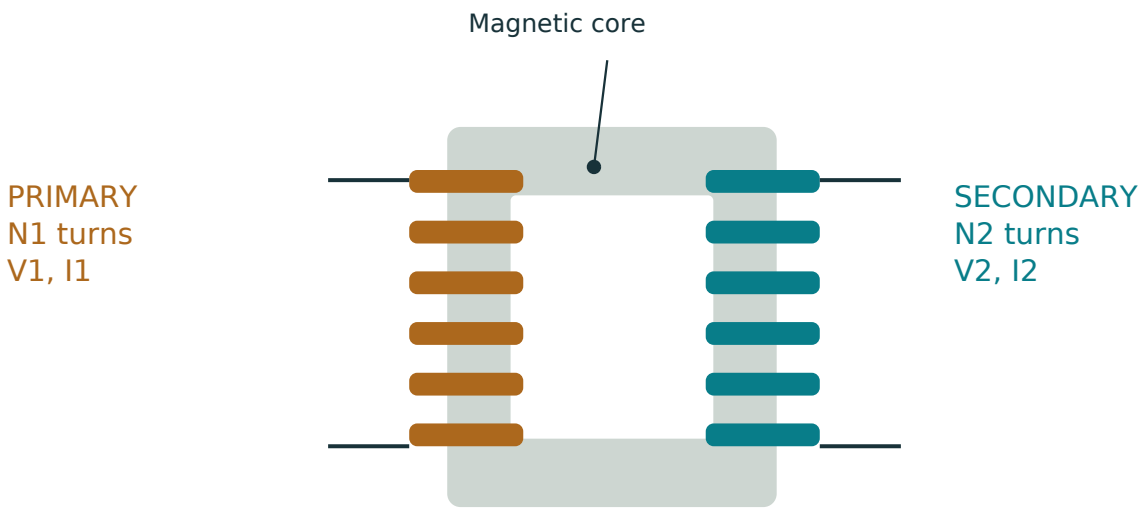
Current is one input to equipment selection. Installation method, conductor temperature, voltage drop, fault withstand, harmonic content and protective coordination also matter. A 100 kW IT load may have upstream cooling and conversion loads outside its measurement boundary.

FOR AN UNBALANCED SYSTEM

Calculate each phase’s active power from compatible voltage and current signals and add the results. A single average current placed into the balanced formula can hide the most heavily loaded phase.

A transformer couples windings through flux

Changing magnetic flux links the windings. The turns ratio sets an ideal voltage ratio; current changes in the opposite direction.



Separate windings shown. An autotransformer has a shared winding and does not provide the same separation.

Use conservation with the ideal ratio

V2/V1 = N2/N1 and I2/I1 = N1/N2 for ideal magnitudes under the stated load conditions. A 10:1 step-down changes 1,000 V and 1 A into 100 V and 10 A at unity power factor: 1,000 W on both sides before losses.

WHAT THE CORE DOES NOT DO

The transformer does not create energy or normally change frequency. Applying steady DC to a conventional AC transformer does not give steady transformed output and can cause excessive current. Real designs have loss, leakage impedance and saturation limits.

An actual transformer’s insulation system, earthing and ratings must be checked; a symbol alone is not a safety guarantee.

Higher voltage can reduce transfer losses

Compare equal active power at different voltages while holding power factor and conductor resistance fixed. Real network design also includes insulation and reactive effects.

Assumed balanced transfer	400 V LL	11 kV LL
Active power / PF	100 kW / 1	100 kW / 1
Line current	144.34 A	5.249 A
Resistance per phase	0.10 Ω	0.10 Ω
$3I^2R$ conductor loss	6,250 W	8.264 W

Read the comparison honestly

The model holds receiving-terminal voltage, delivered power and resistance fixed. Raising voltage by 27.5 reduces current by 27.5 and resistive loss by $27.5^2 = 756.25$. It does not include transformers, line charging, voltage limits or changes to conductor design.

Voltage drop depends on impedance too

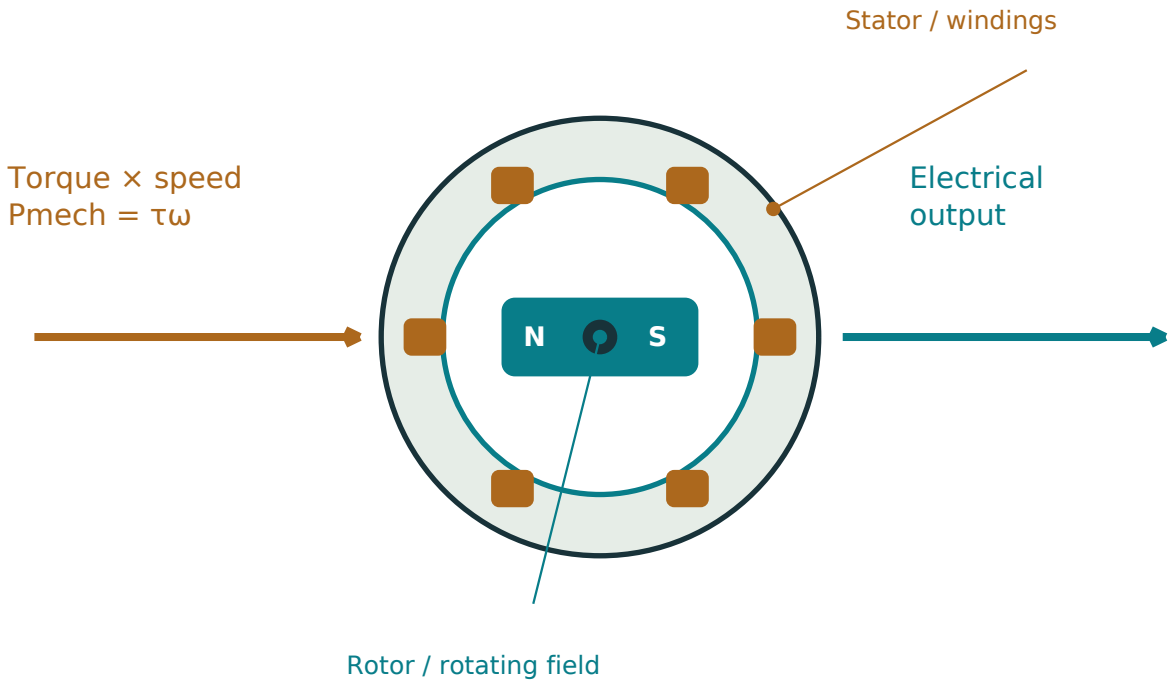
A simplified DC or unity-PF resistive comparison gives $\Delta V \approx IR$ per conductor path as defined. In AC networks, reactance and phase angle also contribute. Voltage regulation is therefore not solved by a power-loss calculation alone.

TRANSFORMER LOSS HAS TWO MAIN FAMILIES

Winding heating generally increases with current squared. Core loss depends strongly on voltage, frequency and core design and can remain while a transformer is energised at little load. Cooling must remove both.

A generator converts torque into electrical power

A turbine supplies mechanical power to a rotor. Magnetic interaction with stationary windings enables electrical energy transfer.



Relate poles, speed and frequency

For a synchronous machine, mechanical speed ns = 120f/p rpm, where p is the total number of magnetic poles. At 50 Hz, a four-pole rotor runs at 1,500 rpm; a twelve-pole rotor runs at 500 rpm. Electrical angle and mechanical angle differ by the number of pole pairs.

Separate torque control from excitation

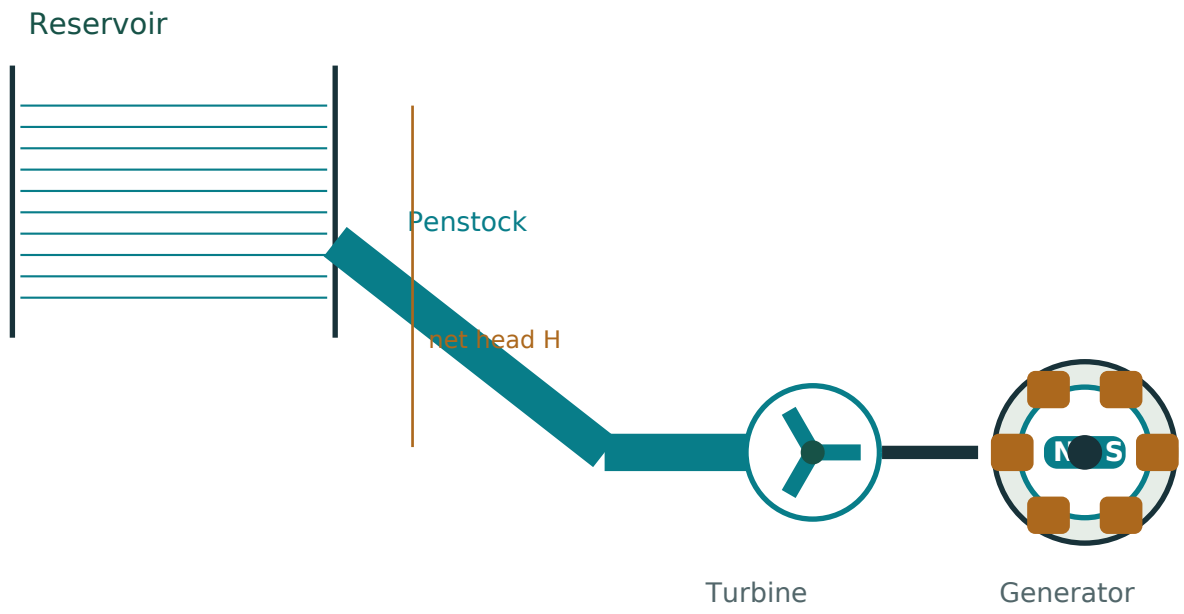
In a grid-connected synchronous generating unit, turbine input primarily influences active-power exchange; excitation strongly influences reactive-power exchange and terminal voltage. These are coupled physical controls with limits, not two independent knobs in every condition.

A ROTATING MACHINE IS NOT AN IDEAL SOURCE

Its capability depends on rotor, stator, thermal, magnetic and stability limits. A kW rating alone cannot specify its acceptable operating envelope.

Hydro connects water, torque and the grid

The reservoir and waterway provide hydraulic energy. The turbine, shaft, generator and transformer each impose their own constraints.



Estimate hydraulic conversion

A first model is Pelec = ρgqHη. Here q is water flow in m³/s, H is net head in metres, ρ is water density, and η combines the modelled turbine-generator efficiency. Use net head after waterway losses, not automatically the elevation difference on a map.

HYPOTHETICAL UNIT

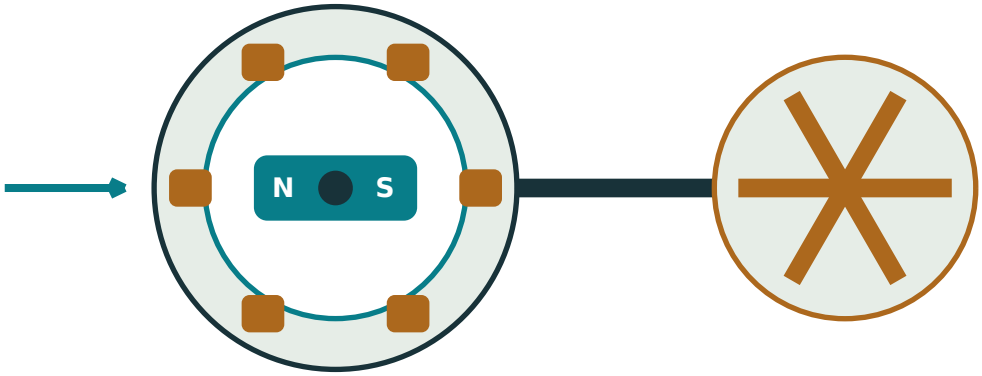
With ρ = 1,000 kg/m³, g = 9.81 m/s², q = 10 m³/s, H = 100 m and η = 0.90, electrical power is 8.829 MW. One hour at that constant output delivers 8.829 MWh; it does not describe the reservoir’s full stored energy.

Why this matters to your Tasmania direction

Changing output also involves water dynamics, ramp limits, vibration and plant constraints. This small equation is an entry point, not a verified model of a Hydro Tasmania unit.

A motor turns electrical input into shaft power

A motor is a physical load with starting behaviour, losses and mechanical dynamics. A nameplate’s shaft kW is not necessarily its electrical input.



Calculate efficiency at a stated operating point

If shaft output is 15 kW and motor efficiency is 90%, electrical active input is 15/0.90 = 16.67 kW. At 400 V line-to-line and PF = 0.85, a balanced input draws about 28.30 A. Drive losses are extra if the measurement boundary includes a separate drive.

Speed is a machine-specific relationship

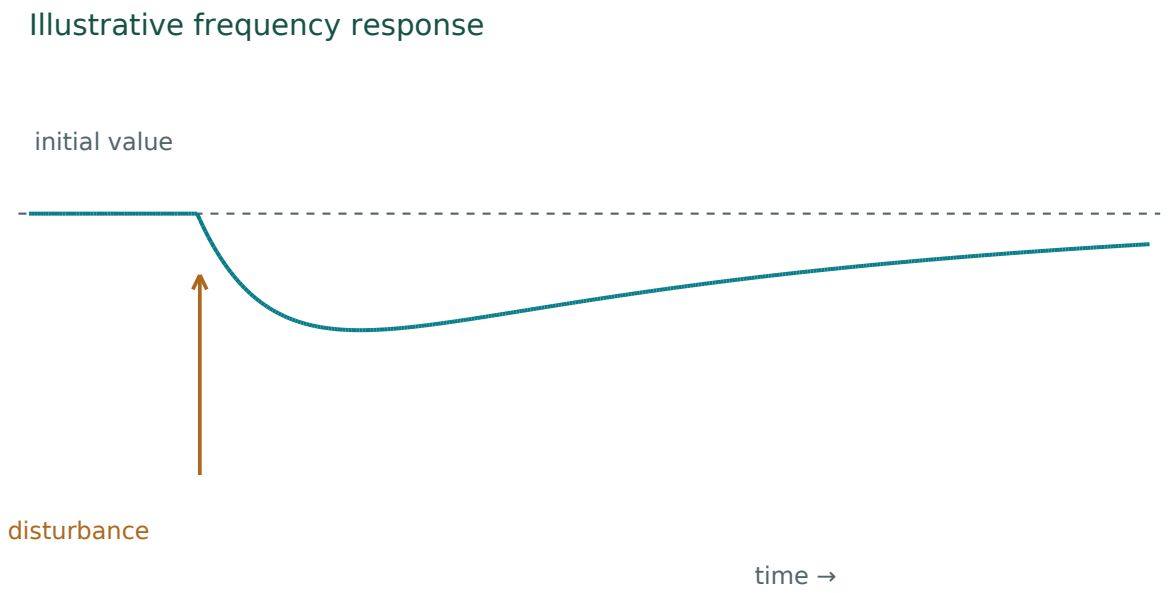
A four-pole induction motor at 50 Hz has a 1,500 rpm synchronous field. If its rotor runs at 1,455 rpm, slip is (1,500 - 1,455)/1,500 = 3%. A variable-frequency drive changes the supplied frequency and voltage through power electronics. A brushless DC fan uses its own commutation electronics.

CONNECT TO COOLING WITHOUT SKIPPING THE MECHANICS

Cooling depends on airflow, pressure and thermal paths. Motor power, rotational speed and useful cooling are related measurements, not interchangeable readings.

Frequency reveals a dynamic power imbalance

In a simplified synchronous system, an active-power deficit initially takes energy from rotating masses. Frequency falls until responses and loads change the balance.



The balance includes losses and transfers

At a defined system boundary, generation plus imports minus exports must cover demand, electrical losses and changes in stored energy. If a synchronous generator suddenly exports more electrical power than its turbine supplies, the difference initially reduces rotor kinetic energy.

Responses operate on different time scales

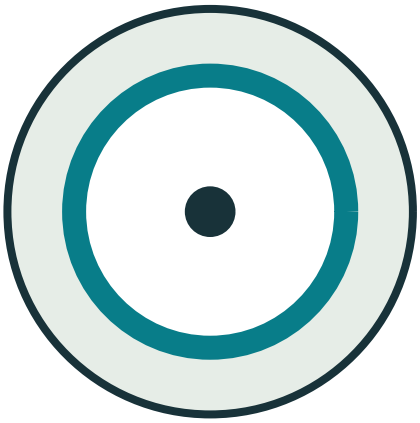
Inertial response begins through physical energy exchange. Governors, fast inverter controls, storage, responsive loads and later dispatch can change active power. Response limits, energy availability, delays and location all matter. Frequency is not a direct meter of total demand.

VOLTAGE AND FREQUENCY ARE DIFFERENT VARIABLES

Voltage behaviour depends strongly on local impedance, reactive power and control. Frequency reflects system dynamics. A voltage sag need not imply the same event as a frequency excursion.

Inertia slows change; it does not refill itself

Rotating kinetic energy can resist rapid frequency change. Sustained recovery still needs a new active-power balance.



$$E_{rot} = \frac{1}{2} J \omega^2$$

J: moment of inertia, kg·m²
ω: mechanical speed, rad/s

H = stored kinetic energy
÷ rated apparent power

Use one consistent base

The inertia constant H is often expressed in seconds because MJ/MVA has units of seconds. For a simplified coherent synchronous equivalent, initial df/dt ≈ f0 ΔP/(2HSbase). Define ΔP as mechanical input minus electrical output, use the same power base and neglect damping and controls for this initial estimate.

INITIAL-SLOPE EXAMPLE

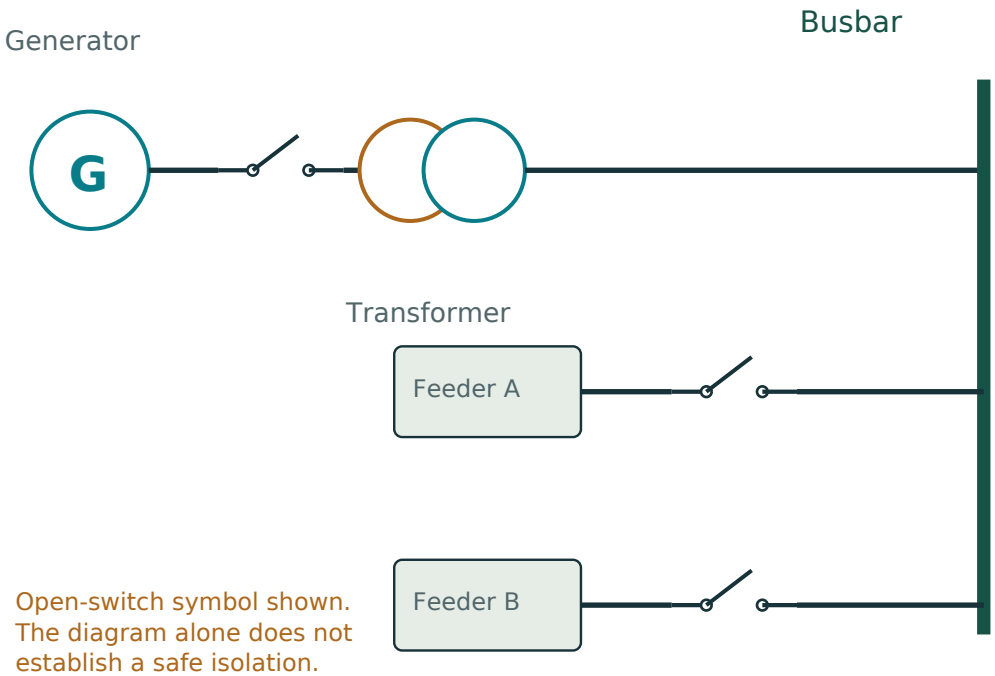
Assume f0 = 50 Hz, H = 4 s and Sbase = 100 MVA. A 10 MW deficit gives ΔP = -10 MW and initial df/dt ≈ -0.625 Hz/s. This is a short-time estimate, not a prediction of frequency after one second or a Tasmanian system model.

Inverter response is designed, not just spinning

Grid-forming controls and fast response can support stability through different mechanisms and limits. Do not equate a battery’s MWh rating with synchronous inertia.

Read a single-line diagram as a network

One drawn line can represent a three-phase circuit. The drawing shows connectivity and equipment boundaries, not every physical conductor.



Trace the actual path

Identify sources, busbars, transformers, breakers, feeders and loads. Read voltage levels and equipment identifiers. A closed path can transmit power; an open path changes connectivity. Interconnected networks split flows according to impedances, voltage magnitudes and phase angles, not a software-selected itinerary.

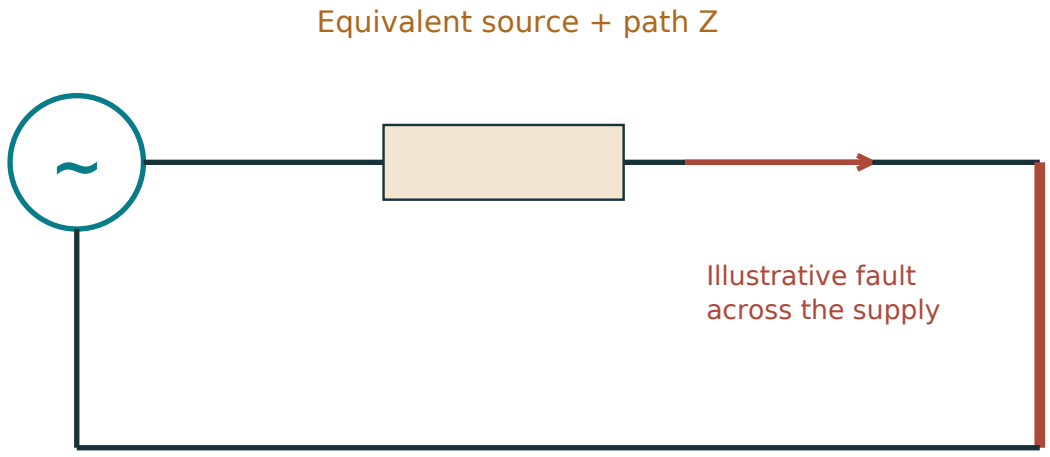
Find the boundary for a proposed controller

Where is the meter? Which load sits behind it? Can a second source backfeed that bus? Which authorised equipment owns switching and protection? Lupine must use a documented interface and accurate topology; it cannot infer these facts from a neat drawing.

Functional single-line illustration, not a switching schedule or installation design.

A fault changes the circuit, not just the load

Fault current is constrained by source and network impedance. The current a load normally draws is not the current a fault can produce.



Start with a deliberately small calculation

In an idealised single-phase equivalent, a 230 V RMS source behind $|Z| = 0.20\ \Omega$ gives a prospective symmetrical RMS current of about 1,150 A for a zero-impedance fault. That is not an installation fault study: actual conditions include impedance angle, asymmetry, sources, earthing and protection timing.

Thermal and mechanical stress rise rapidly

Conductor heating during a short interval is related to $\int i^2 dt$. Electromagnetic forces also increase strongly with current. A breaker must be able to interrupt the prospective current at its location, not merely carry the normal load. Arc hazards and DC interruption require their own analysis.

DO NOT REPRODUCE THIS EXPERIMENT

Fault diagrams are calculation models. Do not short a supply, battery, capacitor bank or wall outlet to “see what happens”.

Protection must detect and interrupt the right fault

Protection is engineered around electrical hazards and equipment limits. Supervisory software can request an action; it does not replace the protective function.

Function	What it addresses	What it does not establish
Overload protection	Excess load current and related heating over time.	Protection from every electric-shock scenario.
Short-circuit protection	High fault current, with suitable interruption capability.	Safe isolation just because a handle says OFF.
Residual-current protection	An imbalance in current through monitored live conductors.	Overload protection unless combined in an appropriate device.
Relay + breaker scheme	A protection relay detects defined conditions and trips rated switchgear.	That a network message or acknowledgement proves contacts opened.

Selectivity keeps a local fault local where possible

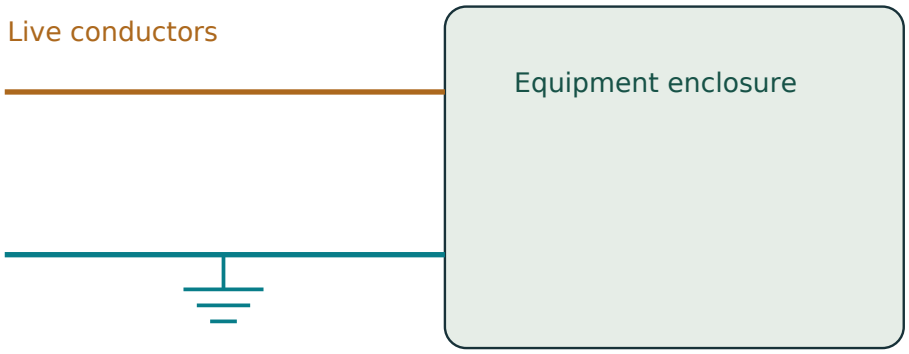
Downstream and upstream devices must coordinate over relevant fault levels and operating conditions. A downstream device may clear first; upstream backup remains available. This requires real device curves and studies. A generic timing sketch cannot set a relay or prove selectivity.

FOR LUPINE

Protective trips, interlocks and local equipment limits remain authoritative. A supervisory request must not bypass them, conceal them or automatically re-energise equipment after a trip.

Earth, neutral and OFF mean different things

A reference conductor, a protective conductor and a verified isolation are distinct concepts. They cannot be substituted for one another.



Protective bonding path shown conceptually. No neutral-earth link location is specified.

An RCD observes a difference

A residual-current device compares the currents through its monitored live conductors. Some current returning by another path can create an imbalance and trigger it. A dangerous active-to-neutral contact may remain balanced, so an RCD is not protection against every possible shock. Overcurrent protection is a separate function.

A software stop is not an isolation

A stopped motor, zero-power reading or accepted OFF command does not prove the absence of dangerous voltage. Multiple sources, stored energy, automatic restart and failed contacts must be considered. Physical isolation and verification are work for appropriately qualified people following an approved procedure.

YOUR HOME LEARNING BOUNDARY

Use drawings and synthetic data here. Continue only the USB-level physical project in Part 02. Do not open switchboards, remove protective covers or attach test leads to mains.

Choose measurements for the question

A five-minute average, a one-second trend and a fast waveform capture describe different aspects of the same system.

Question	Evidence needed	Slow-average limits
How much energy?	Integrated active power over a defined interval and boundary.	Short excursions and their timing.
How heavily loaded?	RMS current by phase, voltage, PF and appropriate time windows.	Peak shape and phase imbalance if only totals are kept.
What changed during a fault?	Suitable waveform/event capture, triggers and synchronised context.	Subcycle behaviour and precise sequence.
Did a command work?	Instruction identity, acknowledgement, fresh physical feedback and limitations.	Causality, contact state or actuator outcome on its own.

Sampling has a physical front end

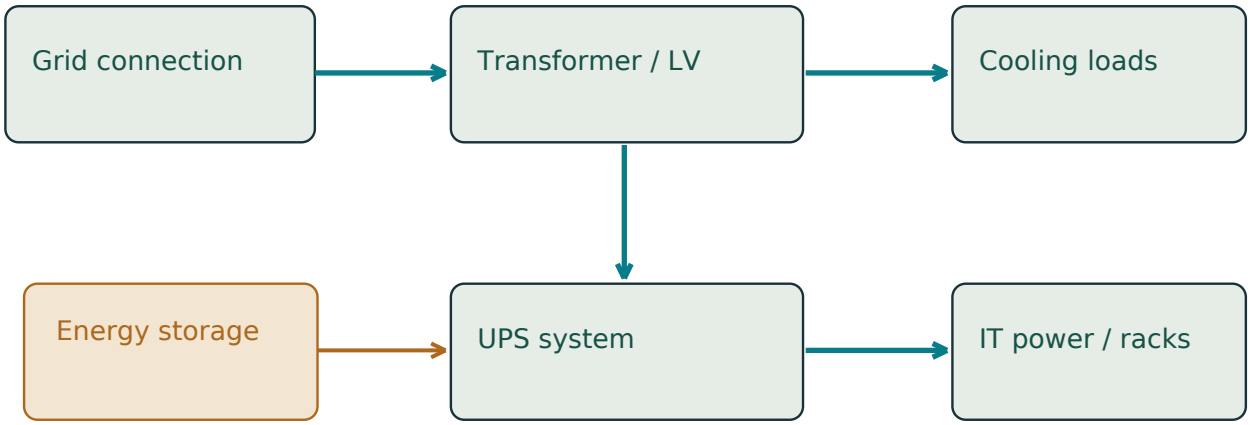
For a band-limited signal, sampling must exceed twice the highest retained frequency; practical systems also need anti-alias filtering, timing quality and sensor bandwidth. A 50 Hz fundamental does not make 100 samples/s a sufficient general power-quality measurement system.

TIME STAMPS NEED SEMANTICS

Record capture window, receive time, units, calibration status, source, uncertainty and aggregation method. A new network packet may contain an old measurement.

Follow the data-centre power path

A conceptual site turns grid power into useful computing and heat. Actual designs vary; this is not a description of a Firmus facility.



Topology is illustrative. Backup generation, transfer paths and bypasses are omitted.

The input power depends on the whole boundary

IT equipment draws power through conversion stages. Pumps, fans and cooling equipment draw their own power. UPS and distribution losses add heat. Some cooling load is motor-driven; IT input often uses power electronics. Each has its own waveform and response to voltage disturbances.

A SMALL ENERGY BALANCE

Assume 100 kW reaches IT, with 20 kW cooling, 5 kW conversion/distribution losses and no other loads. Site active input is 125 kW. Over the same stable interval, PUE = total facility energy / IT energy = 1.25. It does not establish carbon intensity, water use or useful compute.

Part 04 develops heat paths and cooling; Part 07 places the full system in Tasmania’s context.

Experiment: build synthetic AC measurements

Use Python 3 on your computer. This creates mathematical waveforms only; it has no device connection and controls no equipment.

Save as ac_lab.py. The sample window contains ten complete 50 Hz cycles at 10,000 samples/s. The third-harmonic case has sinusoidal voltage and a deliberately distorted current.

```
import math

f, fs, count = 50.0, 10000.0, 2000
angle = [2*math.pi*f*n/fs for n in range(count)]
v = [math.sqrt(2)*10*math.sin(x) for x in angle]

def rms(values):
    return math.sqrt(sum(x*x for x in values)/len(values))

def report(name, current):
    p = sum(a*b for a, b in zip(v, current))/count
    s = rms(v)*rms(current)
    pf = p/s if s else None
    print(name, 'V=', round(rms(v), 4),
          'I=', round(rms(current), 4),
          'P=', round(p, 4), 'S=', round(s, 4),
          'PF=', None if pf is None else round(pf, 4))

resistor = [math.sqrt(2)*math.sin(x) for x in angle]
lagging = [math.sqrt(2)*math.sin(x-math.pi/3)
           for x in angle]
distorted = [math.sqrt(2)*(math.sin(x)+0.5*math.sin(3*x))
             for x in angle]

report('resistor', resistor)
report('60-degree lag', lagging)
report('third harmonic', distorted)
report('zero-current model', [0.0]*count)
```

THE OUTPUT IS SIMULATED

All voltage and current samples come from equations. Zero current is deliberately generated here; it is not a replacement for missing sensor data. Unknown measurement values require an explicit invalid state.

Predict the output before running it

Use the RMS and average-power definitions. Then compare the printed result with your hand calculation.

Case	Vrms	Irms	P / S	PF
Resistor	10 V	1 A	10 W / 10 VA	1.0000
60° current lag	10 V	1 A	5 W / 10 VA	0.5000
Third harmonic	10 V	1.1180 A	10 W / 11.1803 VA	0.8944
Zero-current model	10 V	0 A	0 W / 0 VA	undefined

Change one variable and explain the result

Amplitude: double every current sample. Current, P and S double; PF stays the same.
Phase: change the lag from 60° to 30°. P becomes 10 cos30° = 8.6603 W and PF becomes 0.8660.
Harmonics: remove the 0.5 sin(3x) term. The distorted case becomes the original resistor case.

Run deliberately imperfect windows

Change count to 1,950: the window contains 9.75 cycles, so finite-window results no longer match the ideal complete-cycle averages exactly. Reducing the sample rate without preserving the model bandwidth can create aliasing. These effects belong in your methodology note.

WHAT YOU HAVE ESTABLISHED

You reproduced calculations for known synthetic inputs. You have not validated a meter, inverter, UPS, motor or real grid waveform.

Make a reference sheet from one real document

Choose a freely available manufacturer datasheet for a motor, UPS or transformer. You can do this exercise with no instruments.

Record	What to write
Identity	Manufacturer, exact model, document revision, URL, date accessed and page.
Electrical interface	AC or DC; phase count; frequency; voltage; which values are line, phase, peak or RMS.
Power boundary	Input kW/kVA, output kW, PF and efficiency. State if a number is maximum, rated, typical or conditional.
Constraints	Ambient range, cooling arrangement, permitted loading, protection requirements and operating modes.
Observability	Available measurements; units; update interval; alarms; communication interfaces and known limits.
Unknowns	Every field you could not establish. Leave it unknown instead of filling it with a plausible value.

Turn one unknown into an engineering question

For example: “Does this register report an instantaneous value, an RMS window or an average?” Or: “Does the accepted OFF command prove the output opened?” Put the question and its source in Lupine’s problems log.

This is document interpretation. It authorises no installation, switching or equipment access.

Ten checks before moving on

Write your assumptions and units. Questions 8-10 require explanation as well as a number.

1. An ideal sine has a 24 V peak. What is its RMS voltage? What is the RMS of a ±24 V square wave?

2. At 50 Hz, what is the period? How many milliseconds correspond to a 30° lag?

3. A series model has R = 6 Ω and net X = +8 Ω. With 20 V RMS applied, find |Z|, I and P.

4. A single-phase load takes 2 kW at 230 V RMS and PF = 0.80. Find RMS current and apparent power.

5. A balanced 30 kW load uses 400 V line-to-line at PF = 0.90. Find line current.

6. An ideal 20:1 step-down supplies 50 V at 4 A. Find the primary voltage, current and active power for a unity-PF load.

7. A hydro model has q = 2 m^3/s, net H = 50 m and η = 0.85. Estimate electrical output using ρ = 1,000 and g = 9.81.

8. A six-pole synchronous generator operates at 50 Hz. Find speed. Does a 1 MW active-power increase tell you its new frequency?

9. Voltage is sinusoidal; fundamental current is in phase; third-harmonic current RMS is half the fundamental. Is true PF one?

10. A controller says OFF and site kW is zero. What can you conclude about safe isolation and whether the command caused the change?

Worked answers: 1-5

Compare the method and the assumptions, not only the final number.

1 Sine RMS and square-wave RMS

24/√2 = **16.97 V RMS** for the sine. The symmetric ±24 V square wave is **24 V RMS**, because its squared value is always 576 V². Both have zero signed mean over a complete cycle.

2 Period and time shift

T = 1/50 = 0.020 s = **20 ms**. A 30° lag is (30/360) × 20 = **1.667 ms** at this frequency.

3 Series impedance and heating

|Z| = √(6² + 8²) = **10 Ω**. I = 20/10 = **2 A RMS**. P = I²R = 4 × 6 = **24 W**. The angle is atan2(8, 6) = 53.13°, giving PF = 0.60 under the sinusoidal model.

4 Single-phase loading

I = 2,000/(230 × 0.80) = **10.87 A RMS**. S = P/PF = **2.50 kVA**. This is terminal loading, not an assertion about conversion efficiency.

5 Balanced three-phase current

IL = 30,000/(√3 × 400 × 0.90) = **48.11 A**. V is line-to-line and I is line current. Using phase-to-neutral voltage in this formula without changing the relation gives the wrong result.

Worked answers: 6-10

The last answers distinguish calculations, observations and proof of a physical condition.

6 Ideal transformer

Primary voltage is 20 × 50 = **1,000 V**. Primary current is 4/20 = **0.20 A**. At unity PF, power is **200 W** on either side before loss. Do not transfer this ideal model directly to a real product.

7 Hydraulic conversion

P = 1,000 × 9.81 × 2 × 50 × 0.85 = **833,850 W = 833.85 kW**. This assumes the stated net head, flow and combined efficiency remain valid at that operating point.

8 Mechanical speed and frequency

ns = 120 × 50/6 = **1,000 rpm**. The power increase alone does not determine a new frequency. You need system power balance, controls, inertia and time. A connected unit can increase output while frequency remains close to its target.

9 True power factor

THDi = 0.5 and PF = $1/\sqrt{(1 + 0.5^2)}$ = **0.8944**. The fundamental displacement factor is one; the total waveform’s true PF is lower because the harmonic adds RMS current without adding average power under the assumed sinusoidal voltage.

10 OFF is not verified isolation

You have a reported command state and a power observation within their limits. Neither proves absence of voltage, open contacts or causation. The load might already have stopped; another source or stored energy may remain. Qualified isolation verification is a separate process.

Reference: formulas with conditions

Use RMS values unless peak is explicitly stated. Lower-case q here means water flow; upper-case Q means reactive power.

Relationship	Condition or unit
$T = 1/f$; $\omega = 2\pi f$	T in s, f in Hz; ω in rad/s.
$x_{rms} = \sqrt{[\Sigma(x^2)/N]}$	Equal sample spacing and a specified window.
$V_{rms} = V_{peak}/\sqrt{2}$	Pure sine with no DC offset.
$X_L = 2\pi fL$; $X_C = 1/(2\pi fC)$	Ideal elements at one sinusoidal frequency.
$\lvert Z\rvert = \sqrt{[R^2 + (X_L - X_C)^2]}$	Simple series R-L-C model.
$P = \text{mean}[v(t)i(t)]$; $S = V_{rms} I_{rms}$	Single-phase waveform definitions.
$PF = P/S$	Undefined when S = 0.
$P = VI\cos\varphi$; $Q = VI\sin\varphi$	Sinusoidal single-phase voltage and current.
$P = \sqrt{3}\, V_{LL} I_L PF$	Balanced three-phase conditions.
$V_2/V_1 = N_2/N_1$; $I_2/I_1 = N_1/N_2$	Ideal transformer magnitudes.
$P_{loss} = I^2R$; three phases: $3I^2R$	Specified conductor or per-phase resistance.
$n_s = 120f/p$	Synchronous speed, rpm; p = pole count.
$P_{mech} = \tau\omega$; $\eta = P_{out}/P_{in}$	Torque in N·m; speed in rad/s; like power units.
$P_{hydro} = \rho g q H \eta$	Net head H; q in m^3/s; output in W.
$E_{rot} = \frac{1}{2}J\omega^2$; $H = E_{rot}/S_{base}$	State mechanical speed and consistent base.

Current, loss and fault estimates are learning calculations, not equipment ratings or an installation design.

Words to recognise in technical documents

Return here when a datasheet or engineering discussion uses an unfamiliar term.

RMS Root mean square; a magnitude calculated from squared values over a stated window.

Phase angle Relative position of same-frequency sinusoidal components under a stated reference.

Reactance Frequency-dependent imaginary part of impedance in a sinusoidal model.

Active power / P Net rate of electrical energy transfer, in W.

Reactive power / Q Sinusoidal quadrature power, in var; sign convention must be stated.

Apparent power / S Voltage RMS × current RMS, in VA for a single-phase boundary.

True power factor Active power divided by apparent power, including waveform distortion.

Harmonic / THD Integer-multiple frequency component / a stated distortion ratio.

Line / phase Terminal-to-terminal network values / winding or phase-element values.

Rotor / stator Rotating / stationary portions of a machine.

Slip Difference between synchronous field and induction-rotor speeds, divided by synchronous speed.

Inertia / H Mechanical resistance to speed change / stored-energy-to-rated-power inertia constant.

Busbar / feeder Common electrical connection / circuit supplying another part of the network.

Fault level Available fault-current or related short-circuit-power measure at a stated point.

Selectivity Coordination intended to disconnect the relevant faulted section while retaining other supply.

RCD / isolation Residual-current protection device / an established separation from energy sources.

PUT THREE LABELS ON YOUR NOTES

CALCULATED: follows from stated assumptions. SIMULATED: produced by a model. MEASURED: acquired by an instrument with a defined boundary. None automatically means independently verified.

Primary references behind this volume

These are free technical references, not additional required textbooks. The explanations, examples and diagrams in this guide are original.

OpenStax / Simple AC circuits
Ideal elements, RMS and phase relationships.

OpenStax / Series RLC circuits
Complex impedance and sinusoidal series models.

OpenStax / AC power and transformers
Power definitions and ideal transformer relations.

OpenStax / Transformers
Turns ratios and transmission-loss explanation.

Schneider Electric / Harmonics and PF
True power factor versus displacement power factor.

Schneider Electric / Short-circuit current
Fault-current studies and equipment selection context.

Schneider Electric / Protection
Protection and coordination principles.

US Department of Energy / Hydropower
Water flow, head, turbine and generator functions.

AEMO / Engineering roadmap reports
Inertia in the NEM explained and related technical work.

US DOE / Electrical Science, Volume 4
AC motor speed and slip fundamentals.

WorkSafe Tasmania / RCDs
Residual-current protection and its limitations. Site-specific legal details are not reproduced here.

Scope and review status

Edition 01 / 17 October 2026. Original educational material prepared with ChatGPT. Numerical examples and waveforms are assumed or simulated, not site measurements. No physical AC testing or independent engineering peer review is claimed.

Eight parts: from foundations to field evidence

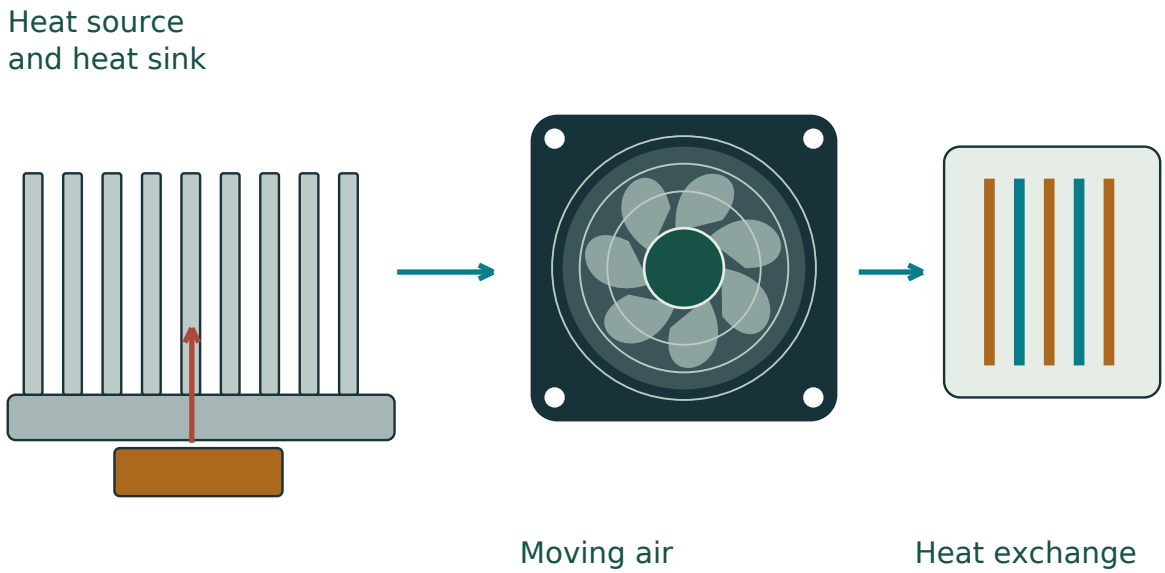
This is the complete planned sequence. Finishing a guide builds knowledge; equipment readiness and commercial demand require separate evidence.

Part	Subject	Concrete learning outcome
01	Electricity you can measure	Use units, circuits, power and measurement limits.
02	Components, sensors and microcontrollers	Acquire and preserve a real low-voltage sensor reading.
03	AC power, machines, networks and protection	Read an electrical system and calculate its basic loading.
04	Heat, fluid flow and cooling	Trace thermal resistance, airflow and liquid-cooling energy balances.
05	Batteries, inverters and power electronics	Understand storage boundaries, conversion and operating limits.
06	Reliable control, communications and evidence	Connect sensing to execution, uncertainty, persistence and reconciliation.
07	Tasmania’s grid and AI data centres	Map local infrastructure and investigate specific engineering gaps.
08	Lupine prototype and field validation	Integrate a bounded test system and evaluate it with real users.

After Part 03, five parts remain. Read pages 03-03 to 03-05 first; then work through the calculation examples before the computer exercise. Part 04 is the next volume.

HEAT, FLUID FLOW & COOLING

Part 04 / Follow energy from a warm component through air or liquid to its final destination.



Functional illustrations of heat-transfer stages, not one complete cooling circuit.

The important result is a physical heat path

Temperature is an observation; heat transfer is an energy rate. A useful cooling design explains the source, its thermal resistance, the moving fluid, the final heat sink and the changing conditions along the path. Every assumed number in this guide stays labelled as a model.

WHAT YOU WILL BE ABLE TO DO

Calculate basic heat and flow balances; read fan and pump duty points; explain liquid loops, dew point and refrigeration; and document one small thermal-response experiment.

The route from heat to evidence

This part stands on the electrical power and sensor foundations from Parts 01-03. No new textbook is assigned.

Pages	Study question
3-5	How do temperature, heat rate and stored thermal energy differ?
6-11	How do conduction, convection, radiation and thermal dynamics work?
12-17	What do flow, pressure, resistance and fan/pump curves describe?
18-23	How do fluid heat balances, heat exchangers and refrigeration connect?
24-28	How do real cooling paths, coolant properties and failure evidence fit together?
29-32	What can I measure at home, and what can I simulate for free?
33-37	Can I solve the checks and use the formulas and vocabulary?
38-40	Where do the references lead, and what belongs in Lupine next?

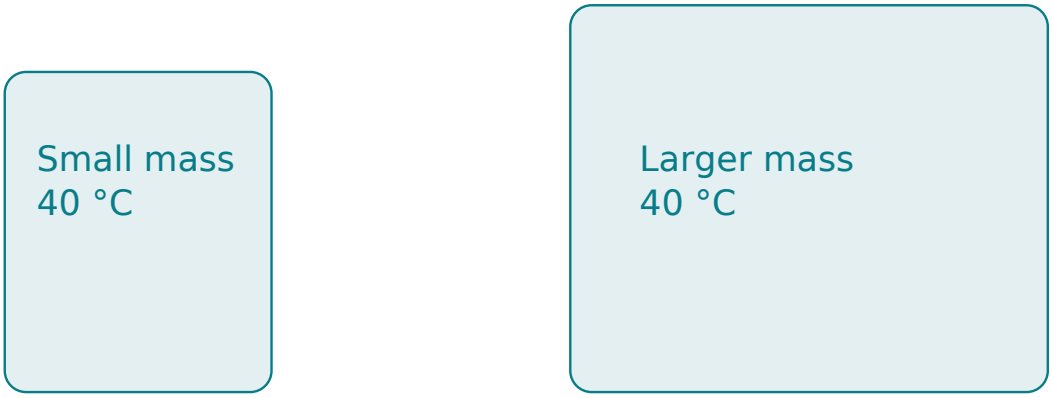
Keep physical work small

The optional experiment uses the USB-powered temperature logger from Part 02 and warmth from your fingers. No heater, liquid circuit, refrigeration work, modified mains equipment or opened computer cooler is required.

Page labels 04-01 to 04-40 remain usable after merging. Four planned parts follow this one.

Temperature is not an amount of heat

Temperature describes thermal state. Heat is energy transferred because of a temperature difference, and heat rate is that transfer per second.



Same temperature does not mean the same thermal energy relative to a reference.

Use three different quantities

Temperature T: measured in °C or K.
Transferred energy Q: measured in joules (J).
Heat-transfer rate Qdot: measured in watts, where 1 W = 1 J/s.
In equations below, “Qdot” means a rate; it is not a volume flow. We use Vdot for volume flow.

Kelvin matters for absolute-temperature formulas

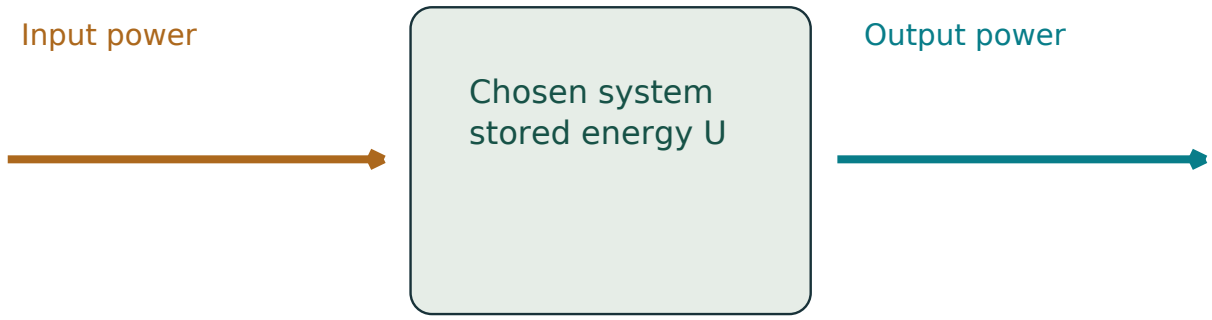
T[K] = T[°C] + 273.15. A temperature difference of 10 °C equals a difference of 10 K. Ratios and fourth powers of absolute temperature must use kelvin; 40 °C is not twice the absolute temperature of 20 °C.

NAME THE THERMAL REFERENCE

An object can be warmer than room air and cooler than a chip. State both ends of a temperature difference before claiming heat will move along a particular path.

Start every system with an energy balance

Choose a boundary and account for input, output and storage. A steady temperature requires the net accumulation to be zero, not all energy flows to be zero.



dU/dt = total energy rate in - total energy rate out

Put heat and electrical work in their right places

For a simple component model, electrical dissipation Pin adds internal energy and heat loss Qdotout removes it: dU/dt = Pin - Qdotout. At steady state the two rates match. During warm-up, input exceeds outward heat flow; some energy is stored in the component.

WORKED BALANCE

A model receives 100 W of electrical dissipation and currently loses 70 W to its surroundings. Stored energy increases at 30 J/s. Over 10 s at those unchanged rates, it gains 300 J. A constant 100 W input does not imply temperature stays constant.

Connect this to computing

Within an appropriate steady thermal boundary, almost all electrical energy used by computing equipment eventually becomes heat. Include energy leaving by other paths or entering storage when the boundary requires it.

Heat capacity sets the energy to warm a mass

Specific heat is a material property under stated conditions. Thermal capacitance includes the amount of material.

$$Q = mcp\,\Delta T$$

$$Cth = mcp\ [J/K]$$

m: mass in kg cp: specific heat in J/(kg·K)

No phase change; properties treated as constant over the interval.

Water example with every assumption visible

Assume 0.50 kg of water, cp = 4,180 J/(kg·K), initially uniform in temperature. A 5 K rise requires Q = 0.50 × 4,180 × 5 = 10,450 J. An ideal net 100 W input would take 104.5 s. Real containers, heat losses and mixing change the observed time.

A heat reservoir is not an infinite sink

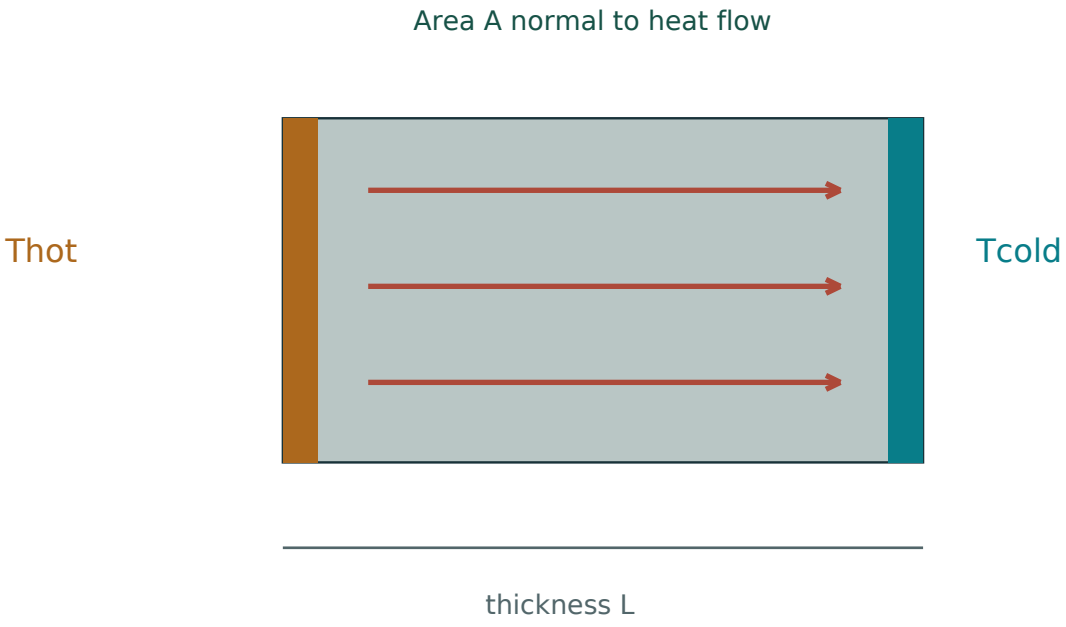
Thermal mass can delay a temperature rise but cannot remove heat indefinitely. A liquid loop with no effective heat rejection eventually warms up. Phase change can store or release additional energy; then sensible heat mcpΔT alone is insufficient.

KEEP CAPACITY AND RESISTANCE SEPARATE

Thermal capacitance Cth describes energy storage per kelvin. Thermal resistance Rth describes a temperature difference per watt of heat flow. Both influence transients, but they are different physical quantities.

Conduction moves heat through material

A temperature gradient drives heat flow. A thin, wide, conductive path has less thermal resistance than a thick, narrow, insulating path.



Use the plane-wall model

For one-dimensional steady conduction with constant conductivity, Qdot = kA(Thot - Tcold)/L. Equivalently Rth = L/(kA), with units K/W. Conductivity k has units W/(m·K). A device’s full thermal path usually contains several regions and interfaces.

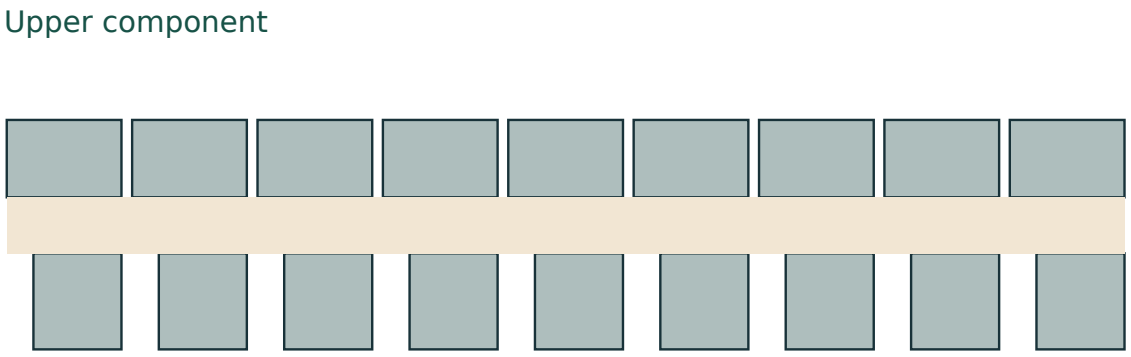
WORKED LAYER

Assume L = 1 mm = 0.001 m, k = 5 W/(m·K) and A = 4 cm² = 0.0004 m². Rth = 0.50 K/W. At 20 W through this layer, its temperature drop is 10 K. This excludes spreading and contact resistance.

The drawing is a flat-wall model, not a chip package or an installation specification.

Interfaces can dominate a thermal path

Two surfaces that look flat touch over only part of their microscopic area. Gaps, pressure, thickness and materials change heat transfer.



Exaggerated section. Actual roughness, bond line and pressure are product-specific.

An interface material is not magic insulation removal

A thermal interface material can replace poorly conducting air gaps, but it has finite resistance. Too thick a layer can add resistance; poor mounting can leave gaps. Tightening force, flatness, electrical isolation, ageing and compatibility all belong in a real assembly specification.

Read the test conditions behind a thermal number

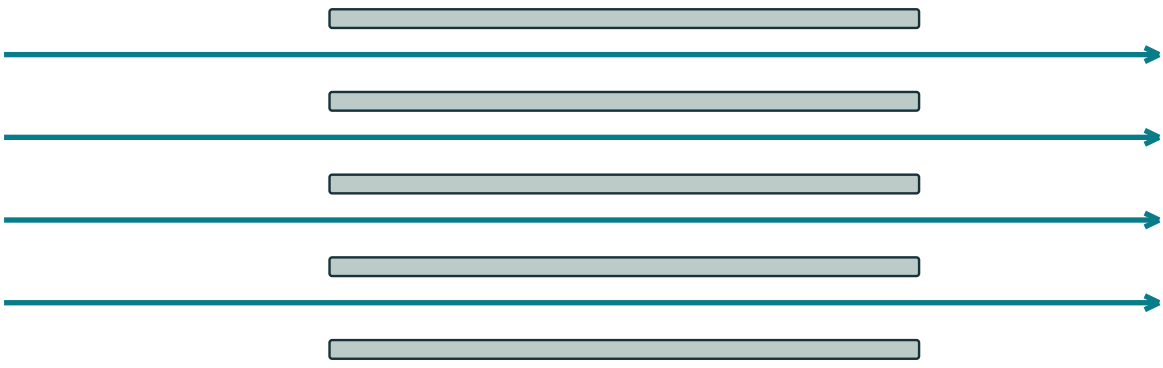
A datasheet’s thermal resistance depends on boundary conditions. Junction-to-case, junction-to-board and junction-to-ambient values are not interchangeable. Do not add overlapping test-path numbers as though they were disjoint resistors. Identify the actual path first.

YOUR FIRST CHECK

Before proposing a faster fan, ask whether heat reaches the heat sink effectively. A poor interface can leave a hot component beside a comparatively cool heat sink.

Convection connects a surface to moving fluid

Fluid motion carries energy, while heat still crosses a near-wall thermal boundary. Natural and forced convection need different models.



Top view: flow between fins; surface Ts → fluid Tfluid

Use a coefficient with conditions attached

A common model is Qdot = hA(Ts - Tfluid). Here h is a heat-transfer coefficient in W/(m²·K), not a universal material constant. It depends on geometry, flow, fluid properties and the chosen reference temperature. Fin area is not automatically fully effective area.

WORKED SURFACE MODEL

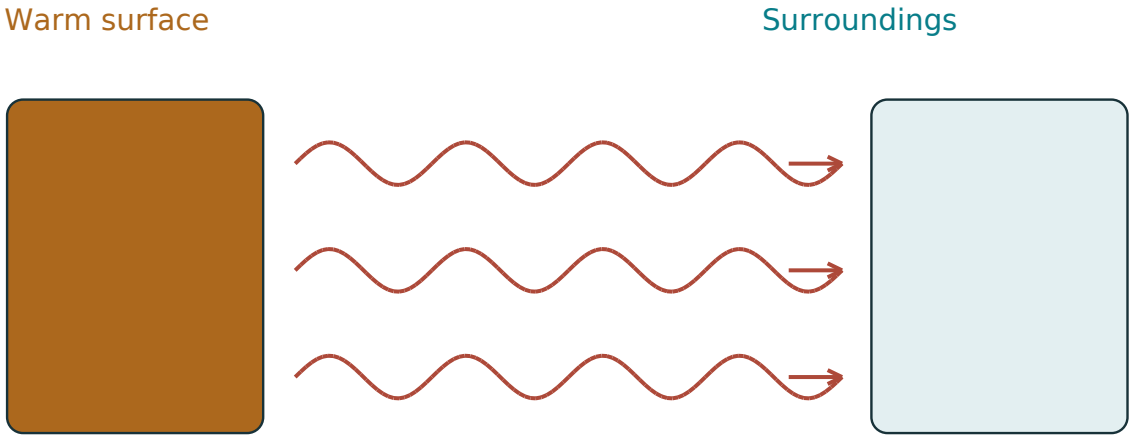
Assume h = 25 W/(m²·K), effective area A = 0.10 m² and surface-to-air difference 20 K. Qdot = 50 W and Rconv = 1/(hA) = 0.40 K/W. These assumed values do not establish a real heat sink’s rating.

More airflow does not guarantee a proportional gain

Flow bypass, recirculation, fin pressure loss and interface resistance can limit the benefit. Measure the relevant temperatures and flow boundary.

Radiation exchanges heat without moving air

Objects emit and absorb thermal radiation. Net exchange depends on absolute temperature, surface properties and geometry.



Both sides radiate. Arrows show the direction of net transfer in this example.

Apply the simplified enclosure formula carefully

For a small diffuse grey surface facing much larger approximately uniform surroundings, $Q_{net} = \varepsilon \sigma A(T_s^4 - T_{sur}^4)$. Temperatures must be in K and $\sigma \approx 5.67 \times 10^{-8}$ W/(m²·K⁴). Other geometries need view factors and surface exchange models.

NUMERICAL EXAMPLE

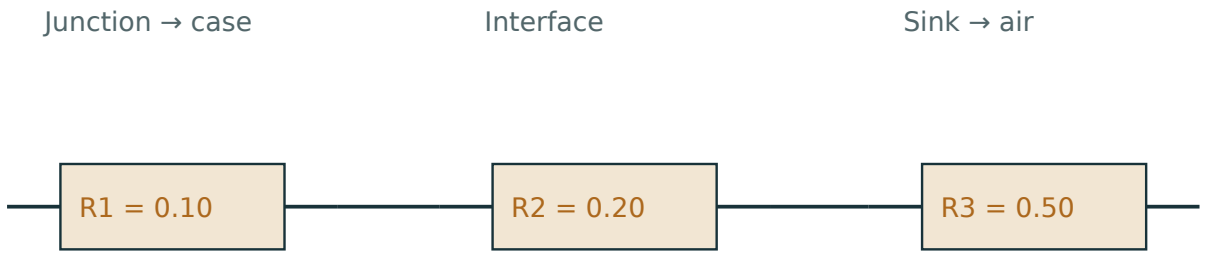
Assume $\varepsilon = 0.90$, $A = 0.10$ m², $T_s = 50$ °C = 323.15 K and $T_{sur} = 20$ °C = 293.15 K. Net radiation is approximately 18 W. Air convection may occur at the same time and must be included separately.

Visible colour is not a reliable infrared specification

Emissivity depends on wavelength and surface condition. A shiny metal surface can also mislead an infrared thermometer unless emissivity and reflections are handled correctly.

Add thermal resistances along a defined path

A thermal network is a useful approximation when the path and boundary conditions are clear. Branches and heat spreading require more than one series chain.



Illustrative disjoint path segments; resistance units are K/W.

Calculate the temperature rise, then check every limit

Assume all 80 W follows this path with Rtotal = 0.10 + 0.20 + 0.50 = 0.80 K/W. The rise is 64 K. At 25 °C ambient the model junction temperature is 89 °C. This is a calculated temperature, not an assertion that the component is permitted to run there.

FIND THE LARGEST CONTRIBUTOR

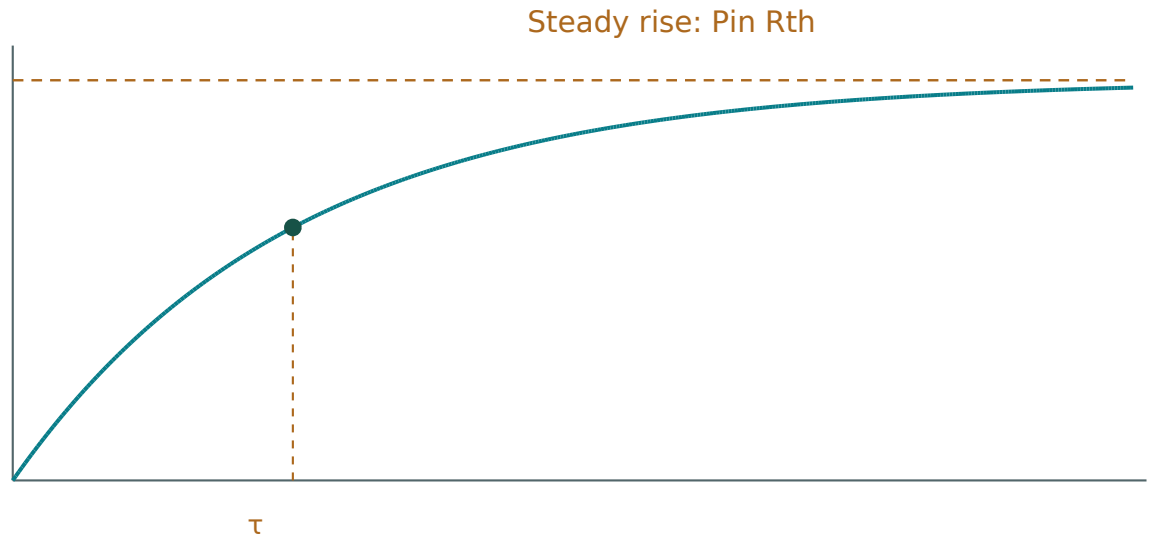
If the sink-to-air term falls from 0.50 to 0.25 K/W while the others stay fixed, Rtotal becomes 0.55 K/W and the rise becomes 44 K. The new model junction temperature is 69 °C. Improvement still depends on airflow and actual boundary conditions.

Parallel paths change the result

Heat may also leave through the board, case or nearby structures. A simple series sum is valid only for the path represented by its input data.

Thermal mass makes the response take time

A first-order lumped model treats one object as approximately uniform in temperature. It is a starting hypothesis, not a universal description of electronics.



Time → At one time constant, 63.2% of the final rise is reached.

Use resistance and capacitance together

Cth dT/dt = Pin - (T - Tamb)/Rth. With constant properties and input, τ = Rth Cth. From initial ambient temperature, T(t) = Tamb + Pin Rth[1 - exp(-t/τ)]. After heat input stops, the excess above ambient decays exponentially in this model.

WORKED TIME SCALE

Assume Rth = 8 K/W and Cth = 20 J/K. Then τ = 160 s. With Pin = 2 W, the eventual rise is 16 K. After 160 s it is about 10.11 K, not the full 16 K. Three time constants give about 95% of the final rise.

Test whether one state is enough

A sensor sheath, contact layer and object can have different temperatures and time constants. A visually smooth curve alone does not prove a one-node model.

Fluid properties shape the cooling problem

Air, water and coolant mixtures carry heat differently and require different pressure and material conditions. State the fluid and temperature range.

Quantity	Meaning	SI unit
Density ρ	Mass per volume.	kg/m³
Specific heat cp	Energy per mass per temperature rise under the stated conditions.	J/(kg·K)
Dynamic viscosity μ	Resistance to shear; strongly temperature-dependent for many fluids.	Pa·s
Conductivity k	Ability to conduct heat through the fluid itself.	W/(m·K)
Volume flow Vdot	Volume crossing a section per second.	m³/s
Mass flow mdot	Mass crossing a section per second.	kg/s

Use consistent property assumptions

This guide’s simple examples use water ρ = 1,000 kg/m³ and cp = 4,180 J/(kg·K), and air ρ = 1.20 kg/m³ and cp = 1,005 J/(kg·K). These are rounded illustrative constants, not a property table for every pressure or temperature.

CHECK THE ACTUAL COOLANT

Glycol concentration, additives, temperature and contamination can change heat capacity, viscosity, freeze protection and compatibility. Do not silently substitute pure-water properties.

Continuity connects flow, area and velocity

In a steady incompressible path without branches or leaks, the same volume flow crosses each section. A smaller tube means faster average velocity at that flow.



Vdot = A × mean velocity; mdot = ρ Vdot

Convert litres per minute before calculating

1 L/min = 0.001/60 = 1.6667 × 10^-5 m³/s. In an 8 mm internal-diameter tube, A = π(0.008)²/4 = 5.0265 × 10^-5 m², giving mean velocity 0.332 m/s. At 4 mm diameter, area is one quarter and velocity is four times greater: 1.326 m/s.

Do not confuse a velocity reading with total flow

A point reading may not equal cross-sectional mean velocity. Turbulence, bends, nonuniform profiles and sensor installation affect the conversion. At branches, mass flows sum; different branches need not share equally.

FOR COMPRESSIBLE FLOW

Mass conservation still applies, but volume flow changes when density changes. Specify actual conditions when comparing air-flow data from different sources.

Pressure and pump head are related, not identical

Pressure is force per area. Head expresses fluid energy per unit weight and can include pressure, elevation and velocity terms.

$$\Delta p = \rho g H$$

$$P_{hydraulic} = \Delta p \, \dot{V}$$

$$P_{electrical} = P_{hydraulic} / \eta_{total}$$

Consistent pump boundary; η_{total} includes the efficiencies being modelled.

Work a pump estimate in SI units

For water with $\rho = 1{,}000$ kg/m³, 5 m of differential head corresponds to 49,050 Pa = 49.05 kPa. At 0.001 m³/s, hydraulic power is 49.05 W. With assumed wire-to-water efficiency 50%, electrical input is 98.10 W. The difference is loss, not additional useful hydraulic power.

A closed loop is not a continual hill climb

In a filled circulation loop returning to the same point, net elevation change around the loop is zero. The pump still overcomes friction and component losses. Absolute pressure, filling, expansion, air removal and cavitation remain important. An open lift between reservoirs has a different system head.

READ PRESSURE CORRECTLY

Distinguish gauge, absolute and differential pressure, and state the sensor tap locations. One discharge-pressure number does not establish flow.

Flow resistance consumes pressure difference

Pipes, bends, valves, filters and cold plates resist flow. Restrictions can change both total flow and how it divides between parallel paths.



At a given flow, the pump must provide the total circuit pressure loss.

Know the shape of the approximation

For a simple fixed geometry in a turbulent regime, a useful local approximation is $\Delta p \approx K \dot{V}^2$. Laminar straight-pipe loss is proportional to flow instead. Fluid viscosity, roughness, diameter, fittings and changing valve position determine whether a chosen curve remains valid.

HYPOTHETICAL RESISTANCE

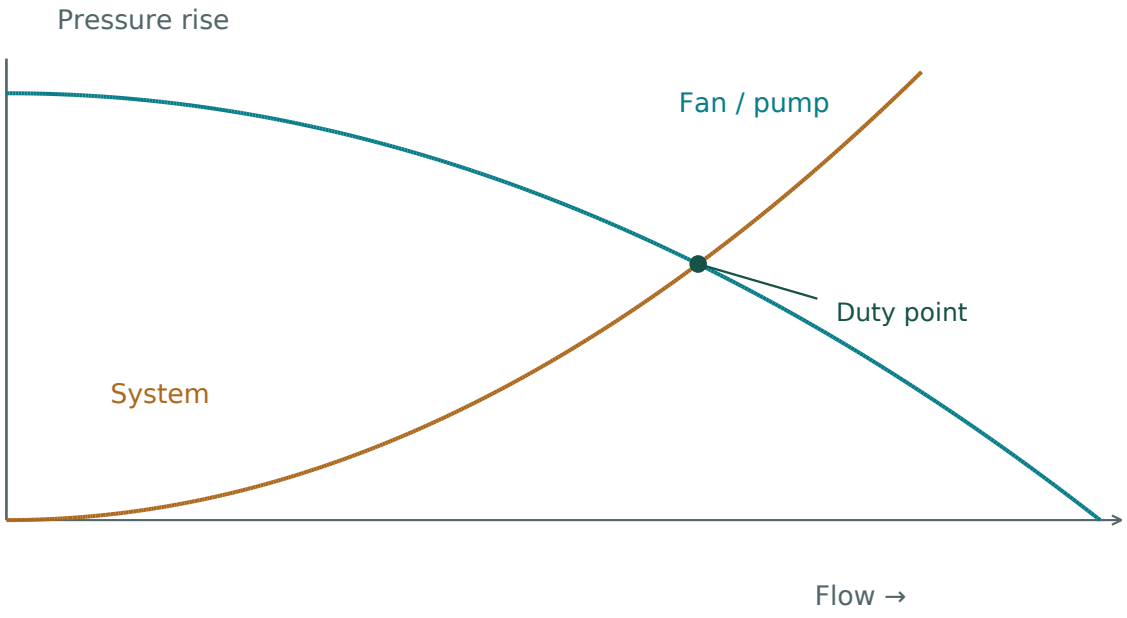
If a circuit needs 20 kPa at 2 L/min and obeys the square-law approximation, it needs 80 kPa at 4 L/min. Hydraulic power rises from 0.667 W to 5.333 W: eight times as much. Motor and pump efficiency are not included.

A failure clue needs context

At unchanged flow a growing filter pressure drop can indicate restriction. At unchanged pump command, the same restriction may instead reduce flow. Interpret the variables together.

The operating point is where two curves meet

A fan or pump has a performance curve. The connected system has a resistance curve. Their intersection determines the steady duty point.



Maximum pressure and free flow are different endpoints

The maximum flow of many fans is quoted near zero pressure rise. Maximum static pressure is near zero flow. A heat sink or radiator usually operates between those endpoints. Comparing only one endpoint can select the poorer fan for the real restriction.

Read the manufacturer’s conditions

Check rotational speed, fluid density, inlet arrangement, measurement standard and acceptable operating region. Fan instability, pump cavitation, minimum flow and efficiency impose limits. A smooth generic curve here is explanatory, not a product characteristic.

A COMMAND IS NOT A DUTY POINT

Requested PWM, RPM feedback, pressure and measured flow each describe a different stage. Preserve each observation with its own units and limits.

Speed changes pressure and power unequally

Affinity laws estimate corresponding points for geometrically unchanged equipment under similarity assumptions. Real efficiencies and system constraints still matter.

Speed ratio r = n2/n1	Approximate ratio
Flow at corresponding points	$\dot{V}_2/\dot{V}_1 \approx r$
Head or pressure rise, fixed density	$H_2/H_1 \approx r^2$
Shaft power, similar efficiency	$P_2/P_1 \approx r^3$

Work the ratios before claiming energy savings

For r = 0.80, flow ratio is 0.80, head ratio is 0.64 and shaft-power ratio is 0.512. If a friction-dominated system follows the corresponding duty points and efficiencies stay similar, a 100 W shaft-power requirement becomes about 51.2 W. Measured electrical input need not scale exactly the same way.

Static head and controls can change the story

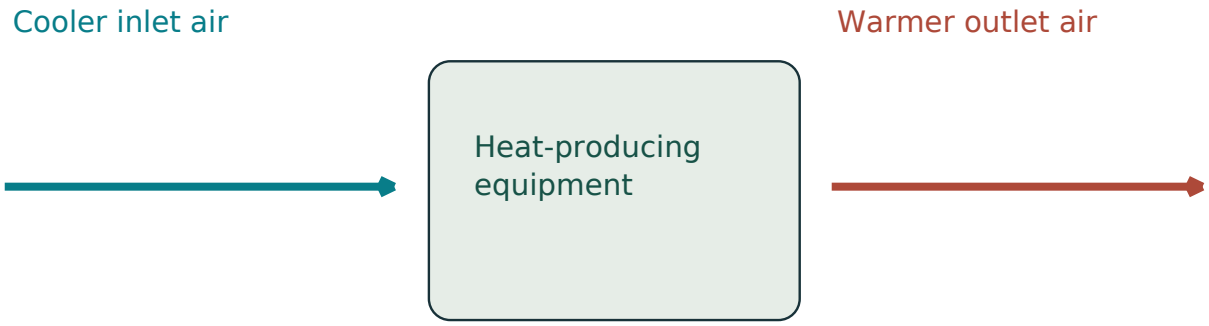
An open lift may stop delivering useful flow below a minimum speed. A valve-controlled or changing system may not follow the original curve. Do not turn the cubic estimate into a guaranteed energy-saving claim. Thermal limits can also prevent the slower-flow operating point.

SERIES AND PARALLEL ARE NOT FREE MULTIPLIERS

At a common flow, series devices add pressure capability; at a common pressure, parallel devices add flow capability in ideal combinations. The actual new duty point still follows the combined and system curves.

Air carries heat through a temperature rise

For sensible heating with no condensation, a steady fluid balance links mass flow, specific heat and inlet/outlet temperature difference.



$$Qdot = mdot\ cp\ (Tout - Tin)$$

Estimate a required air flow

For a hypothetical 10 kW heat load and a 10 K rise, with cp = 1,005 J/(kg·K), mass flow is 10,000/(1,005 × 10) = 0.9950 kg/s. With assumed air density 1.20 kg/m³, volume flow is 0.8292 m³/s = 2,985 m³/h. This assumes the measured air stream captures the whole heat load.

A bigger temperature rise is not always better

With the same heat and less flow, the outlet becomes warmer. Local hot spots or inlet limits may be exceeded before the bulk outlet seems alarming. Mixing and leakage can also invalidate a simple inlet/outlet average.

USE A MASS-FLOW-WEIGHTED TEMPERATURE

A single probe in a nonuniform exhaust is not necessarily the stream’s mean temperature. State probe placement, flow distribution and uncertainty before reporting heat removed.

Water moves the same heat at a smaller volume flow

The useful comparison is volumetric heat capacity ρcp, with pump and system constraints included. Liquid cooling still needs a destination for the heat.

Rounded example property	Air	Water
Density ρ	1.20 kg/m³	1,000 kg/m³
Specific heat cp	1,005 J/(kg·K)	4,180 J/(kg·K)
ρcp	1,206 J/(m³·K)	4,180,000 J/(m³·K)

Calculate the fluid flow for a stated temperature rise

For 100 kW and ΔT = 10 K, the water model needs mdot = 100,000/(4,180 × 10) = 2.392 kg/s. At 1,000 kg/m³ that is 0.002392 m³/s, or 143.54 L/min. A 1 kW branch at the same rise needs 1.435 L/min under the same assumptions.

THE ENERGY TRANSPORT COMPARISON

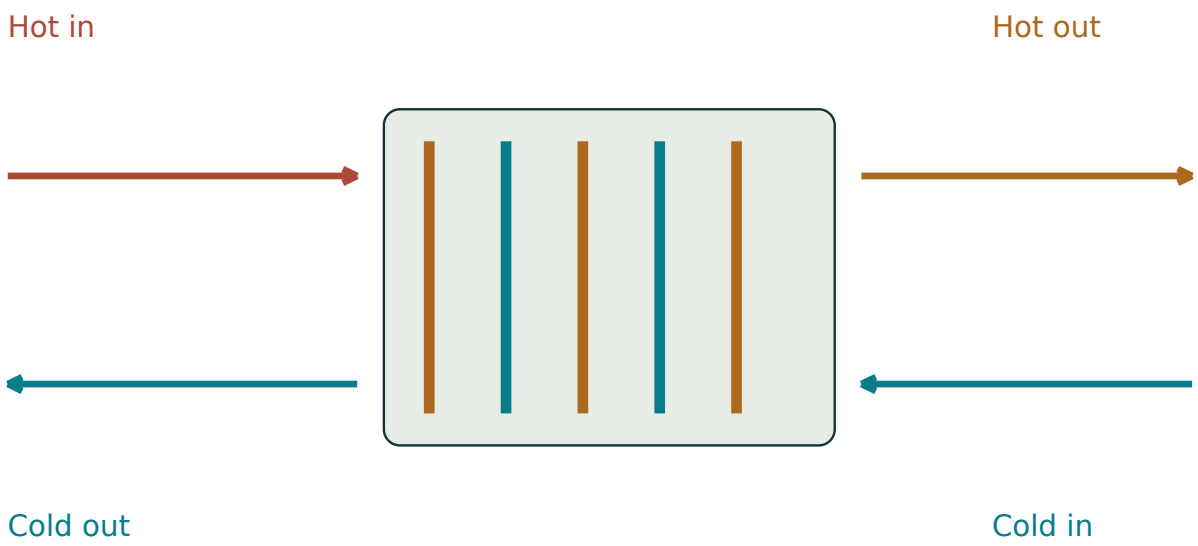
At equal volume flow and temperature rise, these assumed water properties carry about 3,466 times the sensible heat of these assumed air properties. This is not a claim of 3,466 times better system efficiency or chip cooling.

What must be added to the model

Cold-plate resistance, pump power, pressure limits, coolant condition, leak handling, heat-exchanger approach and final heat rejection all remain. A smaller volume flow does not remove those requirements.

A heat exchanger transfers heat between streams

A wall separates the fluids in a typical indirect exchanger. It transfers energy while preserving separate fluid circuits if it remains intact.



ΔTlm = (ΔT1 - ΔT2)/ln(ΔT1/ΔT2); for equal differences, use their common value.

Check both energy balances

Neglecting storage and heat loss to the room, Qdothot = mdoth cp,h(Th,in - Th,out) equals Qdotcold = mdotc cp,c(Tc,out - Tc,in). Different mass flows or heat capacities give different temperature changes. For counterflow, ΔT1 = Th,in - Tc,out and ΔT2 = Th,out - Tc,in. The streams move in opposite directions.

COUNTERFLOW EXAMPLE

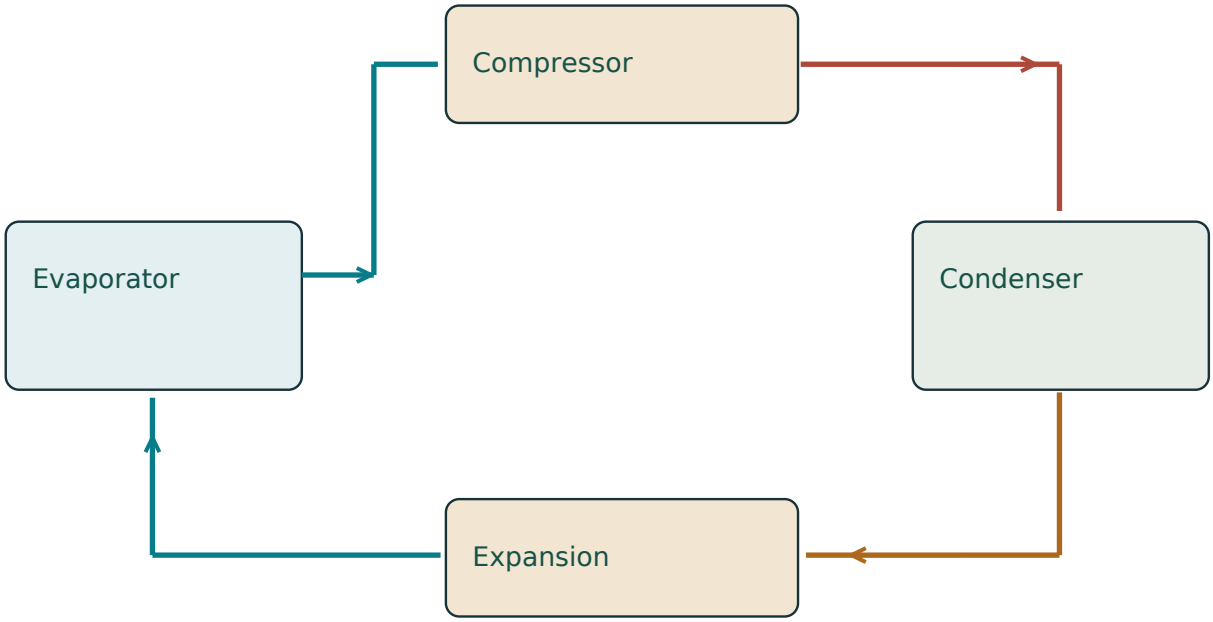
Assume hot water falls from 40 to 35 °C while cold water rises from 25 to 30 °C, both at 1 kg/s and cp = 4,180 J/(kg·K). Heat transfer is 20.9 kW. Both end differences are 10 K; UA = Qdot/ΔTlm = 2,090 W/K in this ideal case.

The approach temperature is a real limit

Finite area and conductance require a temperature difference. Pressure drop and fouling also matter; a heat exchanger does not deliver arbitrary cooling at zero approach.

Refrigeration uses work to move heat uphill

A vapour-compression cycle can move heat from a colder region to a warmer one. The hot side rejects the removed heat plus compressor work.



Keep cooling power separate from electrical input

Cooling COP = Qdotcold/Pelectric for the chosen boundary. If 100 kW is removed with 25 kW electrical input, cooling COP = 4 and hot-side rejection is 125 kW in a loss-accounted steady model. COP above one is possible because heat is moved, not created from work alone.

A coolant pump is not a compressor

The compressor raises refrigerant-vapour pressure; condensation rejects heat, expansion lowers pressure and evaporation absorbs heat. A single-phase water loop merely transports sensible heat unless it connects to equipment providing another temperature lift.

LEARN THE CYCLE WITHOUT OPENING ONE

Use diagrams and manufacturer documentation. Do not open a sealed cooler or handle refrigerants for this learning project.

The last heat sink sets the cooling opportunity

Heat rejection can use outdoor air, evaporation, another water system or a useful heat customer. Each imposes temperature and capacity limits.

Method	Physical route	Key conditions
Dry cooler	Warm fluid transfers heat through a coil to outdoor air.	Dry-bulb temperature, approach, area, fan power and fouling.
Evaporative rejection	Evaporation allows cooling towards outdoor wet-bulb conditions.	Wet bulb, water quality, makeup, blowdown and hygiene.
Economiser mode	Suitable outdoor conditions reduce or avoid compressor cooling.	Hours of suitable conditions, approach, humidity and control transitions.
Heat reuse	A separate user receives useful heat directly or through a heat pump.	Required delivery temperature, demand schedule, distance and reliability.

A cool climate is an input, not a whole design

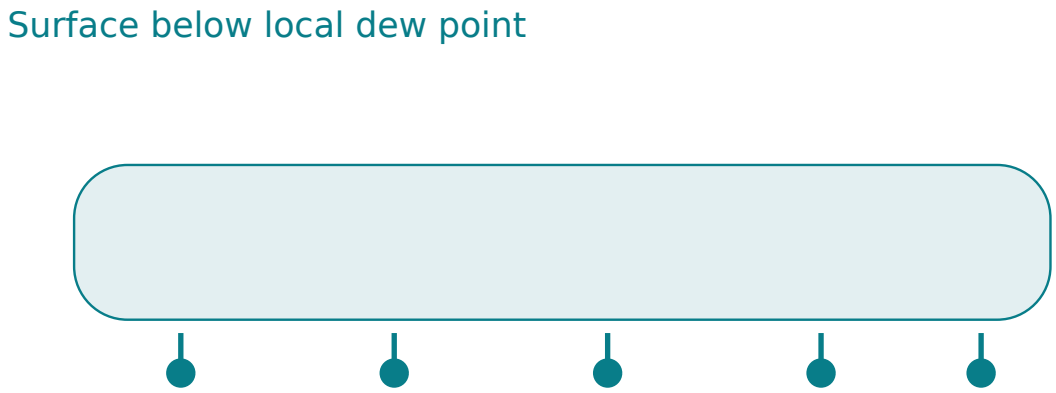
Hobart’s weather can make heat rejection promising, but a design needs hourly conditions, worst cases, equipment limits and local constraints. Annual average air temperature cannot establish year-round cooling capacity or prove that no compressor will ever be required.

FREE COOLING STILL USES RESOURCES

Fans, pumps, controls, filters and maintenance remain. Compare system energy and water across the same boundary and time interval.

Condensation depends on dew point at the surface

Air can deposit liquid water on a sufficiently cold surface. Room temperature by itself does not tell you whether a pipe will sweat.



Illustrative condensation. The water comes from the surrounding air.

Compare local surface temperature with local dew point

Assume room air has a known dew point of 16 °C. A pipe surface at 12 °C is below it, so condensation is possible. A surface at 20 °C is above that value. Real control needs a documented margin for sensor error, variation, cold spots and changing humidity; these example numbers do not specify that margin.

Humidity sensing adds another measurement chain

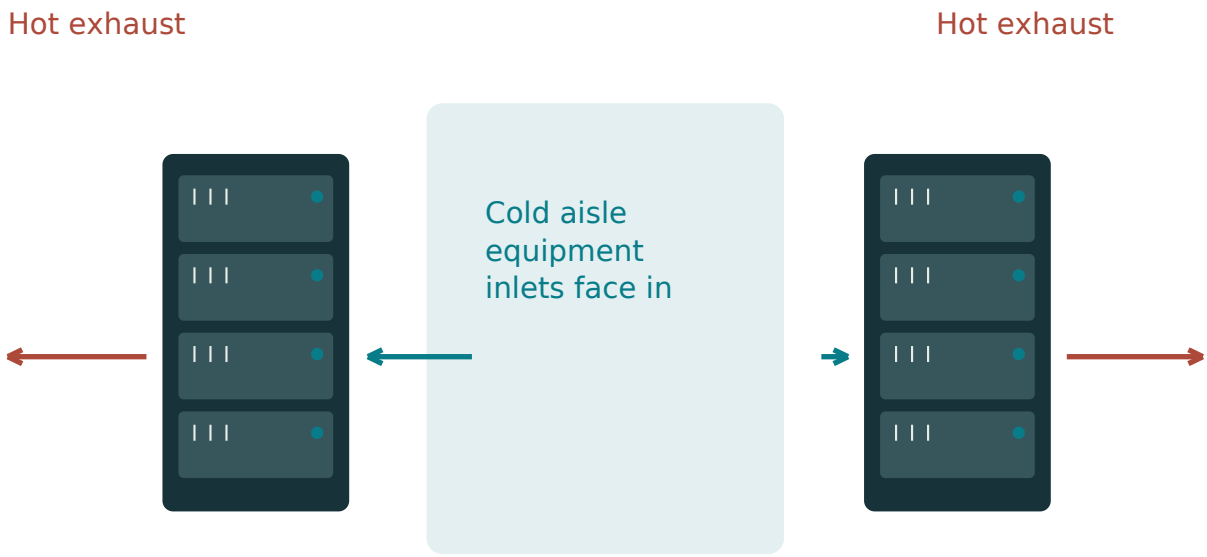
Relative humidity changes with temperature even at unchanged moisture content. A temperature-only DS18B20 cannot measure humidity or independently establish dew point. Surface temperature can differ from bulk coolant temperature because insulation, flow and local heat transfer matter.

LIQUID COOLING IS NOT ALWAYS COLDER

A warm-water loop may avoid condensation and improve rejection opportunities if equipment supports its temperatures. Chip limits, approach temperatures and workload still set the permitted envelope.

Rack airflow must reach the equipment inlet

A room average can hide a hot inlet. Containment, recirculation and bypass determine how much supplied air actually cools the load.



Topological illustration. Actual racks, airflow direction and room arrangement must be checked.

Avoid two kinds of wasted air path

Recirculation: hot exhaust reaches the equipment inlet.
Bypass: supplied cool air returns without passing through the intended equipment.
Both can distort bulk temperature measurements. Blank panels, sealed openings and containment can change the path, but must follow the facility’s engineering design.

Measure the point that protects the load

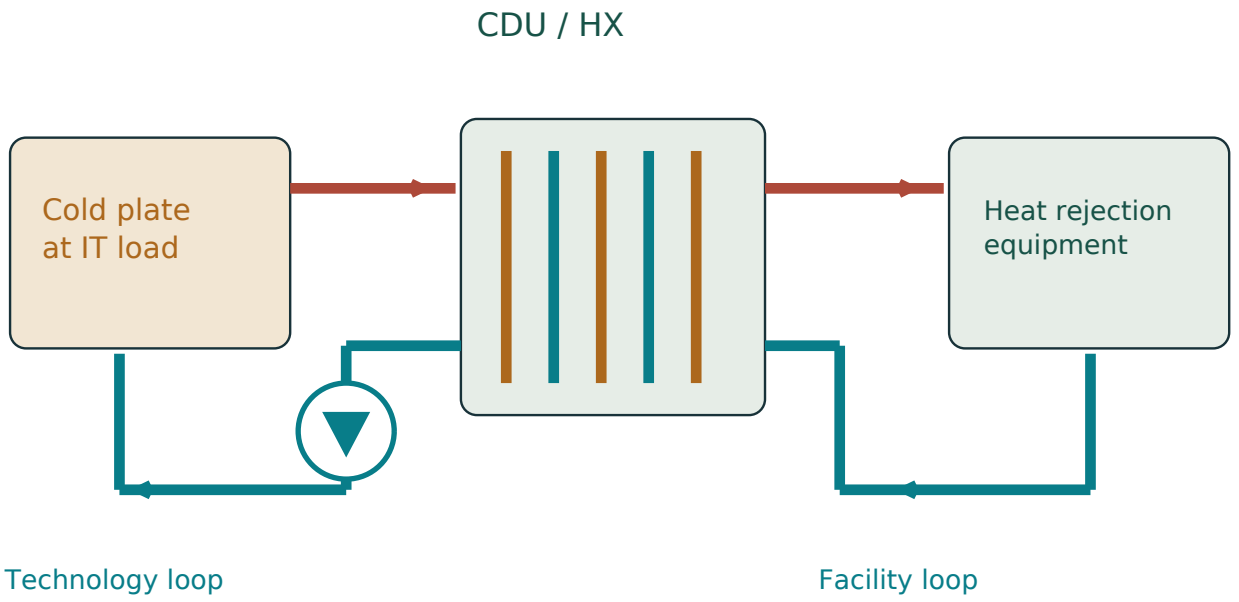
Use the equipment’s required inlet reference, not a convenient wall sensor alone. Record height, rack, side, time and workload. Air distribution, pressure differences and equipment fan behaviour may change when doors or panels move.

CRAH AND CRAC ARE DIFFERENT TERMS

A computer-room air handler commonly uses a chilled-water coil; a computer-room air conditioner commonly uses refrigerant cooling. Real product arrangements vary.

Liquid cooling uses linked, separate circuits

A typical liquid-to-liquid CDU transfers heat between a technology loop and a facility loop. The drawing below is one possible arrangement, not a Firmus site design.



Trace heat and fluid as different paths

Coolant absorbs heat in the cold plate and returns warm to the CDU heat exchanger. Heat crosses the separating wall into the facility circuit. The facility removes it to an external sink. The two fluids do not normally mix; the diagram omits expansion, fill, filtration, valves and detailed protection.

The CDU provides a controlled boundary

Depending on design, it can manage circulation, temperature, pressure, monitoring and fluid separation. A CDU is not automatically a chiller. Some heat can remain in memory, power conversion or other air-cooled parts, so a direct-to-chip loop need not capture every watt of IT heat.

Loop colours mean relatively warmer return and cooler supply; they do not specify absolute temperatures.

Coolant choice is also a materials problem

The fluid touches seals, tubing, metals, pumps, sensors and heat exchangers. Thermal performance is only one selection criterion.

Question	Why it changes the design
What fluid and concentration?	Density, heat capacity, viscosity, freezing behaviour and service requirements change.
Which wetted materials?	Corrosion, galvanic couples, seal swelling and compatibility can create failures.
How is it kept clean?	Particles, biological growth, deposits and fluid ageing can affect flow and heat transfer.
What pressure and temperature limits?	Hoses, connectors, cold plates and expansion provisions must remain within their ratings.
Which cooling method?	Single-phase direct-to-chip and immersion are different architectures. A dielectric-fluid claim is not permission to immerse arbitrary electronics.

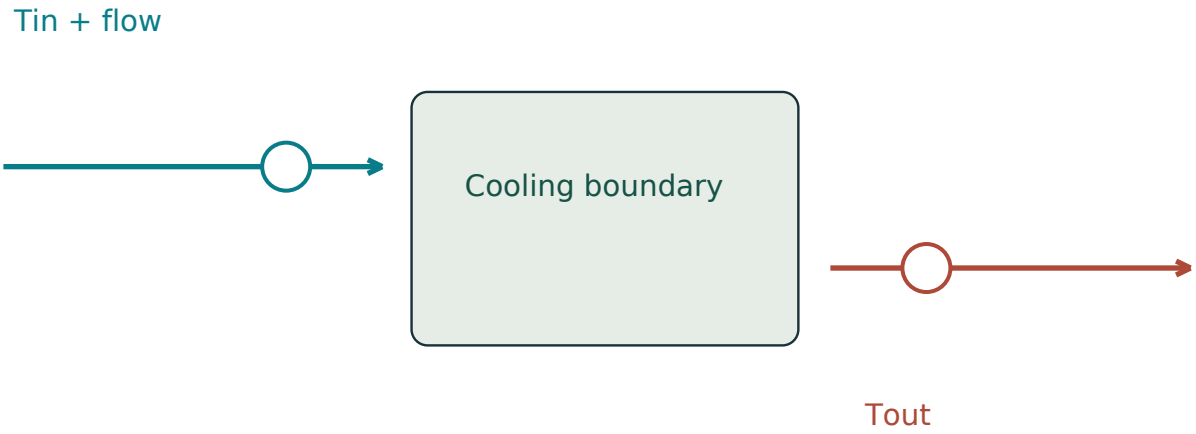
A datasheet claim needs its boundary

Deionised water can acquire ions in service; it should not be treated as permanent electrical insulation. Conductivity, inhibitors, filtration and fluid condition must follow the actual system supplier’s requirements.

This volume assigns no fluid mixing, oil immersion, pressure testing or opened cooling loop at home.

Heat removed needs more than one temperature

A fluid heat balance combines flow, temperature difference and fluid properties. Error in any term can dominate the answer.



Use compatible capture times and the correct fluid properties.

Propagate a simple uncertainty bound

Assume water flow is 0.100 kg/s and measured ΔT is 1.0 K. Qdot = 0.100 × 4,180 × 1.0 = 418 W. If each temperature has an independent stated bound ±0.5 K, a conservative worst-case ΔT bound is ±1.0 K. That is as large as the measured difference, before flow or property error.

Separate absolute accuracy from paired measurement

Calibrated matched sensors can improve differential measurement, but you must establish that performance. Do not assume two identical model numbers have zero relative offset. Clock mismatch, transport delay, mixing and sensor response also matter.

AVOID FALSE PRECISION

“418.000 W” is not justified by a 1 K difference with large uncertainty. Record raw readings, calibration evidence, synchronisation and the calculation method.

Failure diagnosis requires competing explanations

Start from observations and keep uncertainty. Similar temperature symptoms can come from different physical failures.

Observed evidence	Possible explanations	Useful next evidence
Pump command accepted; temperature rising	No flow, insufficient flow, hotter supply, larger load, poor thermal contact.	Fresh flow, pressure difference, supply temperature and load.
Fan RPM normal; component hot	Blocked fins, bypass, recirculation, loose sink, sensor error.	Air path and inlet temperature; contact and sensor checks.
Return ΔT unexpectedly small	Low heat load, high flow, bypass mixing, offset or reversed probes.	Paired temperatures, flow, load and actual topology.
Leak alarm asserted	Actual leak, condensation, sensor contamination or sensor fault.	Local protection response and authorised inspection; preserve the alarm.

Put the action boundary in the record

Who owns protective shutdown? Which measurements can Lupine trust, for how long and within what limits? What is the agreed safe action if evidence disappears? Define these with equipment engineers; do not invent a universal response from one table.

A diagnostic hypothesis is not a verified cause. A successful communication exchange is not proof of cooling.

Home experiment: observe a dry sensor recovering

Use the Part 02 logger and its verified USB-only connections. This measures the response of your sensor assembly, not a chip, cooling loop or person.

Step	Action and purpose
1 / Establish baseline	Put the dry sensor on a nonconductive support, away from sunlight, vents and the board’s heat. Keep the board and connections stationary. Log several minutes.
2 / Apply gentle warmth	If the sensor package can be held without contacting bare leads or disturbing wiring, hold only the sensing body briefly between clean, dry fingers. Otherwise skip physical warming and use the simulation.
3 / Release and mark	Return it to the same support. Mark the release event and continue logging until the trend approaches its earlier baseline. Keep the electronics dry and leave no test unattended.
4 / Repeat	Repeat with the same placement and timing. Record handling differences, drafts and any acquisition errors. Do not discard a run only because it differs from your prediction.

Do not call the result a body-temperature measurement

The package is exchanging heat with fingers, leads, support and room air. Its response includes mounting, sensor conversion and thermal lag. There is no known constant heat input and no established uniform object temperature.

No water, ice, hair dryer, heated resistor, soldering, fan modification or opened equipment is part of this exercise.

Turn the experiment into a useful record

Keep raw data, events and interpretation separate. Reproducible notes are a more durable asset than one attractive temperature graph.

Section	Record
Identity	Run ID, date, firmware, code revision, sensor ROM, package and board model.
Setup	Photograph or sketch; support; probe and board positions; room context; logging interval.
Events	Start, warming begins, release, end; how event times relate to sensor timestamps.
Raw samples	Original serial records, sequence, value, units, acquisition state and errors. Keep failures.
Plot	Time since release versus reported temperature. Mark gaps and events; do not join missing data as known readings.
Interpretation	Baseline stability, repeatability, peak after handling and recovery shape. State confounders and unknowns.

Use a time constant only when the model is supported

For ideal exponential recovery, excess temperature falls to 36.8% after one τ. A stable baseline, a meaningful initial excess and adequate signal-to-noise are needed to estimate it. Your experiment may not satisfy those conditions. Near baseline, small offsets can dominate.

A measured recovery curve does not separately identify Rth and Cth without additional independent information.

Model a known thermal system for free

Run this with Python 3 on your computer. It creates synthetic temperature data from a one-node model; no hardware or network is connected.

Save as thermal_lab.py. Heating is on for 300 s, then off for 300 s. The update is the exact first-order solution over each interval with constant input and parameters.

```
import math

ambient = 22.0            # degC
resistance = 8.0          # K/W
capacity = 20.0           # J/K
step = 1.0               # s
on_power = 2.0            # W
switch_time = 300.0       # s
end_time = 600.0          # s

temperature = ambient
factor = math.exp(-step/(resistance*capacity))
print('time_s,temperature_degC,input_W,kind')

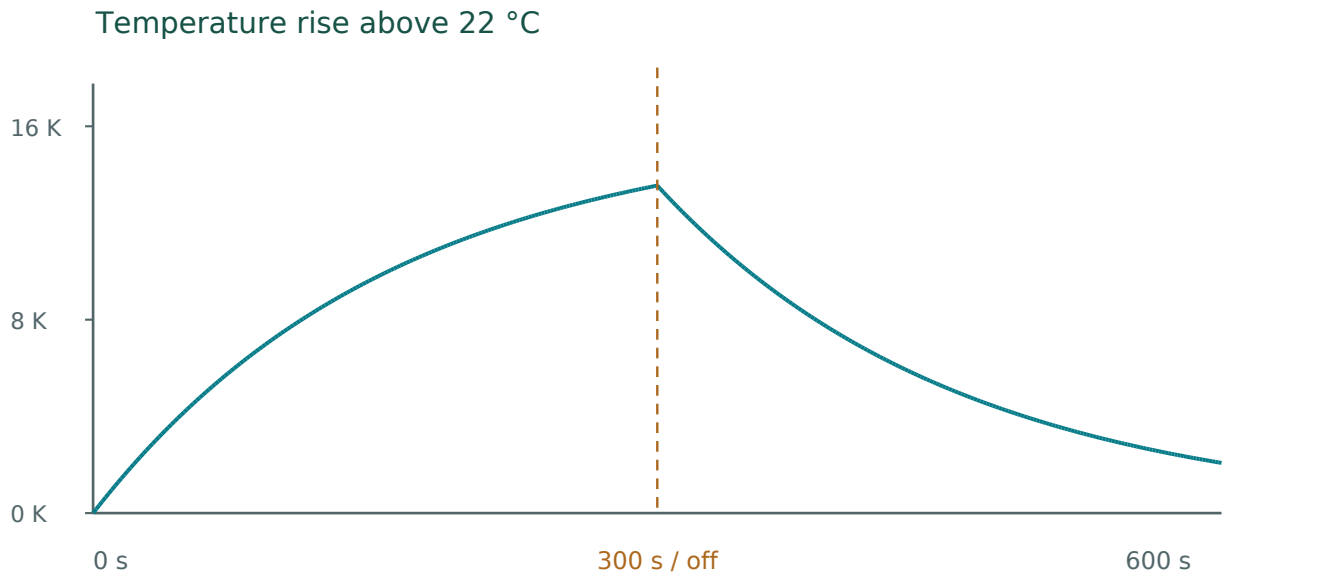
for n in range(round(end_time/step) + 1):
    time_s = n*step
    power = on_power if time_s < switch_time else 0.0
    if n % round(60/step) == 0:
        print(f'{time_s:.0f},{temperature:.4f},'
              f'{power:.1f},simulated')
    if time_s >= end_time:
        break
    equilibrium = ambient + power*resistance
    temperature = equilibrium + (temperature-equilibrium)*factor
```

WHAT THE TIMESTAMPS MEAN

Each row reports temperature at that time. Input_W describes the interval beginning then, so the 300 s row has power 0 W while the temperature still reflects the previous heating. Parameters are invented, not fitted to your sensor.

Read the model before fitting anything real

The synthetic curve shows storage and heat rejection with known parameters. It is a teaching result, not a validated forecast for a server.



Time	Model temperature	Reason
0 s	22.0000 °C	Starts at assumed ambient.
160 s	32.1139 °C	One time constant into heating.
300 s	35.5463 °C	Heating switches off here.
600 s	24.0774 °C	Still above ambient after recovery.

Change one assumption and predict the effect

Double Cth: the same eventual rise is approached more slowly. Halve Rth: the eventual rise halves and τ halves. Raise ambient by 5 K: every absolute model temperature shifts up by 5 K when properties stay fixed.

Ten checks before the next part

Write units, model conditions and evidence limits. The last three checks ask you to reason from observations.

1. A 0.25 kg water mass warms by 8 K with cp = 4,180 J/(kg·K). Find the energy gained, ignoring its container.

2. A layer has L = 2 mm, k = 4 W/(m·K) and A = 10 cm². Find Rth and the temperature drop at 30 W.

3. A path has Rth = 0.60 K/W and carries 50 W. At 24 °C ambient, estimate the hot-node temperature.

4. A one-node model has Rth = 5 K/W, Cth = 40 J/K and constant input 3 W. Find τ and the steady temperature rise.

5. A water stream removes 2 kW with a 5 K rise. Use cp = 4,180 and ρ = 1,000 to find kg/s and L/min.

6. A pump raises pressure by 60 kPa at 2 L/min. Find hydraulic power and electrical input at 40% wire-to-water efficiency.

7. A chiller removes 80 kW with cooling COP = 4. Find electrical input and hot-side heat rejection for this boundary.

8. Room dew point is 15 °C and a pipe surface is 12 °C. Is condensation possible? Does this determine a universal safe coolant setpoint?

9. A fan reports normal RPM while the component temperature rises. Give three possible explanations and one useful extra measurement.

10. You measure inlet 30.0 °C and outlet 30.5 °C with separate ±0.5 K bounds. How strong is the inferred heat-flow claim?

Worked answers: 1-5

Match the equation to the physical boundary before substituting numbers.

1 Sensible heat in water

Q = mcpΔT = 0.25 × 4,180 × 8 = **8,360 J**. This assumes no phase change and constant cp; container energy and losses were explicitly excluded.

2 Conductive thermal resistance

A = 10 cm² = 0.001 m² and L = 0.002 m. Rth = L/(kA) = 0.002/(4 × 0.001) = **0.50 K/W**. At 30 W, ΔT = QdotRth = **15 K** across this layer only.

3 Steady temperature rise

Rise = 50 × 0.60 = **30 K**. Hot-node temperature = 24 + 30 = **54 °C**. This does not prove the component permits that temperature or that the series model captures every path.

4 Time constant and eventual rise

τ = Rth Cth = 5 × 40 = **200 s**. Steady rise = 3 × 5 = **15 K**. A temperature 15 K above ambient is approached gradually, not reached instantly.

5 Required liquid flow

mdot = 2,000/(4,180 × 5) = **0.09569 kg/s**. Vdot = mdot/ρ = 0.00009569 m³/s, or **5.742 L/min**. The fluid balance assumes the full heat reaches that measured stream.

Worked answers: 6-10

The evidence questions are as important as the arithmetic. Keep unknowns explicit.

6 Pump power boundary

2 L/min = 0.00003333 m³/s. Phyd = 60,000 × 0.00003333 = **2 W**. Electrical input = 2/0.40 = **5 W** for the assumed combined efficiency. The 3 W difference is loss.

7 Refrigeration energy balance

Electrical input = 80/4 = **20 kW**. Hot-side rejection = 80 + 20 = **100 kW**. Additional external pumps or fans must be added if they lie outside the stated COP boundary.

8 Dew-point comparison

Yes, condensation is possible because 12 °C is below the stated 15 °C dew point. No universal coolant setpoint follows: local surfaces, humidity changes, uncertainty and the equipment envelope require a documented margin.

9 Normal RPM does not prove cooling

Possible explanations include blocked fins, inlet recirculation, a poor interface, greater heat load or sensor error. Useful evidence could be a fresh inlet-temperature measurement, airflow evidence or measured load. Choose it to distinguish the hypotheses.

10 A small temperature difference with large bounds

Measured ΔT is **0.5 K**; the conservative worst-case bound from the stated sensor bounds is **±1.0 K**. Even its sign is not established by those bounds alone. Flow and property error add uncertainty. Better paired calibration and measurement design are needed before a strong heat-flow claim.

Reference: heat, flow and cooling formulas

Qdot denotes heat rate, Vdot volume flow and mdot mass flow. These names avoid using one symbol Q for several different quantities.

Relationship	Conditions / units
T[K] = T[°C] + 273.15	Use absolute K in radiation and absolute ratios.
Q = mcpΔT; Cth = mcp	Sensible heat, constant properties, no phase change.
dU/dt = energy rate in - rate out	State the full boundary and energy paths.
Rcond = L/(kA); Qdot = ΔT/Rth	1D steady flat-wall model for Rcond.
Qdotconv = hA(Ts - Tfluid)	h and effective area depend on the conditions.
Qdotrad = εσA(Ts⁴ - Tsur⁴)	Small grey surface in large uniform surroundings.
Cth dT/dt = Pin - (T - Tamb)/Rth	One lumped thermal state, constant parameters.
τ = Rth Cth; ΔTsteady = Pin Rth	First-order constant-boundary model.
Vdot = A vmean; mdot = ρ Vdot	Specify density, actual area and mean velocity.
Δp = ρgH; Phyd = Δp Vdot	H here is differential pump head in m.
Pelectrical = Phyd / ηtotal	Specify wire-to-water or another efficiency boundary.
Qdot = mdot cp(Tout - Tin)	Steady sensible-fluid heat balance, with sign stated.
QdotHX = UA ΔTlm	Exchanger model and proper mean ΔT.
COPcool = Qdotcold/Pelectric	Heat moved / electrical input at a stated boundary.
Qdothot = Qdotcold + Pelectric	Steady refrigeration energy balance for that boundary.

Useful conversions: 1 L/min = 1.6667 × 10^-5 m³/s; 1 kPa = 1,000 Pa; 1 kWh = 3.6 MJ.

Words to recognise in cooling documents

Keep the distinction between a physical property, a system state and a controller command.

Thermal resistance Temperature difference per heat-transfer rate, in K/W, under specified conditions.

Thermal capacitance Stored-energy change per temperature rise, in J/K.

Thermal interface Contact region between parts; its condition can strongly affect the heat path.

Convection coefficient h, a condition-dependent surface-to-fluid heat-transfer coefficient.

Mass / volume flow kg/s / m^3/s; connected through density.

Pump head Fluid energy per unit weight, expressed in metres; distinct from elevation alone.

Duty point Intersection of device performance and connected-system characteristics.

Cavitation Formation and collapse of vapour cavities when local pressure conditions permit; a pump hazard.

Approach temperature A stated temperature difference between relevant streams or sink temperatures.

LMTD / UA Log-mean temperature difference / overall conductance times area in an exchanger model.

COP Coefficient of performance; specify heating or cooling and the electrical boundary.

Dry bulb / wet bulb Air temperature / a psychrometric temperature associated with evaporative cooling conditions.

Dew point Temperature at which air at the stated moisture content becomes saturated on cooling.

CDU / TCS Coolant distribution unit / technology cooling system, with architecture-specific functions.

Direct-to-chip / immersion Cold plates at components / equipment in a suitable fluid; different cooling architectures.

Recirculation / bypass Warm exhaust returning to inlets / supplied air missing the intended equipment path.

METRICS NEED BOUNDARIES

PUE compares facility energy with IT energy. It does not by itself establish water use, emissions, heat-reuse value or useful computing output.

References: heat and fluid foundations

Original explanations, diagrams and numerical cases are provided here. These primary references support further checking; no additional textbook reading is required.

OpenStax / Heat and specific heat
Sensible heat, heat capacity and calorimetry.

OpenStax / Mechanisms of heat transfer
Conduction, convection and radiation.

OpenStax / Fluid dynamics
Volume flow, mass flow and continuity.

OpenStax / Viscosity and turbulence
Viscosity and the limits of simple flow-resistance models.

OpenStax / Refrigerators and heat pumps
Cooling and heating COP and energy balances.

US DOE / Pumping system sourcebook
Head, fluid properties, circuit requirements and system performance.

Grundfos / Pump curves
Head-flow relationships and combined performance curves.

Grundfos / How to read a pump curve
Duty point, efficiency and cavitation-related limits.

Grundfos / Pump hydraulics training
Curve interpretation, hydraulic power and affinity-law foundations.

WHY THE EXAMPLES USE ROUNDED CONSTANTS

They make the reasoning reproducible. For an actual design, use the fluid, material, geometry, temperature and pressure properties applicable to that equipment.

References: data-centre cooling systems

Manufacturer examples illustrate principles. They do not establish that a particular product, coolant or architecture is suitable for your project.

US DOE FEMP / Data-centre design guide

2024 overview spanning IT, airflow, cooling, electrical systems, heat recovery and metrics.

US DOE FEMP / Cooling water efficiency

Heat-rejection routes, air management and economiser context. No savings percentages are assumed in this guide.

ASHRAE Handbook / Data centres, Chapter 20

Liquid-cooling loop boundaries, coolant compatibility and dew-point considerations. Access arrangements may vary.

Noctua / Fan performance comparison

Why the pressure-flow curve and actual restriction determine the operating point.

Noctua / Fans in series and parallel

Combined curves and practical installation limitations.

US National Weather Service / Dew point

Dew point, condensation and the distinction from relative humidity.

Edition and review status

Edition 01 / 17 October 2026. Prepared with ChatGPT. Numerical cases and thermal curves are assumed or simulated unless described as your future measurements. Code is tested against analytical results; no hardware or industrial cooling system has been tested for this volume. No independent engineering peer review is claimed.

Bring the thermal evidence back to Lupine

Your next capability is to preserve observations and their limits well enough that a physical outcome can be assessed honestly.

Lupine layer	What this volume adds
Aiva	Physical constraints come before scoring: temperature, flow, pressure, dew-point margin and equipment limits require justified values.
Rail	A cooling request must be separated from acknowledgement, actual actuator state and fresh thermal evidence. Delays and uncertain outcomes remain explicit.
Cloked	Preserve raw measurements, timestamps, setup, calibration status, units and corrections. Record integrity does not make a wrong sensor reading physically valid.

Three things to keep from Part 04

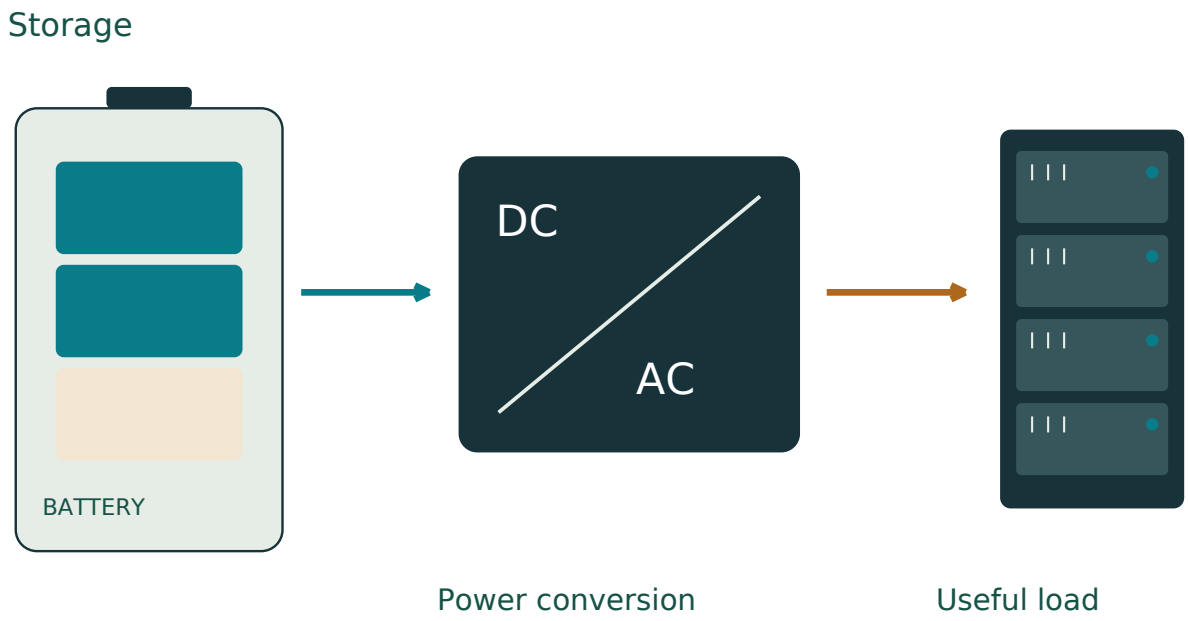
A reference sheet: the equations and definitions on pages 04-36 and 04-37.
A reproducible result: the synthetic thermal model, plus the optional dry-sensor recovery record.
A question worth investigating: one observed or documented cooling failure, with the missing evidence clearly named.

NEXT: PART 05 / BATTERIES, INVERTERS AND POWER ELECTRONICS

The next volume connects electrical storage, conversion, limits and heat to the same measurement discipline. Four planned parts remain after this one: 05, 06, 07 and 08.

BATTERIES, INVERTERS & CONVERSION

Part 05 / Understand stored energy, power limits and the electronics connecting storage to useful loads.



Functional illustrations. Actual systems add protection, controls, isolation and measurement.

Energy availability is only one condition

A battery must hold enough energy, deliver the required power and remain inside its permitted operating envelope. The converter must handle voltage, current, waveform and heat. A controller must know which limits and measurements are authoritative before requesting any action.

WHAT YOU WILL LEARN

Capacity and runtime; SoC and SoH; battery management; DC-DC conversion; inverter switching; grid interfaces; UPS paths; and a bounded storage model with explicit energy accounting.

Read from the cell to the system boundary

Parts 01-04 supplied circuits, sensors, AC and heat. This part connects those foundations to storage and power conversion.

Pages	Question to answer
3-8	What is stored, what do ratings mean and how are battery states estimated?
9-13	Which current, temperature, charging and protection limits constrain use?
14-21	How do capacitors, DC-DC stages, bridges and rectifiers move energy?
22-26	How do inverters, UPS systems and local loads connect to a wider grid?
27-30	How do I calculate efficiency, runtime, heat and credible measurements?
31-33	How can I simulate bounded storage and check every energy transfer?
34-38	Can I solve the checks and use the formulas and glossary?
39-40	What are the primary references and the next Lupine capability?

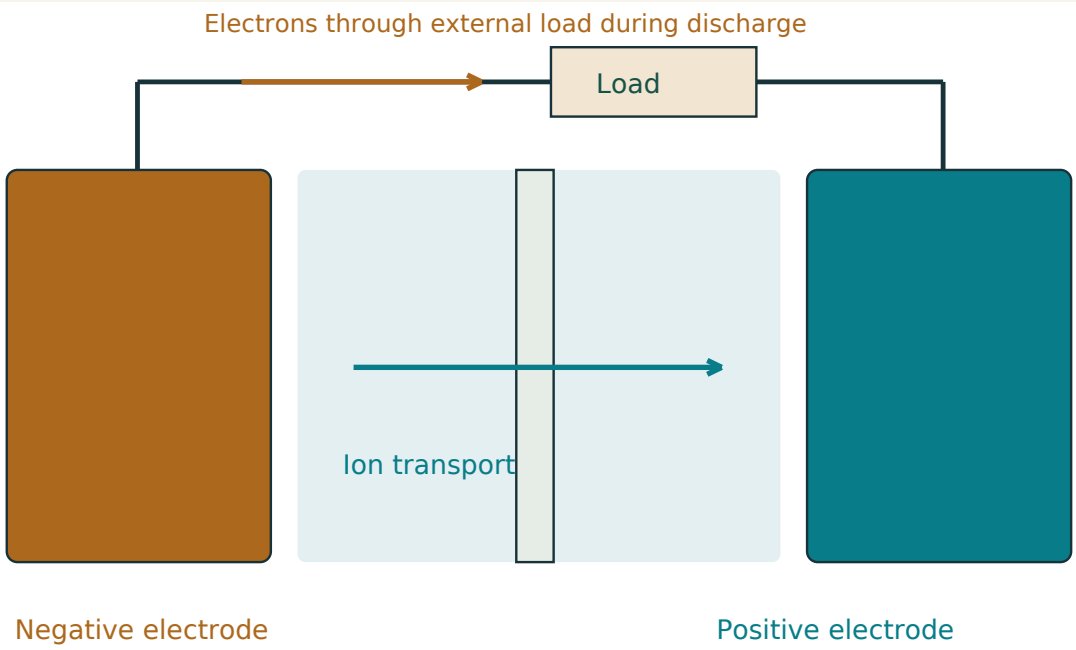
No battery build is assigned

Use calculations, a computer model and external product documentation. Keep physical work at the USB sensor level from earlier parts. Do not open packs, assemble loose cells, short terminals, build mains converters or connect homemade electronics to a grid.

Page labels run 05-01 to 05-40. Three planned parts follow this volume.

A cell stores energy through chemical state

Electrochemical reactions connect the external electrical circuit to ion movement inside the cell. Stored charge and stored energy are related, not identical.



Discharge transfers chemical energy to the circuit

In a rechargeable lithium-ion cell, ions move between electrode materials while electrons travel through the external path. A separator helps prevent electronic contact between electrodes while permitting ionic transport through the electrolyte. Charging drives the reverse overall process with an external energy source.

Chemistry determines the voltage and permitted treatment

Electrode and electrolyte choices change voltage, capacity, power capability, thermal behaviour and ageing. Terms such as LFP and NMC identify chemistry families, not one universal charge voltage or safe operating range. Use the exact cell and pack specification.

A DIAGRAM IS NOT A CELL-CONSTRUCTION GUIDE

This is a functional section with no material recipe. The words anode and cathode can depend on reaction convention; the electrode signs here describe discharge.

Amp-hours and watt-hours answer different questions

Amp-hours describe charge capacity under stated conditions. Watt-hours describe energy. A voltage reference is needed to connect them.

$$1\ \text{Ah} = 3{,}600\ \text{C}$$

$$\text{Energy} \approx \text{nominal voltage} \times \text{Ah}$$

$$\text{Exact electrical energy: } E = \int V(t)\, I(t)\, dt$$

Use seconds to obtain joules; divide joules by 3,600 for watt-hours.

Work an external-label example

A hypothetical pack rated 12.8 V nominal and 100 Ah corresponds to about 1,280 Wh = 1.28 kWh nominal. This approximation does not establish its usable AC output. Terminal voltage changes with load and state; capacity ratings also depend on temperature, rate, cut-off and test method.

THE POWER-BANK TRAP

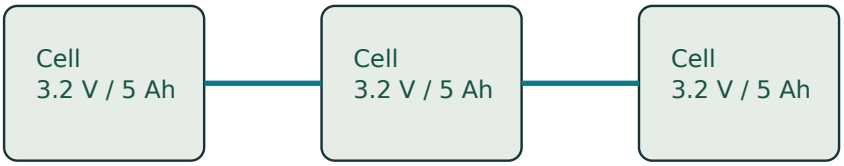
Assume a 5,000 mAh rating refers to a 3.7 V internal cell: nominal energy is 18.5 Wh. At an assumed 90% conversion efficiency to 5 V output, about 16.65 Wh is available, equivalent to 3,330 mAh at 5 V before other limits. Do not compare mAh at different voltages directly.

Read the voltage attached to the capacity

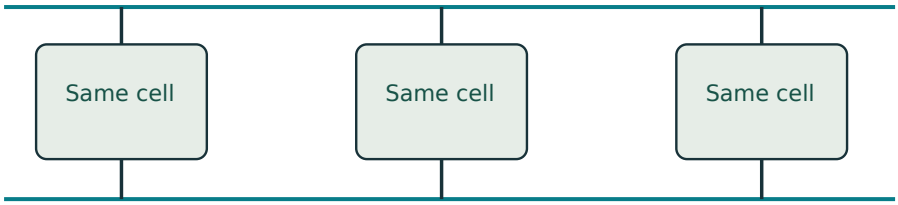
A product can quote cell capacity and USB output ratings on the same label. Preserve which boundary each value belongs to; do not silently attach 5 V to a cell-level mAh figure.

Series adds voltage; parallel adds charge capacity

These are ideal relationships for matched cells in a designed pack. They explain ratings, not how to assemble a battery at home.



Series: 9.6 V, 5 Ah ≈ 48 Wh



Parallel: 3.2 V, 15 Ah ≈ 48 Wh (matched-cell model)

Generalise the arithmetic

For Ns cells in series and Np identical strings in parallel, nominal voltage is NsVcell and charge capacity is NpAhcell. Nominal energy is NsNpVcellAhcell. A hypothetical 4s2p arrangement of 3.2 V, 5 Ah cells gives 12.8 V, 10 Ah and 128 Wh from eight cells.

The real pack is constrained by its components

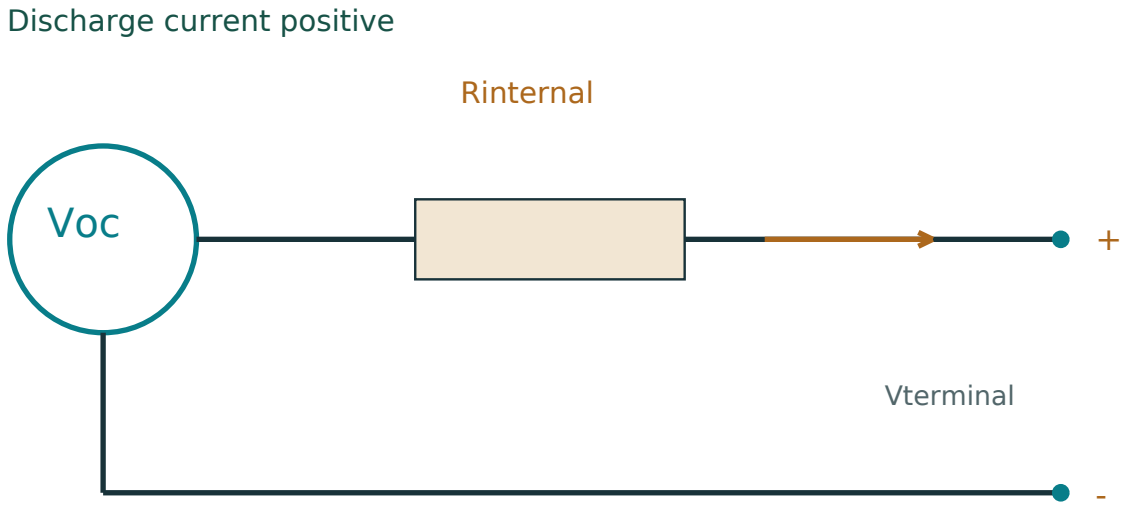
Series groups carry the same string current, but their voltages can diverge. Parallel paths may not share current equally. Differences in resistance, state, temperature or damage matter. A pack needs engineered interconnections, monitoring, fault handling and compatible protection.

PACK RATINGS ARE NOT CELL RATINGS MULTIPLIED BLINDLY

The usable envelope depends on the limiting group and the assembled system. This guide assigns no cell pairing, spot welding or pack modification.

Terminal voltage falls under discharge load

A first battery equivalent circuit combines an open-circuit voltage source and internal resistance. It is useful for intuition but leaves out important dynamics.



Calculate voltage sag and resistive heat

In this simple discharge model Vterminal = Voc - IRinternal. Assume Voc = 12.8 V, Rinternal = 0.040 Ω and I = 10 A. Terminal voltage becomes 12.4 V. Internal resistive heat is I^2R = 4 W, and terminal power is 124 W. The source-side rate is 128 W in this model.

Resistance is not a fixed battery fingerprint

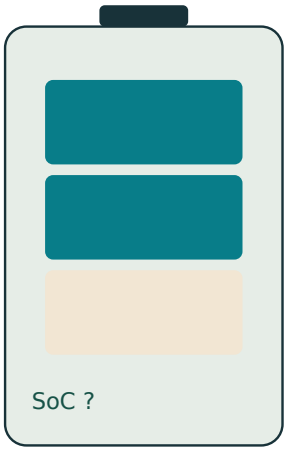
Temperature, SoC, age, time scale and measurement method affect the apparent impedance. Electrochemical polarisation and relaxation add dynamics. A single terminal-voltage reading cannot independently establish SoC or remaining energy.

CONSTANT POWER CAN INCREASE CURRENT

At a regulated 120 W terminal output, current would be 10 A at 12 V but 12 A at 10 V. A voltage decline can therefore increase current and heating until a limit intervenes.

State of charge is an estimate with a reference

SoC commonly expresses remaining charge relative to a defined usable or full-charge capacity. The denominator, method and uncertainty matter.



$$SoC_{next} \approx SoC_{now} - I\,\Delta t / C_{ref}$$

I in A; Δt in hours;
Cref in Ah; SoC as a fraction.

Discharge-positive convention;
charging reverses the sign.

Coulomb counting integrates current

If a 100 Ah reference starts at 80% and supplies a constant 10 A for 2 h, an ideal charge-count model reaches 60%. It assumes a correct starting state, adequate capacity reference and valid current integration. Charge efficiency, self-discharge and operating limits can require correction.

SMALL OFFSET, LONG TIME

A current offset of +0.05 A integrated for 24 h produces 1.2 Ah of false discharged charge: 1.2 percentage points on a 100 Ah reference. More decimal places in the SoC display do not remove that error.

Voltage and models can help, within conditions

Rested voltage, temperature and impedance models may constrain the estimate, but some chemistries have a flat voltage region. SoC is not directly observed by a simple voltmeter.

State of health needs a defined metric

A battery can lose usable capacity and gain resistance. One SoH number need not describe both changes or every service requirement.

Metric	What it indicates
Capacity-based SoH	Measured full-charge capacity / chosen reference capacity, under comparable tests.
Resistance or power capability	Voltage sag, heating and deliverable power under defined conditions.
Calendar and cycle history	Exposure, throughput and temperature history; these are explanatory inputs, not health proof by themselves.
Available energy now	Depends on current state, limits, temperature and load as well as long-term degradation.

Compare like tests before claiming degradation

If a reference capacity is 100 Ah and a comparable controlled test now gives 85 Ah, capacity-based SoH is 85%. Different cut-off voltage, temperature or discharge rate could change the result. That 85% does not automatically mean 85% of every original power capability remains.

AGEING IS A DESIGN INPUT

Cycle depth, time at state, temperature and charging conditions affect different cells differently. An assumed “number of cycles” without a test definition is not a lifetime guarantee.

C-rate expresses current relative to capacity

The C-rate shorthand helps compare loading, but it is not a universal permission to charge or discharge at that rate.

Current = C-rate × capacity in Ah

100 Ah at 0.5C → 50 A

5 Ah at 2C → 10 A

Ideal time at 0.5C is 2 h; actual duration depends on conditions and cut-off.

Power and energy are separate constraints

A hypothetical 10 kWh system with a 5 kW converter can sustain 5 kW for up to 2 h only before reserve, losses and other limits. A 20 kW demand cannot be met just because enough energy is stored. Conversely, a high-power small-energy buffer may meet a short burst but not an extended outage.

A current limit can shrink the power envelope

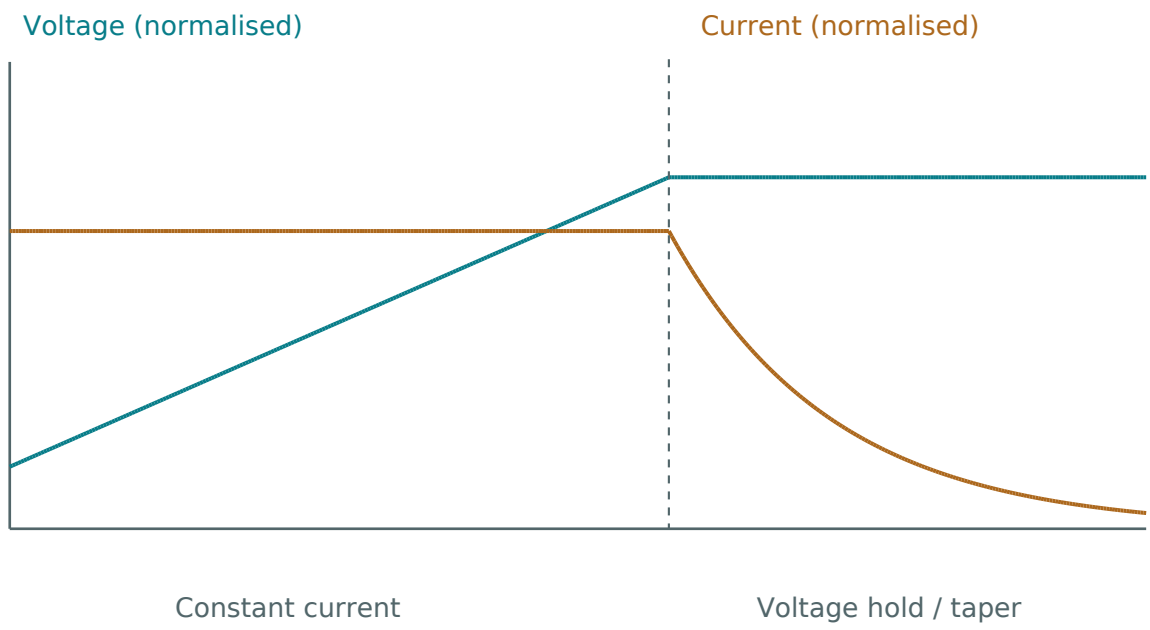
If the maximum discharge current is 100 A and terminal voltage is 40 V, terminal power cannot exceed 4 kW under that limit. At a different voltage, temperature or SoC the permitted power may change. Charge and discharge limits can be different.

READ CONTINUOUS AND PULSE RATINGS SEPARATELY

A short-duration rating requires its duration, repetition, recovery, temperature and protection conditions. It is not a continuous operating value.

Charging follows a chemistry-specific profile

Many lithium-ion charging systems use conditioning, constant current, constant voltage and termination. Exact thresholds come from the supported cell and pack design.



Constant voltage does not mean charging is complete

After the voltage-regulation level is reached, current typically tapers. Completion depends on specified termination logic and conditions, not voltage alone. The charger may also inhibit charging outside temperature limits or detect a fault. Conditioning requirements depend on the actual design.

Do not generalise one lithium profile to every battery

Different lithium chemistries, lead-acid and other systems have different limits and charge-management requirements. Low temperature, imbalance or an aged cell can constrain charging. Use a compatible finished system and its approved charger; software guesses are not an acceptable substitute.

THIS CHART HAS NO DEVICE SETPOINTS

Curves show stages only. Neither their heights nor the stage boundary specify an actual voltage, current or timing for a cell.

A BMS monitors and constrains the battery

Battery-management functions vary by product. Identify what the system measures, estimates, balances, limits and physically disconnects.

Function	Purpose and limitation
Cell/group voltage	Find groups nearing permitted limits. Pack total alone can conceal one weak group.
Current + temperature	Support protection, estimation and thermal limits. Sensor coverage and response still matter.
SoC / SoH estimation	Maintain a model from measurements and history; not direct proof of chemical state.
Balancing	Reduce state differences between series groups within the design’s capability; not a repair for a damaged cell.
Protection / disconnection	Request or perform protective actions using the designed devices, thresholds and architecture.

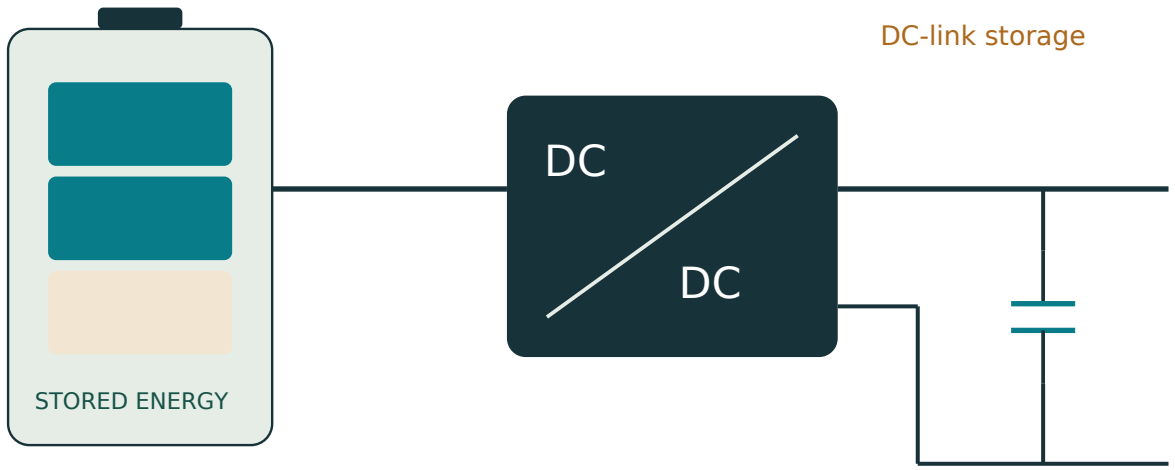
The BMS is not the whole safety case

Fuses, contactors, insulation, enclosure, cooling, fault containment and independent protections may be required by the actual system. A communications register called “BMS OK” cannot establish every cell, connection and actuator is safe.

Lupine should treat authoritative battery limits and protective trips as constraints, not scores to trade off.

A disconnect command is not a de-energised system

Battery systems can retain energy when equipment is off. Protection, isolation and measurement must be designed for the actual source and circuit.



No isolation or discharge path is implied by this functional drawing.

High current matters even below mains voltage

A shorted low-voltage pack can drive hazardous current through a tool, wire or failed component. DC interruption, available fault current and stored capacitor energy must be considered. A protection rating must match the actual circuit; a generic relay or software timeout does not establish it.

Fault response belongs to the engineered system

Overcharge, internal damage, excessive heat or faults can have serious consequences, including thermal propagation in some battery systems. Cell chemistry affects behaviour but does not remove the need for protection. Re-energising after a trip requires an authorised, evidence-based procedure.

YOUR LEARNING BOUNDARY

No loose-cell handling, battery opening, intentional short circuit, high-energy capacitor build or improvised charger is assigned. Work from documents and the simulation in this volume.

Available energy is not available power

Define a feasible operating envelope before choosing a setpoint. Several independent constraints must all remain satisfied.



Schematic envelope only: no real device curve or numerical thresholds are shown.

Combine the constraints instead of averaging them

A request must satisfy energy reserve, available capacity, current limits, cell voltages, temperature limits, converter capability, ramp rate and the authorised operating mode. A high score on one dimension cannot compensate for violating another. Limits may change between request and execution.

HYPOTHETICAL RESERVE CHECK

A 10 kWh reference contains 6 kWh in the simplified energy model. With a 2 kWh lower bound, only 4 kWh may be removed internally. At 90% discharge conversion efficiency, at most 3.6 kWh reaches the AC boundary before that reserve, ignoring auxiliary loads.

A 60% reported SoC is not automatically a calibrated 6 kWh measurement; the example defines its own reference.

A DC-link capacitor buffers a short mismatch

A capacitor exchanges electrical energy as its voltage changes. It can smooth a short disturbance but cannot supply arbitrary energy while holding perfectly constant voltage.

$$E_{cap} = \tfrac{1}{2}CV^2$$

$$\Delta E = \tfrac{1}{2}C(V_{high}^2 - V_{low}^2)$$

Ideal hold-up time ≈ usable ΔE / power

Constant load power, specified voltage range, and neglected losses for this estimate.

Calculate how short the buffer can be

Assume C = 1 mF = 0.001 F and an allowed fall from 400 to 350 V. $\Delta E = 0.5 \times 0.001 \times (400^2 - 350^2) = 18.75$ J. At 1 kW constant output, ideal hold-up is 18.75 ms. This is a high-voltage calculation example, not a construction task.

Precharge limits connection stress

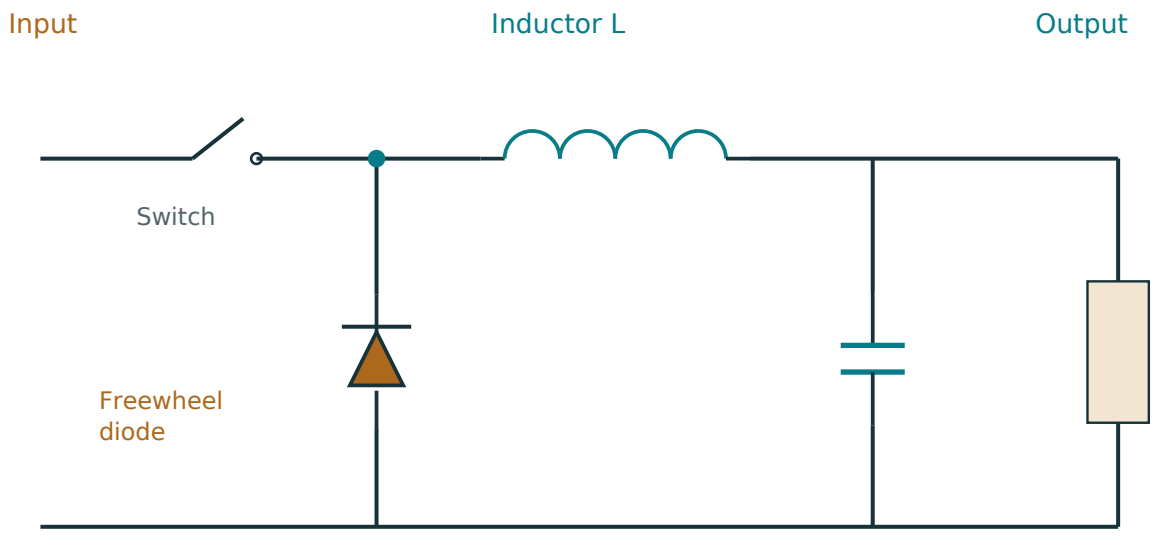
Connecting a discharged capacitor directly to a stiff source can create large inrush current. Engineered precharge raises its voltage before normal connection. The sequence depends on measured voltage, current, time and contactor state. A timer alone does not prove precharge succeeded.

ENERGY CAN REMAIN AFTER POWER IS REMOVED

An accepted OFF command does not establish a discharged DC link. Voltage absence and isolation verification require suitable equipment and authorised procedures.

A buck converter lowers a DC voltage

An inductor, switches and capacitors transfer energy in switching cycles. A feedback controller adjusts operation within limits.



Conceptual power stage. Driver, controller and many practical components are omitted.

Use volt-second balance in continuous conduction

For the ideal buck in steady continuous-conduction operation, Vout ≈ D Vin, with duty fraction D. Assume 12 V input and 5 V output: D ≈ 5/12 = 0.4167. This relationship changes with losses, operating mode and control limits; it is not a complete regulator design.

Inductor volt-second balance: D(Vin - Vout) + (1 - D)(-Vout) = 0.

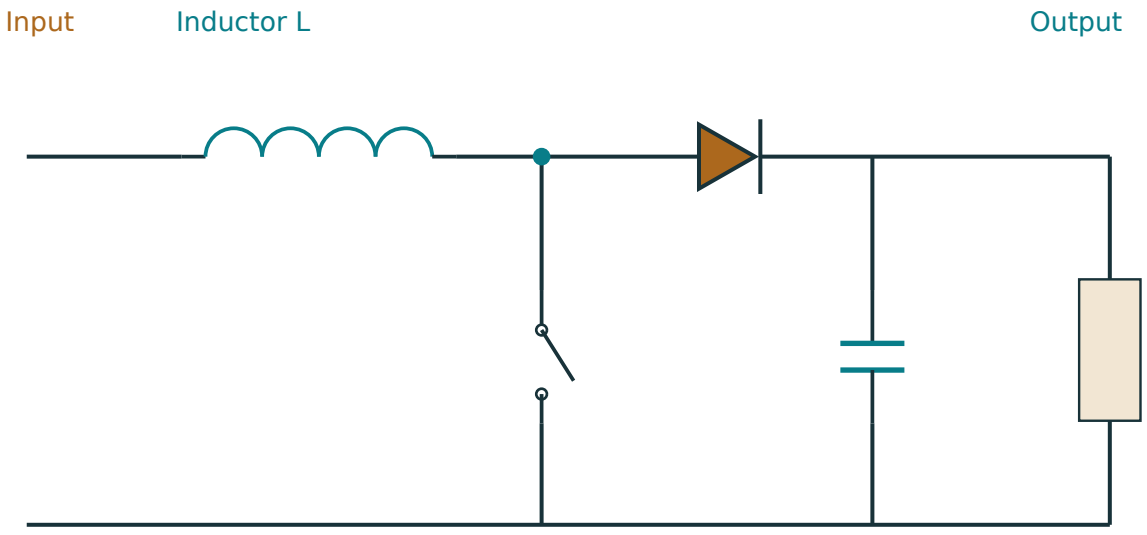
The inductor current keeps flowing between switch events

When the high-side switch conducts, the source feeds the inductor and load. When it opens, the diode provides the continuing current path from return to the switching node. A synchronous design uses a controlled transistor in place of that diode, with additional timing requirements.

No component values or wiring recipe for a physical converter are assigned here.

A boost converter raises a DC voltage

Increasing output voltage requires increased input current for a given output power, once losses are included. It does not multiply energy.



Conceptual nonsynchronous power stage; no gate drive or full protection design is shown.

Follow the two intervals

With the switch on, current builds in the inductor while the diode isolates the output from the switching node. With the switch off, the inductor drives current through the diode to the output. In ideal steady continuous conduction, Vout ≈ Vin/(1 - D).

Inductor volt-second balance: D Vin + (1 - D)(Vin - Vout) = 0.

VOLTAGE GAIN DOES NOT MEAN POWER GAIN

An ideal 5 V to 12 V boost needs D ≈ 1 - 5/12 = 0.5833. For 12 W output at 90% efficiency, input power is 13.333 W and input current at 5 V is 2.667 A. The converter dissipates 1.333 W under these assumptions.

A boost topology does not necessarily block an input-to-output fault path. Actual isolation and protection need checking.

Switching efficiency is a balance of losses

Power semiconductors, magnetics, conductors and capacitors all dissipate heat. Higher switching frequency can reduce some component sizes while increasing other losses.

Loss family	First model / limitation
Conduction	For a resistive switch model, $P \approx I_{rms}^2 R_{DS(on)}$. Temperature and drive voltage change resistance.
Switching transitions	Energy per turn-on and turn-off × switching frequency; device, voltage, current and drive conditions matter.
Gate drive	A rough driver power estimate is $Q_g V_{drive} f_s$. This is not total converter switching loss.
Magnetics + capacitors	Copper, core and ESR losses depend on waveform, frequency, temperature and construction.

Calculate heat at an explicit output point

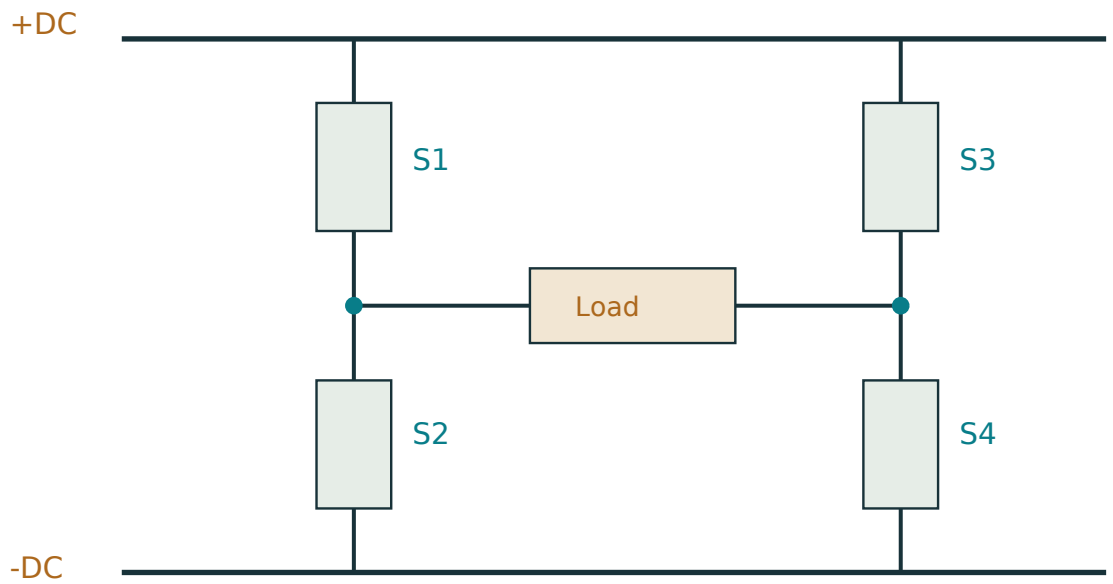
For 1 kW output at 96% efficiency, input is 1,000/0.96 = 1,041.67 W and total loss is 41.67 W. The converter’s thermal path must remove that loss under actual ambient, flow and mounting conditions. Efficiency measured at another load need not apply here.

RECONNECT TO PART 04

If a lumped loss source has 0.8 K/W to ambient, 41.67 W implies about 33.3 K rise in that model. Real semiconductors have multiple thermal paths and limits.

An inverter bridge reverses the load voltage

A full bridge can connect either side of a load to either DC rail. The switch states determine polarity; the controller must prevent conflicting conduction.



Read the diagonal pairs

With ideal S1 and S4 on, the left load terminal is positive relative to the right. With S3 and S2 on, polarity reverses. This basic switching action can create an alternating output. Practical devices also have reverse-current paths, parasitics and gate-drive requirements omitted here.

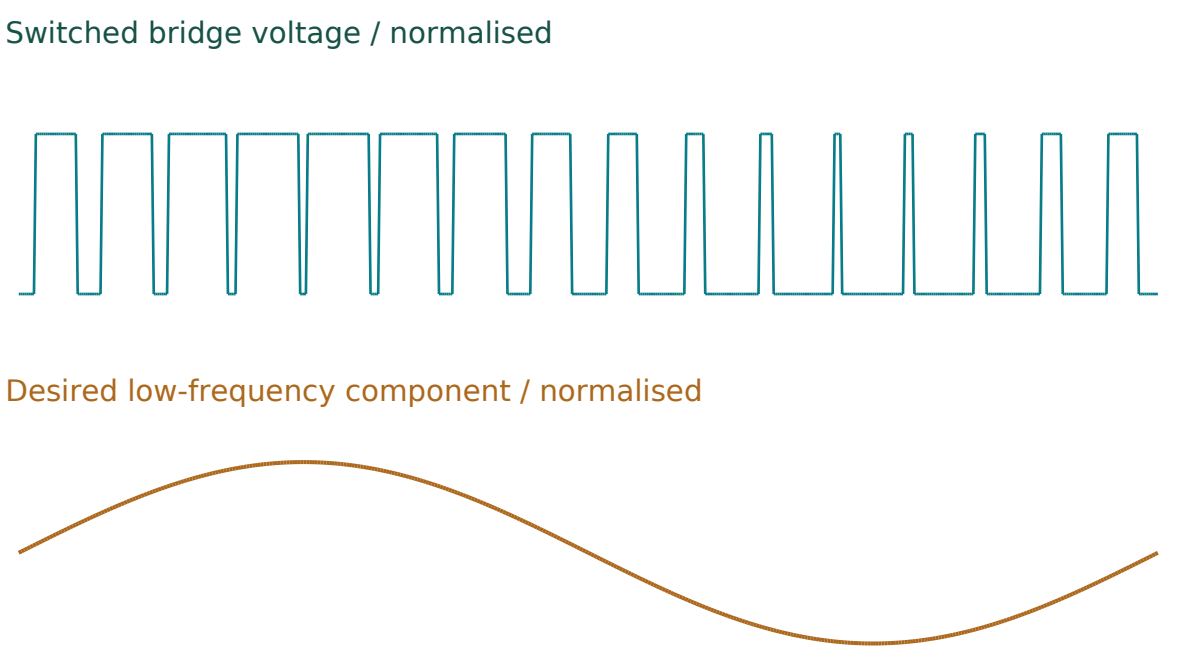
One bridge leg must not short the DC rails

Simultaneous conduction of upper and lower devices in the same leg can create shoot-through. Gate timing, dead time, interlocks and hardware protection are engineered around the actual devices. A microcontroller loop alone does not establish safe switching.

This bridge is a learning schematic, not a complete circuit or a gate-sequence implementation.

PWM shapes an average; filtering shapes the waveform

A switched bridge does not inherently produce a clean sine wave. Modulation, filters, feedback and the connected load determine the delivered waveform.



Separate illustrative views, not a measured waveform or a filter-design result.

Two frequency scales coexist

The desired AC output may be near grid frequency while switching happens much faster. An output filter attenuates unwanted switching components within its design envelope. Filter resonance, load changes, dead time and control dynamics affect the result. One duty fraction cannot describe all output quality.

Control must be stable across the permitted conditions

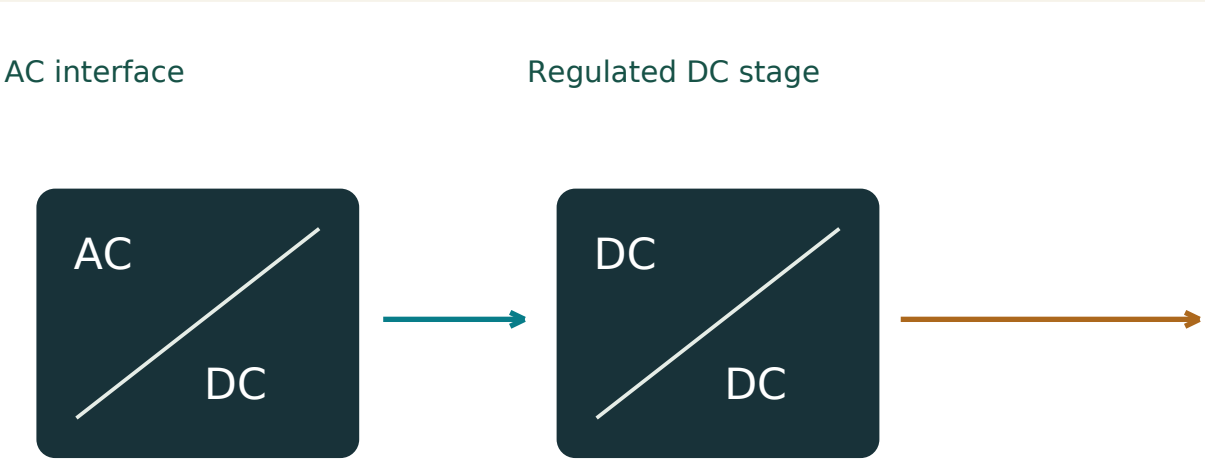
Feedback compares measurements with references and changes modulation. Current limits and protective functions can override the reference. Sampling, computation, filtering and switching add delay. A well-behaved waveform in one simulation does not establish stability for every real network.

FREQUENCY SUPPORT STILL NEEDS ENERGY

Changing an inverter’s control signal does not create sustained active power. The DC source, reserve, converter and operating limits must support it.

Rectification and PFC shape the input side

Rectifiers convert AC to DC. Power-factor correction can shape current draw, but neither the words “rectifier” nor “PFC” specify complete waveform performance.



Functional stages only. Actual active front ends can combine these functions.

A capacitor-input rectifier can draw pulses

When a rectifier charges a DC capacitor near the AC peaks, input current can be concentrated into short pulses. RMS current and harmonic content then matter. Part 03’s true power factor P/(VrmsIrms) is the appropriate boundary metric, not a fundamental phase angle alone.

Active PFC uses controlled current shaping

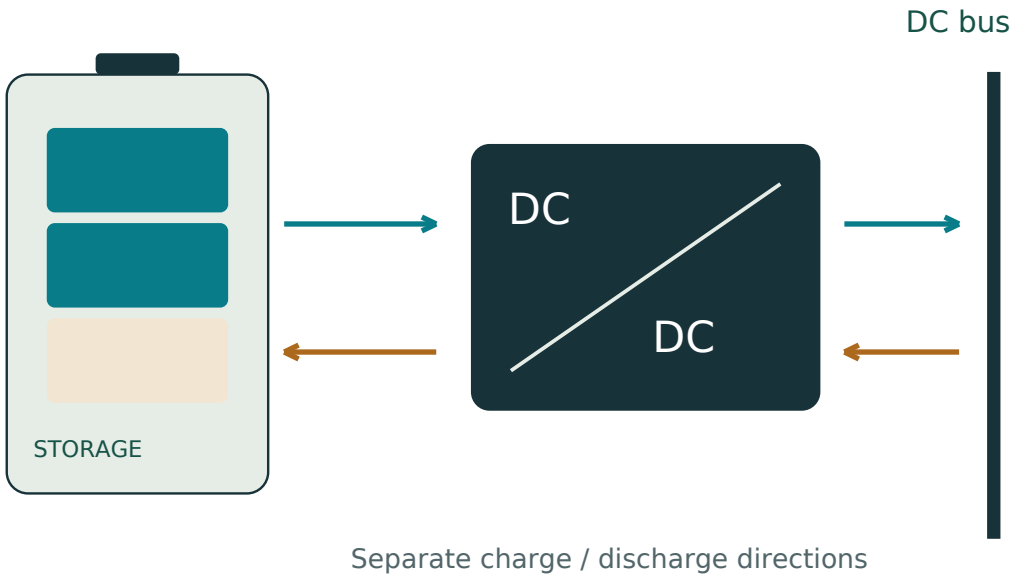
A common single-phase architecture controls current to follow an input-voltage-related reference while regulating a DC bus. Actual quality depends on load, input conditions, control and limits. PFC, efficiency, electrical isolation and bidirectional operation are separate attributes.

CHECK THE EXACT OPERATING POINT

A high PF figure at full load does not prove the same PF at light load or during a transient. Record voltage, power, mode, measurement window and waveform limits.

Bidirectional conversion needs a defined sign

A suitable converter can charge storage from a bus and discharge back to it. Sign conventions and boundaries must survive every command and record.



Separate charge / discharge directions

Choose the boundary first

In this guide’s storage simulation, Pac > 0 means discharge delivered to the external AC boundary. Pac < 0 means AC input used to charge storage. Internal energy then changes at -Pac/ηdis during discharge and at (-Pac)ηchg during charge. Time is in hours when using kW and kWh.

CHECK BOTH DIRECTIONS

Delivering 1 kWh AC at 90% discharge efficiency removes 1/0.90 = 1.111 kWh internally. Absorbing 1 kWh AC at 90% charge efficiency adds 0.900 kWh internally. The difference in each direction is conversion loss for this boundary.

A capable topology is not a permission to export

Connection mode, equipment limits, protections and the authorised interface still govern operation. A bidirectional power stage alone does not establish grid compatibility.

Grid-following and grid-forming describe control

Both use power electronics and must respect current and energy limits. The distinction is how their AC-side control relates to the network waveform.

Concept	Useful description
Grid-following	Typically synchronises to an existing voltage waveform and controls injected current or power relative to it.
Grid-forming	Establishes or regulates a voltage waveform with defined dynamics while interacting with other sources and loads.
Islanded operation	Requires the whole local system to sustain voltage, frequency, power balance and protection after separation.
Black start	Requires a designed energisation sequence, available energy, current capability and compatible equipment; not every grid-forming unit supports it.

Avoid treating a label as a stability certificate

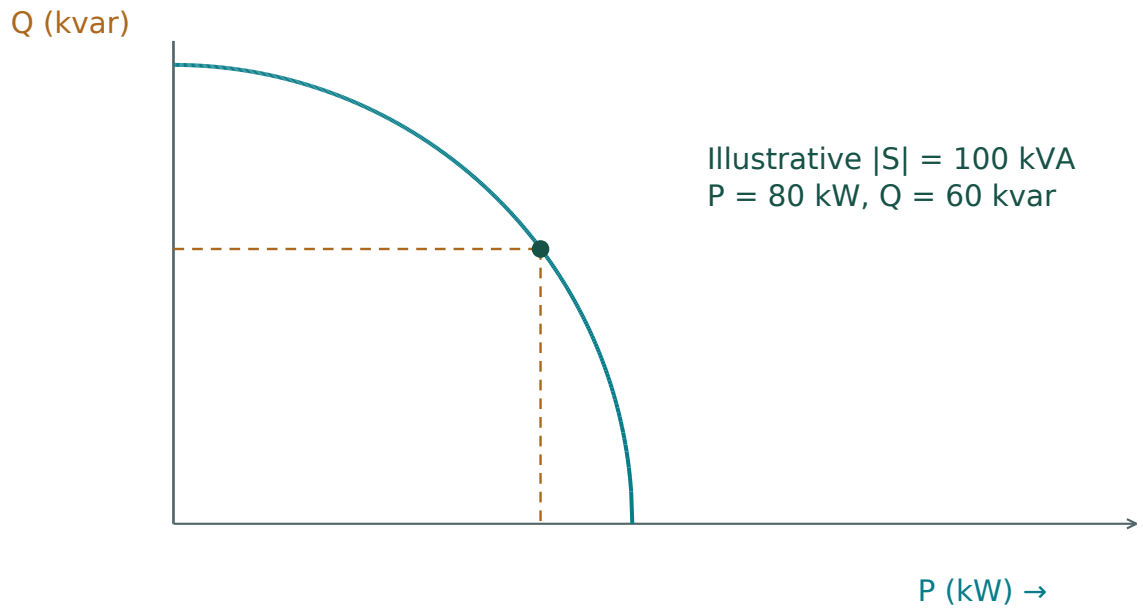
Grid-forming implementations differ. Current limiting, weak-grid interactions, protection and transition behaviour require validated models and equipment testing. A fixed-frequency waveform generator is not the same as a proven grid-forming power system.

CONNECT BACK TO INERTIA

Fast controlled active-power response can support frequency, but its mechanism and limits differ from a synchronous rotating mass. Identify energy source, control, delay and duration.

Reactive power shares the converter’s capability

An inverter’s active-power, reactive-power and current limits interact. A kW rating does not automatically remain available at every reactive-power request.



One quadrant shown. Sign conventions and real equipment envelopes must be checked.

Use the circle only under its stated conditions

For sinusoidal operation with an apparent-power bound Smax, $P^2 + Q^2 \leq Smax^2$ is a simple capability limit. At Smax = 100 kVA and Q = 60 kvar, active power is at most 80 kW. DC source power, current, voltage, temperature and control can impose tighter limits.

Low voltage can make a current limit bind sooner

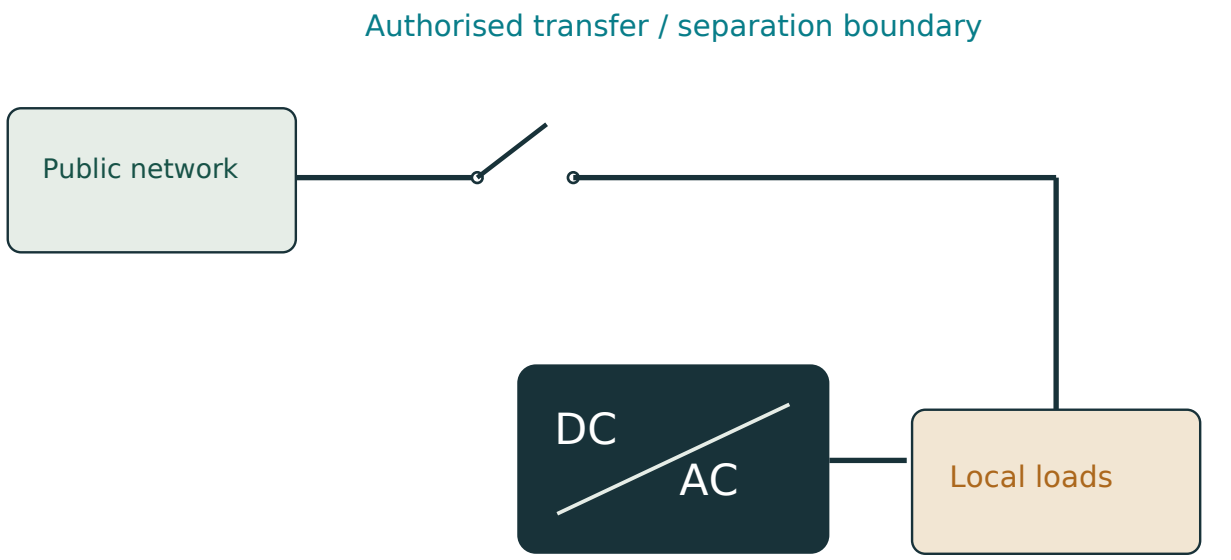
A balanced three-phase converter limited to 100 A at 400 V line-to-line has about 69.28 kVA available from $\sqrt{3}VI$. At 320 V and the same current, this falls to about 55.43 kVA. Fault and transient controls can impose different behaviours.

CLIPPING IS A PHYSICAL OUTCOME TO RECORD

Preserve requested P/Q, applied limits, actual output, mode and alarms. A request acknowledgement does not establish the requested operating point was reached.

Intentional backup needs an engineered boundary

Supplying a local load during an outage is different from energising an unintended island. Separation, earthing, controls and protection must be coordinated.



Functional topology only. Switch position does not verify isolation or specify neutral/earth treatment.

Anti-islanding is a protective function

Unintended island detection and disconnection are part of an actual grid interface. Some systems support intentional backup through a designed transfer arrangement. Ride-through behaviour, reconnection conditions and islanded protection must match the equipment and connection requirements.

Export control is a different function again

A site may limit net export using a meter at a defined connection point. Latency, missing measurements, multiple sources and changing loads affect that loop. Zero reported export does not prove physical separation, and a software power limit does not replace protective anti-islanding.

NO GRID-CONNECTION EXPERIMENT IS ASSIGNED

Study approved product documentation and public engineering material. Do not backfeed a socket, bypass protection or build an improvised transfer arrangement.

A UPS keeps a load supplied through specific paths

Standby, line-interactive and online double-conversion systems have different normal and backup paths. The mode and product determine the actual behaviour.

Topology	Normal path	What to check
Standby	Input normally feeds the load through the designed path; transfer to inverter backup on qualifying disturbance.	Transfer conditions, output waveform, overload and runtime.
Line-interactive	Normal path commonly includes voltage regulation, with inverter backup when needed.	Regulation range, transfer behaviour, battery and output limits.
Online double conversion	Rectifier supplies a DC link; inverter normally supplies the load in double-conversion mode.	Bypass and efficiency modes, overload behaviour, redundancy and actual transfer conditions.

A bypass is a different supply condition

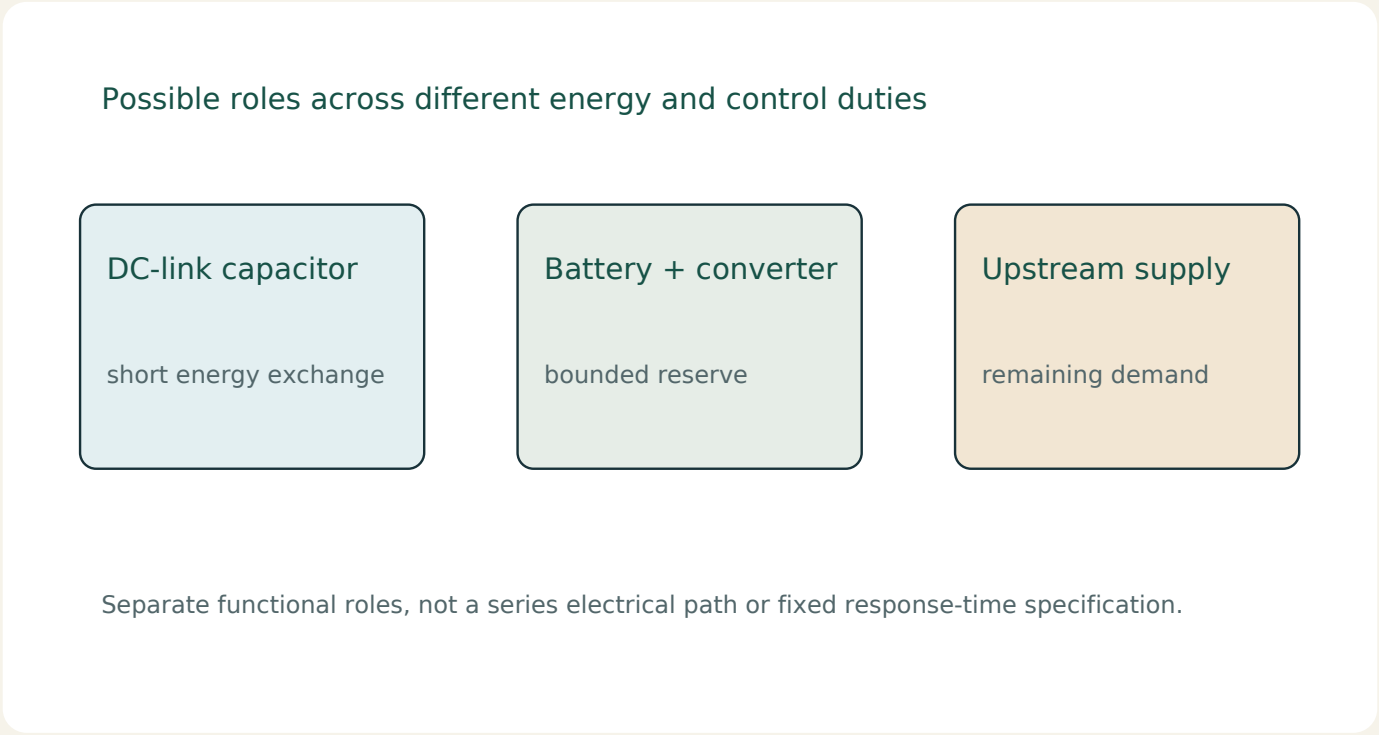
A static or maintenance bypass can route input to output around conversion stages. That can preserve supply while changing conditioning or resilience. Do not infer the load is on the inverter just because the UPS front panel is powered. Observe the reported mode and its evidence limits.

A UPS IS NOT AUTOMATICALLY A GRID-SERVICE BATTERY

Using stored energy for another task can consume backup reserve or violate permitted operation. It needs supported equipment, an explicit reserve policy and authorised interfaces.

Buffers can separate load changes across time scales

Capacitors, batteries and upstream supplies can share a disturbance over different durations. Their controls and limits determine how much smoothing actually occurs.



Calculate both the power and energy requirement

A hypothetical 100 kW excess load for 2 s requires 200 kJ = 0.05556 kWh at the load boundary. At 90% discharge conversion efficiency, internal energy removal is about 0.06173 kWh. Small energy does not make the 100 kW power requirement disappear.

A repeated burst needs a recovery strategy

Storage must recharge between events within permitted current, reserve and thermal limits. A smoother can shift stress to the charging interval or reach its bounds. Evaluate sustained patterns, not one attractive short trace. Converter current and control interaction can limit the benefit.

THE TASMANIA CONNECTION IS AN INVESTIGATION

These principles help examine AI-load behaviour. They do not prove a specific local plant problem or a novel Lupine product; actual traces, interfaces and engineering review are still needed.

Runtime and round-trip efficiency need boundaries

Use consistent stored-energy references and account for reserve, conversion and auxiliaries. Nominal capacity divided by load is only a first ideal estimate.

$$E_{load} \approx \eta_{dis} \times \text{usable internal energy}$$

$$\text{Runtime} \approx E_{load} / \text{constant load power}$$

$$\text{Round-trip efficiency} = \text{recovered energy} / \text{input energy}$$

Same boundary, specified initial/final state, and accounted auxiliaries.

Work a backup estimate

Assume 10 kWh internal reference capacity, starting at 90% of that reference, and stopping at 20%. Available internal energy is 7 kWh in this simplified model. At 92% discharge efficiency and a constant 2 kW AC load, neglecting auxiliaries, runtime is $7 \times 0.92/2 = 3.22$ h.

TWO DIRECTIONS COMPOUND LOSS

If charge efficiency and discharge efficiency are both 90%, ideal round-trip efficiency across the same external boundary is $0.90 \times 0.90 = 81\%$. Charging storage with 10 kWh can return only 8.1 kWh in that model, before auxiliary and standing losses.

What can make a real runtime shorter

Power-dependent efficiency, lower actual capacity, battery temperature, a changing load, unavailable reserve or shutdown thresholds can dominate. A test should preserve its complete conditions.

Thermal limits can constrain repeated power

Battery and converter losses warm the system. Available cooling and thermal state can reduce permitted power before the stored energy is exhausted.

Source of heat	First estimate and limitation
Battery resistance	I^2R is a useful loss term, but real electrochemical heat can include other terms and vary with state.
Converter loss	Pin - Pout at a measured operating point and boundary; not automatically all in one semiconductor.
Auxiliary loads	Fans, pumps, control and heaters draw energy; state which side of the meter supplies them.
Repeated events	Thermal storage accumulates during bursts and recovers between them. One event may not represent the sustained duty.

A small current change can strongly change heat

For a fixed illustrative 0.020 Ω resistance, 20 A gives 8 W and 40 A gives 32 W. Doubling current quadruples this loss term. The actual resistance may also rise or fall with state, so a complete thermal model needs appropriate data.

LIMITS MAY BE DYNAMIC

Record the current permitted power and its timestamp. An old “5 kW available” value can be wrong after a temperature change or a protective event.

Measure the outcome at the correct boundary

A storage system reports commands, limits, states and measurements. They are different evidence types and may refer to different physical points.

Evidence	What it can establish	What remains open
Command reply	A device reports receipt or acceptance under its protocol.	Whether output changed, when, and by how much.
AC active-power reading	Measured net transfer at its stated AC boundary and window.	Battery DC current, stored-energy accuracy or another site load.
Battery V / I / temperature	Observed terminal and sensor states within their limits.	Every cell’s condition or a validated energy inventory.
SoC / available power	An estimate and a limit report from the device.	Method, age, uncertainty, reference and current applicability.

Freshness is part of the acceptance criterion

Keep device time, receive time, aggregation window, units, sign convention, sequence and quality. A communication reconnect can deliver an old reading. A zero measurement differs from a missing value, and a clipped response differs from a successful full setpoint.

A meter trend can support consistency with a command; it does not by itself prove causation.

Document study: decode one finished product

Choose a public datasheet for a sealed power bank, portable power station, UPS or inverter. You do not need to buy, open or run it.

Field	What to capture
Identity	Manufacturer, exact model, revision, source URL, date and page.
Energy rating	Wh or kWh, nominal versus usable, voltage reference for Ah, test conditions.
Power capability	Input/output limits; continuous versus surge; duration; AC/DC boundary and derating.
Conversion + modes	Supported input/output paths, charging, backup, bypass and export capabilities actually documented.
Limits + protection	Temperature range, battery restrictions, alarms, local protections and permitted interfaces.
Unknowns	Any missing efficiency, runtime curve, reserve definition, timestamp semantics or control behaviour.

Turn one ambiguity into a precise question

For example: “Is the energy rating cell nominal energy or usable AC output under a specified test?” Or: “When discharge is refused, can the measured AC output already be nonzero?” Put the source and unanswered question into Lupine’s log.

The deliverable is a one-page reference sheet. Label calculated values separately from manufacturer claims.

A bounded storage model makes limits explicit

The next page implements one energy inventory with charge/discharge power caps and a reserve. It is not a cell, BMS, inverter-dynamics or grid-stability model.

Assumption	Value / convention
Internal energy bounds	2 kWh minimum; 8 kWh maximum.
Starting stored energy	5 kWh for each independent test case.
External power sign	Positive = discharge to AC; negative = AC charging input.
Power caps	2 kW discharge; 1 kW charge magnitude.
Conversion efficiencies	ηdis = 0.90; ηchg = 0.90, constant.
Interval model	Requested power is held for the interval; reported power is interval average. Bounds can shorten service within it.

Clip delivered energy, not the energy inventory after the fact

For discharge, available external energy is (E - Emin)ηdis. For charge, accepted external energy cannot exceed (Emax - E)/ηchg. Apply power caps too. Then compute the new internal state and conversion loss from the energy actually transferred.

UNKNOWN DATA IS OUTSIDE THIS MODEL

All inputs here are known synthetic numbers. A physical interface needs invalid-data handling, authoritative limits and protection beyond these arithmetic checks.

Run the energy-accounting model

Use Python 3 on your computer. Save as storage_lab.py. No battery, network or device interface is used.

```
import math

LOW, HIGH = 2.0, 8.0       # internal kWh
DIS, CHG = 2.0, 1.0        # external kW limits
ETA_D, ETA_C = 0.90, 0.90

def transfer(energy, request_kw, hours):
    if not all(math.isfinite(x) for x in
                (energy, request_kw, hours)):
        raise ValueError('Nonfinite input')
    if not LOW <= energy <= HIGH or hours <= 0:
        raise ValueError('State or duration outside model')
    if request_kw >= 0:
        offered = min(request_kw, DIS)*hours
        delivered = min(offered, (energy-LOW)*ETA_D)
        removed = delivered/ETA_D
        new = energy-removed
        loss = removed-delivered
        average_kw = delivered/hours
    else:
        offered = min(-request_kw, CHG)*hours
        accepted = min(offered, (HIGH-energy)/ETA_C)
        added = accepted*ETA_C
        new = energy+added
        loss = accepted-added
        average_kw = -accepted/hours
    return new, average_kw, loss

for request, hours in [(3.0, 1.0), (2.0, 2.0), (-2.0, 4.0)]:
    new, average, loss = transfer(5.0, request, hours)
    print(request, hours, '->', round(new, 4),
          round(average, 4), round(loss, 4))
```

READ THE THREE RETURN VALUES

New internal energy in kWh; signed interval-average external power in kW; conversion loss in kWh. A limited interval-average result is not proof of a constant delivered power trace.

Check the ledger against known results

Each printed case starts independently at 5 kWh. The code does not carry one case’s state into the next.

Request / interval	New E	Mean AC P	Loss	Limit
+3 kW / 1 h	2.7778 kWh	+2.0000 kW	0.2222 kWh	Power cap
+2 kW / 2 h	2.0000 kWh	+1.3500 kW	0.3000 kWh	Reserve
-2 kW / 4 h	8.0000 kWh	-0.8333 kW	0.3333 kWh	Both

Reconcile every joule across the boundary

Discharge: internal energy removed = AC energy delivered + loss.
Charge: AC energy accepted = internal energy added + loss.
In case 2, 3 kWh is removed, 2.7 kWh is delivered and 0.3 kWh is lost. The initial request was for 4 kWh, so 1.3 kWh remains undelivered.

Probe the boundaries before adding complexity

At LOW, positive discharge must return zero transfer; at HIGH, charging must return zero transfer. Zero request leaves energy unchanged. NaN, infinite values, invalid state and nonpositive duration must be rejected. Try each case and explain the result.

A CONSERVING MODEL CAN STILL BE PHYSICALLY INCOMPLETE

This arithmetic omits voltage, cell states, degradation, auxiliary loads, thermal dynamics, ramp limits, control delays and protective behaviour.

Ten checks before the next part

Write units, boundaries and conditions. Use the sign convention from page 05-21 for charge and discharge.

1. A pack is nominally 25.6 V and 50 Ah. Estimate nominal Wh. Does that establish usable AC output?

2. Identical 3.2 V, 10 Ah cells are represented as 4s3p. Find nominal pack V, Ah, Wh and cell count.

3. A 50 Ah charge reference starts at 80% and supplies 5 A for 2 h. Find ideal ending SoC and the error from a +0.10 A offset over 10 h.

4. A simple source has Voc = 24 V and Rinternal = 0.050 Ω. At 20 A discharge, find terminal voltage, output power and resistive heat.

5. An ideal continuous-conduction buck converts 24 V to 12 V. Find D. A boost converts 12 V to 24 V: find its ideal D.

6. A 2 kW converter output is 95% efficient. Find input and loss. What thermal rise follows at an assumed lumped 0.5 K/W?

7. An inverter has a simple 50 kVA sinusoidal capability bound and requests Q = 30 kvar. Find maximum P under that bound alone.

8. A model starts with 6 kWh internal energy, stops at 2 kWh and has discharge efficiency 90%. How long can it supply 1.5 kW AC with no auxiliaries?

9. A unit reports “grid-forming” and 70% SoC. Does that prove it can black-start this facility or meet a specified outage load?

10. Rail receives “accepted: discharge 2 kW”, but a fresh reading shows only 0.5 kW and a new temperature limit. Is the full request verified?

Worked answers: 1-5

Numerical correctness comes after choosing the right reference and model.

1 Nominal energy

25.6 × 50 = **1,280 Wh = 1.28 kWh**. It does not establish usable AC output: reserve, real capacity, voltage profile, conversion, load and auxiliaries can all change the result.

2 Series and parallel arithmetic

Four series groups give **12.8 V**; three matched parallel strings give **30 Ah**. Nominal energy is **384 Wh**, using **12 cells**. This is ideal rating arithmetic, not a pack-building instruction.

3 Charge inventory and offset

10 Ah is removed, or 20 percentage points of the 50 Ah reference. Ending SoC is **60%** in the ideal model. An offset of 0.10 A for 10 h creates **1 Ah** error, or **2 percentage points** on that reference.

4 Voltage sag and heat

Vterminal = 24 - 20 × 0.050 = **23 V**. Output is 23 × 20 = **460 W**. Internal resistive heat is 20^2 × 0.050 = **20 W**. The model source-side rate is 480 W.

5 Ideal conversion ratios

Buck D = Vout/Vin = 12/24 = **0.50**. Boost D = 1 - Vin/Vout = 1 - 12/24 = **0.50**. These are ideal continuous-conduction steady-state relationships, not complete control or component specifications.

Worked answers: 6-10

The final questions distinguish a label, a request and evidence of a physical result.

6 Conversion loss and thermal rise

Input = 2,000/0.95 = **2,105.26 W**. Loss = **105.26 W**. At an assumed lumped 0.5 K/W, steady rise is **52.63 K**. This does not establish actual semiconductor temperature or an acceptable operating point.

7 Shared apparent-power capability

Pmax = $\sqrt{(50^2 - 30^2)}$ = **40 kW**. DC source, voltage, current, temperature and controls may impose tighter bounds. The result assumes sinusoidal conditions and the stated simple circle.

8 Runtime with reserve and loss

Usable internal energy is 6 - 2 = **4 kWh**. AC energy is 4 × 0.90 = **3.6 kWh**. At 1.5 kW, runtime is **2.4 h** under the simplified constant-load assumptions.

9 Labels do not prove a facility capability

No. Black start and outage service need a supported mode, transfer and protection arrangement, energisation/current capability, usable energy, load behaviour and tested sequence. Reported SoC alone is an estimate without enough context.

10 Accepted but limited output

The full 2 kW outcome is not verified. Preserve the accepted request, fresh 0.5 kW reading and new limit. Assess a limited or mismatched result under the agreed semantics; do not bypass the thermal limit or automatically resend the same command.

Reference: storage and conversion formulas

Use consistent internal and external energy boundaries. A fraction SoC of 0.60 means 60%; a percentage point is an absolute change of one percent of the reference.

Relationship	Condition / unit
1 Ah = 3,600 C; 1 kWh = 3.6 MJ	Charge and energy are different quantities.
$E_{nominal} \approx V_{nominal} \times Ah$	Wh when V and Ah are used.
$E = \int VI\,dt$	J with time in s; convert to Wh as needed.
$V_{pack} = N_s V_{cell}$; $Ah_{pack} = N_p Ah_{cell}$	Ideal matched cells and properly designed arrangement.
$V_{terminal} = V_{oc} - IR_{internal}$	Simple discharge-positive equivalent circuit.
$SoC_{next} \approx SoC_{now} - I\Delta t/C_{ref}$	Fraction; I in A, Δt in h, Cref in Ah.
Capacity SoH = $C_{measured}/C_{reference}$	Comparable capacity test conditions.
$E_{cap} = \frac{1}{2}CV^2$	J with capacitance in F and voltage in V.
Buck: $V_{out} \approx D\,V_{in}$	Ideal steady continuous-conduction model.
Boost: $V_{out} \approx V_{in}/(1 - D)$	Ideal steady continuous-conduction model.
$\eta = P_{out}/P_{in}$; $P_{loss} = P_{in} - P_{out}$	Same direction, boundary and operating point.
$P_{conduction} \approx I_{rms}^2 R$	Resistive approximation at actual conditions.
$P^2 + Q^2 \leq S_{max}^2$	Simple sinusoidal apparent-power bound.
$E_{AC,out} = \eta_{dis}\,\Delta E_{internal}$	Specified discharge efficiency, no other loads.
$\Delta E_{internal,in} = \eta_{chg}\,E_{AC,in}$	Specified charge efficiency, no other loads.
$Runtime \approx \eta_{dis}(E_{start} - E_{min})/P_{load}$	Constant load and internal-energy reference.

Thermal rise $\Delta T \approx P_{loss} R_{th}$ is a steady lumped model, not a complete junction-temperature calculation.

Words to recognise in storage documents

A single product can report several reference capacities, modes and power limits. Preserve their exact definitions.

Cell / module / pack Electrochemical unit / assembled group / complete battery assembly, with product-specific usage.

Nominal / usable energy Reference rating / energy available within a defined operating envelope and test.

SoC / SoH Estimated charge state / a defined health metric; neither is a direct measurement of every cell property.

C-rate / DoD Current relative to an Ah reference / depth of discharge relative to a defined capacity reference.

BMS Battery management system; monitoring, estimation and protective functions vary by design.

Contactor / precharge Power-path switching device / controlled initial charging of a capacitive bus.

DC link Intermediate DC energy-transfer bus, often with capacitance.

Buck / boost DC-DC conversion that steps voltage down / up in its applicable operating mode.

PWM / duty fraction Pulse-width modulation / on-time divided by switching period for the specified signal.

PFC Power-factor correction; waveform shaping is distinct from efficiency and isolation.

Grid-following / grid-forming Control related to an existing waveform / voltage-establishing control with defined dynamics.

Islanding / black start Separated local electrical system / energisation from a de-energised state with an engineered sequence.

UPS / bypass Uninterruptible power supply / an alternate power path with its own conditions and limitations.

Derating / clipping Reduced permitted capability / output constrained by a limit instead of following the full request.

Round-trip efficiency Energy recovered divided by input at a consistent boundary and stated state conditions.

ONE HABIT TO KEEP

For every number ask: which boundary, which direction, which operating point, which timestamp and which evidence status?

Primary references and review status

These are checking resources, not additional required textbooks. Text, numerical cases, code and diagrams in this guide are original.

US DOE / Batteries
Electrochemical storage and the external/internal transport distinction.

TI / Battery current measurement
Coulomb counting, offset and measurement considerations.

TI / BMS topologies
Battery measurement and safe-operating-area context.

TI / Charge management example
CC/CV stages and temperature-dependent charging constraints. This is not a universal charging profile.

TI / Buck, boost and buck-boost topologies
Power-stage operation and topology-dependent limits.

TI / DC-link precharge
Inrush and the purpose of an engineered precharge path.

US DOE / Inverters and grid services
Conversion, network interfaces and energy-source requirements.

NREL / PV and battery grid-forming research
Current limits, black-start studies and validated-model requirements.

NREL / Anti-islanding evaluation
Experimental work distinguishing protected grid operation and island conditions.

Eaton / UPS types
Standby, line-interactive and double-conversion path distinctions.

Edition 01 / 17 October 2026

Prepared with ChatGPT. Models are simplified and numerical examples are assumed. The code is checked for bounds and conservation; no physical battery, converter, UPS or grid-interface testing has been performed for this volume. No independent engineering peer review is claimed.

Bring the storage limits back to Lupine

A useful supervisory system preserves constraints, requests and physical outcomes without confusing any one of them for the others.

Layer	A concrete requirement
Aiva	Reject infeasible choices before scoring. Use current power limits, reserve, mode and data validity; preserve the reason for rejection.
Rail	Separate requested, accepted, limited, refused, uncertain and physically observed outcomes under a defined adapter contract. Reconcile before any potentially duplicating retry.
Cloked	Preserve device identity, sign convention, raw measurements, timestamps, limits, assumptions and corrections. An intact record does not prove the stored-energy estimate was correct.

Three things to keep from Part 05

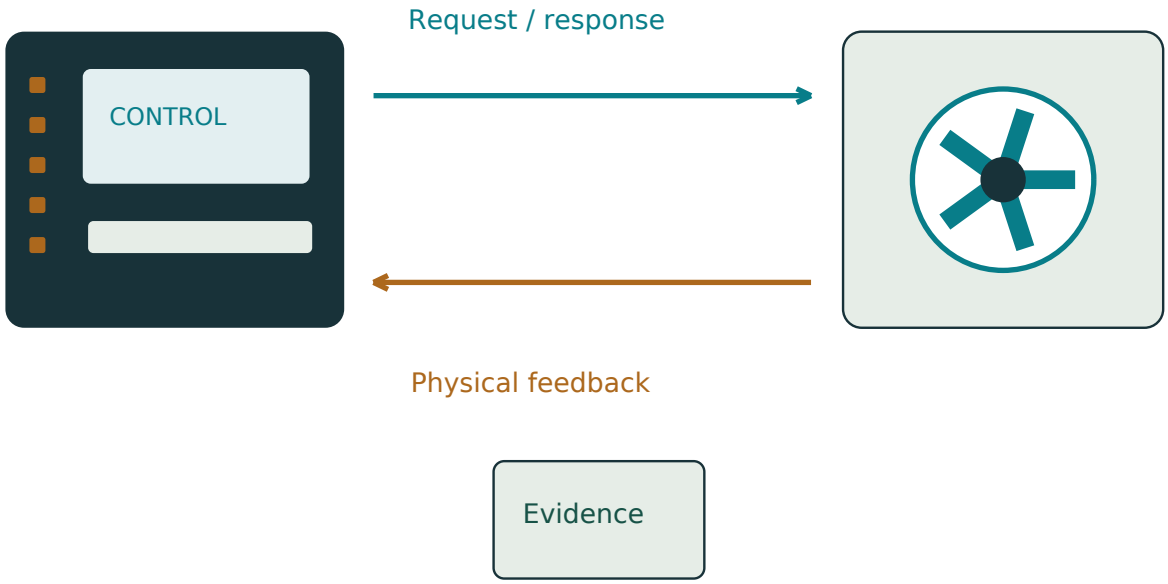
A product reference sheet: one actual public datasheet decoded with explicit unknowns.
A reproducible ledger: the storage simulation, including bounds, losses and undelivered energy.
A precise problem: one real or documented failure where the available evidence cannot distinguish the important outcomes.

NEXT: PART 06 / RELIABLE CONTROL, COMMUNICATIONS AND EVIDENCE

The next part connects device protocols, timing, persistence, fault injection and reconciliation to a bounded physical interface. Three planned parts remain: 06, 07 and 08.

CONTROL, COMMUNICATIONS & EVIDENCE

Part 06 / Connect a decision to a physical outcome without hiding uncertainty, failures or the limits of measurement.



Functional illustration. Network feedback is evidence about a sensor, not a direct guarantee of physical truth.

Make uncertainty an explicit engineering state

A controller observes, decides, requests and checks. Each step can fail differently. The useful capability is to preserve what is known, what is uncertain and what must be reconciled before acting again. Fast electrical protection remains with its engineered local system.

WHAT YOU WILL BE ABLE TO DO

Explain feedback and timing; decode example device data; specify a command contract; reason about retries and recovery; and run deterministic evidence-classification fault cases.

From feedback physics to reliable execution

The first five parts established the physical quantities and limits. This volume asks how software can interact with them honestly.

Pages	Question to answer
3-7	Which controller owns what, and how do feedback, limits and delays behave?
8-10	When is a measurement usable, fresh and correctly identified?
11-15	What do networks and device protocols actually guarantee?
16-22	How do commands, uncertain outcomes, reconciliation and ownership fit together?
23-27	What survives a restart, and what can records, security and watchdogs establish?
28-33	How do I specify failure cases and run a fault-classification exercise?
34-38	Can I answer the checks and use the engineering reference?
39-40	What are the primary sources and the next Lupine deliverables?

Teaching contract, not a repository status report

These are teaching examples. They do not claim that the current Lupine repository already implements the persistence, communication or device interfaces described here.

No physical command, network scan, equipment access or new hardware is required for this part.

Keep protection, local control and supervision distinct

Different layers have different responsibilities and timing. A general-purpose laptop supervisor must not silently replace a device’s protective functions.

Layer	Primary job	Example
Protection / interlock	Detect defined hazardous or prohibited conditions and act through the engineered protective path.	Converter overcurrent trip or local temperature interlock.
Local controller	Regulate the physical process inside its authorised operating envelope.	A drive’s current loop or a CDU’s local flow/temperature control.
Supervisor	Coordinate slower requests, enforce policy, record limits and reconcile evidence.	Request an allowed power setpoint and check fresh output.
Human engineering authority	Define boundaries, acceptance tests and permissions; resolve exceptions beyond the automated contract.	Approve a real device test plan and response to a protective trip.

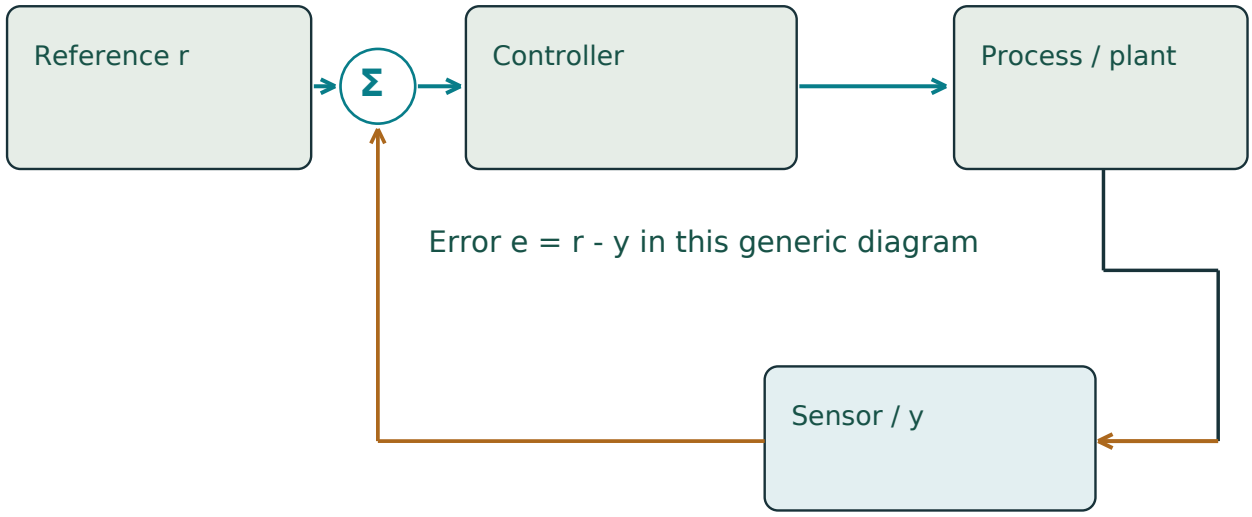
Lupine begins at a bounded supervisory interface

Use a documented device API or simulator with explicit permitted actions. Local limits and trips remain authoritative. A fast response claim must include the full sensing, communication, computation and actuation path, not just how quickly Python returns.

The correct division depends on the actual equipment and engineering design, not this generic table alone.

A feedback loop compares a reference with evidence

The controller changes an input to a physical process. A sensor reports an observation, which is fed back for the next decision.



Sign and gain must match the physical process. Sensor, controller and actuator all have limits.

Control depends on the sign of the physical response

Increasing heater power usually raises temperature. Increasing cooling effort usually lowers it under suitable conditions. Applying the same gain sign to both can create positive feedback. Identify the direction and dynamics of the process before choosing a control law.

Open loop lacks that corrective comparison

A fixed fan command can be open loop with respect to temperature, even if the fan regulates its own motor speed internally. The phrase “closed loop” needs the controlled quantity and the actual feedback path. An RPM reading is not temperature feedback.

OBSERVATION HAS A MEASUREMENT CHAIN

Calibration, location, response time, timestamps, units and quality belong in the loop model. Feedback from the wrong point can regulate the wrong outcome.

Proportional control responds to current error

A P controller maps an error into an effort. Gain has units, and saturation prevents the requested effort from exceeding its permitted range.

$$e_{cool} = T_{measured} - T_{set}$$

$$u_{request} = u_{base} + K_p e_{cool}$$

$$u_{applied} = \mathrm{clamp}(u_{request}, u_{min}, u_{max})$$

Cooling-specific sign convention: positive excess temperature requests more cooling.

Calculate an illustrative request

Assume Tset = 35 °C, measured temperature = 40 °C, baseline effort 20% and Kp = 8 percentage points per kelvin. The request is 20% + 8 × 5 = 60%. At a 90% upper limit, a 12 K error would request 116% but apply only 90%. These are hypothetical numbers, not fan settings to deploy.

More gain is not always better

A larger gain can reduce error in some systems while also increasing overshoot, noise sensitivity or instability when delays and dynamics matter. A constant nonzero error may remain with proportional control. Predict and test the actual model; do not treat a gain table as a universal tuning recipe.

EFFORT IS NOT THE CONTROLLED RESULT

60% PWM does not prove 60% speed, flow, cooling capacity or temperature reduction. Each conversion has its own physical relationship and limitations.

Integral action remembers persistent error

Accumulating error can remove an offset in a suitable stable loop. It also introduces state that must be managed when the actuator cannot follow the request.

$$I_{state,next} = I_{state} + K_i \, e \, \Delta t$$

$$u = u_{base} + K_p \, e + I_{state}$$

State, sampling interval and units are part of the algorithm.

For cooling error e in K, Ki can have units effort/(K·s).

Work one update with the sample interval

Assume Ki = 0.5 percentage points/(K·s), e = 4 K and Δt = 2 s. The integral contribution increases by 4 percentage points. Forgetting Δt changes the effective controller when the sample rate changes. A reboot or mode switch also raises a question: what should happen to the accumulated state?

Saturation can cause windup

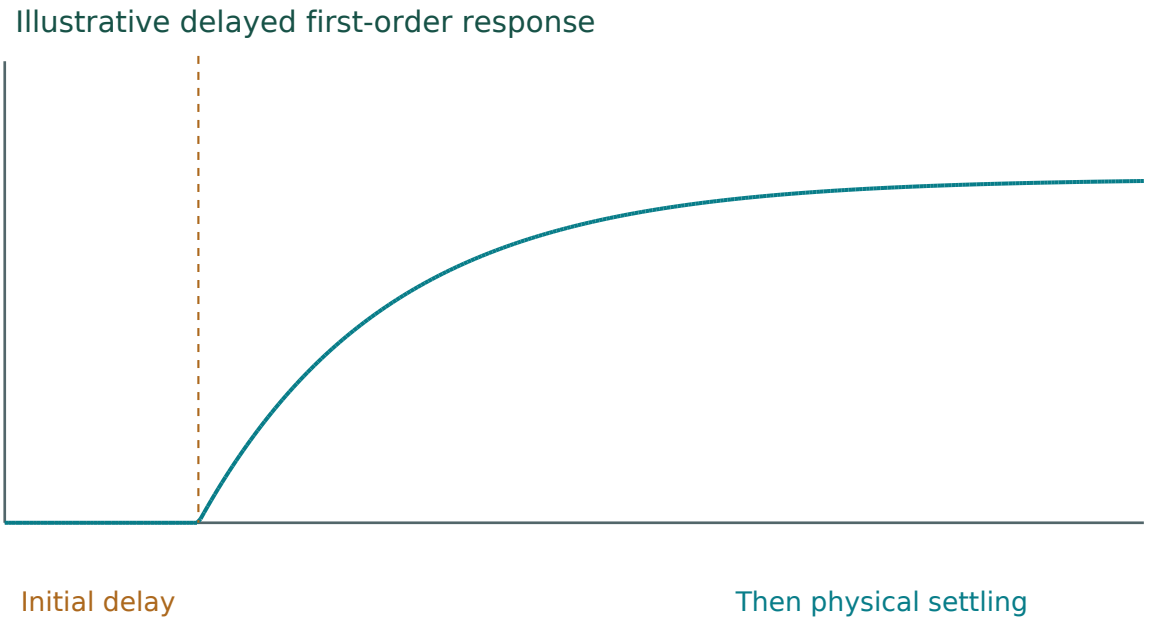
If effort is capped while the integrator keeps growing in the same direction, recovery can be delayed after error reverses. Conditional integration, back-calculation or tracking can address this in a designed controller. Manual/automatic transitions and changing downstream limits need attention too.

DERIVATIVE ACTION IS NOT A FREE UPGRADE

A derivative term responds to rate of change but can amplify measurement noise. Filtering and tuning add design choices. This part does not prescribe a PID for real equipment.

Delay, hysteresis and settling time are different

Waiting for a response, avoiding chatter and deciding when to evaluate an outcome solve different problems. None should be hidden inside a guessed sleep.



Budget the whole evidence path

Delay includes acquisition, transport, queueing and actuation. Settling time describes the response approaching a defined region after the action. A simple hysteresis example turns cooling on above 40 °C and off below 38 °C, retaining the previous state between. That reduces chatter but does not by itself prove stability or safety.

TIMING EXAMPLE

Assume a 750 ms sensor conversion, 100 ms communication allowance and 50 ms processing allowance: at least 900 ms is budgeted before additional queueing or physical settling. It cannot justify a 100 ms end-to-end thermal control claim.

Thresholds and time budgets here are illustrative; actual values require measured behaviour and engineering limits.

A sample is useful only for a defined decision

A valid number can be the wrong measurement, too old or too uncertain. Usability combines identity, meaning, quality and timing.

Check	Ask before using the value
Identity + location	Which device, physical sensor, channel, boundary and boot/session produced it?
Meaning	What units, sign, scale, range and aggregation window apply?
Acquisition quality	Was this read successful? Is the value finite? Is a sensor fault or missing field being substituted?
Timing	When was it captured, when was it received, and can the relevant clocks be compared?
Decision fitness	Is the error, response time and capture position good enough for this particular acceptance test?

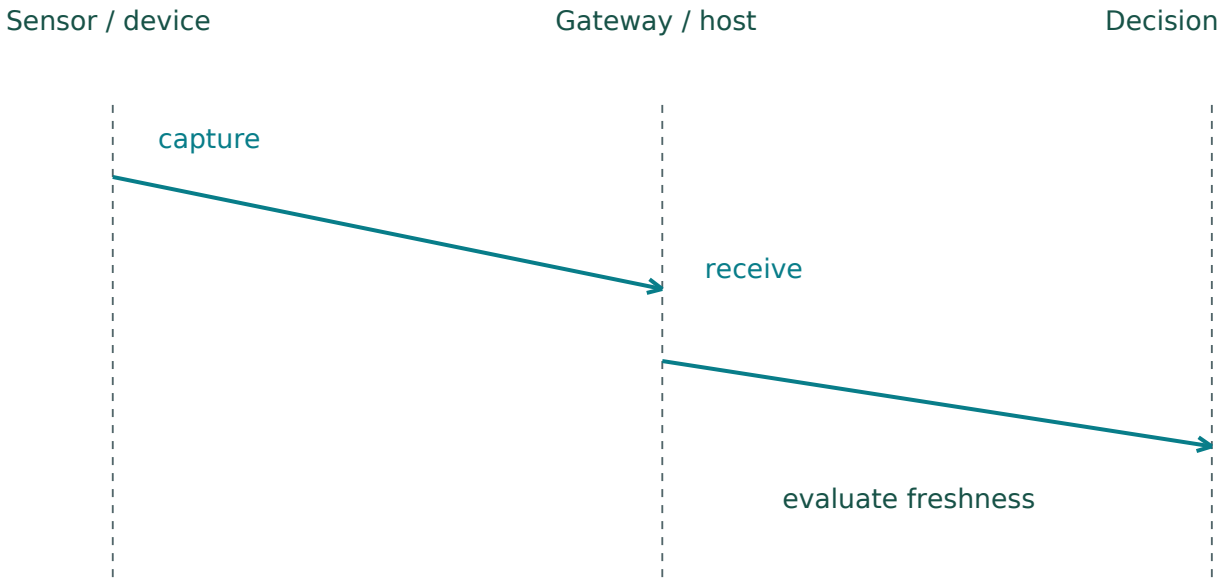
A fresh value is not necessarily accurate

A new reading from a miscalibrated sensor can still be wrong. A calibrated sensor at the wrong location can still answer the wrong question. An unaltered record preserves the observation; it does not repair the measurement chain.

Missing data remains absent. A legitimate zero remains a numeric zero with its own valid measurement context.

Keep event time and receive time separate

Different clocks have different purposes. A current packet can carry an old sample, and timestamps from separate devices cannot simply be subtracted without a clock model.



Transport adds delay. Receive time is not automatically sample time.

Choose a clock suited to the comparison

UTC helps correlate events across systems when synchronisation and uncertainty are known. A monotonic clock helps measure elapsed intervals in one compatible clock domain. Its arbitrary origin is not UTC; compare values only under a valid domain and reset model. Microcontroller ticks may wrap or reset.

A SIMPLE AGE TEST

With comparable timestamps, now = 12.0 s and sample = 11.5 s gives age 0.5 s. If maximum age is 1.0 s, this passes that gate. A future-dated sample, a different boot epoch or unknown clock offset requires explicit handling rather than a negative age silently passing.

State the timestamp semantics

Capture start, conversion completion, device publication and host receipt can differ. Use source documentation and preserve the distinction.

Define a record before collecting thousands of them

A small explicit schema makes units, missing values and revisions reviewable. Raw values and derived verdicts belong in separate fields.

Field group	Illustrative contents
Source	device_id, channel, sensor_id, firmware, boot_id, sequence.
Quantity	value, unit, sign_convention, scale, location, measurement_boundary.
Timing	source_time, clock_domain, clock_uncertainty, receive_time, capture_window.
Quality	read_ok / error / unavailable, calibration reference and diagnostic flags.
Provenance	schema_version, adapter_version, raw payload reference, conversion method.
Derived assessment	verdict, reason, policy_version, assessment_time and links to the input evidence.

Prefer a correction to a silent rewrite

If scale or meaning was wrong, retain the original record and add a documented correction or a new derived view. Keep the original interpretation version. A later reader should be able to reconstruct what the controller knew when it acted.

Choose a stable schema for a bounded use case; the table is a learning template, not a universal industrial standard.

A working network does not prove a working process

Electrical links, transport connections and application semantics are separate layers. Troubleshooting should identify which layer supplied the evidence.

Layer	Example	What success does not prove
Physical / link	Ethernet, Wi-Fi or an RS-485 physical interface.	That the application accepted a command or that the device is healthy.
Addressing + transport	IP address and TCP connection; serial framing for a different link.	That the remote application persisted or executed the request.
Application protocol	Modbus request/response or MQTT publication.	That a physical output changed or a sample was newly captured.
Device contract	Register meaning, command identity, modes, limits and outcome rules.	That sensors are accurate or that the physical process met every requirement.

Start with a known local endpoint

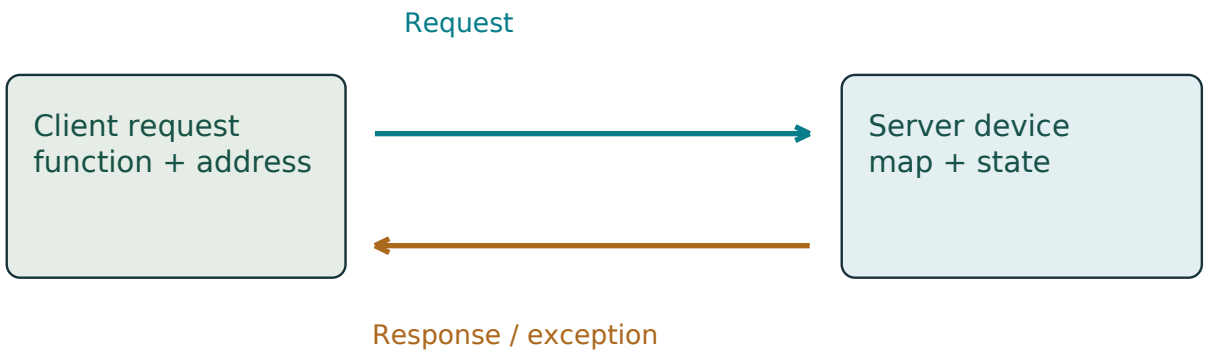
For practice use an in-process fake or a simulator bound to a local test environment. Real industrial access requires explicit authorisation, device identity and supported interfaces. Do not discover your first register map by writing arbitrary addresses.

RS-485 IS NOT MODBUS

RS-485 describes an electrical interface. Modbus RTU defines a serial protocol framing and application behaviour; Modbus TCP uses a different transport arrangement.

Modbus moves typed-by-contract register values

The protocol provides functions and addressable data. The device map supplies much of the physical meaning, including units, scaling and side effects.



Illustrative exchange. Neither message is direct proof of a physical actuator outcome.

Read the map before choosing a function

Common functions read holding or input registers and write permitted values. Register offsets in a PDU and human labels such as “40001” can use different conventions. Confirm function, offset, unit identifier, access permission, update rate and supported multi-register behaviour from the exact manual.

A transport transaction ID is not an execution guarantee

Modbus TCP uses a transaction identifier to match responses. That alone is not a durable idempotency key or a device command ledger. A successful write response has defined protocol meaning; it does not guarantee that the physical setpoint was achieved or remains applied.

READ-ONLY FIRST

Use simulator reads and decode checks before any write-capable adapter. Classic register access needs an appropriate protected network or supported security architecture.

Decode signs, scale and missing-value sentinels

The same 16 raw bits can represent an unsigned integer, a signed integer, a flag set or part of a larger number. The map decides.

Raw / rule	Decoded meaning in this example
0xFF9C as unsigned 16-bit	65,436; not the intended signed value here.
0xFF9C as signed 16-bit	65,436 - 65,536 = -100.
Scale exponent = -1	Physical value = raw_signed × 10^-1 = -10.0.
Map says unit is W	Interpretation becomes -10.0 W under the map’s sign convention.
Reserved / invalid sentinel	Reject or mark unavailable before treating it as a physical measurement.

Multi-register values need an ordering contract

Modbus specifies byte encoding within protocol fields, but a device’s 32-bit value across multiple registers still needs the correct layout and word order from its map. Reading a changing pair separately can also produce an incoherent value unless the device provides suitable snapshot semantics.

SUNSPEC IS A MODEL LAYER, NOT A MAGIC PROBE

SunSpec models can define named points, types and decimal scale factors. Discover the supported model/version and invalid encodings; do not assume an arbitrary inverter implements a particular point or write action.

MQTT delivery and physical execution are different

Publish/subscribe messaging can distribute state and commands. Broker delivery guarantees do not by themselves create a durable actuator outcome.

Feature	Protocol meaning / design consequence
QoS 0	At-most-once delivery at the relevant MQTT delivery scope; a message can be lost.
QoS 1	At-least-once delivery semantics; an application must handle possible duplicates.
QoS 2	Exactly-once message-delivery semantics at the specified protocol scope; not exactly-once physical actuation across an application crash.
Retained message	A new subscriber can receive an earlier stored publication. Arrival now does not establish capture now.
Session / expiry / will	Useful lifecycle tools, but availability and freshness still need application-specific semantics.

A command topic needs a command contract

Include identity, permitted target, expiry, generation and outcome rules when the design requires them. A stale retained state may be appropriate for a dashboard with age displayed, yet unsuitable as permission to actuate. Do not use a broker acknowledgement as a sensor reading.

This guide assigns no connection to a public or industrial broker; the exercise below is entirely local.

An adapter translates a device contract into evidence

A protocol client is not yet an equipment adapter. The adapter must preserve the device’s actual semantics instead of inventing a stronger promise.

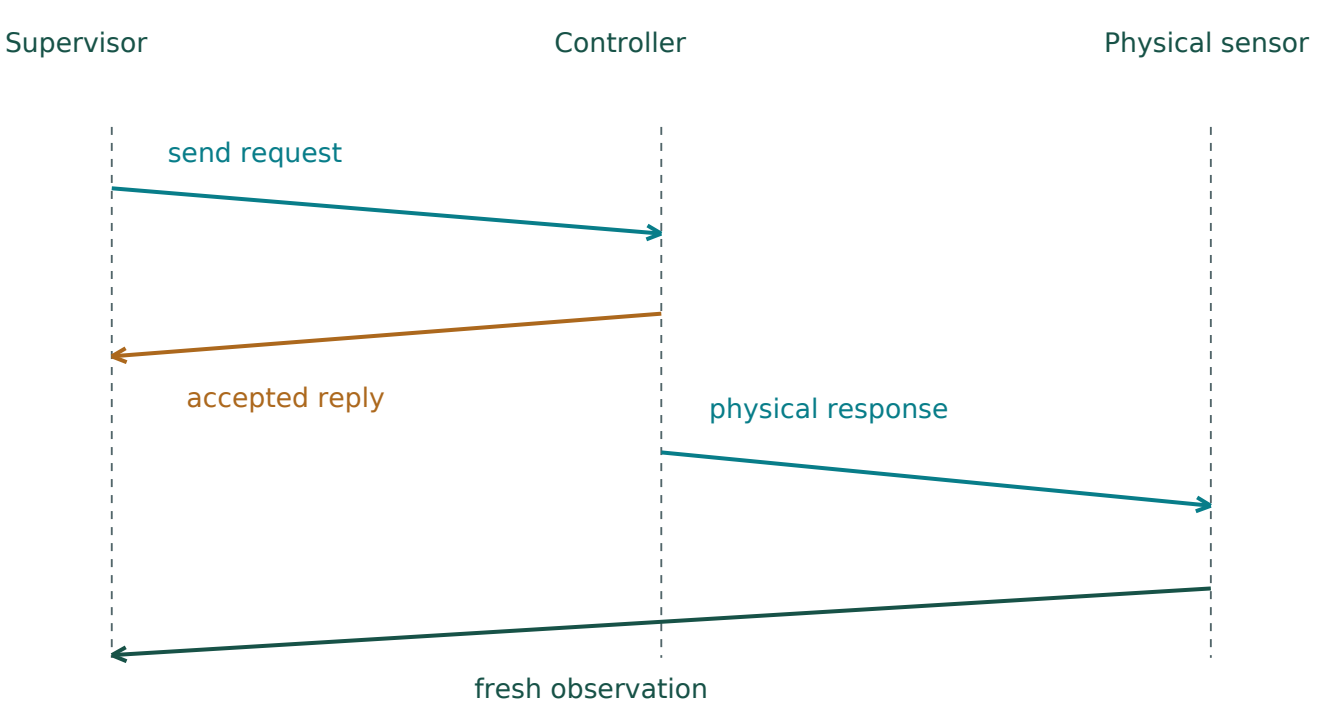
Contract element	What must be established
Identity + version	Device, firmware, map revision, channel and boot/session identity.
Action semantics	Absolute setpoint, incremental action, pulse, start/stop or energy transfer; permitted modes and side effects.
Limits + ownership	Who may write, which local controls can override, and how current limits are obtained.
Acknowledgement	Received, queued, accepted, applied, refused or completed: the exact meaning and identity correlation.
Recovery + expiry	Whether duplicates are recognised, how long identity survives, and whether late commands can still execute.
Measurement evidence	Units, signs, capture time, aggregation, quality, uncertainty and authoritative feedback points.

Unknown is a valid contract entry

If the device cannot answer a question, preserve the limitation and constrain the use case. A generic wrapper cannot manufacture the missing guarantee.

Sent, accepted and observed are different events

Transport success is about a communication step. Physical success is an assessment using an explicit objective and usable evidence.



Define success in the units of the physical objective

For a teaching 100 W target, a fresh valid reading of 101 W may match a ±2 W tolerance after settling. An accepted reply with 40 W feedback does not establish that objective. A setpoint register reading back 100 W may show a stored request, not actual output power.

A match is still bounded evidence

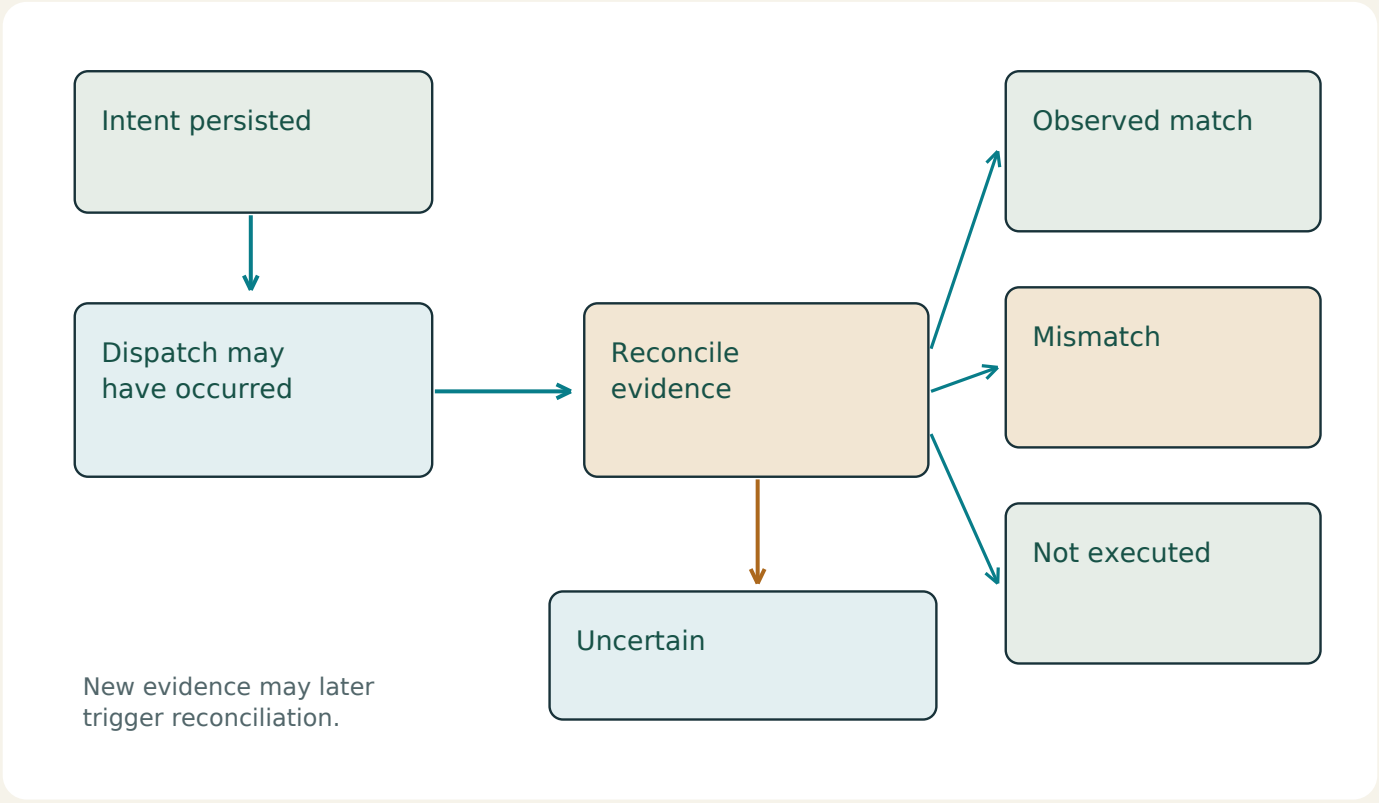
A reading close to the target supports “observed match within this measurement and timing contract”. It does not prove causation, calibration, safe isolation or every hidden state. If the output was already at the target, the value alone does not show that this command caused it.

CHOOSE HONEST STATE NAMES

The exercise uses observed_match, observed_mismatch, not_executed and uncertain. These are teaching labels, not claims about the current names in your repository.

State should represent what evidence justifies

Do not treat every timeout as failed or every reply as completed. Uncertainty must survive until new evidence resolves it under the contract.



Persist transitions with the evidence that supports them

Keep the instruction identity, previous state, new state, reason, input evidence and assessment version. Directly editing an uncertain outcome to failed without new evidence creates permission to retry from an unsupported assertion. State transitions must enforce the contract, not just a UI choice.

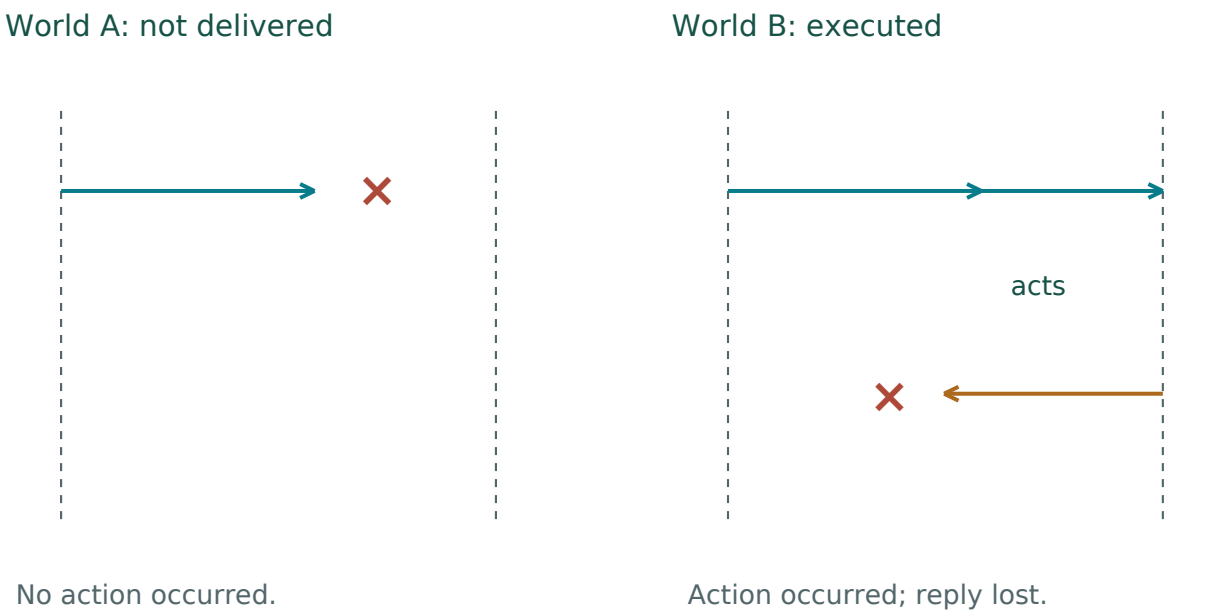
Decision and permission are separate

An outcome classification does not itself authorise a new command. Current limits, ownership, mode, expiry, interlocks and the retry policy still apply. Even a conclusive non-execution result can become irrelevant if the original objective has expired.

Illustrative state relationships; no automatic resend edge is implied.

A lost reply leaves two physically different worlds

The same timeout can follow a request that never arrived or an action that executed successfully. The sender cannot infer which one from silence alone.



Timeout is a statement about the waiting policy

It says that a qualifying reply was not obtained within a budget. It does not prove the receiver rejected the instruction, the actuator stayed unchanged or a delayed message cannot arrive later. Recording timeout as physical failure collapses distinct states and can make retries harmful.

THE RETRY CONSEQUENCE

If the action means “deliver another 1 kWh”, blind retry can duplicate energy transfer. If it is an absolute setpoint, reassertion may still conflict with a newer command, reset a timer or violate a changed mode. Identify actual semantics first.

A new TCP connection or MQTT session does not erase the history of a request that may already have executed.

Reconciliation asks what happened to this instruction

Use command-correlated records and fresh physical feedback. A status lookup must have a defined consistency and lifetime before it can settle uncertainty.

Evidence found	Honest interpretation
Accepted for the same ID + usable matching output	Observed objective match within stated limits; causation is still bounded.
Accepted for the same ID + usable output outside tolerance after settling	Observed mismatch. Record clipping, overrides or failures; do not call full success.
ID not found in a cache	Still uncertain if it can arrive late, be pending elsewhere or have lost history after restart.
Authoritative final no-execution result with late execution fenced out	Can support non-execution under that exact contract, provided other evidence does not conflict.
Stale, absent, conflicting or wrong-epoch evidence	Preserve uncertainty and record what evidence is missing.

A negative answer needs a fence

A fence can be a device-enforced expiry, generation or cancellation barrier that prevents an old instruction from executing later. “Not present at the moment I checked” is a weaker claim and cannot silently become “will never execute”.

If the interface cannot resolve uncertainty, stop the dependent action and use the agreed escalation path.

Idempotency concerns repeated effects, not repeated text

Repeating an instruction is safe only under a defined identity, state and side-effect contract. An identical payload is not enough.

Action shape	Possible repetition problem
Set absolute power to 100 W	Could overwrite a newer 50 W request, refresh an expiry, restart a ramp or act in a changed mode.
Increase by 100 W	Each execution can increment again unless the receiver deduplicates the action.
Deliver 1 kWh	Each execution can add another energy transfer, even when the message bytes match.
Same durable command ID	Useful only if the receiver preserves ID-to-outcome semantics for the required scope and lifetime.

Do not overclaim exactly once

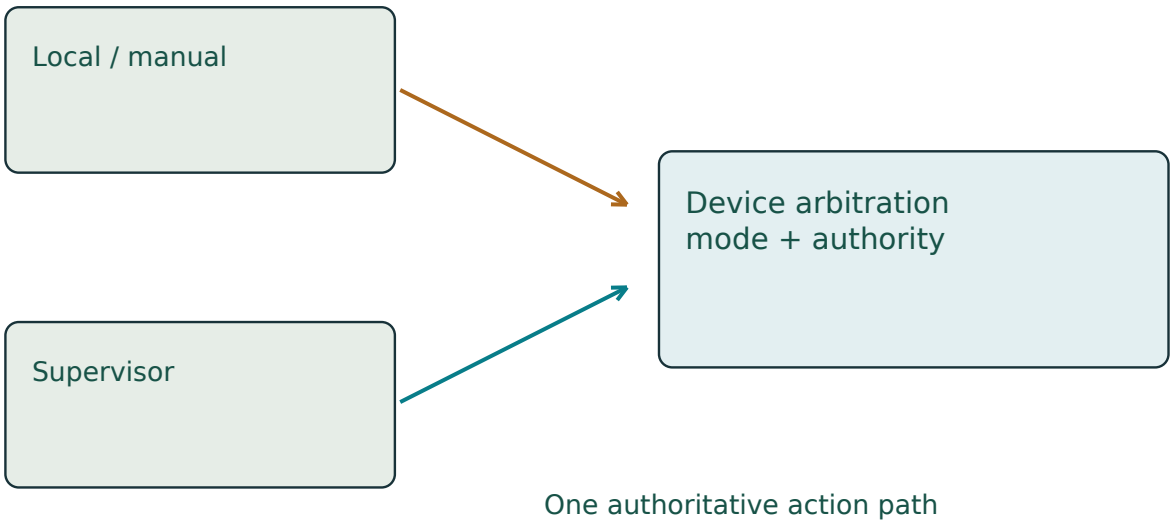
A durable receiver ledger can deduplicate identities, but the physical action and the ledger update may have their own crash gap. A database transaction on the sender does not atomically include a remote motor, inverter or relay. End-to-end guarantees require a real device contract and recovery design.

FOR THIS LEARNING EXERCISE

The classifier sends no commands and implements no retry engine. Its verdicts are not permissions. It helps you test whether evidence justifies an outcome claim.

Two controllers need an ownership rule

A local control panel, site controller and Lupine may all be able to change state. Without an explicit authority model, each can undo another’s legitimate decision.



A lease alone may not fence an old writer

A controller can lose its lease or network while an old request remains in flight. If the actuator accepts that request later, a new owner may be overwritten. A supported fencing mechanism lets the receiver reject stale generations. It must be enforced at the authoritative action point, not merely stored in a laptop variable.

Record mode changes as events

An accepted request can later be overridden by a local trip, a manual operator or another authorised controller. Preserve the request’s identity and the change in mode or generation. A power mismatch should not be hidden just because the original acknowledgement was valid.

FOR AN UNSUPPORTED DEVICE

If there is no enforceable ownership contract, restrict the integration or keep it observational. Do not claim exclusive control from a friendly UI label.

Check constraints again at the point of action

A decision based on a valid limit can become infeasible before execution. Limits, ownership and data validity have to remain part of the request contract.

Gate	A concrete question
Measurement validity	Are the needed state and limit readings fresh, correctly identified and usable?
Physical envelope	Are requested voltage, current, power, flow, temperature and reserve inside the actual permitted region?
Authority + mode	Does this controller still own the action in this mode and generation?
Time + objective	Has the instruction expired, been superseded or become unnecessary?
Device enforcement	Can the receiver reject a late or infeasible request using authoritative current limits?

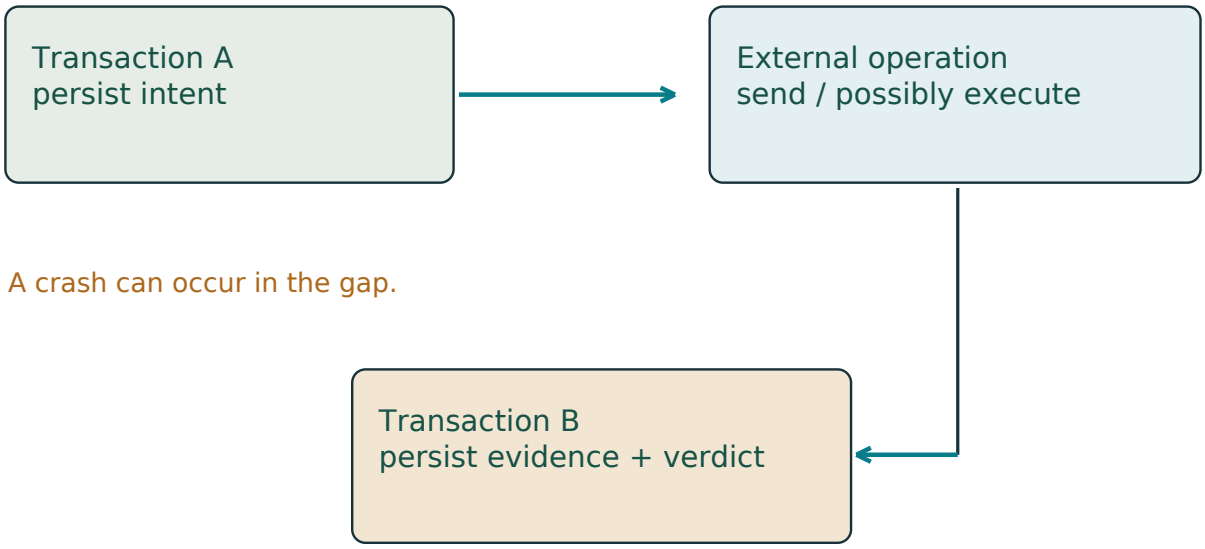
A concrete race condition

At t0 a battery reports 2 kW discharge allowed. At t1 temperature reduces that to 0.5 kW. At t2 a previously calculated 2 kW request arrives. The local controller must enforce its valid envelope; Lupine must record the limited or refused outcome rather than bypass the new limit.

Constraints are pass/fail conditions. A high optimisation score cannot cancel a violated physical limit.

Persist intent before a send can become ambiguous

A durable local record lets a restarted process discover unresolved instructions. It does not make the remote physical action transactional with the database.



A database transaction can protect local consistency

For a suitable transactional store, write the instruction, parameters and dispatch status with unique identity and explicit schema. Commit before the action can be issued. Store each evidence-based transition atomically. Durability still depends on configuration, storage behaviour, error handling and the defined failure model.

The external action remains a separate event

The process may crash after a device acts but before Transaction B commits. On recovery that instruction is unresolved, not automatically failed. A local transaction cannot roll back an already moved actuator or transferred energy. It enables honest recovery work; it does not remove it.

LOGGING IS NOT JUST PRINTING

An in-memory list or buffered console line is not proof that restart recovery can find the intent. Test the actual persistence boundary you claim.

Restart recovery is a state-reconstruction problem

The process’s last memory is gone. Durable intent, device history and fresh measurements must be brought back together before new dependent work proceeds.

Crash cut	What recovery can honestly know
Before committed intent	Under a strict persist-before-send path, no send was authorised through that path. The implementation must enforce this invariant.
After intent, before known send result	Dispatch may or may not have occurred. Preserve the unresolved instruction and reconcile it.
After device action, before local verdict commit	Physical effect may exist while local state is still unresolved. Do not resend from an empty in-memory queue.
After verdict commit	Reload the evidence and its version. Current process state or limits can still have changed since that verdict.

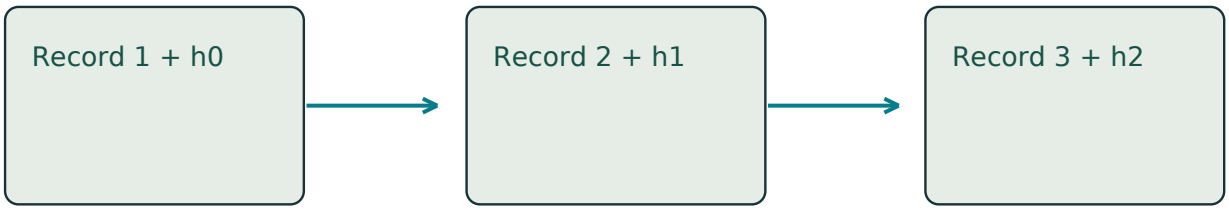
A device reboot can invalidate a lookup assumption

If its command-ID ledger is volatile, an empty history after reboot is not proof the earlier action did not occur. Track the device epoch and the documented history lifetime. If evidence is lost, preserve uncertainty rather than reconstructing a convenient success or failure.

The exercise later in this guide checks verdict logic only; it does not implement or test a durable recovery engine.

Evidence integrity is not physical validity

A hash can bind bytes. A signature can bind a claim to a key. Neither proves that a temperature sensor was calibrated or that an actuator behaved as reported.



hn = Hash(previous digest || defined record bytes)

Use unambiguous framing and a defined serialisation/version.

An unanchored chain can be rewritten consistently

Someone who can replace the records and recompute every later hash can create a chain that still verifies internally. A separately protected checkpoint or signed/anchored digest can make comparison meaningful within its threat model. Integrity also needs identity, key management, completeness and ordering assumptions.

Define the exact bytes that are hashed

JSON property order, number representation and encoding can change hashes. A defined canonicalisation scheme can support repeatable hashing; “sort keys” alone does not prove compliance with a cross-language canonical JSON specification. Keep raw payloads and schema versions where needed.

WHAT CLOKED MUST NOT CLAIM

A preserved record can still contain a wrong measurement, an unsupported causal claim or an invented input. Cryptographic integrity and engineering validity are separate checks.

Security must protect the action path and availability

An industrial interface needs authenticated identity, bounded permissions and controlled change. Encryption alone does not decide whether a request is physically appropriate.

Design property	Concrete implication
Authentication	Know which endpoint or principal produced a request or response; protect credentials and keys.
Authorisation	Grant only the required device/channel actions and modes. Read-only learning should remain read-only.
Segmentation	Separate test, supervisory and equipment networks with deliberately permitted paths.
Input + replay handling	Validate values, identities, expiry and generations at the authoritative receiver where supported.
Availability + change control	Plan resource limits, updates, backups and recovery around the process’s reliability and safety needs.

Transport protection does not fix an unsafe command

TLS can protect a communication channel within its configuration and trust model. It does not prove that a signed-in controller selected an allowed setpoint, used a fresh sensor or understood a register’s units. Application constraints and local protective functions still apply.

No intrusive security testing or industrial network probing is assigned. Use your local simulator and public documents.

Watchdogs detect a symptom, not every failure

A heartbeat can indicate liveness of one task or link. The appropriate response to its loss depends on the physical process and the engineered local design.

Observation	What remains unknown
Heartbeat arrived	The process may still be using stale data, applying the wrong units or failing to actuate.
Heartbeat absent	Device failure, network loss, overload or a delayed message can look similar.
Process restarted	Old commands, integrator state, device mode and physical output may still need reconciliation.
Watchdog output asserted	The actual fallback action and its suitability must be specified and verified locally.

There is no universal “turn everything off” fallback

Cooling loss might require a local run-on action, while another fault might require a protective stop. The correct response is specific to the equipment and hazards. Do not implement a general-purpose supervisor that overrides local protection or re-energises after a trip without the authorised procedure.

TEST LIVENESS AND CORRECTNESS SEPARATELY

A healthy heartbeat plus a frozen measurement is a useful fault case. A restarted UI displaying “connected” is not evidence that control authority or physical output is known.

Build a fault matrix before a happy-path demo

For each injected failure, define the expected evidence, forbidden claim and any forbidden next action. Test invariants as well as individual verdicts.

Injected condition	Expected discipline
Reply lost after action	Uncertain until command-correlated status and fresh feedback resolve it; no blind duplicate action.
Stale or missing sensor	No substituted zero and no verified physical outcome.
Duplicate / delayed request	Apply the documented receiver identity, generation and expiry rules; do not assume the network removes duplicates.
Device reboot / history loss	Notice the epoch change; do not turn an empty ledger into proof of non-execution.
Local limit or mode change	Preserve clipping, refusal or override; current local constraints remain authoritative.
Crash at each persistence boundary	Reload intent and evidence; prove the restart path does not issue an unsupported resend.

Keep the scope attached to the result

A unit-test pass supports its tested logic and assumptions. It does not establish hardware response, storage durability or untested network behaviour.

Write a one-page interface specification first

Use a fake device before negotiating a real equipment integration. The specification should make it possible for another person to disagree with your assumptions.

Section	What to write
Purpose + boundary	One physical objective; measurement point; permitted actions; what remains outside the scope.
Identity + authority	Device, firmware, epoch, owner, mode and instruction identity rules.
Limits + timing	Hard bounds, expiry, settle interval, capture requirements, maximum age and clock assumptions.
Outcome criteria	Exact evidence for match, mismatch, definitive non-execution and uncertainty; reasons to reject evidence.
Recovery + retry	Duplicate handling, late-message barrier, restart behaviour and what happens when evidence cannot be recovered.
Test + review	Named fault cases, expected records, limitations and the person responsible for real-equipment approval.

For the next exercise, all readings use a single invented clock domain and boot epoch. The negative receipt is explicitly stronger than “not found”: it asserts final non-execution with late execution prevented by the teaching contract.

The classifier has a deliberately small contract

It judges supplied synthetic evidence. It does not drive a plant, authenticate a device, persist state or implement retry permission.

Assumption	Teaching value
Instruction identity	cmd-7, issued at fixture time 10.0 s.
Device + epoch	sim-1, boot-1; both receipts and samples must match.
Measurement + clock	W; fixture-1 clock for every compared time.
Physical range + target	0-200 W; default target 100 W; tolerance ±2 W.
Time window	now = 12.0 s; sample at or after 11.0 s; maximum sample age 1.0 s; no future times.
Receipt meanings	accepted or fenced_no_execution. Any bare not_found or unknown kind remains uncertain.
Match meaning	Observed value within tolerance, after time and quality gates. It does not prove causation.

Why the one-second settling gate is labelled assumed

It is invented for this test fixture. Parts 02-05 explain why actual delay, dynamics and measurement uncertainty must be established before choosing a real acceptance rule.

Classify evidence without sending anything

Save this page as evidence_lab.py using Python 3, then append the scenarios on the next page. All times and messages are invented.

```
import math

META = {'device': 'sim-1', 'epoch': 'boot-1',
        'unit': 'W', 'clock': 'fixture-1'}

def number(x):
    return type(x) in (int, float) and math.isfinite(x)

def judge(reply, sample, target=100.0, now=12.0):
    if not number(target) or not 0 <= target <= 200:
        return 'uncertain'
    if not number(now) or now < 11.0:
        return 'uncertain'
    if not isinstance(reply, dict) or not isinstance(sample, dict):
        return 'uncertain'
    if reply.get('cmd') != 'cmd-7':
        return 'uncertain'
    for record in (reply, sample):
        if any(record.get(k) != v for k, v in META.items()):
            return 'uncertain'
    rt, st, value = reply.get('t'), sample.get('t'), sample.get('value')
    if not all(number(x) for x in (rt, st, value)):
        return 'uncertain'
    if not (10.0 <= rt <= now and 11.0 <= st <= now):
        return 'uncertain'
    if now-st > 1.0 or sample.get('quality') != 'ok':
        return 'uncertain'
    if not 0 <= value <= 200:
        return 'uncertain'
    matches = abs(value-target) <= 2.0
    if reply.get('kind') == 'accepted':
        return 'observed_match' if matches else 'observed_mismatch'
    if reply.get('kind') == 'fenced_no_execution':
        return 'uncertain' if matches else 'not_executed'
    return 'uncertain'
```

The function trusts the teaching meaning of a fenced_no_execution receipt. It cannot create that guarantee. Authentication, clock translation and authoritative status retrieval are outside this example.

Inject receipts, readings and identity failures

Each case is independent. Append below page 06-31 and run python evidence_lab.py. No sockets, real clocks or equipment are used.

```
def receipt(kind='accepted', **changes):
    return META | {'cmd': 'cmd-7', 'kind': kind,
                   't': 10.2} | changes

def reading(value=100.0, **changes):
    return META | {'value': value, 't': 11.5,
                   'quality': 'ok'} | changes

cases = [
    ('healthy', receipt(), reading(101), 100),
    ('output unchanged', receipt(), reading(0), 100),
    ('reply lost', None, reading(100), 100),
    ('stale sample', receipt(), reading(t=10.5), 100),
    ('sensor absent, target zero', receipt(), None, 0),
    ('negative conflicts', receipt('fenced_no_execution'),
     reading(100), 100),
    ('final non-execution', receipt('fenced_no_execution'),
     reading(0), 100),
    ('cache not found', receipt('not_found'), reading(0), 100),
    ('wrong command', receipt(cmd='cmd-old'), reading(), 100),
    ('device reboot', receipt(), reading(epoch='boot-2'), 100),
    ('NaN sample', receipt(), reading(float('nan')), 100),
    ('future sample', receipt(), reading(t=13.0), 100),
    ('valid zero', receipt(), reading(0), 0),
]

for name, reply, sample, target in cases:
    print(name, '->', judge(reply, sample, target))
```

WHY ZERO AND NONE ARE DIFFERENT

The valid-zero case has a real synthetic numeric sample and passes its gates. The absent-sensor case has no observation at all. The function never turns absence into a zero reading.

Expected outcomes and their limits

Predict these results before running the program. The useful work is explaining why each gate exists, not memorising the names.

Case	Expected verdict
healthy	observed_match
output unchanged	observed_mismatch
reply lost	uncertain
stale sample	uncertain
sensor absent, target zero	uncertain
negative conflicts	uncertain
final non-execution	not_executed
cache not found	uncertain
wrong command	uncertain
device reboot	uncertain
NaN sample	uncertain
future sample	uncertain
valid zero	observed_match

Make the claim no stronger than the test

You have exercised a pure classifier against supplied records and assumed semantics. The test does not demonstrate an actuator, a late-message fence, clock synchronisation, persistent recovery or calibration. Those require separate implementations and acceptance tests.

THE NEGATIVE-CONFLICT CASE IS DELIBERATELY CONSERVATIVE

The target may already have been present before the command, but the fixture does not resolve that history. It returns uncertainty rather than authorising a retry from conflicting evidence.

Ten checks before the next part

State assumptions, physical boundaries and evidence limits. The last questions concern actions that the classifier itself never takes.

1. Cooling error is +3 K. Baseline effort is 20% and Kp is 8 percentage points/K. Find the request before a 90% upper limit.

2. Ki is 0.25 percentage points/(K·s), error is 4 K and Δt is 2 s. How much integral contribution is added?

3. A comparable sample time is 11.4 s and now is 12.0 s. Find age. What changes if the sample came from an unknown clock epoch?

4. A signed 16-bit raw value is -250 with decimal scale factor -1. Find the physical number. What must be checked before assigning a unit?

5. An MQTT QoS 2 exchange completed. Does this prove a remote pump physically executed exactly once through a process crash?

6. A Modbus write reply is valid, but a fresh flow reading stays outside the objective after the allowed settling time. What is justified?

7. A timed-out command is not found in the device cache. Name two reasons why this may still not establish final non-execution.

8. The process restarts with committed intent but no verdict. Can an empty in-memory queue authorise a resend?

9. A hash chain verifies after someone rewrites its records and recomputes every later hash. What was missing, and does a checkpoint prove sensor accuracy?

10. The classifier returns not_executed. Is a new action automatically authorised, even if the device mode or physical limit changed?

Worked answers: 1-5

The arithmetic is small. The units and the claimed boundary are the important part.

1

Proportional cooling request

Request = 20 + 8 × 3 = **44%**. It is below 90%, so this gate does not clip it. It remains a control-effort request, not a measured speed, flow or cooling rate.

2

Integral update

Added contribution = 0.25 × 4 × 2 = **2 percentage points**. The time interval belongs in the calculation. The controller still needs saturation handling, valid feedback and an appropriate state policy.

3

Age and clock validity

Age = 12.0 - 11.4 = **0.6 s** in the comparable-clock case. With an unknown epoch or clock relationship, that subtraction is not justified. A fresh receive time cannot repair unknown capture time.

4

Signed scale conversion

Physical number = -250 × 10^-1 = **-25.0**. The exact map must supply unit, sign convention, valid range and invalid/sentinel encodings. A decimal calculation alone does not tell you whether the quantity is W, °C or another unit.

5

Protocol delivery is not exactly-once actuation

No. QoS 2 concerns its defined MQTT delivery scope. The application, its persistent state, the actuator and a crash boundary need their own execution contract. Do not attach a physical guarantee to a broker acknowledgement.

Worked answers: 6-10

Keep uncertainty and permission separate, including after an apparently conclusive verdict.

6

Protocol reply versus physical evidence

The write exchange succeeded under its protocol semantics, but the flow objective is **not verified**. With usable feedback and a complete acceptance contract, record an observed mismatch or limited outcome. Preserve local limits and diagnostics.

7

Not found is weaker than fenced non-execution

The message may still arrive late, be queued outside that cache, or have executed before volatile history was lost during reboot. A final negative result needs the actual contract to rule out these alternatives and prevent late execution.

8

Restart does not erase an unresolved action

No. Reload the durable instruction and reconcile it. The device might already have acted before the process lost memory. An empty local queue supplies no evidence about the physical outcome.

9

Internal consistency versus protected history

A separately protected checkpoint, trusted signature/anchor or comparable independent reference was missing from the threat model. Such evidence can support integrity within its assumptions. It **does not establish calibration or physical truth** of the recorded measurement.

10

A verdict is not an action permission

No. A new request still needs a current objective, valid measurements, ownership, mode, physical limits, expiry and the authorised retry policy. The classifier does not implement these action gates or a real late-message fence.

Reference: rules to carry into an interface review

These are engineering prompts and teaching equations. They are not a substitute for a device’s specification or protective design.

Relationship / rule	Condition / interpretation
Generic error e = r - y	Choose control polarity to match the actual plant.
Cooling error ecool = T - Tset	Illustrative convention: hotter means more cooling effort.
u = ubase + Kp e	Kp has effort/error units; apply valid limits.
ΔIstate = Ki e Δt	Includes actual interval and suitable anti-windup design.
Age = now - capture_time	Only with comparable clocks, domains and epochs.
abs(value - target) ≤ tolerance	Validate identity, quality, time and uncertainty.
Physical = signed_raw × 10^sf	Only after correct type, sentinel and map interpretation.
Timeout ≠ physical failure	The outcome can remain unknown after an action.
Not found ≠ will never execute	Requires history and a late-action fence.
Protocol success ≠ observed objective	Protocol reply is not physical feedback.
Persist intent before possible send	Enforce the ordering; recover unresolved work after restart.
Uncertain does not become failed by editing	Require new evidence and controlled reconciliation.
Verdict ≠ permission to retry	Recheck current authority, objective, envelope and policy.
Hash consistency ≠ trustworthy history	Compare a protected reference under a trust model.
Record integrity ≠ measurement validity	Check calibration, location, timing and meaning.

No timeout, tolerance, gain or cryptographic mechanism becomes valid solely because it appears in this table.

Words to recognise in control and evidence work

Use precise terms when discussing an interface with an equipment engineer.

Plant / setpoint Physical process being controlled / desired value under the chosen control law.

Feedback / feedforward Correction using observed output / action using a known reference or disturbance model.

Saturation / windup Actuator or controller limit / inappropriate accumulation of integral state while the limit binds.

Hysteresis / settling State-dependent switching thresholds / response entering a defined acceptable region over time.

Clock domain / epoch A valid timestamp reference / an identity for a reset or session boundary.

Freshness / validity Age acceptable for the decision / evidence fit for its stated interpretation and limits.

Adapter / register map Device-specific translation and contract / address, type and physical-meaning specification.

Receipt / observation Report about handling an instruction / report from a measurement chain.

Idempotency / deduplication Repeated effect equivalent under a contract / recognising repeats within defined scope and lifetime.

Reconciliation / fencing Resolving an instruction from evidence / preventing superseded or invalid late actions at the receiver.

Durability / atomicity Survival under a defined failure model / all-or-none changes inside the transaction boundary.

Checkpoint / canonicalisation Separately retained reference / defined repeatable byte representation for records.

Authentication / authorisation Establishing identity / deciding which actions that identity is permitted to perform.

Watchdog / heartbeat A liveness-monitoring mechanism / a signal about one defined component’s liveness.

Fault injection / invariant Deliberately introduced failure / a property that must remain true across the tested cases.

FOUR EVIDENCE LABELS

Assumed: chosen input. Simulated: produced by a model. Measured: acquired through an instrument. Verified: a stated claim checked against a defined method and limits.

Primary references and review status

These sources support checking specific principles and protocol details. No additional paid textbook or software is required for the exercise.

University of Michigan CTMS / PID
Feedback, controller terms and limits of generic gain guidance.

MathWorks / Anti-windup example
Why actuator saturation requires controller-state treatment.

Python / Time documentation
Monotonic clock semantics and elapsed-time limitations.

Modbus Organization / Application protocol
Functions, addressing and field encoding.

Modbus Organization / TCP guide
Transaction context and response correlation.

SunSpec / Device information models
Types and decimal scale-factor semantics.

OASIS / MQTT 5.0
Delivery scope, retained publications and session semantics.

SQLite / Atomic commit
Local transaction and crash-recovery assumptions; not remote actuator atomicity.

NIST / SP 800-82 Rev. 3
OT security with process reliability and safety considerations.

RFC 8785 / JSON canonicalisation
Defined serialisation for repeatable cryptographic operations.

Edition 01 / 17 October 2026

Original material prepared with ChatGPT. Numbers and device messages are assumed. Code is tested with synthetic evidence; no real protocol, persistence engine, actuator or control system has been validated for this volume. No independent engineering peer review is claimed.

Bring a reviewable contract back to Lupine

The immediate deliverable is a small, explainable specification and test set. Production capability requires implementing and validating each claimed boundary.

Layer	What this volume asks you to specify
Aiva	A feasible choice based on valid current observations, limits and authority. Keep hard constraints separate from ranking.
Rail	Durable intent; device-correlated outcomes; explicit uncertainty; controlled reconciliation; and no retry permission from an unsupported failure assertion.
Cloked	Versioned records, raw evidence, timestamps, reasons and integrity checks with an explicit trust model. Keep physical validity as a separate claim.

Three artefacts to retain

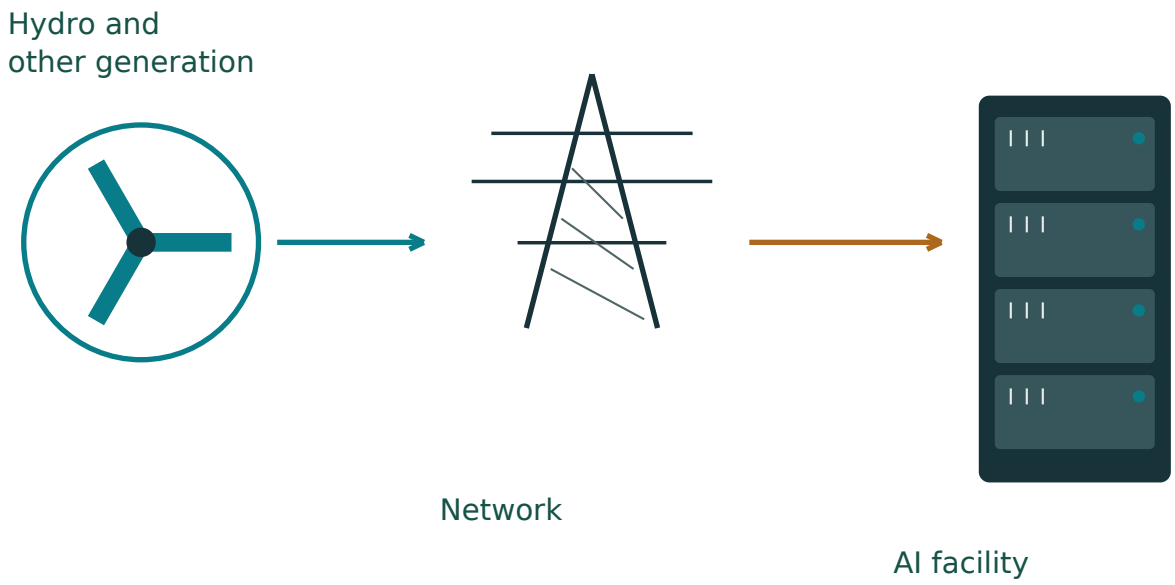
One interface brief: the contract on page 06-29, with every unknown visible.
One explainable test run: the 13 classifier scenarios, your predictions and corrections.
One gap list: actual implementation work still needed for communication, persistence, clock handling, ownership and physical acceptance.

NEXT: PART 07 / TASMANIA'S GRID AND AI DATA CENTRES

The next volume places the physical and software foundations in the local infrastructure context. Two planned parts remain: Part 07 and Part 08, the Lupine prototype and field-validation guide.

TASMANIA’S GRID & AI DATA CENTRES

Part 07 / Connect your engineering foundations to a real region, its infrastructure and questions worth investigating.



A conceptual power path, not an actual electrical connection or site single-line diagram.

Your direction becomes an evidence-led investigation

The aim is a technical life and a future Lupine energy-infrastructure company, beginning in Hobart. Learn how the local system works, find a specific physical or interface failure, and earn access to the evidence needed to test it. Published ambitions and measured operating results remain different evidence types.

EDITION SNAPSHOT / 17 OCTOBER 2026

Project statements are checked against primary public sources. Status and dates can change. Calculations and the code fixture are explicitly hypothetical; no confidential plant or Firmus telemetry is used.

From local context to a testable question

The earlier parts taught physics and reliable execution. This volume supplies an infrastructure map, source discipline and a practical investigation route.

Pages	Question to answer
3-10	How do geography, hydro, networks, interconnection and security fit together?
11-16	What is an AI factory, and what do current local project statements establish?
17-20	How do power, heat, workload flexibility and connectivity interact?
21-25	Which public data can I use, and what conclusions can it support?
26-29	Who can help check the engineering, and how do I define one real problem?
30-33	How do I audit intervals, missing values and energy coverage on a computer?
34-37	Can I answer the checks and keep the core quantities straight?
38-40	Where are the sources, and what should go into the final prototype part?

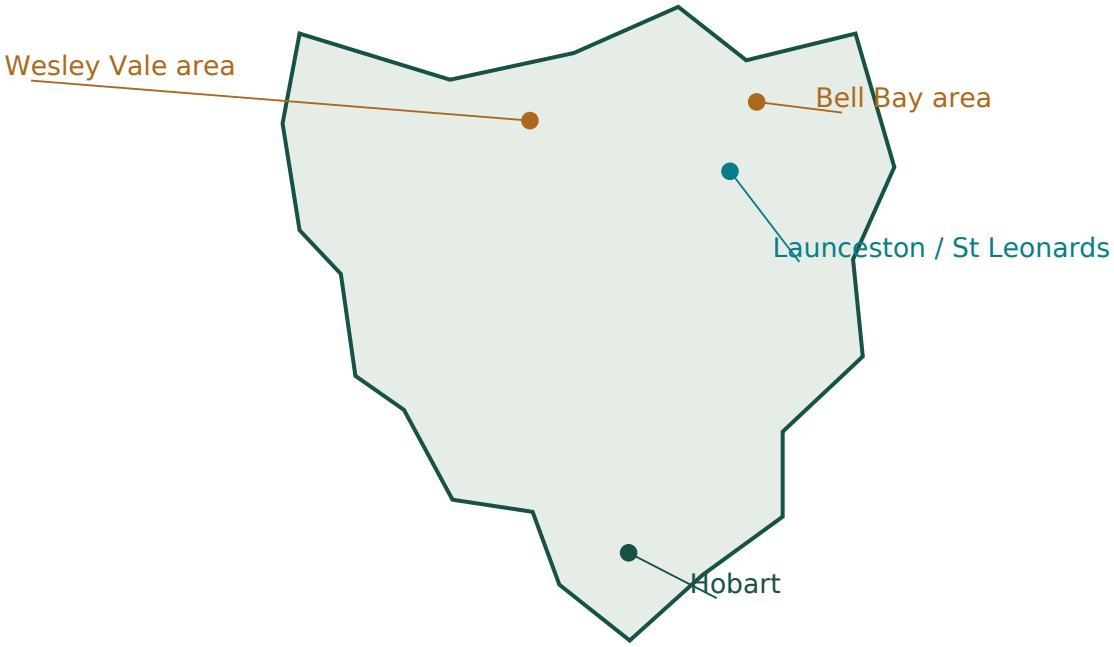
No finance or trading assignment

This is an electrical, thermal, computing and control investigation. Public market-system data is used to understand physical quantities and evidence limits; no electricity-price forecasting, investment analysis or trading project is assigned.

The references use S01-S24 identifiers. They are clickable on pages 07-38 and 07-39.

Orient yourself without inventing a network map

Tasmania’s geography helps you locate institutions and projects. A geographic sketch does not establish electrical paths, connection capacity or site access.



Stylised orientation only: not to scale, no asset coordinates or transmission routes implied.

Hobart is your base; the northern sites are a separate geography

Firmus’ community page identifies northern Tasmanian development locations and describes engineering/software staff in both the north and south. It does not imply that a Hobart resident can access a live AI facility or that a particular technical vacancy is available. [S07]

Use official maps for actual infrastructure

TasNetworks’ planning material is the starting point for transmission and distribution context. Read map legends, voltage levels, existing/proposed line styles and publication dates. A line passing near a site does not prove an available or approved connection. [S01]

Visit only public or specifically authorised places; drawings are not directions to restricted electrical assets.

Separate the organisations by their engineering role

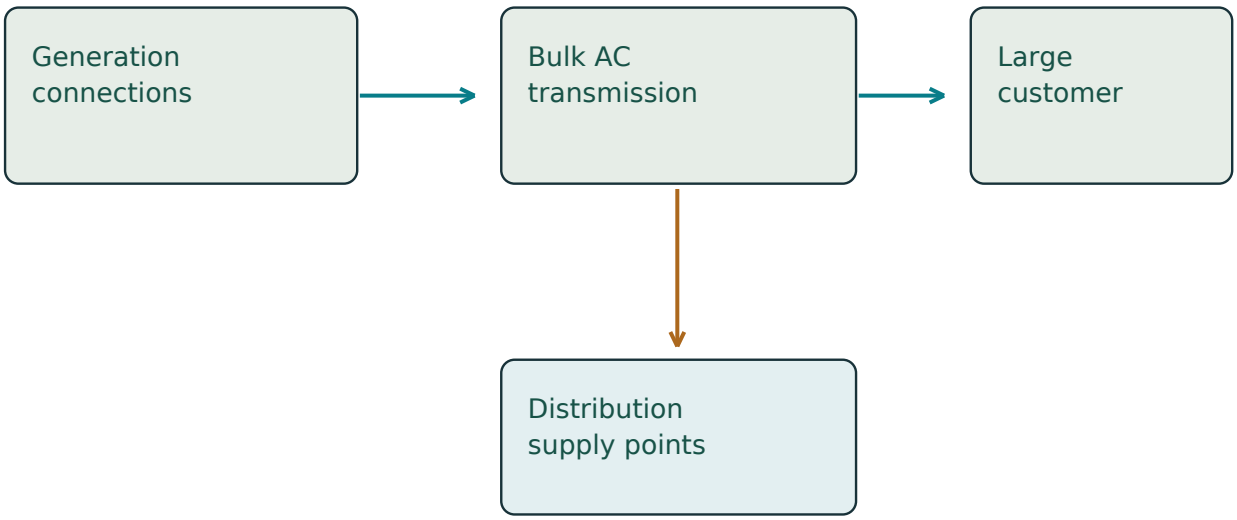
A generator, network owner, system operator, project developer and equipment owner each know different parts of the physical system.

Organisation	Relevant technical role
Hydro Tasmania	Hydropower generation, water/storage assets and future generating/storage proposals; the energy-supply interface is distinct from planning approval. [S02, S06]
TasNetworks	Transmission and distribution planning and asset/network interfaces in Tasmania. Its APR and supplementary files expose useful planning questions. [S01]
AEMO	NEM-wide power-system coordination, dispatch information and security planning; public regional data is not a customer’s private waveform trace. [S13, S14, S18]
APA / Basslink	Owner/operator information on the existing Tasmania-Victoria HVDC interconnector. [S03]
Marinus Link / project partners	Delivery information for new interconnection, converters and related infrastructure; construction progress differs from commissioned capability. [S04, S05]
Firmus / equipment suppliers	Facility, compute, cooling and control design claims; access to real measurements must be established with the relevant owner. [S07-S10, S23, S24]

No organisation listed here has agreed to advise Lupine or provide data. Treat this as a role map, not a promised partnership.

The network has location-specific transfer limits

A statewide energy surplus does not guarantee spare capacity at a particular substation, line or customer connection.



Functional hierarchy, not TasNetworks' actual topology or a connection design.

Read the local voltage hierarchy

The 2025 TasNetworks APR describes a 220 kV bulk network, with 110 kV corridors and peripheral connections, and substations linking transmission, distribution and directly connected customers. That describes structure, not a blanket ability to add any load anywhere. [S01]

Capacity is constrained by more than conductor heating

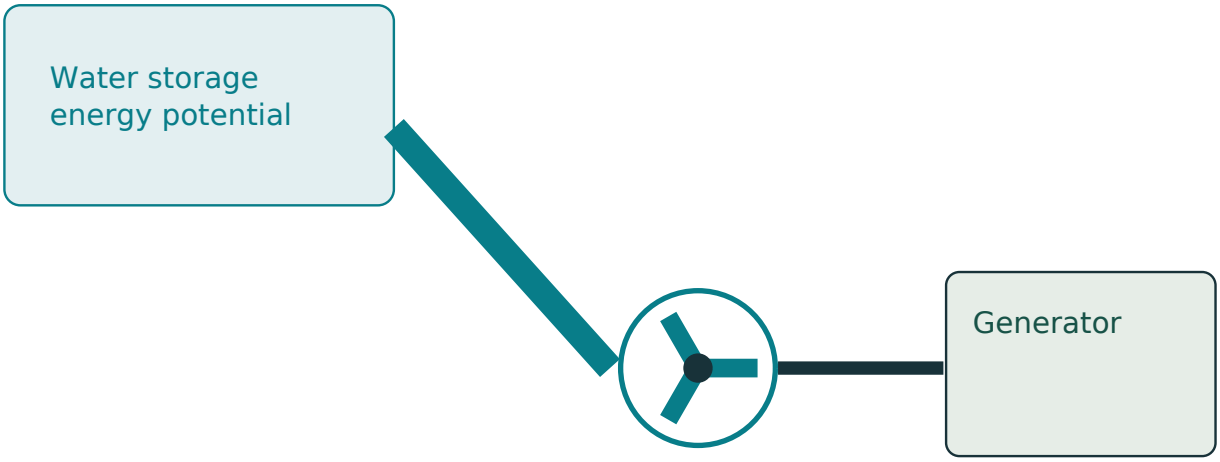
Thermal ratings, voltage, stability, protection, credible contingencies, planned outages and equipment condition can limit transfer. A connection study needs the actual network configuration and load behaviour. The public planning report is a starting context, not permission to connect.

USE THE SUPPLEMENTARY FILES CAREFULLY

TasNetworks publishes planning attachments including substation forecasts, load profiles and limitations. Preserve their year, scenario and definitions before quoting a number. [S01]

Hydro links stored water to time-varying power

Water in storage provides energy potential; turbine-generator units provide power capability. The same stored energy can support different durations at different output levels.



Waterway → turbine → shaft → generator; losses and operating limits apply.

Separate water availability from instantaneous capability

Rainfall, inflows, storage levels, head and environmental requirements affect available hydraulic energy. Unit availability, permitted operating regions, waterways and network constraints affect deliverable power. Hydro Tasmania’s public system material describes the schemes; a private unit model is a different evidence level. [S02]

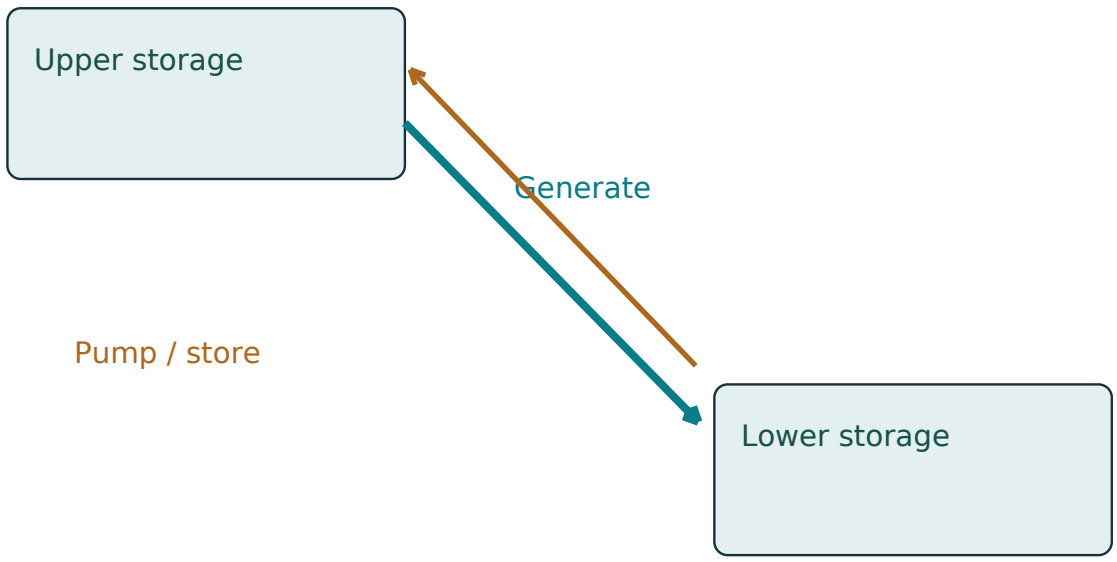
A HYPOTHETICAL POWER-ENERGY COMPARISON

An ideal 1,000 MWh usable inventory can supply 100 MW for 10 h or 250 MW for 4 h, only if the plant can actually deliver that power and the stated inventory is usable. It does not imply a 1,000 MW unit.

The simplified reservoir arithmetic excludes changing head, losses, inflows and plant-specific constraints.

Pumped storage shifts energy; it does not create it

Pumping consumes electricity to move water uphill. Generation later returns part of that energy through the turbine-generator path.



Opposite operating modes. Round-trip losses require more input than recovered output.

Keep Cethana’s proposal separate from the existing scheme

Hydro Tasmania describes a proposed Cethana pumped-hydro development using the existing lake as lower storage, with a new upper storage and underground station. Its published concept is 750 MW with 20 hours of storage. Treat these as project-design claims, not operating capacity. [S12]

CALCULATE THE DESIGN-LEVEL ENERGY

750 MW × 20 h = 15,000 MWh = 15 GWh at the stated duration/rating interpretation. It is not a measured stored-energy reading. With an assumed 80% round-trip efficiency, delivering 15 GWh would require 18.75 GWh input across the same external boundary.

The 80% value is invented for this example, not an asserted Cethana specification.

Basslink exists; Marinus adds future interconnection

Both use HVDC concepts from Part 05. Their ratings, stages, availability and dynamic operating limits must be kept separate.

Asset / project	Primary-source statement
Basslink	APA describes an existing 500 MW nameplate HVDC connection between George Town and Loy Yang. Actual transfer is constrained by current conditions and limits. [S03]
Marinus Stage 1	The developer’s overview describes an initial 750 MW stage. A 16 October 2026 update reports site establishment and preparation for major construction. It is not commissioned transfer capacity. [S04, S05]
Marinus Stage 2	The overview describes a second 750 MW stage at a later date. Do not treat the combined 1,500 MW programme as already available. [S04]
Related network works	TasNetworks describes north-west line/substation works, including a new Heybridge switching station to support the Marinus connection. [S11]

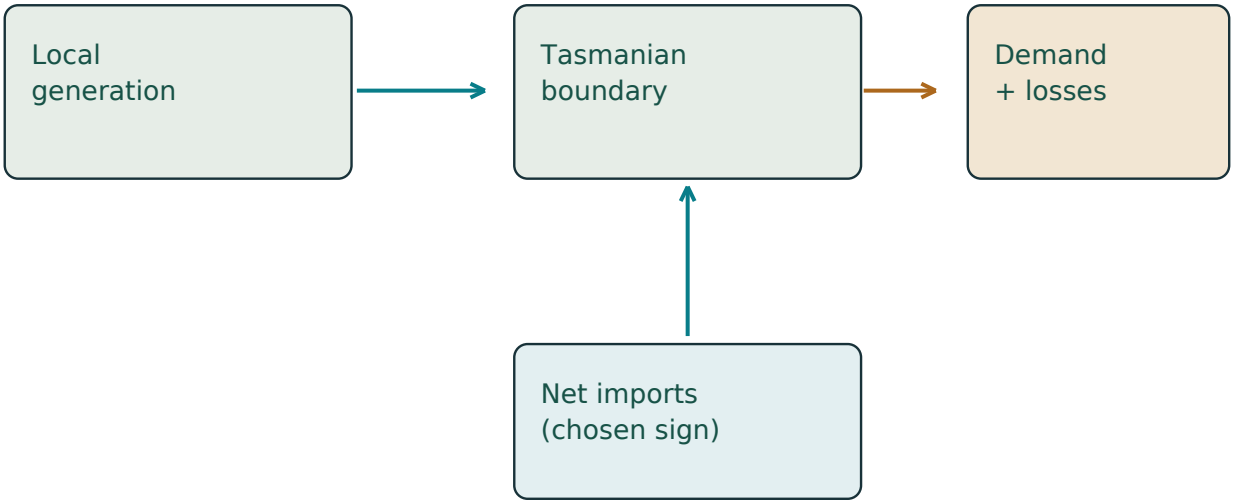
An HVDC link does not simply merge two AC rotors

Converters regulate exchange between AC systems through a DC connection. Controls, source availability, limits and contingencies determine behaviour. The loss of an importing link is a different disturbance from the loss of an exporting link.

Status snapshot checked 17 October 2026. Later project announcements require a new dated update.

A regional power balance needs a sign convention

Generation, demand, interchange, losses and storage must refer to compatible boundaries and times. Published fields may not align with your first intuitive definition.



Simplified steady balance: generation + imports = demand + losses.

Work an invented regional example

Assume local generation 1,200 MW, imports 150 MW and demand 1,300 MW, with no modelled storage change. The remaining 50 MW is the assumed loss/residual term. If the same 150 MW were actually an export under the source’s sign convention, the interpretation would be entirely different.

Do not use the residual as automatic fault evidence

Different timestamps, source revisions, embedded generation, scheduled loads or inconsistent sent-out/gross definitions can create an apparent imbalance. Before blaming a sensor or plant, reconcile the actual definitions. Interconnector dashboard signs are explicitly documented by AEMO. [S13]

A RATING IS NOT A FLOW READING

500 MW nameplate capacity does not mean Basslink is transferring 500 MW, in the direction you assumed, at the moment in question.

Security is more than annual energy adequacy

A power system can have enough energy over a year and still face a local voltage, frequency, fault-response or transfer problem at a particular moment.

Question	Evidence and concept
Energy adequacy	Generation and usable storage over time, inflows, demand, imports and relevant scenarios.
Frequency response	Active-power imbalance, inertia, responsive sources/loads, delay, limits and duration.
System strength	Ability to maintain/control a stable voltage waveform at the relevant location, including after disturbances. [S18]
Network transfer	Topology, line/transformer limits, voltage, stability and credible contingencies.
Restoration + protection	Energisation sequences, fault detection, interruption, backup and supported islanding arrangements.

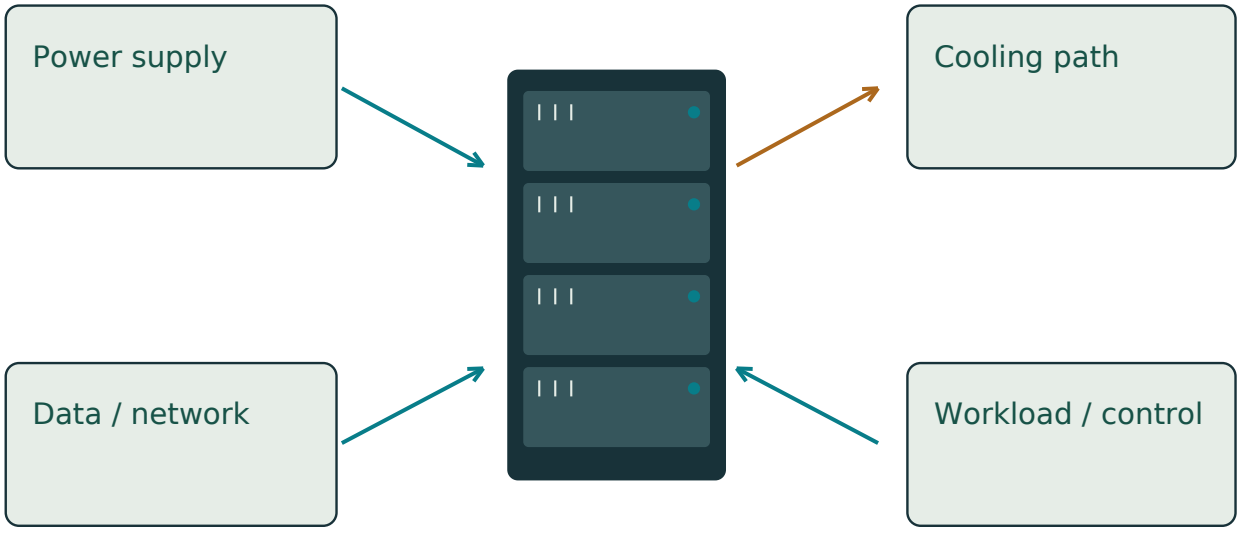
AEMO’s public assessments frame questions, not a private model

System-security planning considers strength, inertia and network-support needs. A specific Tasmanian problem requires the relevant current assessment and network/equipment detail. A generic statement that AI loads vary cannot establish a named generator’s shaft risk. [S18]

Use Part 03 for machine and frequency foundations; use Part 06 for the evidence needed before making an outcome claim.

An AI factory couples computing to physical services

Accelerators need electrical power, memory, data movement and heat removal. Useful computing output depends on the full system and workload, not GPU nameplate capacity alone.



Training and inference create different service constraints

Training may involve repeated computation, synchronisation and checkpointing. Inference serves requests whose latency and availability requirements can limit flexibility. Neither is universally interruptible. GPU, memory, network and software bottlenecks change useful output per unit energy. NVIDIA’s AI-factory material describes an integrated stack. [S23]

Read accelerator metrics with a workload definition

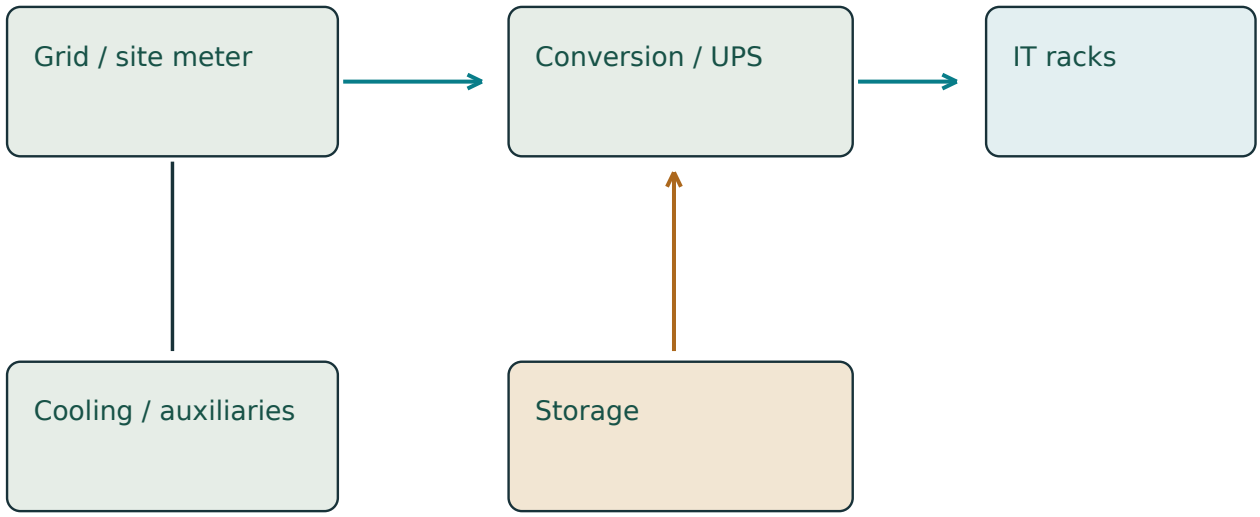
Tokens per second, training time, energy per completed task and utilisation each answer different questions. Specify model, precision, batch size, accuracy/quality, hardware and boundary. A low facility overhead ratio does not establish useful computing efficiency.

YOU DO NOT NEED A GPU CLUSTER TO LEARN THE INTERFACES

You can study public architecture documents and practise measurement, timing and failure semantics with the small apparatus and simulators already introduced.

Trace the facility electrical path to its meters

A site connection supplies more than chip power. Storage, conversion, cooling and distribution can sit on different sides of a reported measurement.



Illustrative architecture. It is not a Firmus electrical drawing or a guarantee of topology.

Use matched boundaries for a PUE calculation

PUE is facility energy divided by IT energy over the same defined interval. In an invented steady case, 100 MW IT plus 20 MW other site demand gives 120 MW total and a power-ratio proxy of 1.20. The corresponding energy ratio is 1.20 only when the same conditions and boundaries apply throughout the interval.

Storage can move the site-meter result without changing IT work

A battery supplying 10 MW might temporarily reduce net grid draw while the IT load stays unchanged. That does not automatically reduce facility energy used, since recharge and conversion losses must be included consistently. Report grid exchange separately from load and stored-energy change.

Name the meter, aggregation window, auxiliary loads and storage treatment before publishing any efficiency figure.

Trace the thermal path to the final heat sink

Direct-to-chip liquid cooling moves heat efficiently only if the full route - from component interface to external rejection - works within its envelope.



Heat crosses boundaries; the two liquid circuits need not mix.

A conceptual heat path, not a documented layout of any Tasmanian site.

Keep cooling architecture and performance claims separate

Firmus' Wesley Vale FAQ distinguishes immersion use in Singapore from planned direct-to-chip cooling in Tasmania for next-generation platforms. That is a developer description of intended architecture, not a measured thermal map or proof of a specific site's current operation. [S24]

Use Part 04 to turn the claim into engineering questions

Which fraction of heat is captured in liquid? What are the supply/return temperatures, fluid properties and flow bounds? Where is final rejection? How are condensation, leak detection, residual air heat and loss of flow handled? Which measurements are available and trustworthy?

A LIQUID LOOP IS NOT AN INFINITE ENERGY SINK

Thermal mass delays warming; steady operation still requires heat rejection. A cool climate or a CDU label cannot establish year-round thermal capacity.

Do not merge different public capacity numbers

A site’s IT rating, a supply agreement and a multi-site future request are different quantities. Their dates, status and boundaries belong beside the numbers.

Published figure	What the source actually describes
84 MW	Firmus’ Southgate page labels this as Launceston critical IT load capacity. It is a developer-published project figure. [S08]
104 MW	Hydro Tasmania’s public Firmus page describes a three-year Aurora retail agreement for St Leonards. It is a supply-arrangement figure, not an IT-meter reading. [S06]
444 MW	That Hydro page describes a future long-term request for proposed Tasmanian factories. Its displayed 27 August 2026 update says no decision had been made and an outcome was expected by year end. [S06]
Current construction / proposals	Firmus’ community page says the Launceston anchor is under construction and identifies Bell Bay and Wesley Vale as expansion plans. These statements are not commissioning evidence. [S07]

Why 104 / 84 is not a measured PUE

The arithmetic is about 1.238, but the sources do not establish matched facility and IT energy readings for one interval or the same operating state. Preserve the numbers with their own meanings. Check for newer dated documents before relying on project status.

Source snapshot: 17 October 2026. The page’s internal update date is not the date of an independent site inspection.

Flexible connection is a physical contract

A non-firm or constrained connection can require demand to reduce or disconnect under specified conditions. It needs reliable authority, action and restoration semantics.



Reduce → confirm outcome → restore only under the agreed conditions

Functional sequence. It does not specify actual TasNetworks protection or a Firmus interface.

One concrete public proposal to study

Firmus' Wesley Vale R&D; Q&A; describes a proposed maximum 26 MW non-firm arrangement, including demand reduction/disconnection when required and restoration under agreed conditions. It proposes combining battery storage with power control and workload orchestration. These are company-described plans, not your measured results. [S09]

The technical questions are specific

What signal is authoritative? What is the required reduction, deadline and baseline state? Does a reply mean received, queued or applied? How is actual site draw checked? What prevents premature restoration after a lost message, reboot or ownership change? Those questions connect directly to Part 06.

The private connection agreement, limits and acceptance criteria have not been obtained for this guide.

Study the existing controls before proposing another

Grid-responsive compute and battery-backed sites are active engineering areas. Broad category overlap is evidence to investigate, not a reason to invent an originality claim.

Public example	What it establishes / what it does not
Firmus engineering principles	The company describes an integrated proprietary layer for telemetry, cooling, GPU orchestration and grid interaction. It does not disclose every implementation or unresolved failure. [S10]
Wesley Vale R&D; proposal	The public plan explicitly links batteries, workload management and flexible network behaviour. It shows this broad combination is already being investigated. [S09]
NVIDIA AI-factory stack	Supplier material combines compute, networking, software and physical infrastructure. It does not prove a particular Lupine test gap is already solved. [S23]
Lupine investigation	Compare an exact failure and required evidence against documented functions and real-user experience. “Our architecture transfers” alone does not establish demand.

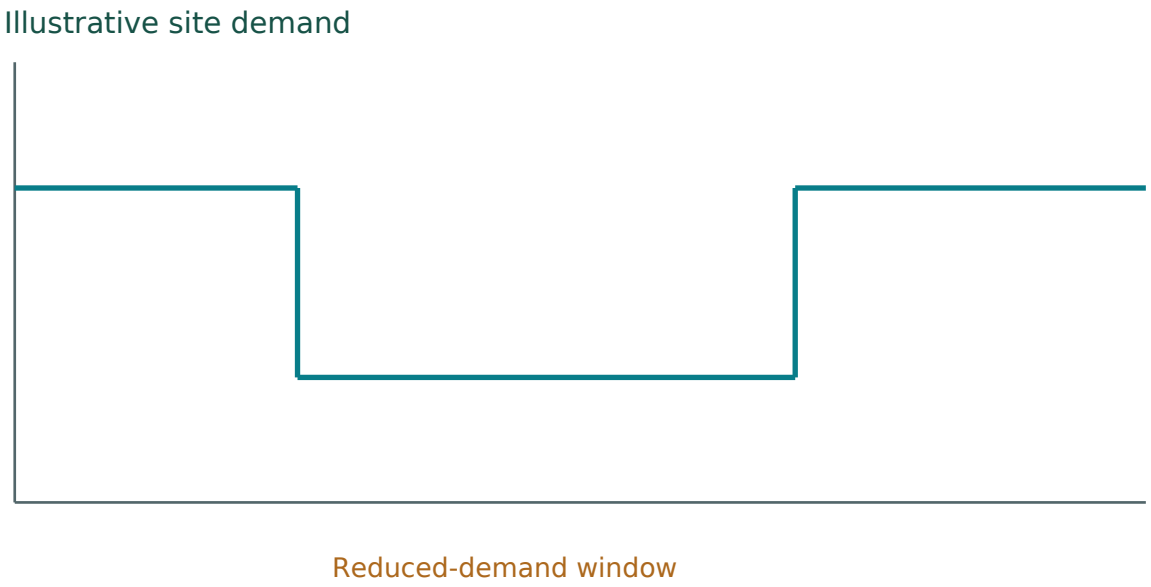
A reasonable initial technical wedge to test

A small, hardware-agnostic test harness for uncertain supervisory outcomes could be useful if real engineers identify a missing capability. Treat this as a hypothesis. Access, existing tools and practical value must be checked before extending the implementation.

No guarantee of novelty, adoption, hiring or commercial success is implied by this proposed investigation.

A flexible load needs a power-and-time specification

A promise to “reduce demand” is incomplete without magnitude, deadline, duration, recovery and the effect on useful work.



Calculate the energy displaced in a bounded example

Assume a site reduces grid demand from 20 MW to 12 MW for 15 min. The reduction is 8 MW and the avoided grid energy during that window is 8 × 0.25 = 2 MWh. If a battery alone supplies the difference at 90% discharge efficiency, it loses 2.222 MWh internally, before auxiliaries.

There are different ways to meet the same meter target

Reducing computation, shifting a job, changing cooling demand and discharging a battery have different effects on service, temperature and later demand. A site-meter reduction alone does not identify the mechanism or prove total energy savings. Rebound and recharge belong in the evaluation.

A FLEXIBILITY ENVELOPE IS MULTIDIMENSIONAL

Record allowed MW, duration, ramp, reserve, thermal margin, workload constraints and restoration conditions. A single “flexible” flag is not enough.

Thermal storage gives time, not unlimited flexibility

Changing electrical or cooling power changes energy balances. A controller must preserve equipment limits during the response and its recovery.

Accumulation rate = heat in - heat out

dT/dt ≈ heat deficit / effective Cth

Time margin ≈ Cth × allowed ΔT / deficit

Lumped, constant-deficit estimate; local hot spots and dynamics can invalidate it.

Work a deliberately simplified thermal buffer

Assume an effective thermal capacitance of 5 MJ/K and a 1 MW heat-removal deficit. The lumped temperature rises at 0.2 K/s. A 2 K allowance would last about 10 s if all assumptions held. The value is not a real data-centre limit and does not establish a safe cooling shutdown.

Compute reduction and cooling reduction are not equivalent

Lower compute power can reduce heat generation, but coolant and surfaces respond with delays. Reducing a pump may instead harm heat transfer while the chips still dissipate power. Local control, temperature observations and permitted operating envelopes remain authoritative.

BRING THE PHYSICS INTO THE INTERFACE

A command acknowledgement cannot confirm thermal safety. Define the measurements, delays, limits and independent protective behaviour needed for the actual device.

Weather helps size cooling; averages can mislead

Dry-bulb temperature, humidity/dew point, wet-bulb conditions and equipment approach affect different heat-rejection methods.

Evidence	Useful engineering question
Dry-bulb air temperature	What rejection temperature is possible through a dry cooler with a finite approach and permitted fan power?
Dew point / humidity	Could a cold pipe or surface condense water under local conditions and sensor uncertainty?
Wet-bulb conditions	What evaporative cooling opportunity and water-management constraints exist?
Hourly / seasonal extremes	How often can a proposed operating envelope be met, and what handles the remaining conditions?
Water-use boundary	Is a claim about makeup, withdrawal, consumption, process water or the cooling loop? Over what period and weather?

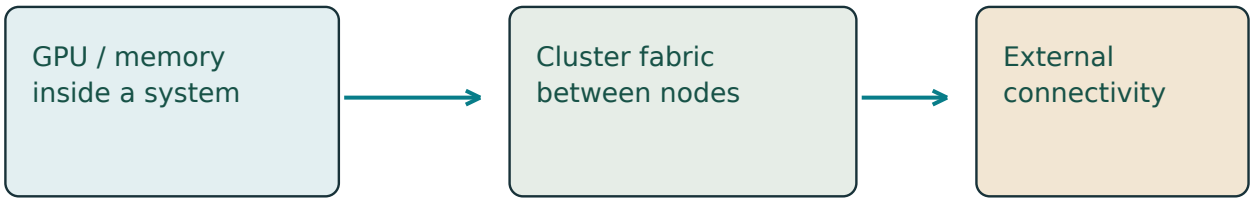
Use Bureau of Meteorology data with its quality context

Public observations and climate statistics can support a first investigation, but station, period and quality status matter. Daily maximum/minimum data cannot reconstruct every hourly cooling condition. A Hobart station is not automatically representative of a northern industrial site. [S19]

Do not convert a developer’s water-minimisation claim into an independently measured WUE or a water entitlement.

Data movement is part of the energy problem

An AI job can be limited by communication as well as arithmetic. A fibre announcement, a GPU interconnect and an application’s end-to-end latency describe different layers.



Bandwidth, latency, synchronisation and data placement all affect useful work.

The drawing is a conceptual hierarchy, not a physical cable route or site topology.

Coupled bottlenecks change what a power reduction costs

A waiting accelerator may use less power yet deliver little useful work. A network-limited training step can respond differently from a compute-limited one. Workload flexibility must be measured against completed work, quality and service requirements, not only GPU utilisation. [S23]

Ask which measurement boundary the claim uses

Rack, cluster, data hall and internet-path counters represent different traffic and timing. Public connectivity announcements establish developer intentions; they do not establish available route diversity, commissioned throughput or your access to diagnostic data.

A USEFUL LOW-BUDGET TASK

Draw a public architecture’s power, heat and data boundaries on one page. Mark each measured, specified, inferred or unknown field instead of filling gaps with plausible details.

Start with free primary information

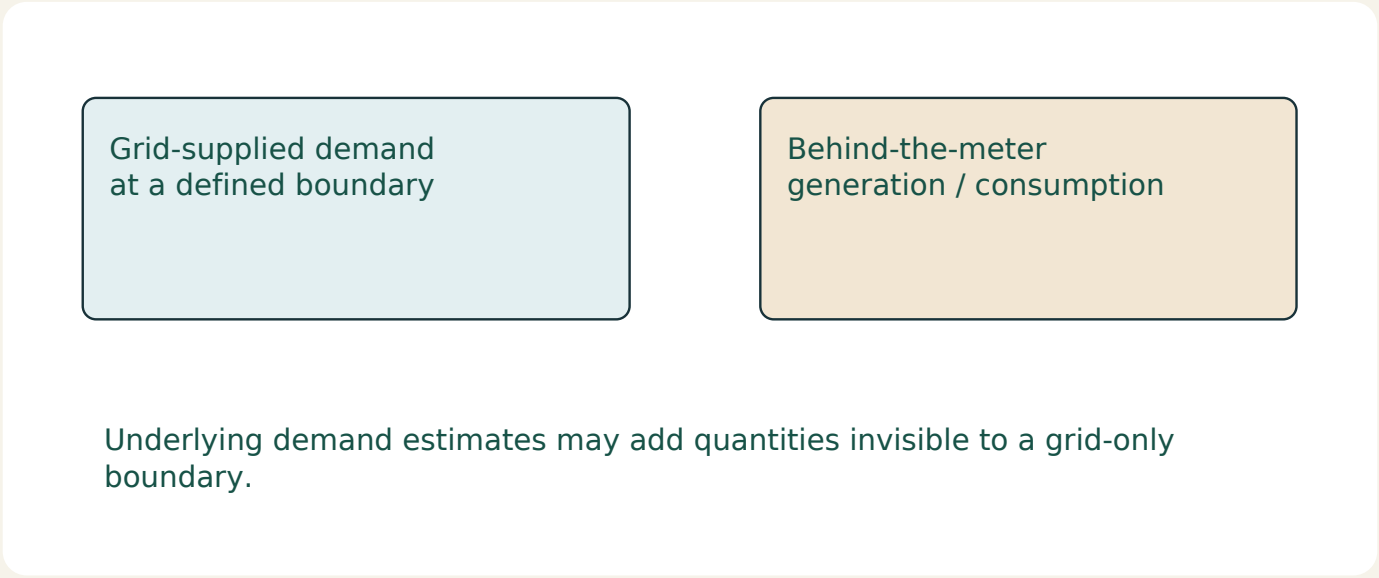
Different sources answer different engineering questions. Downloading a public file gives you evidence to inspect, not automatic permission to infer a site’s behaviour.

Source	Useful for / important limit
TasNetworks APR [S01]	Network structure, planning scenarios and published limitations. Check attachment year and scenario; it is not a live connection assessment.
AEMO dashboard [S13]	Orientation to demand, dispatch and interconnector displays. Read each chart definition; a screen image loses much of the underlying metadata.
AEMO files + dictionary [S14, S15]	Machine-readable regional dispatch data and table definitions. Preserve raw files, headers, keys and revision information.
Operational demand [S16]	A separately defined demand product. Its boundary and averaging period must be checked before comparison with dispatch fields.
BOM [S19]	Station observations and climate summaries. Daily summaries cannot establish second-by-second cooling performance.
Project and planning pages [S04-S12, S22]	Dated design intentions and approval/construction context. They do not independently establish delivered efficiency or operating availability.

First download task: retain one source file, its exact URL, retrieval time and dictionary version. Explain one field before plotting anything.

Demand is a definition, not one universal number

A regional dispatch quantity, operational demand and an estimate including behind-the-meter generation can describe different boundaries at the same time.



Read the table and field together

The dashboard’s scheduled-demand description and DISPATCHREGIONSUM fields are not interchangeable labels. Read the current dictionary, including dispatchable-load treatment, intervention flags and units. A forecast is not an observation simply because it shares a timestamp. [S13, S15]

Read the time statistic as well

AEMO’s documented DEMANDOPERATIONALACTUAL field represents an average over a 30-minute period; other products can have other resolutions. A five-minute dispatch solution is not automatically a five-minute mean measured load. Do not integrate an instantaneous target and label the result exact metered energy. [S16]

BEFORE COMPARING TWO SERIES

Write boundary + field + unit + interval + statistic + revision policy for each. If any differs, explain the conversion or keep the series separate.

Use two clocks without confusing them

NEM timestamps use fixed AEST, UTC+10. Hobart civil time changes with daylight saving; your computer’s collection clock is a third, separate piece of metadata. [S17]

Same illustrative instant	Clock reading
UTC	16 Oct 2026, 14:05
NEM time / fixed AEST	17 Oct 2026, 00:05 +10:00
Hobart / AEDT on this date	17 Oct 2026, 01:05 +11:00

An interval end is not a collection time

If a documented five-minute mean has interval end 00:05, it describes (00:00, 00:05] under the chosen convention. A download at 00:09 does not move the measurement to 00:09. Store interval start/end and retrieval time separately; record whether the source uses interval-begin or interval-end labels.

Keep offsets explicit and preserve revisions

Use timezone-aware timestamps. Convert the same instant between clocks rather than replacing an offset string. If a source republishes an interval, retain both raw versions and a stated rule for selecting the analytical version; do not silently count both as extra energy.

QUICK CHECK

14:05 UTC + 10 hours = 00:05 the next day in NEM time. In Hobart during AEDT, the same instant is one hour later on the local clock.

Make a small acquisition manifest

A reproducible analysis starts before the chart: retain what arrived, where it came from and which transformation produced your working table.

Record	Example / reason
Source identity	Exact URL, publisher, table/product name, schema/dictionary version and download filename.
Time identities	Source interval and zone; publication/revision time if exposed; separate UTC retrieval time.
Selection rule	Region TAS1 where applicable; applicable run/intervention/version keys; a documented duplicate/revision policy.
Integrity	SHA-256 of retained raw bytes. This detects change relative to that saved digest; it does not establish that the source measurement was correct.
Transformation	Script revision, field mappings, unit conversions, exclusions and any estimated rather than measured quantities.
Limitations	Missing intervals, uncertain field meanings, provisional quality and conclusions the dataset cannot support.

Apply the same discipline to project status

A developer announcement and a dated council planning update are separate records. Keep both; check subsequent decisions instead of turning either into a timeless status claim. [S22]

Write the claim your evidence can support

A beautiful graph can still answer the wrong question. Use this page as a review before publishing any Lupine technical note.

Evidence available	Supported / unsupported inference
Five-minute regional series	Can describe the recorded regional quantity at that resolution. Cannot establish a particular generator shaft's fast oscillation or fatigue.
Regional load changes	Can identify a regional pattern. Cannot attribute the change to Firmus, a control command or one customer without additional evidence.
Developer design efficiency	Can report a dated, attributed design claim. Cannot call it your measured operating PUE or compare unmatched boundaries.
Command + later reading	Can establish a sequence and observed state within sensor limits. Does not automatically prove the command caused the state change.
Hash-linked event records	Can support integrity checking against trusted anchors. Cannot make an inaccurate sensor or a fabricated source record true.
Simulator passes	Can show the implementation passes specified artificial cases. Cannot prove compatibility, timing or safety on real plant.

Keep useful uncertainty visible. Unknown is a legitimate engineering result, not a blank to fill with confidence.

Use local expertise to sharpen your questions

You can begin with free public research pages and technical papers. Contact and equipment access are future possibilities to establish, not prerequisites already granted.

UTAS CREPS
Power systems; power electronics; renewables and thermal research

TasTAFE
Electrical trades, instrumentation, refrigeration and training pathways

Match the question to the discipline before asking for time, data or equipment.

Research fit: UTAS

CREPS publicly describes work in power systems, power-electronics/control, renewable energy and thermal engineering. Use its current staff and publication pages to find relevant work. A question about load-response evidence differs from a question about hydrogenerator dynamics or a cooling loop. [S20]

Practical fit: TasTAFE

TasTAFE describes clean-energy training facilities and trade pathways. Read the current curriculum and entry requirements if you explore formal training. No course place, free tuition, laboratory access or electrical-work authorisation is assumed here. [S21]

A SMALL REQUEST HAS A CLEAR PURPOSE

Show one diagram, one evidence limit and one technical question. Ask which assumption is wrong or which public reference to read. Record the answer accurately; do not imply endorsement.

Keep a problem log grounded in evidence

Your log is a record of learning and observed limitations. Forty invented problems are less useful than five precise entries you can trace back to a real source or experiment.

Field	What to write
Event / source	Date, device or document, exact page/field and where the evidence is retained.
Observation	What you directly saw, measured or read. Label documented reports separately from your own measurements.
Expected behaviour	The stated requirement and who needs it. If you inferred the requirement, mark it as a hypothesis.
Failure / uncertainty	What diverged, what remains unknown, frequency if actually recorded, and possible alternative explanations.
Existing handling	What the controller, standard, vendor or research already does. Link the strongest existing approach you found.
Next discriminating test	The smallest test that could disprove your explanation; required access, signal quality and success/failure criterion.

Example hypothesis: after a communications outage, a controller’s status endpoint may report a cached setpoint as actual output. It is not a local industry finding until you establish that behaviour on an identified system.

Choose one gap you can actually investigate

A future company needs a consequential unmet need. A first learning prototype needs a narrow question with accessible evidence. Those are connected milestones, not the same proof.

Hard gate	Pass condition before expanding
Specific failure	An identified event, interface or timing ambiguity, rather than “energy needs better AI”.
Existing alternatives	Documented comparison with current controller features, integration tools and research. No claim that a generic capability is new.
Evidence access	A named public dataset, permitted test device or willing equipment owner with a concrete route to the relevant signals.
Testable result	An outcome you can falsify: e.g. stale telemetry never becomes verified completion under the stated test conditions.
Bounded authority	The experiment’s commands, power, timing and stop conditions are limited; real plant integration has owner-approved engineering review.
User consequence	Someone can explain why this failure matters and how they currently detect/recover it. A pleasant conversation alone does not establish demand.

If equipment is unavailable, build the simulator to learn; label the hardware and commercial hypotheses unresolved.

Turn one question into a two-page problem brief

This brief is a reviewable engineering argument. Its job is to expose assumptions and make the next test concrete, rather than promise an industry-wide platform.

PAGE A / THE PROBLEM

1. System and boundary
2. Concrete failure event
3. Evidence and uncertainty
4. Existing handling
5. Who experiences the consequence

PAGE B / THE TEST

6. Falsifiable hypothesis
7. Signals and equipment access
8. Method and stop conditions
9. Acceptance criteria
10. Limits and next decision

Ask for a technical correction, not approval of your ambition

Present the smallest question the reviewer can assess. For example: “Does this device’s status register report a requested target or measured output, and what timestamp establishes freshness?” The reply can improve your test even if the reviewer has no interest in a product.

Keep evidence and authorship inspectable

Attach source versions, relevant traces, test scripts and a short statement of what you personally ran and can explain. Record disagreement without editing it into endorsement. Do not identify private equipment or share traces without the owner’s permission.

A DECISION AFTER REVIEW

Proceed, narrow the question, obtain missing access, or stop this hypothesis. Stopping a disproved hypothesis is progress in discovering what Lupine should build.

Exercise: audit interval coverage before energy

This is a generic synthetic load fixture. It is not Tasmania data, not AEMO output and not a realistic regional load profile. Every valid value is defined to be a true interval mean.

Interval end / fixed +10:00	Mean MW	Quality
16 Oct 2026 00:05	1,000	ok
00:10	1,100	ok
00:15	blank	missing
00:20	1,200	ok
00:25	0	ok
00:30	1,300	ok

The observation window is 00:00 to 00:30

Six consecutive five-minute intervals are expected. Five have usable readings; one is unknown. A measured zero is usable. A blank marked missing is not zero. An absent expected row will also count as unknown.

DO THE ARITHMETIC FIRST

Known energy = (1,000 + 1,100 + 1,200 + 0 + 1,300) MW × (5/60) h = 383.333 MWh. This is energy over known intervals only. Full-window energy remains unknown.

The next two pages form one Python 3 script using only the standard library. No live download is performed.

Code 1 / validate and preserve unknowns

Save this code and the next page’s fixture in interval_audit.py. The strict +10:00 input convention belongs to this teaching schema; convert real source times correctly before using it.

```
import csv, io, math
from datetime import datetime, timedelta, timezone
ZONE = timezone(timedelta(hours=10))
START = datetime(2026, 10, 16, tzinfo=ZONE)
STEP = timedelta(minutes=5)
EXPECTED = {START + i*STEP for i in range(1, 7)}

def audit(text):
    reader = csv.DictReader(io.StringIO(text))
    if reader.fieldnames != ["end_aest", "power_mw", "quality"]:
        raise ValueError("unexpected header")
    seen, good = set(), {}
    for row in reader:
        if None in row or any(v is None for v in row.values()):
            raise ValueError("malformed row")
        end = datetime.fromisoformat(row["end_aest"])
        if end.utcoffset() != timedelta(hours=10):
            raise ValueError("normalise to fixed +10:00 first")
        if end not in EXPECTED or end in seen:
            raise ValueError("unexpected or duplicate interval")
        seen.add(end)
        raw, quality = row["power_mw"].strip(), row["quality"]
        if quality == "missing" and not raw:
            continue
        if quality != "ok":
            raise ValueError("inconsistent quality/value")
        power = float(raw)
        if not math.isfinite(power) or power < 0:
            raise ValueError("invalid load value")
        good[end] = power
    known = sum(good.values()) * STEP.total_seconds()/3600
    n, total = len(good), len(EXPECTED)
    return dict(valid=n, unknown=total-n, coverage=n/total,
                known_mwh=known,
                valid_mean_mw=sum(good.values())/n if n else None,
                full_mwh=known if n == total else None)
```

This rejects negative loads because this fixture models consumption only; signed interconnector flows need a different schema.

Code 2 / run the synthetic fixture

Append this fixture below the previous page. Run: python3 interval_audit.py. Preserve both the script and its console output as a simulated exercise.

```
DATA = """end_aest,power_mw,quality
2026-10-16T00:05:00+10:00,1000,ok
2026-10-16T00:10:00+10:00,1100,ok
2026-10-16T00:15:00+10:00,,missing
2026-10-16T00:20:00+10:00,1200,ok
2026-10-16T00:25:00+10:00,0,ok
2026-10-16T00:30:00+10:00,1300,ok
"""
result = audit(DATA)
for key, value in result.items():
    print(key, value)
```

A real AEMO adapter is a separate engineering task

It must identify the correct table, headers, units, region, timestamps, interval statistic and revision/intervention keys. Validate the adapter against the current dictionary and a small independently checked sample. This script does not unzip, parse or validate a NEMweb report. [S14, S15]

Do not silently change the meaning of power

For this fixture, integration is exact by definition because values are interval means. For a dispatch target or instantaneous reading, multiplying by interval duration is an approximation. State that assumption and avoid labelling the result exact measured energy.

FREE, REPEATABLE AND LIMITED

No equipment, API key or paid software is needed. Passing this exercise establishes data-handling behaviour for a defined fixture, not physical plant performance.

Expected results and deliberate failure tests

The numbers below follow from the synthetic fixture. Alter the inputs and predict the result before rerunning the script.

Output	Expected result
valid / unknown	5 / 1 intervals
coverage	0.833333... = 83.333...% of the window
known_mwh	383.333333... MWh
valid_mean_mw	920 MW over the five known intervals
full_mwh	None: full-window energy is unknown

Mutation checks worth keeping

Change a valid value to NaN: reject it. Duplicate an interval: reject it. Remove the missing row: coverage stays 5/6. Replace the missing row with 900,ok: all six intervals are valid, known and full energy both become 458.333 MWh. Keep the valid zero: it still counts toward coverage.

What this deliberately does not do

The toy validator has one hard-coded window and one load-only schema. It does not authenticate a source, infer missing energy, handle arbitrary sampling periods or select revisions from real market files. Extend those only after defining and testing their semantics.

YOUR EXPLANATION SHOULD FIT IN TWO SENTENCES

Coverage is incomplete, so full-window energy is unknown. The known energy total is useful only when its coverage and interval-mean assumption are reported beside it.

Ten questions before moving to the prototype

Answer in your own words, then check the next two pages. These test physical boundaries and evidence judgement rather than memorisation of company announcements.

1. Why can a region have enough annual energy yet be unable to connect a large new load at a particular substation?

2. A proposed store is rated 750 MW for 20 h. What energy does that imply, and why is charging energy larger?

3. Why should proposed Marinus capacity not be added to currently operating interconnection capacity?

4. Why does 104 MW retail supply divided by 84 MW critical IT not establish operating PUE?

5. A load drops from 20 to 12 MW for 15 minutes. How much energy reduction is that, and what does it fail to prove?

6. What distinguishes a NEM interval timestamp from a Hobart wall-clock timestamp and a retrieval timestamp?

7. Can a five-minute regional series establish that one AI facility caused a generator’s shaft fatigue? Explain.

8. Why do the synthetic zero and missing value have different effects on coverage?

9. What survives if you hash a raw file whose sensor values were inaccurate?

10. What must change before a simulator result becomes evidence about a real controller?

Answers 1-5 / physical boundaries

If an answer differs, locate the assumption that caused the difference. The point is to make your engineering reasoning inspectable.

1 / Energy sufficiency is not local transfer capacity

Annual MWh says little about the instantaneous MW a specific line or transformer can carry. Voltage, stability, protection, contingency conditions and outages also matter. A site needs its own connection assessment.

2 / Power × duration gives energy

750 MW × 20 h = 15,000 MWh = 15 GWh. At an assumed 80% round-trip efficiency, returning 15 GWh requires 18.75 GWh input. The efficiency is illustrative, not a verified Cethana design figure.

3 / Proposed is not commissioned

A project announcement, construction work and an energised asset are different states. Even operating nameplate capacity is not a promise of availability at every instant. Keep stage, date and operating constraints beside a capacity claim.

4 / The numerator and denominator do not form measured PUE

The figures describe different published boundaries or arrangements, not simultaneous total facility energy and IT energy over a common period. Measured PUE needs matching boundaries, units, period and instrumentation.

5 / Two MWh, with important limits

8 MW × 0.25 h = 2 MWh of avoided load relative to the stated comparison. This alone does not prove a causal demand-response baseline, an owner-approved service, battery feasibility or acceptable workload/cooling outcomes.

Answers 6-10 / evidence boundaries

An honest engineering conclusion says what was established, by which observation or test, and what remains unresolved.

6 / Timezone, interval and collection are separate

NEM time is fixed UTC+10. Hobart uses UTC+11 during daylight saving. An interval timestamp identifies the source’s defined period; retrieval time says when you collected it. Convert instants correctly and preserve all three meanings.

7 / No: the resolution and attribution are insufficient

A regional five-minute series cannot resolve fast shaft dynamics or isolate a customer’s waveform. A fatigue assessment needs an appropriate machine/network model, excitation data, mechanical parameters and specialist validation.

8 / Zero is a value; missing is absent knowledge

The zero interval is known to have zero mean consumption in this fixture. It contributes no energy but does contribute coverage. The missing interval contributes neither known energy nor valid coverage; its actual energy is unknown.

9 / Integrity relative to a reference, not truth

A saved digest can help detect later byte changes when compared with a trusted reference. It cannot establish sensor calibration, correct boundaries, truthful publication or an intervention’s causal effect.

10 / Establish a real, permitted device interface

Check protocol and register meanings, device timing, fresh physical feedback and independent limits. Test loss, stale data, restart and reconciliation under an agreed procedure. Passing the simulator remains useful but does not replace those tests.

Keep the quantities and boundaries straight

Use this as a quick reference while reading a planning report or writing your own technical note.

Quantity / relation	Meaning / limit
Power: MW	Rate of energy transfer at a specified boundary; not storage inventory.
Energy: MWh = MW × h	Constant power or interval-mean power integrated over duration; do not mix MW and MWh.
Round-trip efficiency	Returned energy / input energy over a stated cycle and boundary.
PUE = facility energy / IT energy	Matching period and boundaries required. A nameplate ratio is not measured operating PUE.
Heat buffer: dT/dt = Qnet/C	Lumped approximation; C in J/K and Qnet in W give K/s. Local hotspots can violate it.
Coverage = valid duration / expected duration	For equal intervals this equals valid count / expected count. Missing data remains unknown.
Nameplate / available / dispatched	Design rating, currently feasible capability and instructed/realised behaviour are distinct.
Specified / simulated / measured / verified	Label the evidence type and verification limits. No label substitutes for a test description.

Primary sources / infrastructure

Open a source by clicking its title. These are publisher pages, not independent audits of project performance. Read the latest dated statement and retain the version used.

S01	TasNetworks / network planning and APR
S02	Hydro Tasmania / power stations
S03	APA / Basslink asset description
S04	Marinus Link / project overview
S05	Marinus / 16 Oct 2026 construction update
S06	Hydro Tasmania / Firmus proposal
S07	Firmus / Tasmanian community commitments
S08	Firmus / Project Southgate
S09	Firmus / Wesley Vale R&D; questions and answers
S10	Firmus / engineering principles
S11	TasNetworks / building NWTD
S12	Hydro Tasmania / Cethana pumped hydro

Edition reviewed 17 October 2026 (Hobart). Source S06 carries its own 27 August 2026 update; S05 is dated 16 October 2026. Plans can change after publication. A page remaining online is not proof that every statement is still current.

Primary sources / data and learning

Open a source by clicking its title. These are publisher pages, not independent audits of project performance. Read the latest dated statement and retain the version used.

S13	AEMO / NEM data dashboard
S14	AEMO / public dispatch files
S15	AEMO / DISPATCHREGIONSUM dictionary
S16	AEMO / operational demand definitions
S17	AEMO / fixed NEM-time data-model convention
S18	AEMO / system security planning
S19	Bureau of Meteorology / climate data
S20	UTAS / CREPS research and people
S21	TasTAFE / clean-energy training context
S22	Latrobe Council / Wesley Vale project update
S23	NVIDIA / AI-factory architecture
S24	Firmus / Wesley Vale cooling and project FAQ

Edition reviewed 17 October 2026 (Hobart). Source S06 carries its own 27 August 2026 update; S05 is dated 16 October 2026. Plans can change after publication. A page remaining online is not proof that every statement is still current.

Bring Tasmania into your Lupine engineering notebook

Part 07 is complete when you can explain the local system without converting proposals into assets, public aggregates into private telemetry, or simulated behaviour into field validation.

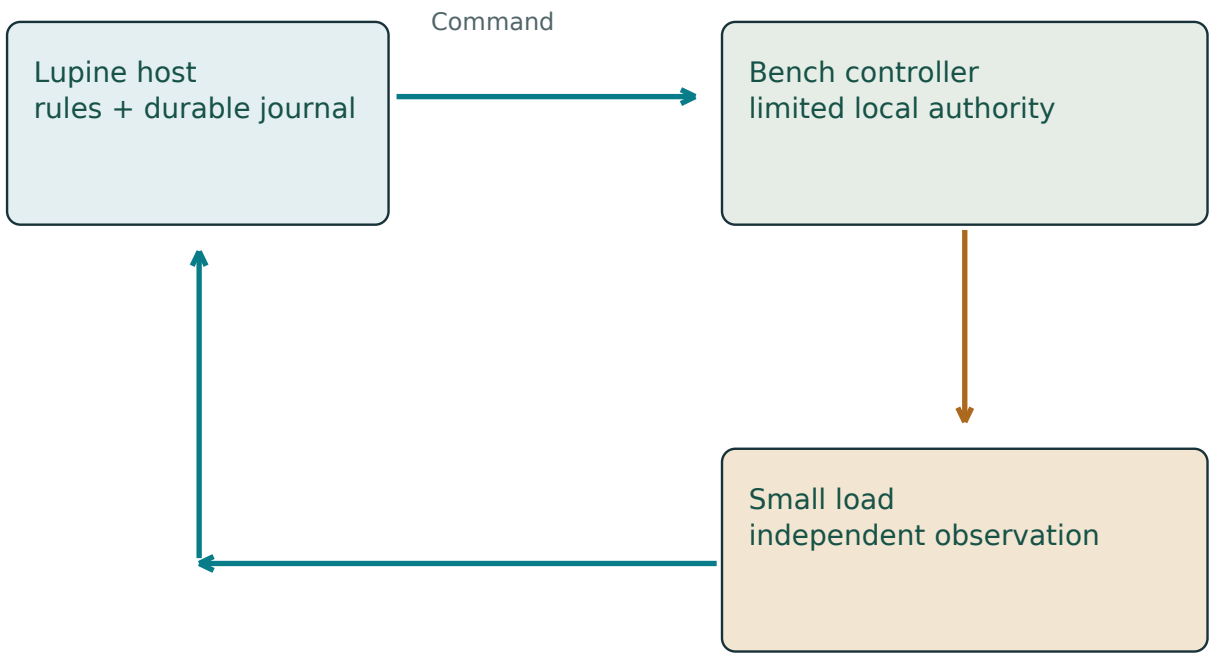
Keep this artefact	Minimum useful content
One system map	Generation, network, interconnection and AI-facility boundaries; geographic sketch separate from electrical topology.
One dated status sheet	Operating, under construction or proposed; source and date beside each capacity figure.
One data manifest	Retained raw source, digest, field definitions, timestamps, selection rule and unresolved limitations.
One completed code exercise	Your prediction, script output and mutation checks; label the fixture simulated.
One problem brief	A specific failure, existing handling, falsifiable test and credible route to evidence or equipment.
One learning explanation	In your own words: power vs energy; interval mean vs target; observation vs cause; integrity vs validity.

One part remains: Part 08 / Lupine prototype and field validation

The final volume joins the bench, reliable execution, evidence records and a bounded route towards an equipment-owner-reviewed pilot. A working prototype begins the investigation; it does not guarantee a market.

LUPINE / BUILD, TEST, EXPLAIN

Part 08 / Prototype and Field Validation. Turn the foundations into a small physical experiment whose behaviour, failures and evidence you can explain.



Physical feedback returns evidence, not certainty without limits.

The first prototype answers one narrow question

Can Lupine distinguish an accepted instruction from an observed physical outcome, preserve uncertainty when communication fails, and recover without silently repeating an unresolved action? Begin with a harmless indicator, then a guarded 5 V fan fixture only after its electrical interfaces are checked.

THE COMPANY IS THE LONG-TERM DIRECTION

This is a learning and validation platform for an energy-infrastructure company. A product opportunity still needs a specific user problem and evidence that existing systems do not handle it adequately.

The route through the final volume

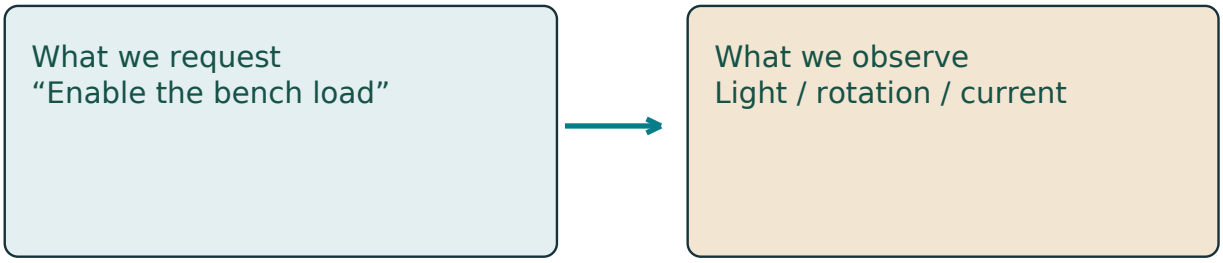
Use the earlier parts as references, not prerequisites to memorise. Work through one gate at a time and keep the evidence that lets you explain your decisions.

Engineering decision	Meaning / evidence
3-5 / Scope and gates	Define the claim, architecture and measurable milestones.
6-10 / Bench foundations	Choose minimal equipment; establish electrical, measurement and timing boundaries.
11-20 / Reliable execution	Write requirements, contracts, reconciliation, restart, evidence and failure tests.
21-24 / Infrastructure interfaces	Learn protocol semantics, read-only integration, security and physical energy limits.
25-30 / Run and reason	Follow a test procedure, report results and run a tested software-only verdict exercise.
31-36 / Build your pathway	Organise the repo and notebook; work towards a reviewed equipment trial and problem brief.
37-40 / Check and continue	Answer the checks, use free primary references and connect the whole eight-part series.

Edition: 17 October 2026, Hobart. Original diagrams and teaching examples. No live equipment was operated to prepare this volume; numerical examples and code cases are simulated unless explicitly described otherwise.

One claim, one fixture, explicit limits

Lupine’s initial task is reliable supervisory execution with evidence. It is not yet a battery optimiser, electrical protection system or commissioned plant controller.



A reported target is not the same variable as a measured physical outcome.

Choose the outcome before choosing the sensor

If the claim is “the indicator is lit”, use an independent light observation, not the GPIO output register. If the claim is “the fan is rotating”, current alone is insufficient: a stalled fan can consume power. If the claim is watts, voltage and current need matching boundaries and timing.

Keep the first action reversible and bounded

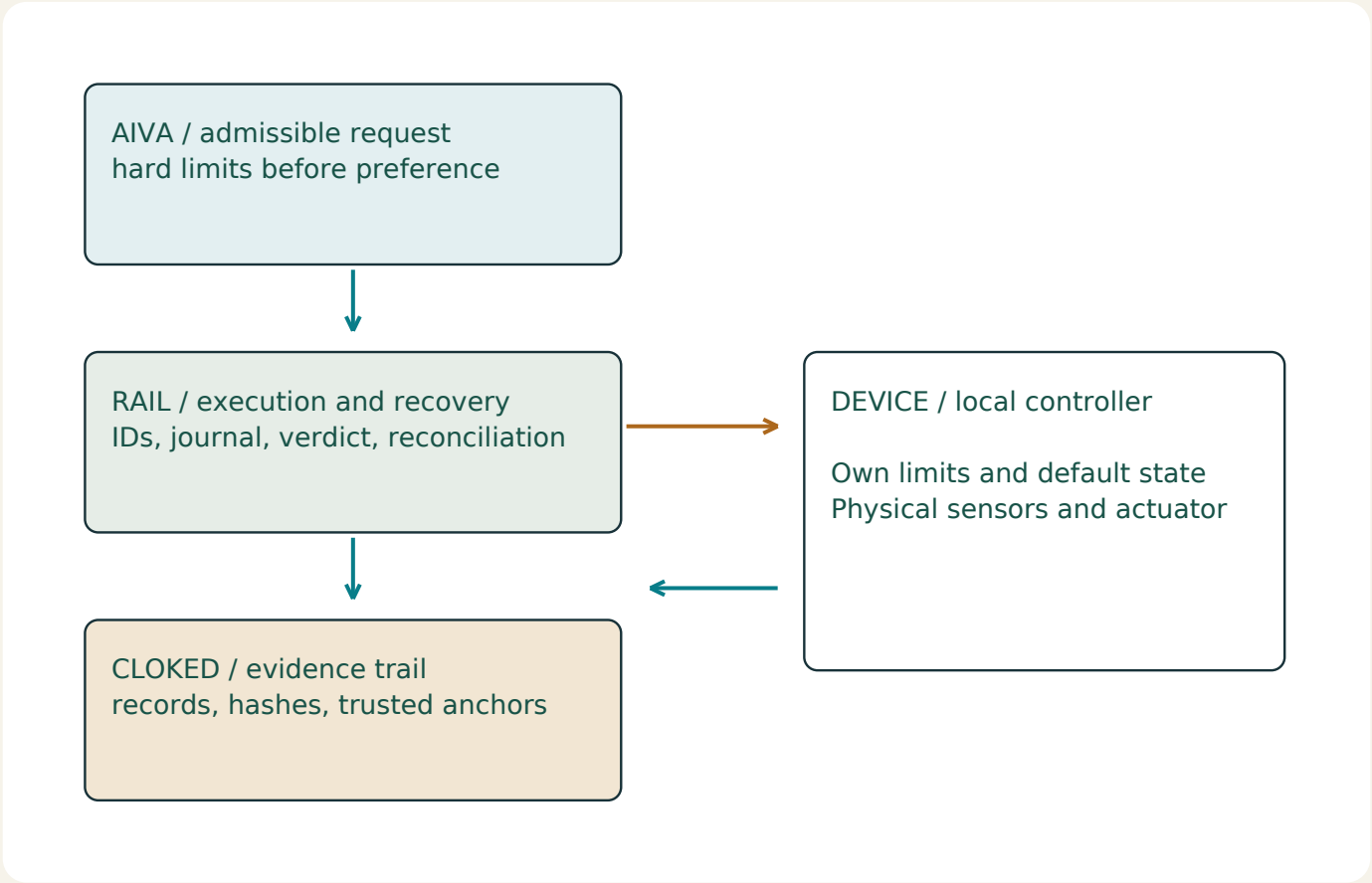
The first indicator carries no cooling responsibility. The optional fan cools nothing essential and stays guarded. A physical supply disconnect remains available. No mains work, opened power supplies, loose lithium cells, grid connection or production equipment belongs in this home experiment.

YOUR FIRST WRITTEN CLAIM

“Under these listed bench conditions, the evaluator reports target-state confirmation only from the required receipt and a fresh qualified observation.” It does not prove causation or industrial safety.

Keep three layers and the device boundary clear

This is a proposed energy-facing architecture. The existing repository must be inspected before implementation; this handbook does not claim the current code already provides it.



Local constraints do not depend on a laptop staying connected

The device must enforce its permitted outputs and timeout policy locally. Lupine can supervise, request and record; it must not replace independent electrical or thermal protection. For this bench the default can be load OFF. A real cooling system may need a completely different safe response.

One controller owns each output

A test harness, UI and background service must not compete to write the same actuator. Assign authority and an explicit handover rule; log which actor issued every permitted command.

Progress by demonstrated gates

A calendar is useful for scheduling practice. Passing a gate requires evidence and understanding, not the passage of a week.

Engineering decision	Meaning / evidence
G0 / explain the rules	Predict every verdict on pages 28-30, run the script and explain any surprise.
G1 / observe a real indicator	Use a board-supported indicator and independent optical observation; record the limits of your observation method.
G2 / establish the load fixture	Check actual component documentation, supply budget, guarded fan and feedback. Establish baseline response without network control.
G3 / execute and recover	Command the fixture; introduce one software-level communication fault at a time. Reconcile by ID and fresh physical evidence.
G4 / survive process restart	Persist unresolved commands; restart without losing the lock or dispatching a second unresolved instruction.
G5 / investigate a real interface	Read a permitted device’s documentation and telemetry. Expand to commands only under an equipment-owner-reviewed procedure.

The software exercise is free. Hardware and access are optional next gates; no purchase is required to begin today.

Buy only what the next gate needs

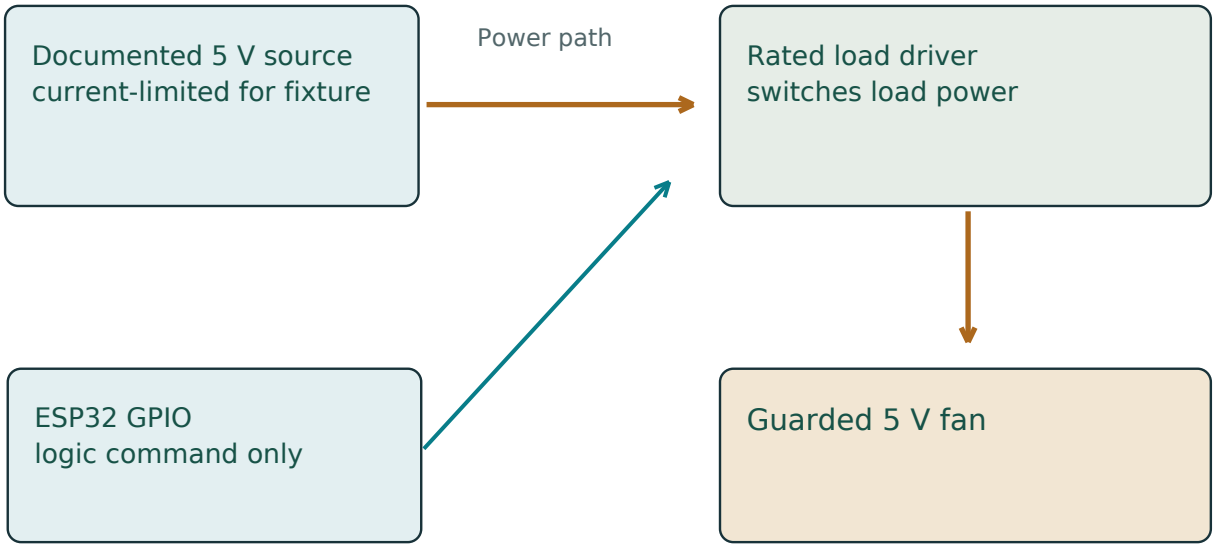
Use an existing computer first. The table is a capability checklist, not a priced shopping list or an assurance that interchangeable modules are compatible.

Engineering decision	Meaning / evidence
Already available / G0	Computer, Python 3 and a folder for code, predictions and results. No cloud service or paid software needed.
G1 / indicator	An ESP32 development board with a documented onboard indicator, USB data cable, and independent observation. Confirm the exact board revision and indicator circuit. [R01]
G2 / physical load	A small guarded 5 V fan, a driver explicitly compatible with the fan and 3.3 V logic, a documented power source and appropriately rated wiring/connectors.
G2 / feedback	A compatible tachometer output or non-contact optical speed measurement if claiming rotation; suitable current/voltage measurement if claiming electrical power.
Check / diagnose	A multimeter used within its range and terminal ratings. Start with voltage/continuity checks; series current measurement requires correct jacks, fuse and range.
Not needed yet	Oscilloscope, mains power analyser, battery pack, industrial PLC, inverter, rack server or paid telemetry platform.

Choose actual part numbers before drawing a wiring diagram. An unverified pinout is a missing requirement, not a minor detail.

Draw the power budget before wiring

A USB-powered development board can contain both a 5 V supply rail and 3.3 V logic. That does not make its GPIO a 5 V motor supply. [R01]



Functional block diagram only. Return paths, protection and pin wiring depend on the selected parts.

An assumed current budget

If a fan needs 0.20 A running and 0.45 A during startup, and the board/sensors need up to 0.25 A, coincident startup demand is 0.70 A. These are invented values. Use actual maximum/startup figures and cable ratings, then verify voltage under load; a nominal supply rating alone is not enough.

Check interfaces with power removed

Confirm driver input voltage, output rating, reset behaviour, grounding/isolation and required transient protection from manufacturer documentation. Do not feed the motor from a GPIO or connect a 5 V signal directly to a 3.3 V-only input. Do not tie unrelated USB 5 V outputs together.

The handbook deliberately leaves pin-level wiring unset until the exact board, driver and fan documentation is known.

Measure the quantity you intend to verify

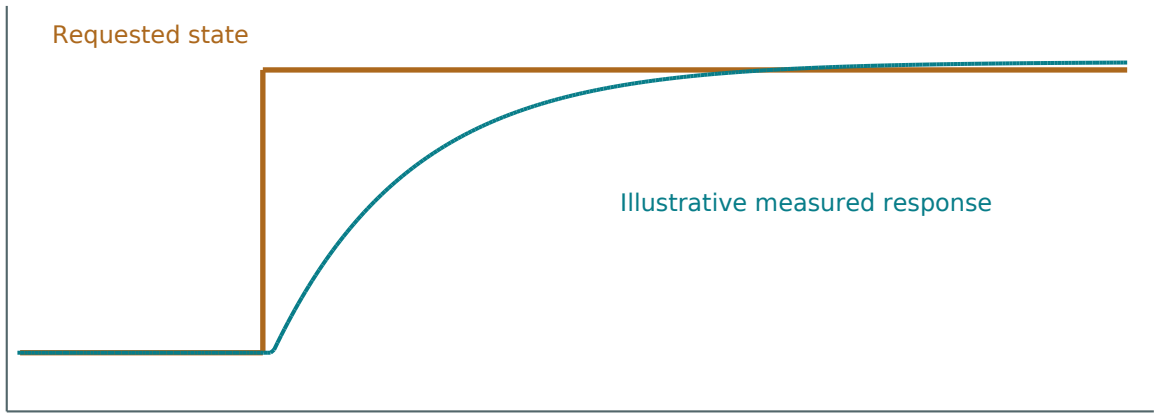
A second observation is useful only when it adds information. Two software fields copied from the same cached setpoint are not independent physical evidence.

Engineering decision	Meaning / evidence
Light output	An optical reading can observe emitted light. Ambient light, alignment, saturation and indicator polarity need checking.
Fan rotation	Tach pulses or optical speed measurement can observe rotation. Verify pulses per revolution, signal voltage and whether missing pulses mean stopped or sensor failure.
Current	Current indicates electrical behaviour; it does not prove airflow or rotation. Keep sensor offset and measurement range visible.
Voltage and power	Power requires suitable voltage/current measurements at the same boundary. If supply voltage is only assumed, label computed watts estimated.
Temperature	A sensor measures its location and responds with delay. One probe cannot establish all component temperatures or safe operating limits.
Command / acknowledgement	These describe communication and intent. They cannot replace a physical measurement simply because they are easy to log.

At G1, a human observation is a manual bench result. Automated Rail verification needs a qualified sensor path, not a claimed automation.

Establish a baseline before injecting failures

The healthy fixture is your reference. Record the same quantities in healthy and faulted runs so that a communication defect is not confused with a weak power source or sensor problem.



Time → / conceptual curve, not recorded fan data

A baseline run sheet

Record device/firmware, supply, sensor method and calibration check, starting state, request ID, timestamps, voltage/current where measured, physical response and stop condition. Take a setup photo showing the wiring labels and guarded load. Save raw data rather than only a chart.

Measure response without assuming it is instantaneous

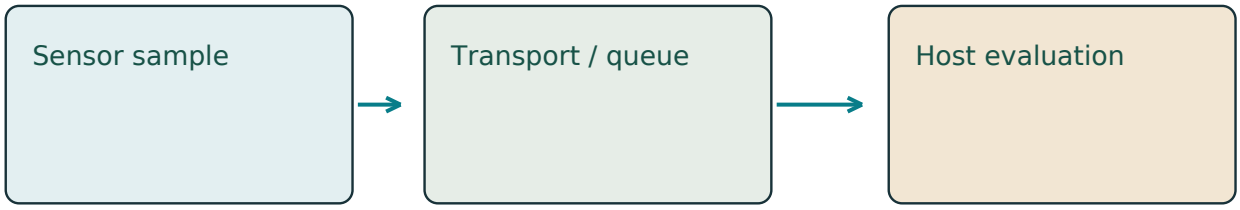
Repeat controlled OFF-to-ON and ON-to-OFF transitions with the fixture unloaded and supervised. Observe startup delay, coast-down and variation. Never jam the rotor to invent a fault; a software adapter can simulate inconsistent or missing evidence without physically damaging a motor.

THE BASELINE IS STILL A BENCH RESULT

Report the actual number of runs and conditions. A handful of successful starts establishes only those observed starts, not a field reliability percentage.

Use a time budget with an uncertainty budget

A fresh arrival is not necessarily a fresh measurement. Separate sensing, communication, decision and actuation delay.



State can change while any of these stages is waiting.

Work out the maximum usable age

An assumed example: a 100 ms sampling interval, up to 80 ms delivery delay and 20 ms handling delay can yield a 200 ms age bound only if those bounds are established and queues/retries cannot add more. A source timestamp from an unsynchronised clock does not establish that bound.

Two distinct clocks

Use monotonic elapsed time for local timeouts within one running host process. Use UTC wall-clock time for cross-system records, with an explicit synchronisation/error policy. A monotonic value from one process or boot is not comparable with another machine’s or a later boot’s value. [R02]

THE EXERCISE USES A SINGLE ARTIFICIAL CLOCK

Pages 28-30 put command and sample times in one simulated clock domain. A device adapter must solve real timestamp provenance before treating those rules as physical verification.

Write requirements that can fail a test

A requirement states observable behaviour and its conditions. “Reliable”, “smart” and “safe” without a test boundary are ambitions, not acceptance criteria.

Engineering decision	Meaning / evidence
REQ-01 / physical evidence	Accepted alone never becomes completed. A qualified fresh physical observation matching the target is required after the response allowance.
REQ-02 / uncertainty	A lost reply, stale sample, wrong ID or unknown sample quality preserves uncertainty instead of substituting zero or success.
REQ-03 / unresolved lock	An unresolved action blocks another automatic action on the same resource, including after host restart.
REQ-04 / no hidden retry	A timeout does not create a new action ID or send the same command again. Reconciliation reads precede any new authorised action.
REQ-05 / local bounds	The controller enforces its own allowed outputs and timeout/default policy without the host connection.
REQ-06 / inspectability	Every verdict retains its input evidence, rule version, reason and limits. A complete rerun can explain the decision.

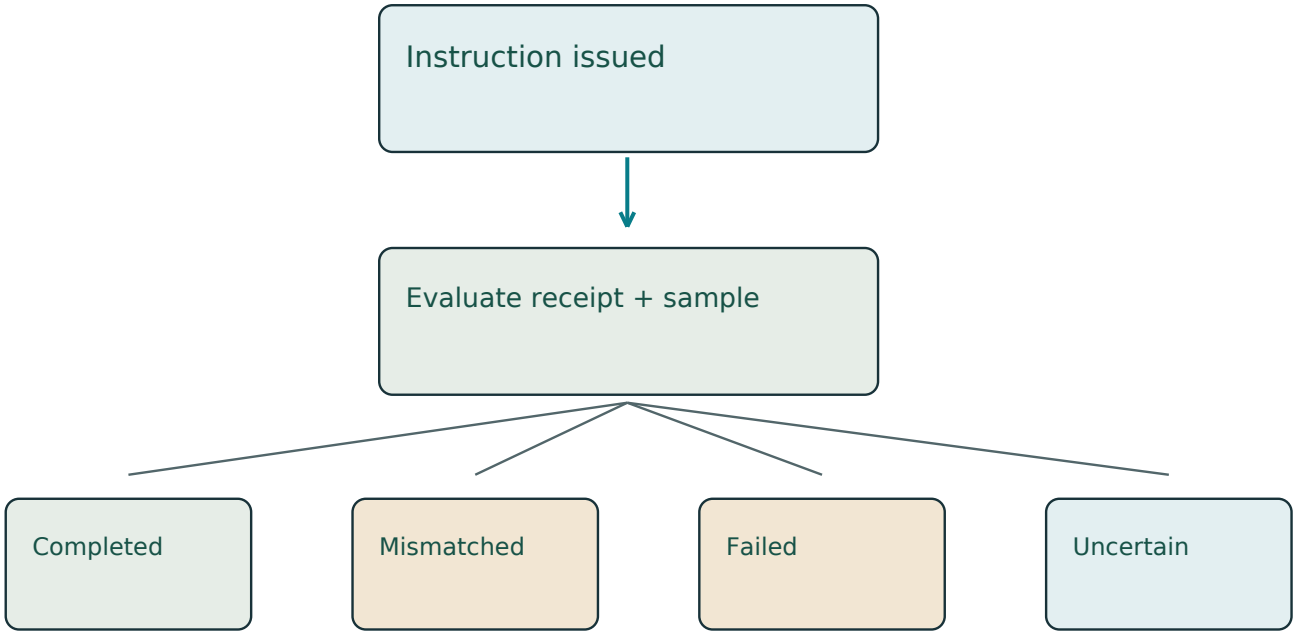
Define the command and observation contract

Your command ID relates a request to a device receipt. A sensor sample needs its own identity and provenance; echoing the command ID into a reading does not prove the action occurred.

Engineering decision	Meaning / evidence
Command	Resource ID; unique command ID; absolute target; allowed bounds; issue/expiry times under a defined clock policy; protocol/schema version.
Receipt	Resource and command IDs; accepted/rejected/unknown; rejection reason; device boot/session identity; accepted target if supported.
Observation	Resource and sensor IDs; sample sequence; acquisition time and clock basis; value and unit; quality; calibration/configuration version.
Status query	Read the recorded command result by ID if supported. Distinguish current setpoint, command history and independently measured output.
Ordering	One owner per resource; command ordering rule; stale-command rejection; explicit behaviour if an ID reappears with a different target.
Expiry	State whether expiry prevents initial execution or also changes an ongoing output. Never assume those are the same feature.

Represent knowledge separately from physical state

Completed means the stated outcome rule passed within its limits. It is not a permanent assertion that the equipment will remain in that state.



Verdicts describe the evidence for this action; the resource has a separate observed state.

Four conservative outcomes for the teaching rule

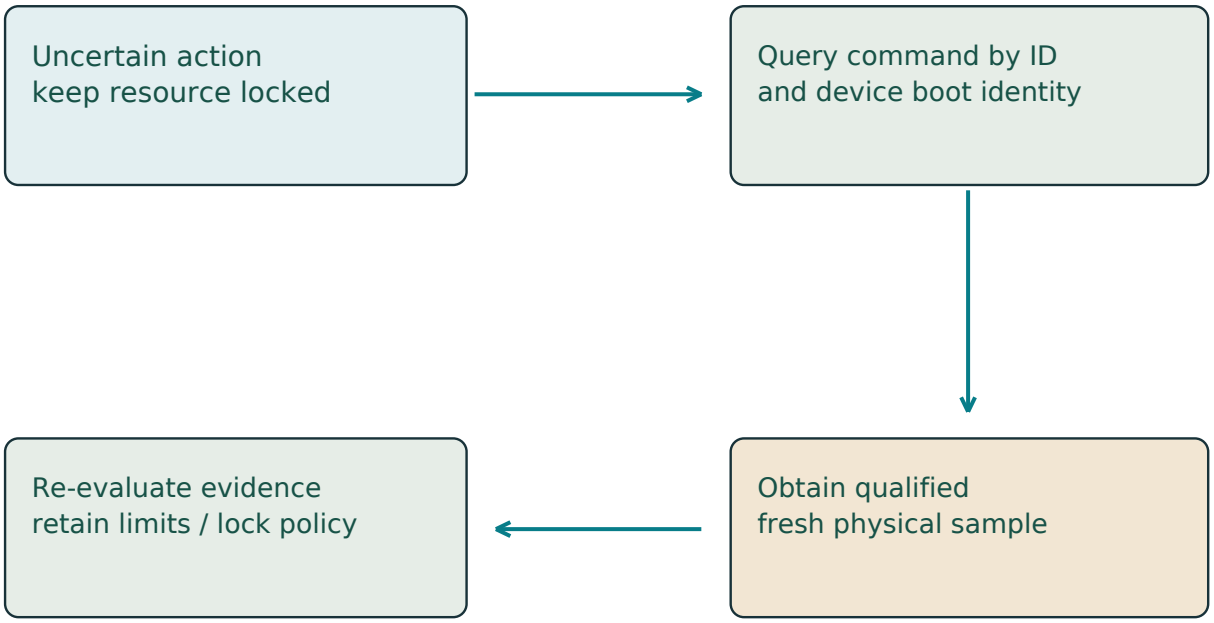
Accepted + fresh qualified target observation becomes completed; accepted + qualified opposite observation becomes mismatched. A correlated rejection with qualified opposite observation becomes failed. Missing, stale, premature or contradictory evidence becomes uncertain. The miniature code implements exactly this policy.

Do not hide the reason behind one word

Record whether uncertainty came from missing receipt, bad identity, stale sample, unsettled response or conflicting evidence. A physical alarm can exist alongside any command verdict. The verdict evaluator does not replace alarm handling.

Reconcile with reads before deciding to act again

The channel can fail after an actuator has changed. Repeating the request immediately can conceal the original outcome or produce a second action.



A lost acceptance reply can remain uncertain

If status by ID later confirms acceptance and the qualified sample matches, the rule can confirm the target state. If the device cannot establish command identity, observing the target alone leaves the action uncertain under this policy. Record useful observations without inventing a receipt.

Set a reconciliation deadline

When the evidence cannot resolve the action, escalate to a documented manual review or leave the resource inhibited. A deadline ends an automatic attempt to learn; it does not convert unknown into failed.

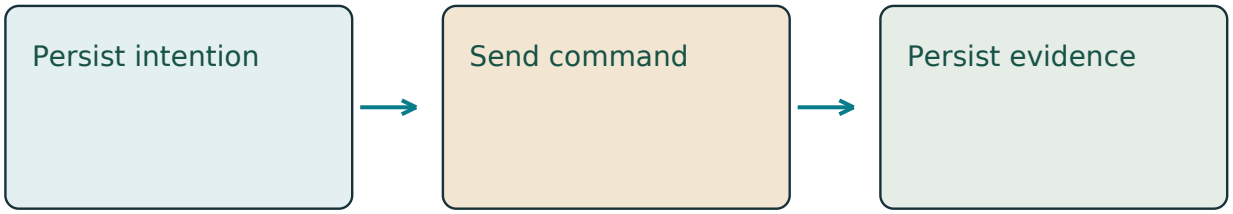
Idempotency reduces duplicate effects, not all risk

An absolute “set state ON” differs from “toggle state” or “deliver one pulse”. Repeating the latter can cause an additional physical action.

Engineering decision	Meaning / evidence
Absolute target	Repeatedly setting the same ON state may converge to one state, but an intervening OFF command, expiry or changed authority can make a delayed ON dangerous.
Increment / toggle / pulse	A duplicate can add energy, reverse state or repeat a movement. Never treat an arbitrary physical action as idempotent.
Device deduplication	The device must remember action IDs, target fingerprints and results for a defined retention window; reject the same ID with different content.
Host retry	Deduplication only helps if the host reuses the original identity and the device still retains it. It is not a reason to retry unresolved actions in this prototype.
New action after failure	An evidence-based failed verdict is necessary context, not blanket permission. Recheck limits, ownership and physical state before authorising another action.
Teaching-code limit	The pure evaluator sends nothing and authorises no retry. A separate executor must implement locks, persistence and reconciliation.

Restart must not erase uncertainty

A crash can occur between saving an intention, transmitting a command, receiving a reply and saving the reply. Treat the device and database as two systems without one shared transaction.



A crash at any boundary can leave a persisted intention with an unknown physical result.

A conservative recovery rule

Commit an unresolved command record before transmission. If that commit fails, send nothing. After a restart, load unresolved records and inhibit automatic writes for their resources. Query and observe; do not infer “not sent” just because no receipt was saved. Even a crash before the send may require reconciliation.

Atomic local storage is not atomic physical execution

A database transaction can keep related local records consistent under documented storage assumptions. It cannot atomically include a network request and a motor. Retain command IDs across process restarts and device boot changes, and test the chosen database settings and recovery paths. [R03, R04]

DO NOT COMPARE OLD MONOTONIC TIMES AFTER REBOOT

Persist wall-clock context and boot/session identity. Recovery needs a fresh timing basis and new observations; the toy single-clock evaluator is not a restart implementation.

Store enough evidence to reconstruct a decision

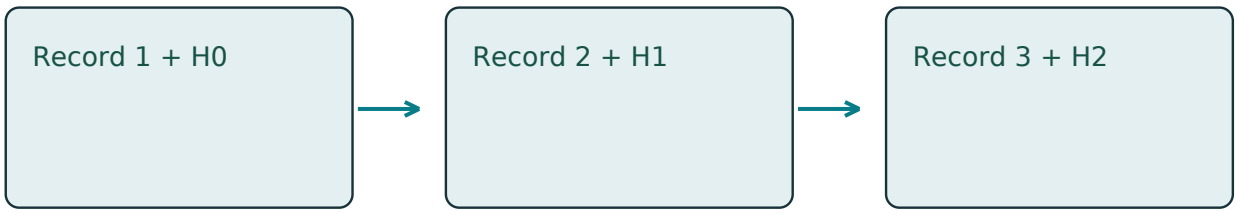
A structured journal should support both machine checks and your own explanation. Store raw input separately from normalized evidence and final verdict.

Engineering decision	Meaning / evidence
Command record	ID, resource, target, schema, limits, rule version, actor, persisted time and unresolved lock state.
Receipt record	Original payload, parsed fields, resource/command identity, boot/session, arrival time and source-validation status.
Sample record	Raw payload, acquisition/arrival times, sequence, sensor identity, quality, value/unit and calibration configuration.
Evaluation record	Exact evidence references, evaluation time/clock domain, verdict, reason and applicable tolerances/settling policy.
Recovery record	Restart identity, unresolved resources, queries issued, reconciliation evidence and manual decisions with reasons.
Storage failure	Disk-full or write errors must be visible. Stop initiating new experimental actions if the required journal cannot be committed; preserve the device’s independent local behaviour.

A screenshot is a useful illustration, not a substitute for raw records, source metadata and the executable rule version.

Cloked: integrity needs an external reference

Hash chaining links each new record to its predecessor. Its useful claim depends on what an attacker can rewrite and where a trusted reference is retained.



Hn = SHA-256(canonical record bytes + previous digest), under a fixed encoding.

Detecting change is not proving truth

Changing one stored record breaks later comparisons if the original later hashes are trusted. Someone able to rewrite every record and recompute the chain can produce an internally consistent replacement. A separately retained trusted head digest, signatures or another trust boundary is needed to detect that replacement.

Anchor and verify with a stated threat model

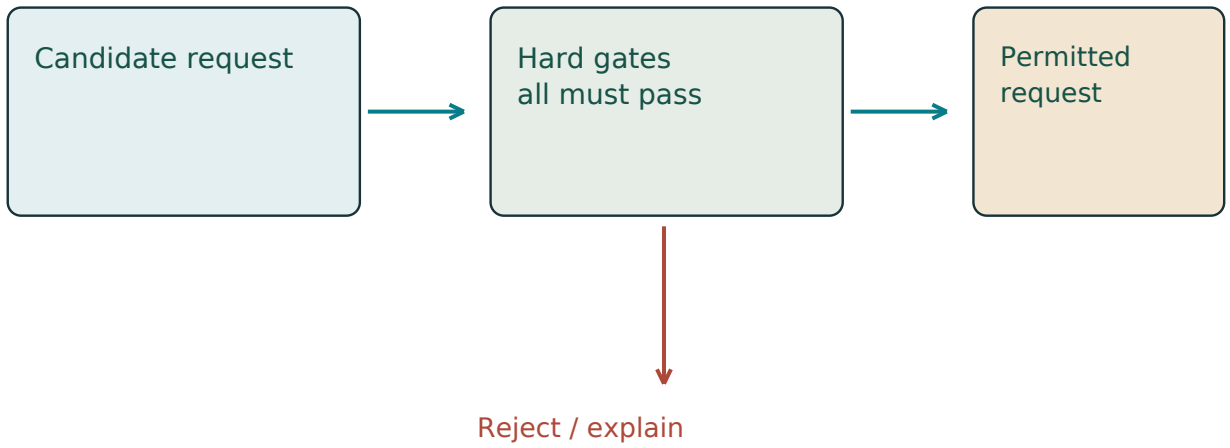
For a bench experiment, export a signed or independently retained manifest after each run and compare it during review. Explain who controls both copies, what truncation looks like, and whether keys can be compromised. Two files under the same unrestricted account are a weak separation, not tamper-proof evidence.

VALIDITY REMAINS A SENSOR AND METHOD QUESTION

No hash establishes calibration, correct timestamps, causal demand reduction or electrical safety. Preserve those limits beside any integrity check. Python provides SHA-256 primitives; it does not supply your trust model. [R05]

Aiva: reject invalid requests before ranking

A weighted preference must never compensate for a violated current limit, expired permission or missing required feedback. The bench needs simple constraints before clever optimisation.



Example hard gates for the harmless fixture

Resource belongs to the experiment; owner is the current controller; no unresolved action; command is absolute and permitted; limits and sensor configuration are known; local readiness is established; required journal write succeeds. Each failure produces a reason, not a lower preference score.

Keep the first policy deterministic

One allowed target and a small explicit rule set are enough. Learning-based control adds training data, validation and failure modes without helping you understand a lost receipt. Optimisation can come later if a real problem requires it; limits remain independently enforced.

A USEFUL TEST

Offer a candidate with excellent preference scores but one violated hard limit. The result must be rejection. If ranking can admit it, the boundary between constraints and preferences is wrong.

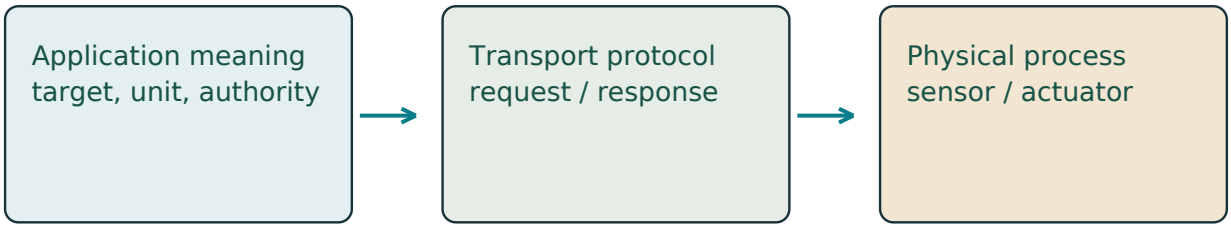
Build a fault matrix before writing the demo

Inject faults in an adapter or simulator. Keep the real load guarded and electrically normal; never short wires, stall the rotor or disable physical protections to create a scenario.

Engineering decision	Meaning / evidence
Acceptance + target sample	Completed, but only after settling and freshness/quality checks. Record that target-state confirmation does not establish causation.
Acceptance + opposite sample	Mismatched after the stated response allowance. Do not keep cycling power to make the plot look healthy.
Reply lost after action	Uncertain. Read status by original ID and collect fresh feedback; no automatic resend.
Receipt says rejected; target observed	Conflicting action evidence: uncertain under the teaching policy. The output may already have been at target.
Stale / missing / future sample	Uncertain; never replace missing data with zero. A future sample time indicates a clock/evidence problem.
Wrong ID / reboot / host crash	Do not use mismatched receipts or old-session timing as verification. Recover locks and reconcile using the relevant identities.

Choose semantics before choosing a protocol

HTTP, serial and Modbus can all carry a target value. None automatically tells you whether the physical outcome occurred, whether feedback is fresh or whether a retry is safe.



Successful transport is one layer of evidence; process outcome belongs to another.

Start with the simplest local interface you can inspect

A USB serial adapter can teach framing, identities, timeouts and stale responses before adding Wi-Fi. Define explicit message boundaries, versioning, maximum sizes and errors. A local HTTP endpoint is an alternative, not a reason to expose a bench controller to the public internet.

Add one source of failure at a time

First test a deterministic fake adapter. Then a real controller with the same normalized contract. Introduce delayed or dropped replies in the host adapter before testing an actual network disconnect. Keep raw protocol messages beside parsed evidence so a normalization defect is visible.

WHAT A TIMEOUT MEANS

The requester did not obtain the expected response within its deadline. The timeout does not establish that the instruction was never received or never executed.

Read a Modbus map as an engineering document

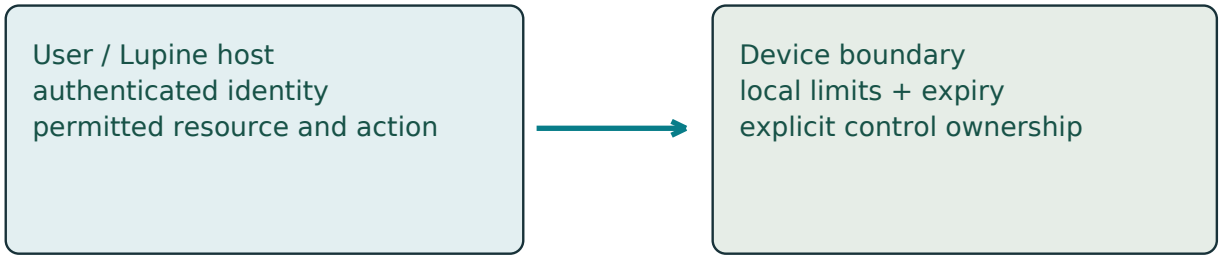
Modbus defines data and function-code transactions. A vendor or SunSpec model gives registers their equipment meaning; compatibility must be checked at both layers. [R06, R07]

Engineering decision	Meaning / evidence
Address convention	Distinguish displayed register numbers from zero-based protocol offsets. Confirm unit/device address, supported functions and model version.
Encoding	Register width, signed/unsigned representation, multi-register word order and byte order. Test a known value before trusting a plot.
Scaling and invalids	Raw 123 with scale factor -1 can mean 12.3 in the stated unit. Check invalid sentinels and scale-factor validity before arithmetic.
Read versus write	A writable target register is not a measured-output register. A successful write response is not physical outcome verification.
Refresh and persistence	Read update period, command expiry, nonvolatile settings, restart defaults and control ownership. These are device-specific requirements.
First integration	Use only documented, permitted read-only access to a non-production test device. Do not probe an organisation’s network or write guessed register values.

A toy register server teaches protocol handling. It does not certify SunSpec conformance or compatibility with a real inverter.

Separate bench access from plant authority

A physical control interface needs identity, bounded permissions and a recovery path. Connecting to a local Wi-Fi network is not a sufficient authority model.



Allow only the required path; keep production networks and cloud exposure out of the bench.

A limited home setup

Use a private test network or USB; disable unnecessary services, avoid port forwarding, protect credentials and keep secrets out of logs and Git. Reject expired, unauthorized or wrong-resource commands. Record operator actions and test what happens when credentials or connectivity fail.

Industrial security is a separate review

NIST’s OT guidance addresses security alongside physical reliability and safety. Traditional Modbus TCP should not be assumed to provide authenticated secure transport. Modbus Security defines a TLS/certificate-based variant; actual device support and deployment still need checking. [R08, R09]

NO SECURITY CLAIM FROM ONE DEMO

A successful certificate handshake, local password or isolated router is one control. It is not penetration testing, an OT security assessment or permission to connect to real infrastructure.

Link control to energy and heat with real units

The fixture can teach measurement and recovery without pretending that a small fan reproduces an AI cooling plant or battery’s dynamics.

Electrical power: $P = V \times I$

Energy: $E = \Sigma P_{mean} \times \Delta t$

Thermal trend: $C \times dT/dt \approx Q_{in} - Q_{out}$

Use a common measurement boundary; the thermal relation is a lumped approximation.

An assumed bench calculation

At measured constant 5.0 V and 0.20 A, input power is 1.0 W. Over 10 minutes at that power the energy is 0.167 Wh. These values are hypothetical. Controller and driver power must be included if your claimed boundary is the whole fixture rather than the fan terminals.

A thermal extrapolation has additional requirements

Fan rotation does not establish airflow, heat transfer or safe component temperatures. Real cooling depends on the full heat path, flow, pressure, temperature limits and response times. An OFF-on-timeout policy acceptable for an unloaded bench fan could be dangerous for essential cooling.

KEEP THE PROTECTION BOUNDARY

OpenEMS itself distinguishes supervisory EMS cycle timing from fast or safety-critical control. Use such architecture as a reference; do not move independent protection into a laptop script. [R10]

Run one experiment as a repeatable procedure

Change one fault at a time. A written procedure is what lets a future reviewer reproduce your result and distinguish an experimental effect from a setup change.

Engineering decision	Meaning / evidence
1 / identify the run	Assign a run ID; record hypothesis, requirement IDs, code revision, device firmware, circuit revision, sensor setup and environmental conditions.
2 / preflight	Inspect with power removed; confirm guarded load, rated connections, local default and accessible disconnect. Check measurement ranges and sensor health.
3 / baseline	Establish starting physical state and fresh readings. Confirm clock/session identities and that no action is unresolved.
4 / action and fault	Persist intention, issue one bounded absolute request, inject one documented adapter-level fault, retain raw messages and physical observations.
5 / recover	Apply the stated reconciliation procedure. Observe whether the resource lock and no-resend requirement hold; stop on unexpected behaviour.
6 / close	Return the fixture to its documented resting state, disconnect power, export evidence and record deviations, result and unresolved questions.

A results sheet should expose the evidence

Use one row per trial and retain the linked raw records. The example below is simulated; it is a format to fill with your own observations, not a measured portfolio result.

Case	Evidence	Verdict
SIM-01	Accepted; target observed after settling; age within limit.	completed
SIM-02	Accepted; opposite state observed after settling; fresh sample.	mismatched
SIM-03	Reply absent; fresh target observation exists.	uncertain
SIM-04	Later status by original ID reports accepted; new qualified target sample.	completed on reconciliation
SIM-05	Rejected; observed target already present.	uncertain: conflict

Keep both first-look and reconciliation results

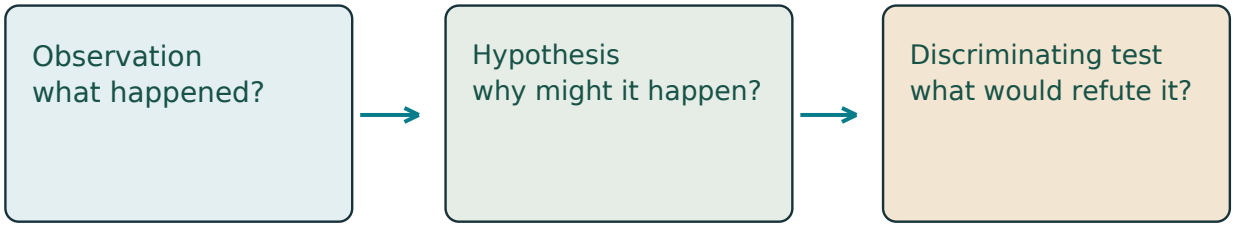
Do not overwrite the initial uncertain verdict with the later completed one. Append the later decision with its new evidence and time. The sequence matters: it shows what was known when, including how long writes were inhibited.

ADD THE MISSING COLUMNS IN YOUR REAL JOURNAL

Run ID, command ID, sample sequence/time/quality, clock basis, reason, physical method, code version and raw-record paths. A compact report table is not the full evidence store.

Analyse failures without polishing them away

Your strongest portfolio note may be the one that explains a false success, a mistaken timestamp assumption or a sensor limitation clearly enough for another engineer to reproduce it.



Separate a software defect, a physical response and an invalid measurement method.

Example: a fan appears slow after a lost reply

The reply loss may be unrelated. A supply voltage drop, sensor alignment change or warm-start condition could affect the observation. Retain baseline voltage, sample age and physical feedback; repeat with the communication fault removed while keeping the physical setup unchanged.

Report counts without inventing reliability

If 20 scripted simulated cases pass, say that. If 12 physical trials pass under one setup, say that. Neither establishes a general failure probability or an industrial availability figure. Discuss missed cases, selection bias, measurement limits and the next test that could change your conclusion.

A USEFUL TECHNICAL NOTE HAS A REPRODUCTION PATH

Question, diagram, method, raw evidence, result, limitations and next test. Distinguish AI-assisted implementation from work you personally predicted, ran and can explain.

Exercise 1 / define the normalized evidence

Pages 28-30 form one software-only Python 3 script. Save as verdict_demo.py. Dataclasses organise inputs; they do not validate a device protocol or authenticate a source.

```
from dataclasses import dataclass
import math

@dataclass(frozen=True)
class Command:
    resource: str
    cid: str
    target: bool
    sent: float

@dataclass(frozen=True)
class Receipt:
    resource: str
    cid: str
    status: str

@dataclass(frozen=True)
class Sample:
    resource: str
    value: object
    at: object
    quality: str = "ok"

def finite_number(x):
    return (type(x) in (int, float)
            and math.isfinite(x))
```

The sample is a qualified Boolean physical-state observation in one synthetic clock domain. No GPIO, network or plant is accessed.

Exercise 2 / implement the conservative rule

Append below page 28. Completed here means the evidence rule confirms the target state within its stated limits; it does not establish causation or authorise a retry.

```
def verdict(cmd, receipt, sample, now, settle=2.0, age=1.0):
    times = (cmd.sent, now, settle, age)
    if (not all(finite_number(x) for x in times)
            or settle < 0 or age < 0 or now < cmd.sent):
        raise ValueError("invalid evaluation time policy")
    if type(cmd.target) is not bool:
        raise ValueError("target must be Boolean")
    if not cmd.resource or not cmd.cid:
        raise ValueError("command identity required")
    if receipt is None:
        return "uncertain", "no receipt"
    if (receipt.resource != cmd.resource
            or receipt.cid != cmd.cid):
        return "uncertain", "receipt identity mismatch"
    if receipt.status not in ("accepted", "rejected"):
        return "uncertain", "unknown receipt status"
    if sample is None or sample.resource != cmd.resource:
        return "uncertain", "no matching physical sample"
    if sample.quality != "ok" or type(sample.value) is not bool:
        return "uncertain", "unqualified physical sample"
    if not finite_number(sample.at):
        return "uncertain", "invalid sample time"
    if not cmd.sent + settle <= sample.at <= now:
        return "uncertain", "premature or future sample"
    if now - sample.at > age:
        return "uncertain", "stale sample"
    matches = sample.value == cmd.target
    if receipt.status == "accepted":
        return (("completed", "target observed") if matches
                else ("mismatched", "opposite observed"))
    return (("uncertain", "rejection conflicts with target")
            if matches else ("failed", "rejected; opposite observed"))
```

No executor, storage, locks, device session checks or sensor qualification algorithm is implemented by this pure function.

Exercise 3 / predict, run and mutate

Append below page 29. Run: python3 verdict_demo.py. First write your prediction for each case; compare it with the output and explain any difference.

```
cmd = Command("bench", "C1", True, 0.0)
accepted = Receipt("bench", "C1", "accepted")
rejected = Receipt("bench", "C1", "rejected")
on = Sample("bench", True, 2.5)
off = Sample("bench", False, 2.5)
cases = [
    ("healthy", accepted, on, "completed"),
    ("no motion", accepted, off, "mismatched"),
    ("reply lost", None, on, "uncertain"),
    ("refused", rejected, off, "failed"),
    ("conflict", rejected, on, "uncertain"),
    ("sensor absent", accepted, None, "uncertain"),
    ("too early", accepted,
     Sample("bench", True, 1.0), "uncertain"),
    ("future", accepted,
     Sample("bench", True, 3.1), "uncertain"),
    ("bad quality", accepted,
     Sample("bench", True, 2.5, "bad"), "uncertain"),
]
for name, receipt, sample, expected in cases:
    result, reason = verdict(cmd, receipt, sample, now=3.0)
    assert result == expected, (name, result, reason)
    print(name, result, reason)

# Same physical sample, now too old:
assert verdict(cmd, accepted, on, now=4.0)[0] == "uncertain"
# Reconciliation supplies the previously missing acceptance:
assert verdict(cmd, accepted, on, now=3.0)[0] == "completed"
# A missing observation cannot confirm an OFF target:
stop = Command("bench", "C2", False, 0.0)
ack = Receipt("bench", "C2", "accepted")
assert verdict(stop, ack, None, now=3.0)[0] == "uncertain"
print("12 checks passed; all evidence was simulated")
```

Now add wrong-ID, wrong-resource, invalid-value and exact-age-boundary cases. Passing these does not test the missing executor.

Restructure by adding a bounded energy module

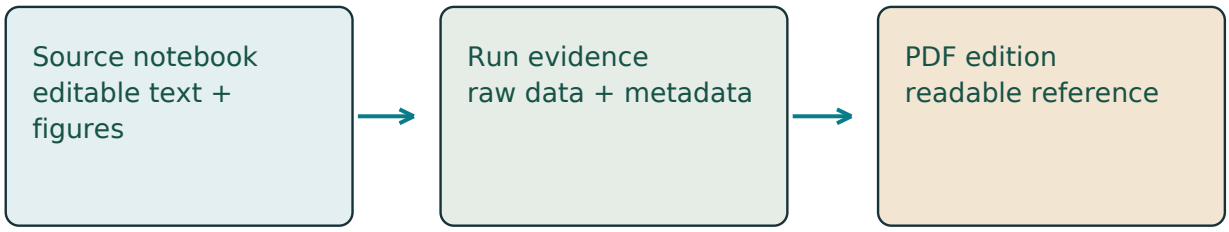
This is a proposed organisation, not a report that your GitHub repository has been inspected or changed. Preserve existing history and working tests while introducing the new direction.

Proposed path	Purpose
docs/energy/mission.md	Energy direction, bench boundaries, current evidence and unresolved product hypothesis.
docs/energy/requirements.md	Numbered outcome, recovery, persistence and device-limit requirements.
src/energy/contracts.py	Typed command, receipt and observation schemas with validation.
src/energy/outcome.py	Pure deterministic evidence rules; no networking or GPIO.
src/energy/executor.py	Durable lifecycle, ownership, unresolved lock, no-resend and recovery orchestration.
src/energy/adapters/	Fake adapter first; later an actual documented serial/network device adapter.
tests/energy/	Behavioural fault cases, parser tests, journal/restart tests and separately marked hardware tests.
experiments/ + evidence/	Versioned procedures and manifests. Store bulky raw runs appropriately; exclude credentials/private traces.

Keep Aiva/Rail/Cloked responsibilities explicit. Avoid moving or deleting the old domain merely to improve the folder names; inspect dependencies and make small reversible changes.

Build your own reference as you learn

The handbook explains the principles. Your notebook records what your actual setup did, how you know and what you personally understand well enough to reproduce.



PDF is the reading copy; retained source and raw evidence keep it correctable.

Use the same seven headings for every experiment

Question; prediction; system diagram; procedure and limits; raw observations; interpretation; correction and next test. Attach a setup photo, code revision, parts/firmware list and measurement units. Keep a short “I can explain this without AI” paragraph in your own words.

Publish versions and corrections honestly

Export a dated PDF after a completed experiment or weekly learning block. Keep its source and the earlier edition. Correct a false claim explicitly; do not rewrite historical raw observations to match a new interpretation. Use simulated, assumed, published and measured labels wherever readers might confuse them.

MAKE IT USEFUL AS A DESK REFERENCE

Include a one-page wiring legend for the selected parts, field definitions, timing policy, verdict table, troubleshooting notes and links to retained records. Index by concept and experiment ID.

Your first week starts with explanation, not shopping

This is a suggested five-session sequence. Shorten or repeat sessions when needed; there is no deadline that makes an unclear concept ready for physical control.

Engineering decision	Meaning / evidence
Session 1 / 45-60 minutes	Read pages 3, 11 and 13. Write the first claim and predict the healthy, mismatch and lost-reply outcomes.
Session 2 / 45-60 minutes	Type pages 28-30 into one script. Run it, save the output and explain why the missing OFF observation remains uncertain.
Session 3 / 45-60 minutes	Change one receipt ID, sample quality and timestamp. Predict each verdict before running. Add one regression case for any mistake.
Session 4 / 45-60 minutes	Draw the host/device/sensor boundaries. Inspect any board you already own; find its exact documentation and indicator pin/polarity. Do not guess.
Session 5 / 45-60 minutes	Write a two-page learning note: what works in simulation, what is not implemented, and what parts/evidence the next gate needs.
Optional weekend review	Explain the demo from a blank page without AI. If you cannot, simplify or repeat the relevant case before adding hardware.

A multi-year direction, with nearer evidence gates

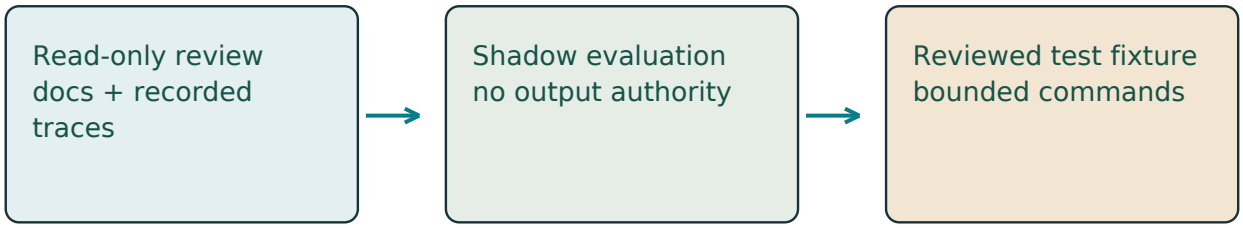
Your goal is expertise and an energy technology company. The timeline below is provisional: a learning schedule and decision structure, not a forecast of employment, funding or commercial success.

Indicative window	Technical outcome / decision
Month 1	Explain and extend the evaluator; establish an indicator baseline; retain a small, correct technical notebook.
Months 2-3	Complete a documented guarded-load fixture if parts permit. Demonstrate fresh feedback, controlled adapter faults and honest uncertainty.
Months 3-6	Build durable execution and restart recovery; learn one documented industrial protocol; write a precise two-page problem brief.
Months 6-12	Seek technical criticism and establish legitimate access to a non-production device or recorded trace. Compare existing solutions and revise the hypothesis.
Year 1-2	If an equipment owner has a specific unresolved issue, run a reviewed bounded trial. Measure usefulness against the current method and decide whether to productise.
Year 2-3+	Develop depth in the chosen electrical, thermal or control specialty; pursue repeatable deployment only if evidence supports demand and maintainability.

Study pace, employment, access and equipment can change these windows. Technical acceptance depends on tests and evidence.

Move from bench to field through bounded access

A field trial begins with an owner’s concrete problem and permission. The bench establishes learning; it does not authorise commands on an inverter, battery, plant controller or cooling system.



Each expansion requires its own evidence, owner agreement and engineering procedure.

Begin with a non-production interface or recorded trace

Ask for the exact failure, current recovery method and allowed data. De-identify private traces and respect access restrictions. A shadow evaluator can compare its verdicts against an engineer’s interpretation without commanding equipment. Establish what your added layer detects that existing tools miss.

A controlled trial needs a concrete change plan

Name the responsible equipment engineer, command boundary, permitted operating envelope, independent protections, monitoring, rollback and stop criteria. Define who can revoke control and who responds to unexpected behaviour. Training, installation and compliance requirements depend on the actual equipment and work.

A GENUINE STOP CONDITION

If device semantics, timestamps, authority or protective boundaries are unclear, remain read-only. The next task is resolving that missing information, not writing around it.

Ask whether the capability deserves to be a product

Reliable execution is useful engineering, but usefulness alone does not establish a startup opportunity. Look for a specific costly failure that people cannot already handle adequately.

Engineering decision	Meaning / evidence
Problem evidence	Identified user, equipment class, failure event, consequence and records showing it happens. Keep frequency/cost unknown unless established.
Current alternative	Describe the controller, gateway, vendor tool, standard or procedure already used. Compare the same failure and operating boundary.
Lupine contribution	A measurable difference: better ambiguity detection, shorter diagnosis or a clearer recovery record under a defined interface. Treat each benefit as a hypothesis until tested.
Feasibility	Data/command access, timing, local limits, installation burden, security and maintainability. A clever algorithm cannot replace an unavailable signal.
Pilot acceptance	Owner-agreed cases, reference judgement, pass/fail thresholds, unexpected-behaviour handling and restrictions on claims.
Decision	Continue, narrow, change the approach or stop this hypothesis. Keep the energy direction while allowing evidence to change the product.

Ten questions to own the prototype

Answer before checking the next page. Use the code and diagrams to explain your reasoning; do not rely on remembering the verdict labels alone.

1. Why is a GPIO output register insufficient evidence that a fan is rotating?

2. The acceptance reply disappears, but a fresh sample shows the target. What does the teaching rule conclude, and what can reconciliation add?

3. Why does a device’s rejection plus an observed target remain uncertain under this policy?

4. Why must a persisted unresolved command block automatic writes after a host restart?

5. Which part of the physical action can a local database transaction not include atomically?

6. What is the difference between an absolute setpoint, a toggle and a pulse when replies are lost?

7. Why can a newly arrived sample still be too old to verify an outcome?

8. Why is a default-OFF unloaded bench fan not a safe-state rule for essential plant cooling?

9. What does a hash chain fail to detect if an attacker can rebuild all records and hashes?

10. What evidence would turn this general capability into a credible Lupine product hypothesis?

Answers / explain the limits as well as the result

If you miss a question, return to its diagram or run a small code mutation. The correction becomes part of your personal reference.

1 / Different variables

The GPIO register reports an electrical/software command. Rotation needs a qualified physical observation; current alone also cannot exclude a stall.

2 / Preserve uncertainty

No receipt means uncertain in this policy. Status by the original ID can later provide acceptance; a new qualified sample can support target-state confirmation.

3 / Conflicting evidence

The physical target may predate or result from another actor. A rejection does not explain that observation; do not turn it into an automatic retry.

4 / Restart preserves responsibility

The first action may have happened before the crash. Losing the in-memory state does not establish that nothing happened.

5 / Two separate systems

A local transaction cannot atomically commit the device’s physical action across a network. The uncertainty window must be managed.

6 / Different duplicate effects

An absolute target aims at a state. A toggle or pulse can create another effect on repetition; even absolute targets need ordering and expiry rules.

7 / Arrival is not acquisition

Queueing or caching can delay an old sample. Freshness needs a trustworthy acquisition time or a validated age bound.

8 / Safe state depends on consequence

Stopping an unloaded fixture is harmless by design. Stopping essential cooling can cause overheating; real plant needs its own reviewed response.

9 / No trusted external anchor

A rebuilt chain can be internally consistent. Detection needs an independently trusted reference or other established trust boundary.

10 / A specific unmet need

An identified user, documented failure, comparison with current handling, permitted evidence access and a measurable benefit under a credible test.

Free primary references for this prototype

Click a title to open the publisher’s reference. Board/device documentation must match the exact hardware and firmware used; these links do not replace a selected-part design review.

R01	Espressif / ESP32 datasheet and GPIO constraints
R02	Python / time: monotonic clock semantics
R03	Python / sqlite3: transactions and persistence
R04	SQLite / atomic commit and storage assumptions
R05	Python / hashlib: SHA-256 primitives
R06	Modbus / application protocol V1.1b3
R07	SunSpec / Modbus models and specifications
R08	NIST / SP 800-82 Rev. 3: OT security
R09	Modbus / security protocol and implementation guides
R10	OpenEMS / Edge architecture and control boundaries

Reading order

Start with R02 and your own board documentation. Use R03-R05 while building persistence/evidence. Read R06-R10 when you can already explain the bench contract and its failure cases. Public references are free; hardware and some vendor tools may not be.

Eight parts complete; your own evidence starts now

You now have a reference series for learning electricity, machines, heat, storage, control and Tasmania’s infrastructure. Expertise comes from using it to predict, build, measure, correct and explain.

Part	Reference topic	Use it when…
01	Electricity You Can Measure	Units, circuits and measurements need checking.
02	Components, Sensors and Microcontrollers	You need to select interfaces and understand a sensor.
03	AC Power, Machines, Networks and Protection	You need the grid and machine foundations.
04	Heat, Fluid Flow and Cooling	You need to trace heat to its final sink.
05	Batteries, Inverters and Power Electronics	Power conversion and storage limits matter.
06	Reliable Control, Communications and Evidence	An instruction or observation becomes uncertain.
07	Tasmania’s Grid and AI Data Centres	You need local context and public-data boundaries.
08	Prototype and Field Validation	You need a bounded experiment and a credible next gate.

Start with one thing you can finish and explain

Today: predict three cases, type the verdict demo and save your first learning note. Next: establish an independently observed indicator. Keep Lupine’s energy direction steady while letting each experiment refine what the company should eventually build.